

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 3 TIMBER AND BAMBOO: 3A TIMBER

BUREAU OF INDIAN STANDARDS

C O N T E N T S

FOREWORD	... 3
1 SCOPE	... 7
2 TERMINOLOGY	... 7
3 SYMBOLS	... 10
4 MATERIALS	... 11
5 PERMISSIBLE STRESSES	... 47
6 DESIGN CONSIDERATIONS	... 49
7 DESIGN OF COMMON STEEL WIRE NAIL JOINTS	... 57
8 DESIGN OF NAIL LAMINATED TIMBER BEAMS	... 65
9 DESIGN OF BOLTED CONSTRUCTION JOINTS	... 69
10 DESIGN OF TIMBER CONNECTOR JOINTS	... 73
11 GLUED LAMINATED CONSTRUCTION	... 78
12 LAMINATED VENEER LUMBER	... 82
13 GLUED FINGER JOINTS	... 83
14 STRUCTURAL USE OF PLYWOOD	... 85
15 TRUSSED RAFTER	... 87
16 STRUCTURAL SANDWICHES	... 90
17 LAMELLA ROOFING	... 93
18 NAIL AND SCREW HOLDING POWER OF TIMBER	... 101
19 PROTECTION AGAINST TERMITE ATTACK IN BUILDINGS	... 104
20 CROSS LAMINATED TIMBER (CLT)	... 105
21 MASS TIMBER	... 110
22 LIFE CYCLE ASSESSMENT (LCA) OF TIMBER FOR USE IN BUILDINGS	... 111
ANNEX A MECHANICAL PROPERTIES OF FOREIGN SPECIES TIMBER	... 114
LIST OF STANDARDS	... 115

National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part C/Subsection 3A) deals with the structural design aspect of timber structures. In this Subsection, the various species of Indian timber, classified into three groups depending on the structural properties influencing the design most, are included.

This Section was first published in 1970 which was subsequently revised in 1983, 2005 and 2016 as a part of National Building Code of India. In the 1983 version of this code, timber was covered under Section 3 of Part 6 under the title 'Wood', which did not cover bamboo. In the version of 2005, the scope of Section 3 was enlarged as Section 3 Timber and Bamboo, which was subdivided into Subsection 3A 'Timber' and Subsection 3B 'Bamboo'. This Subsection pertains to 3A 'Timber'.

In the first revision of 1983, provisions of this Subsection were updated and design of nailed laminated timber beams, were included and information on bolted construction joints was added. In the second revision of 2005, a number of terminologies related to timber for structural purpose were added, strength data of additional species of timber were included, requirements for structural timber and preferred cut sizes thereof were modified, requirements for glued laminated construction and finger joints and also for laminated veneer lumber were introduced. Brief details were included for structural sandwiches, glued laminated beams, lamella roofing, nail and screw holding power of timber, structural use of plywood and trussed rafter. Further, guidelines for protection against termite attack in buildings were also incorporated in the last revision of this Subsection. Also, keeping into consideration the dwindling forest resources, brief details on use of plantation timbers including certain fast growing species and suitable guidelines in terms of their seasoning, sawing, treatment, etc and also use of finger jointing and glued laminated timber was introduced in the last revision so as to effect judicious use of timber.

The third revision of 2016 addressed several significant changes, including modifications to definitions of various terms to align with prevailing engineering practices and the addition of new terminologies related to structural timber. The permissible lateral strength (in double shear) of mild steel common wire was updated to include various new timber species, with modified strength values for certain existing species based on the latest research. Guidance on increased permissible lateral strength for mild steel nails clenched across the grain, as opposed to nails with points cut flush, was incorporated. Design provisions for timber connector joints were further updated, along with data on block shear test results of glued timber joints based on indigenous studies. Additionally, information on the strength properties of glued finger joints from indigenous research was included for design guidance. A new figure illustrating possible orientations of planks in glue-laminated beams (glulam) was added, as well as a design outline for horizontally laminated beams. Updates were also made to the design provisions for trussed rafters, including data on lateral load-bearing strength of common wire nails in plywood-to-wood joints. Provisions for lamella roofing design were detailed and the composite nail/screw-holding power of specific Indian timbers was provided for design guidance.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

As a result of experience gained in implementation of 2016 version and feedback received as well as formulation of new standards in the field and revision/updation of some of the existing standards, a need to revise this Subsection was felt. This revision has, therefore, been brought out to take care of these aspects.

The significant changes incorporated in this revision include the following:

- a) Cross-laminated timber components including material specifications, dimensions, and fire resistance requirements;
- b) Mass timber construction and its components given in [21](#);
- c) Fire safety enhancements and fire-resistant assembly methods for mass timber, ensuring compliance with structural and fire safety standards;
- d) Focus on sustainability with introduction of life cycle assessment (LCA) to evaluate the environmental impact of timber from production to disposal; and

- e) Introduction of mechanical properties of foreign timber species, to support informed design, specification and safe application.

Reference may be made to Part F 'Fire and Life Safety' of this NBCS for fire protection of the timber buildings.

The information contained in this Subsection is largely based on the following Indian Standards:

IS 399 : 1963	Classification of commercial timbers and their zonal distribution (<i>revised</i>)
IS 883 : 2016	Design of structural timber in building — Code of practice for (<i>fifth revision</i>)
IS 1150 : 2000	Trade names and abbreviated symbols for timber species (<i>third revision</i>)
IS 2366 : 1983	Code of practice for nail-jointed timber construction (<i>first revision</i>)
IS 4891 : 1988	Specification for preferred cut sizes of structural timber (<i>first revision</i>)
IS 4983 : 1968	Code of practice for design and construction of nailed laminated timber beams
IS 11096 : 1984	Code of practice for design and construction of bolt jointed timber construction
IS 14616 : 1999	Laminated veneer lumber — Specification

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Subsection is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Subsection.

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Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to ‘**accepted standard**’ in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Subsection, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section.

For example:

a) Good practices [C-3A(1)] refers to the Indian Standard(s) give at serial number (1) of the list of standards given at the end of this Part C/Subsection 3A, that is, IS 707 : 2011 ‘Timber technology and utilization of wood, bamboo and cane — Glossary of terms (*third revision*)’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 3 TIMBER AND BAMBOO: 3A TIMBER

1 SCOPE

1.1 This NBCS (Part C/Subsection 3A) covers the general principles involved in the design of structural timber in buildings, including elements of structures connected by fasteners/fastening techniques.

1.2 This shall not be interpreted to prevent the use of material or methods of design or construction not specifically mentioned herein; and the methods of design may be based on analytical and engineering principles, or reliable test data, or both, that demonstrate the safety and serviceability of the resulting structure. The classification of timber into strength groups should not be interpreted as preventing the use of design data desired for a particular timber or grade of timber on the basis of reliable tests.

2 TERMINOLOGY

For the purpose of this Subsection, the following definitions and those given in the accepted standard [C-3A(1)], shall apply.

2.1 Structural Purpose Definitions

2.1.1 *Beam, Built-up Laminated* — A beam made by joining layers of timber together with mechanical fastenings, so that the grain of all layers is essentially parallel.

2.1.2 *Beam, Glued-Laminated* — A beam made by bonding layers of veneers or timber with an adhesive, so that grain of all laminations is essentially parallel.

2.1.3 *Bressummer* — A beam, across broad opening, sustaining super structure.

2.1.4 *Diaphragm, Structural* — A structural element of large extent placed in a building as a wall, or roof, and made use of to resist horizontal forces such as wind or earthquakes-acting parallel to its own plane.

2.1.5 *Duration of Load* — Period during which a member or a complete structure is stressed as a consequence of the loads applied.

NOTE — Timber is stronger under loads of brief duration than under permanently applied loads.

2.1.6 *Edge Distance* — The distance measured perpendicular to grain from the centre of the connector to the edge of the member.

2.1.7 *End Distance* — The distance measured parallel to grain of the member from the centre of the connector to the closest end of timber.

2.1.8 *Finger Joint* — End joint formed by machining a number of similar, tapered, symmetrical fingers in the ends of timber member, which are then bonded together.

2.1.9 *Finger Jointed Timber* — Timber containing adhesively bonded finger joints spaced at manufacturer specified minimum spacing.

2.1.10 *Fundamental or Ultimate Stress* — The stress which is determined on small clear specimen of timber, in accordance with accepted standards [C-3A(2)]; and does not take into account the effect of naturally occurring characteristics and other factors.

2.1.11 *Glue* — A substance capable of holding materials together by surface attachment.

2.1.12 *Glued Laminated (Glulam)* — A wood structural element comprising of sawn wood lamination bonded together with adhesives and in which the grains of all laminations are parallel longitudinally.

2.1.12.1 Glued laminated, horizontally — Glued laminated timber where gluing planes are perpendicular to the largest dimension of the transverse section, that is, in cross-wise direction.

2.1.12.2 Glued laminated, vertically — Glued laminated timber where gluing planes are perpendicular to the smallest dimension of the transverse section.

2.1.13 Inside Location — Position in buildings in which timber remains continuously dry or protected from weather.

2.1.14 Isoptera — An order of social insects consisting of the termites.

2.1.15 Lamella Roof — A roof or frame, made up of relatively short members, called lamellas, forming a network of framing timbers. This network forms a barrel-like structure of mutually braced and stiffened units, arching over the building between supports, and together with the sheathing forms a diaphragm for resistance of vertical and lateral loads.

2.1.16 Laminations — Thin elements of wood of appreciable width and length which, when superimposed on each other and fastened together with their grains essentially parallel make up cross section of considerable depth or width.

2.1.17 Laminated Veneer Lumber — A structural composite made by laminating veneers, 1.5 mm to 4.2 mm thick, with suitable adhesive and with the grain of veneers in successive layers aligned along the longitudinal (length) dimension of the composite.

2.1.18 Latewood — The portion of the wood that is formed in the later part of growth season and consists of thicker walled cells.

2.1.19 Loaded Edge Distance — The distance measured from the centre to the edge towards which the load induced by the connector acts, and the unloaded edge distance is the one opposite to the loaded edge.

2.1.20 Location — A term generally referred to as exact place where a timber is used in building.

2.1.21 Mechanically Laminated — A wood structural element comprising of laminations which are not glued but which are held together with mechanical fastenings such as nails or bolts and in which all laminations have the grains approximately parallel longitudinally.

2.1.22 Outside Location — Position in buildings in which timbers are occasionally subjected to wetting and drying as in the case of open sheds and outdoor exposed structures.

2.1.23 Permissible Stress — Stress obtained by applying factor of safety to the ultimate stress.

2.1.24 Sandwich, Structural — A layered construction comprising a combination or relatively high-strength facing material intimately bonded to and acting integrally with a low density core material.

2.1.25 Spaced Column — Two column sections adequately connected together by glue, bolts, screws or otherwise.

2.1.26 Stressed Skin — A form of construction in which the outer skin, in addition to its normal functioning of providing a surface covering, acts integrally with the framing members contributing to the strength of the unit as a whole.

2.1.27 Structure, Permanent — Structural units in timber which are constructed for a long duration and wherein adequate protection and design measures have initially been incorporated to render the structure serviceable for the required life.

2.1.28 Structure, Temporary — Structures which are erected for a short period, such as hutments at project sites, for rehabilitation, temporary defence constructions, exhibition structures, etc.

2.1.29 Structural Element — The component timber members and joints which make up a resulting structural assembly.

2.1.30 Structural Grades — Grades defining the maximum size of strength reducing natural characteristics (knots, sloping grain, etc) deemed permissible in any piece of structural timber within designated structural grade classification.

2.1.31 Structural Timber — Timber in which strength is related to the anticipated in-service use as a controlling factor in grading and selection and/or stiffness.

2.1.32 Stud — One of a series of slender wood structural members placed vertically as supporting members in a wall partition, or similar structural unit.

2.1.33 Termite — An insect of the order *Isoptera* which may burrow in the wood or wood products of a building for food or shelter.

2.1.34 Wet Location — Position in buildings in which timbers are almost continuously damp or wet in contact with the earth or water, such as piles and timber foundations.

2.2 Definitions of Defects in Timber

2.2.1 Check — A separation of fibres extending along the grain which is confined to one face of a piece of wood.

2.2.2 Compression Wood — Abnormal wood which is formed on the lower sides of branches and inclined stems of coniferous trees. It is darker and harder than normal wood but relatively low in strength for its weight. It can be usually identified by wide eccentric growth rings with abnormally high proportion of growth latewood.

2.2.3 Dead Knot — A knot in which the layers of annual growth are not completely intergrown with those of the adjacent wood. It is surrounded by pitch or bark. The encasement may be partial or complete.

2.2.4 Decay or Rot — Disintegration of wood tissue caused by fungi (wood destroying) or other micro-organisms.

2.2.5 Decayed Knot — A knot softer than the surrounding wood and containing decay.

2.2.6 Diameter of Knot — The maximum distance between the two points farthest apart on the periphery of a round knot, on the face on which it becomes visible. In the case of a spike or a splay knot, the maximum width of the knot visible on the face on which it appears shall be taken as its diameter.

2.2.7 Discolouration — A change from the normal colour of the wood which does not impair the strength of the wood.

2.2.8 Knot — A branch base or limb embedded in the tree or timber by natural growth.

2.2.9 Knot Hole — A hole left as a result of the removal of a knot.

2.2.10 Live Knot — A knot free from decay and other defects, in which the fibres are firmly intergrown with those of the surrounding wood. (*Synonyms*: 'Intergrown knot'; *cf.* 'Dead knot')

2.2.11 Loose Grain (Loosened Grain) — A defect on a flat sawn surface caused by the separation or raising of wood fibres along the growth rings; *cf.* 'Raised grain'.

2.2.12 Loose Knot — A knot that is not held firmly in place by growth or position, and that cannot be relied upon to remain in place; *cf.* 'Tight knot'.

2.2.13 Mould — A soft vegetative growth that forms on wood in damp, stagnant atmosphere. It is the least harmful type of fungus, usually confined to the surface of the wood.

2.2.14 Pitch Pocket — Accumulation of resin between growth rings of coniferous wood as seen on the cross-section.

2.2.15 Sap Stain — Discolouration of the sapwood mainly due to fungi.

2.2.16 Sapwood — The outer layer of log, which in the growing tree contain living cells and food material. The sapwood is usually lighter in colour and is readily attacked by insects and fungi.

2.2.17 Shake — A partial or complete separation between adjoining layers of tissues as seen in end surfaces.

2.2.18 Slope of Grain — The inclination of the fibres to the longitudinal axis of the member.

2.2.19 Sound Knot — A tight knot free from decay, which is solid across its face and at least as hard as the surrounding wood.

2.2.20 Split — A crack extending from one face of a piece of wood to another and runs along the grain of the piece.

2.2.21 Tight Knot — A knot so held by growth or position as to remain firm in position in the piece of wood; cf. 'Loose knot'.

2.2.22 Wane — The original rounded surface of a tree remaining on a piece of converted timber.

2.2.23 Warp — A deviation in sawn timber from a true plane surface or distortion due to stresses causing departure from a true plane.

2.2.24 Worm Holes — Cavities caused by worms.

3 SYMBOLS

For the purpose of this Subsection, the following letter symbols shall have the meaning indicated against each:

a	=	Projected area of bolt in main member ($t' \times d_3$), mm ²
B	=	Width of the beam, mm
C	=	Concentrated load, N
D	=	Depth of beam, mm
D_1	=	Depth of beam at the notch, mm
D_2	=	Depth of notch, mm
d	=	Dimension of least side of column, mm
d_1	=	Least overall width of box column, mm
d_2	=	Least overall dimension of core in box column, mm
d_3	=	Diameter of bolt, mm
d_f	=	Bolt-diameter factor
E	=	Modulus of elasticity in bending, N/mm ²
e	=	Length of the notch measured along the beam span from the inner edge of the support to the farthest edge of the notch, mm
F	=	Load acting on a bolt at an angle to grain, N
f_{ab}	=	Calculated bending stress in extreme fibre, N/mm ²
f_{ac}	=	Calculated average axial compressive stress, N/mm ²
f_{at}	=	Calculated axial tensile stress, N/mm ²
f_b	=	Permissible bending stress on the extreme fibre, N/mm ²
f_c	=	Permissible stress in axial compression, N/mm ²
f_{cn}	=	Permissible stress in compression normal (perpendicular) to grain, N/mm ²
f_{cp}	=	Permissible stress in compression parallel to grain, N/mm ²
$f_{c\theta}$	=	Permissible compressive stress in the direction of the line of action of the load, N/mm ²
f_t	=	Permissible stress in tension parallel to grain, N/mm ² [extreme fibre stress in bending-tension (f_b) is safe practical estimate of tensile strength due to mode of failure in beams]
H	=	Horizontal shear stress, N/mm ²
I	=	Moment of inertia of a section, mm ⁴
K	=	Coefficient in deflection depending upon type and criticality of loading on beam
K_1	=	Modification factor for change in slope of grain
K_2	=	Modification factor for change in duration of loadings

$K_3, K_4,$ K_5, K_6	=	Form factors
K_7	=	Modification factor for bearing stress
K_8	=	Constant equal to $0.584 \sqrt{\frac{E}{f_{cp}}}$
K_9	=	Constant equal to $\frac{\pi}{2} \sqrt{\frac{UE}{5q f_{cp}}}$
K_{10}	=	Constant equal to $0.584 \sqrt{\frac{2.5E}{f_{cp}}}$
l or L	=	Span of a beam or truss, mm
M	=	Maximum bending moment in beam, N-mm
N	=	Total number of bolts in the joint
n	=	Shank diameter of the nail, mm
P	=	Load on bolt parallel to grain, N
p_1	=	Ratio of the thickness of the compression flange to the depth of the beam
Q	=	Static moment of area above or below the neutral axis about neutral axis, mm ³
q	=	Constant for particular thickness of plank
q_1	=	Ratio of the total thickness of web or webs to the overall width of the beam
R	=	Load on bolt perpendicular (normal) to grain, N
S	=	Unsupported overall length of column, mm
t	=	Nominal thickness of planks used in forming box type column, mm
t'	=	Thickness of main member, mm
U	=	Constant for a particular thickness of the plank
V	=	Vertical end reaction or shear at a section, N
W	=	Total uniform load, N
x	=	Distance from reaction to load, mm
y	=	Factor determining the value of form factor K_4
δ	=	Deflection at middle of beam, mm
θ	=	Angle of load to grain direction
Z	=	Section modulus of beam, mm ³
λ_1	=	Percentage factor for t'/d_3 ratio, parallel to grain
λ_2	=	Percentage factor for t'/d_3 ratio, perpendicular to grain

4 MATERIALS

4.1 Species of Timber

The species of timber recommended for structural purposes are given in [Table 1](#) along with their permissible stresses and properties. For information on mechanical properties of certain foreign timber species, reference may be made to [Annex A](#).

Table 1 Safe Permissible Stresses for the Species of Timber

[Clauses [4.1](#), [4.2](#), [4.7.2\(b\)](#), [5.4.1](#), [5.4.2](#), [5.4.2.2](#), [6.5.8.3.1\(b\)](#), [10.2.3](#) and [15.3](#)]

SI No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
i)	GROUP A:																		
	a) <i>Acacia catectou</i>	Khair (KHA)	Uttar Pradesh	1 009	13.44	20.1	16.8	13.4	1.6	2.2	13.8	12.3	10.1	7.7	6.0	4.9	I	–	A
	b) <i>Acacia chundra</i>	Red kutch	Madhya Pradesh	1 086	16.79	26.5	22.0	17.6	2.2	3.2	17.9	15.9	13.0	10.9	8.4	6.9	–	–	A
	c) <i>Albizia odoratissima</i>	Kala siris (KSI)	Chennai	737	13.54	18.7	15.6	12.5	1.5	2.2	13.3	11.8	9.6	7.3	5.6	4.6	I	e	B
	d) <i>Bruguiera spp.</i>	Bruguiera (BSV) (Mangrove)	Andaman	897	17.68	21.9	18.3	14.6	1.2	1.7	14.3	12.7	10.4	5.5	4.3	3.5	III	–	–
	e) <i>Grewia tiliifolia</i>	Dhaman (DHA)	Chennai	788	14.82	18.3	15.2	12.2	1.3	1.9	12.0	10.7	8.7	6.0	4.7	3.8	III	d	B
	f) <i>Hopea utilis (Balano carpus utilis)</i>	Karung	Chennai	987	16.91	25.1	20.9	16.7	1.5	2.2	16.4	14.6	11.9	9.3	7.3	5.9	–	–	–

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	g) <i>Hopea glabra</i>	Hopea (HOP)	Chennai	1 081	14.79	21.3	17.8	14.2	1.5	2.2	14.5	12.9	10.6	9.9	7.7	6.3	I	–	A
	h) <i>Hopea parviflora</i>	Hopea (HOP)	Chennai	923	13.03	18.6	15.5	12.4	1.3	1.8	13.2	11.8	9.6	9.2	7.3	6.0	I	e	A
	j) <i>Manilota polyandra</i> (Syn. <i>Cynometra polyandra</i>)	Ping (PIG)	Assam	903	13.20	19.1	15.9	12.7	1.3	1.8	1.2	10.4	8.5	5.7	4.4	3.6	III	b	A
	k) <i>Mesua ferrea</i>	Mesua (MES)	Assam	965	16.30	23.3	19.4	15.5	1.2	1.8	15.5	13.8	11.3	5.9	4.6	3.7	I	–	A
	m) <i>Mimusops littoralis</i>	Bullet-wood (BUL)	Southern Andaman	1 103	17.39	22.7	18.9	15.1	1.5	2.1	14.2	12.7	10.4	11.3	8.8	7.2	I	–	A
	n) <i>Pesciloneuron indicum</i>	Ballagi (BAL)	Chennai	1 139	16.29	22.4	18.7	15.0	1.5	2.2	14.7	13.1	10.7	8.7	6.8	5.5	I	e	A
	p) <i>Pterocarpus Scantalinus</i>	Red sanders (MA)	Chennai	1 121	12.73	25.0	20.9	16.7	1.7	2.5	18.1	16.1	13.2	11.8	9.2	7.5	–	–	A
	q) <i>Sageraea elliptica</i>	Chooi (COC)	Andaman	869	15.06	21.5	17.9	14.3	1.1	1.5	12.5	11.1	9.1	5.3	4.1	3.4	–	–	A

Table 1 (Continued)

SI No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
						Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain					
	Botanical Name	Trade Name				Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location	Durability Class ¹⁾	Treat-ability Grade ²⁾	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	r) <i>Stereospermun celonoides</i>	Padri (PAD)	Chennai	731	12.94	19.0	15.8	12.7	1.1	1.6	11.9	10.6	8.7	4.0	3.1	2.6	III	–	B
	s) <i>Vitex altissima</i>	Milla (MIL)	Maharashtra	937	13.01	18.2	15.2	12.1	1.2	1.7	12.6	11.2	9.2	9.5	7.4	6.1	I	e	A
ii)	GROUP B:																		
	a) <i>Albizzia lebbeck</i>	Kokko (KOK)	Andaman	642	11.17	13.4	11.2	9.0	1.1	1.5	9.0	8.0	6.5	4.4	3.4	2.8	I	e	B
	b) <i>Anogeissus latifolia</i>	Dhaura, Axle wood (AXL) (Bakli)	Uttar Pradesh	892	10.55	16.1	13.4	10.7	1.1	1.6	9.1	8.1	6.6	4.7	3.7	3.0	I	e	A
	c) <i>Artocarpus hirsulus</i>	Aini (AIH)	Chennai	600	10.45	15.0	12.5	10.0	0.7	1.1	10.4	9.2	7.5	3.3	2.6	2.1	I	–	B
	d) <i>Acacia nilotica</i>	Babul (BAB)	Uttar Pradesh	797	-	-	12.9	10.3	1.4	2.1	8.9	7.9	6.4	5.2	4.0	3.3	I	b	B
	e) <i>Acacia ferruginea</i>	Safed khair	Maharashtra	993	12.28	23.0	19.2	15.3	1.7	2.4	13.9	12.4	10.1	9.9	7.7	6.3	–	–	–

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
						Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain					
	Inside Location	Outside Location				Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location	Durability Class ¹⁾	Treat-ability Grade ²⁾			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	f) <i>Acrocarpus fraxinifolius</i>	Mundani (MUN)	Chennai	690	12.59	16.1	13.4	10.8	1.2	1.8	10.5	9.4	7.7	4.6	3.6	2.9	III	c	B
	g) <i>Aglaia odulis</i>	Aglaia (AGL)	Assam	815	12.56	18.2	15.2	12.1	1.4	2.0	10.1	8.9	7.3	4.4	3.4	2.8	–	–	A
	h) <i>Anogeissus acuminota</i>	Yon	Orissa	844	11.67	17.6	14.7	11.7	1.3	1.8	10.8	9.6	7.9	5.1	4.0	3.3	–	–	A
	j) <i>Atalanlia monophylla</i>	Jungli-nimbu (JHI)	Orissa	897	10.31	16.7	13.9	11.1	1.5	2.1	11.3	10.0	8.2	6.3	4.9	4.0	–	–	–
	k) <i>Altingia excelsa</i>	Jutili (JUT)	Assam	795	11.37	17.1	14.3	11.4	1.2	1.8	11.0	9.8	8.0	6.8	5.3	4.4	II	e	A
	m) <i>Amoora spp.</i>	Amari (AMA)	West Bengal	625	1.05	13.4	1.1	9.2	0.9	1.3	8.4	7.4	6.0	3.7	2.9	2.4	II	d	B
	n) <i>Bucklandia populnea</i> (Syn. <i>Exbucklandia populnea</i>)	Pipli (PIP)	West Bengal	672	9.89	12.8	10.7	8.6	1.1	1.5	7.9	7.0	5.7	3.5	2.7	2.2	III	e	C
	p) <i>Cassia fistula</i>	Amaltas (AMT)	Uttar Pradesh	865	11.80	19.2	16.0	12.8	1.4	2.0	12.3	10.9	8.9	7.2	5.6	4.6	I	–	A

Table 1 (Continued)

SI No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	q) <i>Carallia lucida</i>	Maniawaga	Assam	748	12.60	18.4	15.3	12.3	1.2	1.7	11.4	10.1	8.3	5.9	4.6	3.8	–	–	–
	r) <i>Canarium strictum</i>	Dhup	Chennai	655	11.86	13.3	11.1	8.9	0.9	1.2	8.1	7.2	5.9	2.8	2.2	1.8	III	–	C
	s) <i>Cassia sienea</i>	Kasod	Madhya Pradesh	820	10.50	15.4	12.8	10.9	1.0	1.4	10.8	9.6	7.9	5.5	4.3	3.5	–	–	-
	t) <i>Casuerina equisetifolia</i>	Casuarina (CAS)	Orissa	769	11.44	14.6	12.2	9.8	1.3	1.8	8.2	7.3	5.9	4.0	3.1	2.5	III	e	A
	u) <i>Celophyllum temculosum</i>	Poon (POO)	Maharashtra	657	9.77	13.4	11.2	9.0	0.8	1.1	8.6	7.7	6.3	2.8	2.2	1.8	II	–	B
	v) <i>Chloroxylon swielenia</i>	Satin wood (CFI)	Madhya Pradesh	865	11.69	18.2	15.1	12.1	1.4	2.0	10.9	9.7	8.0	6.3	4.9	4.0	III	–	A
	w) <i>Cullenia resayoana</i> (Syn. <i>C. excelsa</i>)	Karani (KAP)	Chennai	625	12.43	14.7	12.3	9.8	0.6	0.9	9.0	8.0	6.6	2.7	2.1	1.7	III	b	C
	y) <i>Diploknema butyracea</i> (Syn. <i>Bassia butyrance</i>)	Hill mahua (HMA)	Southern Andaman	780	10.64	15.3	12.8	10.2	1.0	1.5	9.9	8.8	7.2	6.6	5.2	4.2	–	–	–
	z) <i>Dysoxylum malebaricum</i>	White ceda (WCE)	Chennai	745	10.92	13.2	11.0	8.8	1.0	1.4	8.0	7.1	5.8	3.1	2.4	1.9	I	–	B

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	aa) <i>Dipterocarpus grandiflorus</i>	Gurjan (GUR)	Northern Andaman	758	11.71	12.5	10.5	8.4	0.8	1.1	7.9	7.1	5.8	2.7	2.1	1.7	I	—	B
	ab) <i>Dipterocarpus macrocarpus</i>	Hollong (HOL)	Assam	726	13.34	14.5	12.0	9.6	0.8	1.1	8.8	7.9	6.4	3.5	2.7	2.2	III	a	B
	ac) <i>Dichopsis polyantha</i> (Syn. <i>Palaquium polyanthum</i>)	Tali (TAL)	Assam	734	11.24	14.9	12.4	10.0	1.1	1.6	9.9	8.8	7.2	4.7	3.7	3.0	—	—	B
	ad) <i>Dichopsis elliptica</i> (Syn. <i>Palaquium ellipticum</i>)	Pali (PAL)	Chennai	606	11.86	13.9	11.6	9.3	0.7	1.0	8.5	7.5	6.2	2.9	2.2	1.8	I	e	B
	ae) <i>Diospyros micropylla</i>	Ebony (EBO)	Maharashtra	776	12.15	14.2	11.9	9.5	0.9	1.3	8.3	7.3	6.0	3.3	2.6	2.1	—	—	A
	af) <i>Diospyros pyrrhocarpus</i>	Ebony (EBO)	Northern Andaman	843	9.93	13.5	11.2	9.0	1.0	1.4	7.9	7.0	5.7	4.0	3.1	2.5	III	—	A
	ag) <i>Dipterocarpus bourdillonii</i>	Gurjan (GUR)	Kerala	699	12.71	13.6	11.3	9.0	0.7	1.0	7.8	6.9	5.7	2.5	1.9	1.6	—	—	B

Table 1 (Continued)

SI No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	ah) <i>Eucalyptus globulus</i>	Eucalyptus (Blue gum) (BLN)	Chennai	912	14.83	15.9	13.2	10.6	10.3	1.5	9.0	8.0	6.5	3.4	2.6	2.1	I	e	A
	aj) <i>Eucalyptus ougenioides</i>	Eucalyptus	Chennai	853	11.47	16.4	13.6	10.9	1.2	1.7	11.3	10.0	8.2	7.6	5.9	4.8	–	–	–
	ak) <i>Eugenia gardnery</i>	Jaman (JAM)	Chennai	952	11.94	14.8	12.3	9.8	1.1	1.6	9.2	8.2	6.7	5.8	4.5	3.7	III	d	–
	am) <i>Eugenia jambolana</i>	Jaman (JAM)	Uttar Pradesh	778	10.94	16.0	13.3	10.6	1.2	1.7	9.7	8.6	7.1	4.7	3.7	3.0	–	–	–
	an) <i>Gluta travancorice</i>	Gluta (GLU)	Chennai	726	12.73	13.5	11.3	9.0	0.9	1.3	9.0	8.0	6.6	4.0	3.1	2.5	I	–	A
	ap) <i>Grewia veslita</i>	Dhaman (DHA)	West Bengal	758	12.00	15.4	12.6	10.3	1.4	2.0	9.1	8.1	6.6	4.1	3.2	2.6	III	d	B
	aq) <i>Heritiera spp.</i>	Sundri (SUN)	Assam	872	13.37	17.9	14.9	11.9	1.3	1.8	11.0	9.8	8.0	6.5	5.0	4.1	I	–	A
	ar) <i>Kingiodendron pinnatum</i> (Syn. <i>Hardwickia pinnata</i>)	Piney (PIN)	Chennai	617	10.62	13.2	11.0	8.8	0.4	1.3	8.2	7.3	6.0	2.9	2.2	1.8	–	–	B

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	as) <i>Kayea floribund</i>	Karal	Assam	813	10.88	16.8	14.0	1.1	1.1	1.6	10.1	9.0	7.3	4.4	3.4	2.8	III	–	–
	at) <i>Lagerstromia lanceolata</i>	Benteak (BEN)	Chennai	617	10.76	12.7	10.6	8.5	0.8	1.2	8.2	7.3	5.9	3.4	2.6	2.2	I	e	B
	au) <i>Lagerstromia parviflora</i>	Lendi (LEN)	Uttar Pradesh	734	10.97	14.3	11.9	9.5	1.1	1.6	8.7	7.7	6.3	3.7	2.9	2.4	I	e	A
	av) <i>Mimusops elengi</i>	Bakul (BKL)	Chennai	885	12.39	17.3	14.4	11.5	1.3	1.8	11.0	9.8	8.0	5.6	4.3	3.6	I	–	A
	aw) <i>Machilus macrantha</i>	Machilus (MAC)	West Bengal	692	10.00	12.4	10.3	8.3	1.0	1.5	8.2	7.3	6.0	3.5	2.7	2.2	III	e	B/C
	ay) <i>Miliuse tyomentosa</i> (Syn. <i>Saccopapetalum tomentosum</i>)	Hoom (HOO)	Maharashtra	745	11.06	14.8	12.3	9.9	0.9	1.3	9.7	8.6	7.0	3.5	2.7	2.2	III	–	B
	az) <i>Pometia pinnata</i>	–	Andaman	788	12.90	14.3	11.9	9.5	1.1	1.6	9.1	8.0	6.6	4.0	3.1	2.5	–	–	–
	ba) <i>Pterocarpus dolbergioides</i>	Padauk (PAD)	Northern Andaman	721	11.24	17.1	14.3	11.4	1.0	1.5	12.0	10.7	8.7	5.5	4.3	3.5	I	c	B
	bb) <i>Mesua assamica</i>	Kayea	Assam	842	12.83	17.4	14.5	11.6	1.0	1.4	11.7	10.4	8.5	5.3	4.1	3.3	II	e	–

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	bc) <i>Pterocarpus marsupium</i>	Bijasal (BIJ)	Maharashtra	803	10.25	14.9	12.4	9.9	0.9	1.3	9.1	8.1	6.6	4.1	3.2	2.6	I	e	B
	bd) <i>Fraxlnus macrantha</i>	Ash (ASH)	Uttar Pradesh	712	10.69	15.0	12.5	10.0	1.2	1.7	8.5	7.6	6.2	4.3	3.3	2.7	III	–	B
	be) <i>Fraxlnus execlsior</i>	Ash (ASH)	Punjab	719	10.41	14.8	12.3	9.8	1.2	1.7	8.1	7.2	5.8	3.3	2.6	2.1	III	–	B
	bf) <i>Planchonia valida</i> (Syn. <i>P. andamanica</i>)	Red bombwe (RBO)	Andaman	913	13.10	16.1	13.4	10.7	1.0	1.4	10.8	9.6	7.9	4.9	3.8	3.1	III	–	–
	bg) <i>Quercus lamellosa</i>	Oak	West Bengal	87	12.44	14.5	12.1	9.7	1.2	1.7	8.7	7.8	6.4	3.8	2.9	2.4	II	c	A
	bh) <i>Quercus griffithii</i>	Oak	Meghalaya	974	10.06	13.1	10.9	8.8	1.1	1.6	8.0	7.1	5.8	4.6	3.6	2.9	–	–	A
	bj) <i>Quercus incane</i>	Oak	Punjab	1 008	10.82	15.8	13.1	10.5	1.2	1.8	8.7	7.8	6.3	5.0	3.9	3.2	–	–	A
	bk) <i>Quercus lineate</i>	Oak	West Bengal	874	12.63	15.2	12.7	10.1	1.2	1.7	9.6	8.6	7.0	5.3	4.1	3.4	II	c	A

Table 1 (Continued)

SI No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
						Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
	Inside Location	Outside Location				Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location					
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	bm) <i>Quercus semecarpifolia</i>	–	Punjab	834	11.58	15.8	13.1	10.5	1.3	1.8	8.3	7.3	6.0	3.8	2.9	2.4	–	–	A
	bn) <i>Shorea robusta</i> ⁴⁾	Sal (SAL)	Madhya Pradesh	805	12.67	16.9	14.0	11.2	0.9	1.3	10.6	9.4	7.7	4.6	3.5	2.9	I	e	A
	bp) <i>Soymida fabrifuga</i>	Rohini (ROH)	Chennai	1 116	12.22	21.5	17.9	14.4	1.6	2.3	15.0	13.3	10.9	12.9	10.0	8.2	I	–	A
	bq) <i>Shorea talura</i>	–	Maharashtra	721	12.20	16.8	14.0	11.2	1.1	1.6	12.6	11.2	9.2	6.8	5.3	4.3	–	–	–
	br) <i>Plerygota alata</i> (Syn. <i>Sterculia alata</i>)	Narikel (NAR)	Assam	593	10.95	13.4	11.8	8.9	0.8	1.2	8.2	7.3	6.0	2.7	2.1	1.7	III	–	C
	bs) <i>Syzygium cumini</i>	Jaman (JAM)	Assam	841	10.55	14.8	12.4	9.9	1.1	1.6	9.0	8.0	6.5	6.9	5.4	4.4	II	e	A
	bt) <i>Terminalia bellirica</i>	Bahera (BAH)	Uttar Pradesh	729	10.19	13.6	11.3	9.0	1.0	1.4	8.4	7.5	6.1	3.7	2.8	2.3	III	b	B
	bu) <i>Terminalia chebula</i>	Myrobalan (MYR)	–	918	12.37	17.1	14.2	11.4	1.1	1.6	1.2	10.4	8.5	6.7	5.2	4.3	II	c	A

Table 1 (Continued)

SI No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m³	Modulus of Elasticity (All Grades and All Locations) × 10³ N/mm²	Permissible Stress for Grade I N/mm²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	bv) <i>Terminalia citrina</i>	–	Assam	755	11.89	17.1	14.3	11.4	1.1	1.6	10.8	9.6	7.9	5.0	3.9	3.2	–	–	–
	bw) <i>Terminalia manii</i>	Black-chuglam (BCH)	Southern Andaman	822	12.66	16.8	14.0	11.2	1.1	1.6	10.3	9.2	7.5	5.1	4.0	3.2	II	a	B
	by) <i>Tectona grandis</i>	Teak (TEA)	Uttar Pradesh	660	9.97	15.5	12.9	10.3	1.2	1.6	9.4	8.3	6.8	4.5	3.5	2.8	I	e	B
	bz) <i>Terminalia paniculate</i>	Kindal (KIN)	Maharashtra	765	10.57	13.1	10.9	8.7	0.9	1.3	8.6	7.7	6.3	3.6	2.8	2.3	I	c	A
	ca) <i>Alreminalia alata</i>	Laurel (LAU), Sain	Chennai	906	10.54	15.1	12.5	10.0	1.1	1.6	9.4	8.4	6.8	6.2	4.8	4.0	I	b	A
	cb) <i>Terminalia bilata</i>	White-chuglam (WCH)	Southern Andaman	690	12.38	15.5	13.0	10.4	0.9	1.2	9.8	8.7	7.1	3.6	2.8	2.3	III	e	B
	cc) <i>Thespesia populnea</i>	Bhendi (BHE)	Maharashtra	766	10.36	18.9	15.8	12.6	1.3	1.9	11.3	10.0	8.2	4.4	3.4	2.8	–	–	B
	cd) <i>Xylia xylocarpa</i>	Irul (IRU)	Maharashtra	839	11.63	16.2	13.5	10.8	1.3	1.8	10.9	9.7	7.9	7.8	6.0	4.9	I	e	A

Table 1 (Continued)

SI No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	ce) <i>Zanthoxylum budranga</i>	Mullilam (MUL)	West Bengal	587	10.65	14.7	12.2	9.8	0.9	1.2	9.5	8.4	6.9	3.4	2.6	2.1	I	e	B
	cf) <i>Adina oligocephala</i>	-	Arunachal	715	11.17	15.2	12.7	10.1	1.2	1.7	10.3	9.2	7.5	4.0	3.1	2.4	–	–	–
	cg) <i>Castanopsis indica</i>	Chestnut	Meghalaya	688	12.54	14.8	12.3	9.9	1.0	1.4	9.8	8.7	7.1	3.4	2.7	2.2	–	–	B
	ch) <i>Eucalyptus citriodara</i>	Eucalyptus	Nilgiri	831	12.12	17.3	14.4	11.5	1.4	2.0	11.0	9.8	8.0	4.2	3.3	2.7	–	–	–
	cj) <i>Eucalyptus citriodata</i>	Eucalyptus	Ooty	725	9.35	15.4	12.9	10.3	1.0	1.4	8.6	7.6	6.3	3.0	2.4	2.0	–	–	–
	ck) <i>Eucalytus tereticornis</i>	Eucalyptus	Chennai	777	11.05	16.7	13.9	11.1	1.0	1.4	9.7	8.6	7.1	3.4	2.6	2.2	III	e	–
iii)	GROUP C:																		
	a) <i>Tbizia procera</i>	White siris	Uttar Pradesh	643	9.02	13.4	11.2	8.9	1.0	1.4	8.5	7.6	6.2	4.3	3.3	2.7	I	c	B
	b) <i>Artocarpus lakocha</i>	Lakooch (LAK)	Uttar Pradesh	647	6.14	10.0	8.3	6.7	1.0	1.4	5.3	4.7	3.8	2.8	2.2	1.8	I	–	B

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾	
						Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain						
	Inside Location	Outside Location				Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location	Durability Class ¹⁾	Treat-ability Grade ²⁾				
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)			(16)	(17)	(18)	(19)
	c)	<i>Artocarpus hetarophyllus</i> (Syn. <i>A. Integrifolia</i>)	Jack, kathal (KAT)	Chennai	617	9.46	13.9	11.6	9.2	1.0	1.5	9.3	8.3	6.8	4.5	3.5	2.9	I	d	B
	d)	<i>Aphanamixis polystachya</i> (Syn. <i>Amoora rehituka</i>)	Pitraj (PIT)	West Bengal	668	8.98	12.3	10.2	8.2	1.1	1.5	8.0	7.1	5.8	4.0	3.1	2.6	I	–	B
	e)	<i>Adina cordifolia</i> ⁴⁾	Haldu (HAL)	Uttar Pradesh	663	8.54	13.3	11.1	8.9	1.0	1.4	8.7	7.7	6.3	4.4	3.4	2.8	III	a	B
	f)	<i>Anthocephyalus chinensis</i> (Syn. <i>A. Cadamba</i>)	Kadam (KAD)	–	485	1.88	9.7	8.1	5.4	0.7	1.0	5.9	5.3	4.3	1.9	1.5	1.2	III	a	–
	g)	<i>Arlocarpus chaplasha</i>	Chaplash (CHP)	Assam	515	9.11	13.2	11.0	8.8	0.9	1.2	8.5	7.5	6.2	3.6	2.8	2.3	III	d	B
	h)	<i>Acacia leucophloea</i>	Hiwar (HIW)	Madhya Pradesh	737	7.85	13.4	11.2	9.0	1.0	1.5	7.5	6.7	5.4	4.5	3.5	2.8	–	–	A
	j)	<i>Acacia melanoxylone</i>	Black wood	Chennai	630	9.45	13.0	10.8	8.7	1.1	1.5	7.6	6.8	5.5	3.2	2.5	2.0	–	–	–

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
						Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain					
	Inside Location	Outside Location				Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location					
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	k) <i>Acacia mearnsii</i> (Syn. <i>A. mollissima</i>)	Black wattle	Chennai	669	6.10	10.4	8.6	6.9	0.8	1.2	6.0	5.4	4.4	2.3	1.8	1.5	–	–	–
	m) <i>Accer spp.</i>	Maple (MAP)	Punjab, Uttar Pradesh	551	7.35	9.9	8.2	6.5	0.9	1.3	5.5	4.9	4.0	2.1	1.7	1.4	III	–	B
	n) <i>Aegla marmalos</i> (Syn. <i>Intsia bijuga</i>)	Bael (BEL)	Uttar Pradesh	890	8.81	13.5	11.2	9.0	1.4	2.0	8.8	7.8	6.4	6.8	5.3	4.3	III	–	B
	p) <i>Afzelia bijuga</i>	-	Andaman	705	9.16	13.2	11.0	8.8	1.1	1.5	7.9	7.1	5.8	4.0	3.1	2.6	–	–	–
	q) <i>Ailanthus grandis</i>	Gokul (GOK)	West Bengal	404	7.94	8.3	6.9	5.5	0.6	0.8	5.3	4.7	3.9	1.1	0.9	0.7	III	–	C
	r) <i>Anogeissus pendula</i>	Kardhai (KAH)	Uttar Pradesh	929	9.75	17.0	14.2	11.4	1.3	1.8	9.8	8.7	7.1	6.5	5.1	4.2	III	–	A
	s) <i>Areca nut</i>	-	Kerala	833	9.48	15.2	12.7	10.2	1.2	1.6	10.8	9.6	7.8	7.3	5.7	4.7	–	–	–
	t) <i>Albizia lucida</i>	-	Arunachal Pradesh	566	8.51	10.7	8.9	7.1	8.2	1.2	7.3	6.3	5.3	2.3	1.8	1.5	–	–	–
	u) <i>Azadirachta indica</i>	Neem (NEE)	Uttar Pradesh	836	8.52	14.6	12.1	9.7	1.3	1.8	10.0	8.9	7.3	5.0	3.9	3.2	–	–	–

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m³	Modulus of Elasticity (All Grades and All Locations) × 10³ N/mm²	Permissible Stress for Grade I N/mm²											Preservative Characters		Refracteriness to Air Seasoning³)
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class¹)	Treat-ability Grade²)	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	v) <i>Boswellia seriata</i>	Salai (SAA)	Bihar	551	7.21	9.4	7.9	6.3	0.7	1.1	5.5	4.9	4.0	2.1	1.6	1.3	I	e	C
	w) <i>Bridelia retusa</i>	Kassi (KAS)	Bihar	584	9.42	11.6	9.7	7.7	0.9	1.3	7.1	6.3	5.1	4.0	3.1	2.6	I	e	B
	y) <i>Betula Inoides</i>	Birch (BIR)	West Bengal	625	9.23	9.6	8.0	6.4	0.8	1.1	5.7	5.0	4.1	2.2	1.7	1.4	–	–	B
	z) <i>Bischofia javanica</i>	Uriam Bishopwood (URI)	Chennai	769	8.84	9.6	8.2	6.5	0.8	1.1	5.9	5.3	4.3	3.6	2.8	2.3	III	–	A
	aa) <i>Burserra serrata</i> (Syn. <i>Protium serratum</i>)	Muntenga (MUR)	Arunachal Pradesh	756	1.17	15.5	13.3	10.5	0.9	1.3	10.1	9.0	7.4	5.3	4.1	3.4	II	c	–
	ab) <i>Careya arbersa</i>	Kumbi (KUM)	Uttar Pradesh	889	8.37	13.1	10.9	8.8	1.0	1.5	7.7	6.8	5.6	5.3	4.1	3.4	I	e	A
	ac) <i>Cedrus deodara</i>	Deodar (DEO)	Himachal Pradesh	557	9.48	10.2	8.7	7.2	0.7	1.0	7.8	6.9	5.7	2.7	2.1	1.7	I	c	C
	ad) <i>Cupressus torulosa</i>	Cypress (CYP)	Uttar Pradesh	506	8.41	8.8	7.6	6.2	0.6	0.8	6.9	6.2	5.0	2.4	1.8	1.5	I	e	C

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	ae) <i>Castanopsis hystrix</i>	Indian chestnut (ICH)	West Bengal	624	9.85	10.6	8.8	7.0	0.8	1.2	6.4	5.7	4.6	2.7	2.1	1.7	II	b	B
	af) <i>Chukrasia vclutina</i> (Syn. <i>C. Tabularis</i>)	Chickrassy (CHI)	West Bengal	666	8.35	11.8	9.8	7.9	1.1	1.5	7.1	6.3	5.2	3.9	3.1	2.5	II	c	B
	ag) <i>Calophyllum wightianum</i>	Poon (POO)	Maharashtra	689	8.68	13.5	11.2	9.0	1.0	1.4	8.7	7.8	6.4	4.0	3.1	2.5	II	–	B
	ah) <i>Canarium strictum</i>	White dhup	Assam	569	10.54	10.1	8.4	6.7	0.7	1.1	6.2	5.5	4.5	2.1	1.6	1.3	III	–	C
	aj) <i>Chlorophora excelsa</i>	-	Chennai	471	6.57	10.2	8.5	6.8	0.5	0.7	6.4	5.6	4.6	2.0	1.6	1.3	–	–	–
	ak) <i>Cocosnucifera</i>	Coconut (COC)	Kerala	761	7.34	9.2	7.7	6.1	0.7	1.1	9.5	8.4	6.9	3.9	3.0	2.5	–	–	–
	am) <i>Dalbergia latifolia</i>	Rosewood (ROS)	Madhya Pradesh	884	8.39	12.9	10.8	8.6	1.1	1.6	8.0	7.1	5.8	4.2	3.3	2.7	I	–	B
	an) <i>Dalbergia sissee</i>	Sisso (SIS)	Punjab	799	7.14	12.8	10.7	8.5	1.3	1.8	8.2	7.3	6.0	4.2	3.3	2.7	I	e	B
	ap) <i>Dillemia indica</i>	Dillenia (DIL)	West Bengal	617	8.61	12.1	10.0	8.0	0.8	1.2	7.3	6.5	5.3	2.7	2.1	1.7	III	a	B

Table 1 (Continued)

SI No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	aq) <i>Dillenia pentagyne</i>	Dillenia (DIL)	West Bengal	622	7.56	11.8	9.9	7.9	0.9	1.3	7.1	6.3	5.2	3.5	2.7	2.2	III	d	B
	ar) <i>Diospyres melanoxylon</i>	Ebony (EBO)	Maharashtra	818	7.69	10.9	9.1	7.3	0.9	1.2	7.0	6.2	5.1	3.3	2.6	2.1	II	–	A
	as) <i>Duabanga grandiflora</i> (Syn. <i>D. Sonneratioides</i>)	Lampati (LAP)	West Bengal	485	8.38	9.8	8.2	6.5	0.6	0.9	6.4	5.7	4.7	1.8	1.4	1.1	III	c	C
	at) <i>Elesocarpus tuberculatus</i>	Rudrak (RUD)	Chennai	466	8.74	9.7	8.1	6.4	0.7	1.0	6.3	5.6	4.6	2.0	1.5	1.3	–	–	C
	au) <i>Eucalyptus hybrid</i>	Mysore gum (MGU)	Chennai	753	6.00	10.2	8.5	6.8	0.9	1.2	7.3	6.5	5.3	4.0	3.1	2.5	III	e	–
	av) <i>Calitres rhomboidea</i> (Syn. <i>Frenela rhomboidea</i>)	–	Chennai	607	6.48	9.2	7.7	6.1	0.7	1.0	6.9	6.1	5.0	4.0	3.1	2.6	–	–	–
	aw) <i>Garuga pinnata</i>	Garuga (GAU)	Uttar Pradesh	571	7.58	11.7	9.7	7.8	1.0	1.5	7.2	6.4	5.3	3.4	2.6	2.1	I	e	B
	ay) <i>Gmelina arborea</i>	Gamari (GAM)	Uttar Pradesh	501	7.02	9.8	8.2	6.6	0.8	1.2	5.7	5.0	4.1	4.2	3.2	2.7	I	e	B

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	az) <i>Gardonia latifolia</i>	Gardenia (GAI)	Madhya Pradesh	705	7.13	14.1	11.7	9.4	1.2	1.7	8.4	7.4	6.1	4.6	3.6	3.0	—	—	—
	ba) <i>Hardwickis binata</i>	Anjan (ANJ)	Madhya Pradesh	852	6.64	14.1	11.8	9.4	1.3	1.8	9.0	8.0	6.5	7.4	5.6	4.7	I	e	—
	bb) <i>Heloptelea integrifolia</i>	Kanju (KAN)	Uttar Pradesh	592	7.46	12.0	10.0	8.0	0.9	1.3	6.7	6.0	4.9	2.8	2.2	1.8	III	b	B
	bc) <i>Heterophragma rexburghii</i>	Palang (PAL)	Madhya Pradesh	616	8.69	12.3	10.2	8.2	0.7	1.0	7.9	7.0	5.7	3.4	2.6	2.1	—	—	—
	bd) <i>Juglans spp.</i>	Walnut (WAL)	Uttar Pradesh	565	9.00	9.9	8.3	6.6	0.9	1.2	5.8	5.2	4.2	2.2	1.7	1.4	III	—	B
	be) <i>Lagerstrosmia speciosa</i> (Syn. <i>L. flesregihal</i>)	Jarul (JAAR)	Northern Andaman	622	8.53	12.1	10.1	8.1	0.8	1.8	7.7	6.8	5.6	3.4	2.6	2.2	II	e	B
	bf) <i>Lannea grandis</i> (Syn. <i>L.coromandelica</i>)	Jhingan (JHI)	Uttar Pradesh	557	5.63	8.5	7.1	5.7	0.6	0.9	4.9	4.4	3.6	2.2	1.7	1.4	III	e	B
	bg) <i>Leucanena leucocephala</i>	Subabul (SUB)	Uttar Pradesh	673	6.32	11.6	9.7	7.8	1.0	1.5	7.4	6.6	5.4	3.8	3.0	2.4	—	—	—

Table 1 (Continued)

SI No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	bh) <i>Lophopatalum wightianum</i>	Banati (BAN)	Chennai	460	7.33	8.5	7.5	5.6	0.5	0.8	5.3	4.7	3.8	1.8	1.4	1.1	III	–	C
	bj) <i>Madhuca longifolia varlatifolia</i> (Syn. <i>Bassia latifolia</i>)	Mahua (MAU)	Madhya Pradesh	936	8.82	13.0	10.8	8.7	1.0	1.4	7.5	6.7	5.5	6.3	4.9	4.0	I	e	A
	bk) <i>Mangifera indica</i>	Mango, Aam (MAN)	Orissa	661	9.12	12.2	10.2	8.2	1.0	1.4	7.3	6.5	5.3	3.1	2.4	2.0	III	a	C
	bm) <i>Machilus macrantha</i>	Machilus (MAC)	Chennai	521	7.63	10.2	8.5	6.8	0.7	1.0	6.3	5.6	4.6	2.4	1.9	1.5	III	e	B
	bn) <i>Mallotus philippinensis</i>	Raini (RAI)	Uttar Pradesh	662	7.51	10.8	9.0	7.2	1.0	1.4	6.0	5.4	4.4	2.9	2.3	1.8	III	–	B
	bp) <i>Manglietia insignia</i>	-	Assam	449	10.37	10.9	9.1	7.3	0.7	1.0	8.0	7.1	5.8	3.4	2.6	2.1	–	–	–
	bq) <i>Michelia montana</i>	Champ (CHM)	West Bengal	512	8.25	10.9	9.1	7.3	0.7	1.0	6.6	5.9	4.8	2.8	2.2	1.8	I	–	B

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	br) <i>Mitragyna pervifolia</i> (Syn. <i>Stephagyne pervifolia</i>)	Kaim (KAI)	Uttar Pradesh	651	7.82	12.6	10.5	8.4	1.0	1.5	7.9	7.0	5.7	3.7	2.9	2.4	III	b	B
	bs) <i>Michelia excelsa</i>	Champ (CHM)	West Bengal	513	10.12	9.8	8.2	6.5	0.7	1.0	6.1	5.5	4.5	1.6	1.3	1.0	II	e	B
	bt) <i>Miliusa velutnia</i>	Domsal (DOM)	Uttar Pradesh	747	7.92	11.7	9.7	7.8	1.1	1.6	7.0	6.3	5.1	3.7	2.9	2.4	III	–	–
	bu) <i>Morus alba</i>	Mulberry (MUL)	Uttar Pradesh	743	8.20	11.8	9.8	7.9	1.0	1.4	6.6	5.8	4.8	3.8	2.9	2.4	II	–	B
	bv) <i>Morus serrata</i>	Mulberry (MUL)	Himachal Pradesh	657	7.03	10.2	8.5	6.8	0.9	1.3	5.6	5.0	4.1	2.6	2.0	1.6	III	–	B
	bw) <i>Morus laevigata</i>	Bola (BOL)	Andaman	588	8.61	12.3	10.2	8.2	1.0	1.5	7.2	6.4	5.3	3.3	2.5	2.1	–	–	B
	by) <i>Ougeinia eejeinensis</i> (Syn. <i>O. delbergioides</i>)	Sandan (SAD)	Madhya Pradesh	784	8.54	13.3	11.1	8.9	1.2	1.7	8.5	7.5	6.2	5.1	3.9	3.2	I	–	B
	bz) <i>Phoebe hainesiana</i>	Bonsum (BOH)	Assam	566	9.50	13.2	11.0	8.8	0.8	1.2	8.8	7.8	6.4	2.8	2.1	1.8	II	c	B

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	ca) <i>Pinus roxburghii</i> (Syn. <i>P. longifolia</i>)	Chir (CHR)	Uttar Pradesh	525	9.82	8.5	7.3	6.0	0.6	0.9	6.0	5.3	4.4	2.0	1.5	1.3	III	b	C
	cb) <i>Pinus wallichiana</i>	Kail (KAL)	–	515	6.80	6.6	5.6	5.0	0.6	0.8	5.2	4.6	3.8	1.7	1.3	1.0	II	c	C
	cc) <i>Phoebe goalperansis</i>	Bonsum (BOH)	Assam	511	7.65	9.7	8.1	6.5	0.7	1.0	6.6	5.9	4.8	2.2	1.7	1.4	II	c	B
	cd) <i>Parretiopsis jacquementiena</i>	Rohu Parrotia	Himachal Pradesh	761	5.77	12.5	10.4	8.3	1.2	1.7	6.8	6.1	5.0	4.0	3.1	2.5	III	–	B
	ce) <i>Pinus kesia</i> (Syn. <i>Pinus insularis</i>)	Khasi pine (KPI)	North East	513	7.38	8.9	7.4	5.9	0.6	0.7	5.8	5.2	4.3	1.5	1.2	1.0	III	a	B
	cf) <i>Pistacia integerrima</i>	Kikar singhi	Jammu and Kashmir	881	7.32	13.1	10.9	8.7	1.2	1.7	8.0	7.1	5.8	4.3	3.4	2.8	–	–	–
	cg) <i>Podocarpus nerrifolius</i>	Thitmin (THT)	Southern Andaman	533	9.41	12.5	10.4	8.3	6.1	0.9	8.0	7.1	5.8	2.6	2.0	1.6	II	–	–
	ch) <i>Polyalthia fragrances</i>	Debbaru (DEB) (Nedunar)	Maharashtra	752	9.15	11.9	9.9	7.9	0.8	1.2	6.7	6.0	4.9	3.0	2.3	1.9	III	–	B

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m³	Modulus of Elasticity (All Grades and All Locations) × 10³ N/mm²	Permissible Stress for Grade I N/mm²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	cj) <i>Polyalthia coreoides</i>	–	Madhya Pradesh	700	9.29	13.2	11.0	8.8	1.0	1.4	7.1	6.3	5.2	3.2	2.5	2.0	–	–	–
	ck) <i>Prunus napeulensis</i>	Arupati	West Bengal	548	9.41	104.4	8.7	69.6	0.9	1.2	6.7	6.0	4.9	2.4	1.9	1.6	–	–	–
	cm) <i>Pterespermum acerifolium</i>	Hattipaila (HAT)	West Bengal	607	9.55	13.5	11.3	9.0	0.9	1.2	8.7	7.7	6.3	3.2	2.5	2.0	III	c	B
	cn) <i>Qucrcus spp.</i>	Oak	North East	657	11.65	11.4	9.5	7.6	0.8	1.2	6.7	5.9	4.8	2.0	1.6	1.3	II	c	B
	cp) <i>Raderomachera xylocarpe</i> (Syn. <i>Sterosperam xylocarpum</i>)	Vedankonnai	Chennai	696	8.52	13.2	11.0	8.8	1.1	1.5	9.0	8.0	6.6	4.3	3.3	2.7	II	a	–
	cq) <i>Schleichera oleosa</i> (Syn. <i>S. trijuga</i>)	Kusum (KUS)	Bihar	1 032	12.12	15.5	13.0	10.4	1.5	2.1	10.9	9.7	7.9	6.1	4.2	3.9	II	a	A
	cr) <i>Schima wallichii</i>	Chilauni (CHL)	West Bengal	693	9.57	11.1	9.3	7.4	0.9	1.3	6.6	5.9	4.8	2.3	1.8	1.4	III	d	B
	cs) <i>Shotea assamica</i>	Makai (MAK)	Assam	548	9.27	11.1	9.2	7.4	0.9	1.3	7.1	6.3	5.2	2.9	2.2	1.8	III	c	B

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m³	Modulus of Elasticity (All Grades and All Locations) × 10³ N/mm²	Permissible Stress for Grade I N/mm²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	ct) <i>Sonneralia apetale</i>	Keora (KEO)	West Bengal	617	8.63	12.8	10.7	8.5	0.9	1.3	7.4	6.6	5.4	4.8	3.7	3.0	II	–	B
	cu) <i>Stereospermum suaveolans</i>	Padri (PAD)	Uttar Pradesh	721	8.86	13.3	11.1	8.9	0.9	1.3	7.3	7.0	5.7	3.5	2.7	2.2	III	–	B
	cv) <i>Tactona grandis</i>	Teak (TEA)	Madhya Pradesh	617	8.49	12.8	10.7	8.5	0.8	1.3	7.9	7.0	5.7	4.0	3.1	2.6	I	e	B
	cw) <i>Terminalia arjuna</i>	Arjun (ARJ)	Bihar	794	7.71	12.2	10.2	8.2	1.1	1.6	7.4	6.6	5.4	5.2	4.1	3.3	II	b	B
	cy) <i>Terminalia myriocarpa</i>	Hollock (HOC)	Assam	615	9.62	11.9	9.9	8.0	0.9	1.2	7.6	6.7	5.5	2.9	2.2	1.8	III	a	B
	cz) <i>Terminalia procera</i>	White bombwae (WBO)	Northern Andaman	626	8.99	11.8	9.8	7.9	0.9	1.3	7.2	6.4	5.3	3.0	2.3	1.9	III	b	B
	da) <i>Taxus buccata</i>	Yew (YEW)	West Bengal	705	7.79	14.3	11.9	9.5	1.2	1.7	8.7	7.8	6.4	4.7	3.7	3.0	–	–	–
	db) <i>Tamarindus indica</i>	Imli (IML)	Chennai	913	5.63	11.4	9.5	7.6	1.2	1.7	7.0	6.2	5.1	5.3	4.1	3.4	–	–	B

Table 1 (Continued)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	dc) <i>Toena ciliata</i>	Toon (TOO)	Uttar Pradesh	487	6.40	8.7	7.3	5.8	0.7	1.0	5.4	4.8	3.9	2.4	1.8	1.5	II	c	B
	dd) <i>Vateria indica</i>	Vellapine (VEL)	Chennai	535	10.95	11.5	9.6	7.6	0.7	1.1	7.5	6.7	5.5	2.3	1.8	1.4	III	e	C
	de) <i>Aeculas indica</i>	Horse chestnut (HCH)	Uttar Pradesh	484	7.55	8.5	7.1	5.7	0.8	1.1	4.8	4.2	3.5	1.8	1.4	1.1	—	—	B
	df) <i>Borassus flabelsfer</i>	Tad (Palmyra) (TAD)	Arunachal Pradesh	838	8.79	10.5	8.8	7.0	0.7	1.0	10.0	8.8	7.2	4.7	3.6	2.7	—	—	—
	dg) <i>Eucalyptus cemaldulensis</i>	Eucalyptus	Karnataka	804	9.53	12.8	10.6	8.5	0.8	1.1	7.2	6.4	5.2	3.5	2.7	2.2	—	—	A
	dh) <i>Eucalyptus camaldulenis</i>	Eucalyptus	Uttar Pradesh	781	7.03	12.4	10.4	8.3	1.1	1.6	7.9	7.0	5.7	3.5	2.8	2.3	—	—	A
	dj) <i>Eucalyptus pilularia</i>	Eucalyptus	Tamil Nadu	713	9.22	14.8	12.3	11.1	1.0	1.4	8.5	7.6	6.2	2.8	2.2	1.8	—	—	A
	dk) <i>Eucalyptus propingus</i>	Eucalyptus	Tamil Nadu	584	7.93	12.8	10.7	8.5	0.8	1.2	8.0	5.4	4.4	2.5	1.9	1.6	—	—	A

Table 1 (Concluded)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content Kg/m ³	Modulus of Elasticity (All Grades and All Locations) × 10 ³ N/mm ²	Permissible Stress for Grade I N/mm ²											Preservative Characters		Refracteriness to Air Seasoning ³⁾
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress			Shear all Locations		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class ¹⁾	Treat-ability Grade ²⁾	
						Inside Location	Outside Location	Wet Location	Horizontal	Along Grain	Inside Location	Outside Location	Wet Location	Inside Location	Outside Location	Wet Location			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	dm) <i>Eucalyptus saligna</i>	Eucalyptus	Uttar Pradesh	819	8.24	11.5	9.6	7.6	1.5	2.1	8.2	7.3	6.0	6.2	4.8	4.0	—	—	A
	NOTE — Any other species (including foreign species) of timber may be classified into Group A, B, or C based on their safe permissible stresses and other relevant properties, corresponding to the classification table provided for given Indian timber species. The assignment of a timber species to a specific group may be determined through standardized testing and evaluation of mechanical properties, including strength, durability and density, in accordance with the applicable test methods and criteria specified in accordance with accepted standards [C-3A(2)].																		

¹⁾ Classification for preservation based on durability tests, etc.

Class:

- I Average life more than 120 months;
- II Average life 60 months and above but less than 120 months; and
- III Average life less than 60 months.

²⁾ Treatability Grades:

- a Heartwood easily treatable;
- b Heartwood treatable, but complete penetration not always obtained; in case where the least dimension is more than 60 mm;
- c Heartwood only partially treatable;
- d Heartwood refractory to treatment; and
- e Heartwood very refractory to treatment, penetration of preservative being practically nil even from the ends.

³⁾ Classifications based on seasoning behaviour of timber and refractoriness with respect to cracking, splitting and drying rate:

- A Highly refractory (slow and difficulty to season free from surface and end cracking);
- B Moderately refractory (may be seasoned free from surface and end cracking within reasonably short periods, given a little protection against rapid drying conditions);
and
- C Non-refractory may be rapidly seasoned free from surface and end-cracking even in the open air and sun. If not rapidly dried, they develop blue stain and mould on the surface.

⁴⁾ Species thus marked and tested from other localities show higher strength to enable their categorization in higher group.

For Example:

- i) Sal tested from West Bengal, Bihar, Uttar Pradesh and Assam can be classified as Group 'A' species.
- ii) Haldu tested from Bihar can be classified as Group 'B' species.
- iii) Morus laevigate (Bole) of Assam can be classified in Group 'B' species.

4.1.1 Grouping

Species of timber recommended for constructional purposes are classified in three groups on the basis of their strength properties, namely, modulus of elasticity (E) and extreme fibre stress in bending-tension (f_b).

The characteristics of these groups are as given below:

- Group A* — E above 12.6 kN/mm² and f_b above 18.0 N/mm².
- Group B* — E above 9.8 kN/mm² and up to 12.6 kN/mm² and f_b above 12.0 N/mm² and up to 18.0 N/mm².
- Group C* — E above 5.6 kN/mm² and up to 9.8 kN/mm² and f_b above 8.5 N/mm² and up to 12.0 N/mm².

NOTE — Modulus of elasticity given above is applicable for all locations and extreme fibre stress in bending is for inside location.

4.1.2 Timber species may be identified in accordance with good practice [C-3A(3)].

4.2 The general characteristics like durability, treatability and seasoning of the species are also given in [Table 1](#). Species of timber other than those recommended in [Table 1](#) may be used, provided the basic strength properties are determined and found in accordance with [4.1.1](#).

NOTE — For obtaining basic stress figures of the unlisted species, reference may be made to the Forest Research Institute, Dehradun.

4.3 The permissible lateral strength (in double shear) of mild steel common wire nail shall be as given in [Table 2](#) and [Table 3](#) for different species of timber.

4.4 Moisture Content in Timber

The permissible moisture content of timber for various positions in buildings shall be as given in [Table 4](#).

Table 2 Permissible Lateral Strengths (in Double Shear) of Nails 3.55 mm Dia (9 SWG), 80 mm Long
(Clauses [4.3](#), [7.8.2](#) and [8.3.2](#))

Sl No.	Species of Wood		For Permanent Construction Strength per Nail		For Temporary Structures Strength per Nail (for Both Lengthening Joints and Node Joints)
	Botanical Name	Trade Name	Lengthening Joints	Node Joints	
			$N \times 10^2$	$N \times 10^2$	$N \times 10^2$
(1)	(2)	(3)	(4)	(5)	(6)
i)	<i>Albies pirdrow</i>	Fir	8	2	12
ii)	<i>Acacia catecha</i>	Khair	20	17.5	37.0
iii)	<i>Acacia nilotica</i>	Babul	15	11	32.5
iv)	<i>Acrocarpus fraxinifolius</i>	Mundani	18	9.5	19.5
v)	<i>Adina cordifolia</i>	Haldu	23.5	10	22
vi)	<i>Albizia lebbeck</i>	Kokko	20	7	24
vii)	<i>Albizia odoratissima</i>	Kala siris	14	5	22
viii)	<i>Anogeissus latifolia</i>	Axlewood	20	10	29
ix)	<i>Aphanamixis polystachya</i>	Pithraj	19	9	19

Table 2 (Continued)

Sl No.	Species of Wood		For Permanent Construction Strength per Nail		For Temporary Structures Strength per Nail (for Both Lengthening Joints and Node Joints)
	Botanical Name	Trade Name	Lengthening Joints	Node Joints	
			$N \times 10^2$	$N \times 10^2$	
(1)	(2)	(3)	(4)	(5)	(6)
x)	<i>Azadirachta indica</i>	Neem	22	11.5	19
xi)	<i>Balanocarpus utilis</i>	Karung	17.5	21	32
xii)	<i>Bischofia javanica</i>	Uriam	13	6.5	18
xiii)	<i>Betula alnoides</i>	Birch	13	12	22.5
xiv)	<i>Boswellia serrata</i>	Salai	12	5	14.5
xv)	<i>Calophyllum spp.</i>	Poon	16	9	21
xvi)	<i>Canarium euphyllum</i>	White dhup	9	8	10.5
xvii)	<i>Castanopsis hystrix</i>	Indian chestnut	18	10.5	23.5
xviii)	<i>Cedrus deodara</i>	Deodar	9	4	15
xix)	<i>Chukrasia tabularis</i>	Chikrassy	24	8	27
xx)	<i>Cinnamomum spp.</i>	Cinnomon	12	9	13
xxi)	<i>Cullenia rosaana</i> (Syn. <i>Cullenia excalsa</i>)	Karani	11.5	5	19
xxii)	<i>Cupressus torulosa</i>	Cypress	6	5	18
xxiii)	<i>Dalbergia latifolia</i>	Rosewood	19	8.5	23
xxiv)	<i>Dalberia sissoo</i>	Sissoo	14.5	7.5	19
xxv)	<i>Dipterocarpus macrocarpus</i>	Hollong	17	7	20
xxvi)	<i>Dipterocarpus spp.</i>	Gurjan	19	9	19.5
xxvii)	<i>Dillenia pertagyna</i>	Dillenia	16.5	12	16
xxviii)	<i>Diospyros melanoxylon</i>	Ebony	26.5	10	30.5
xxix)	<i>Duabanga grandiflora</i>	Lampati	15	5	12
xxx)	<i>Dysoxylum binectariferum</i>	White Cedar	7.5	5	14.5
xxxi)	<i>Eucalyptus eugenioides</i>	Eucalyptus	17	10	30
xxxii)	<i>Garuga pinnata</i>	Garuga	11.0	7.5	21.0
xxxiii)	<i>Gmelina arborea</i>	Gaman	8	4	9.5
xxxiv)	<i>Grerillea robusta</i>	Silver oak	12.0	7.0	10.5
xxxv)	<i>Grewia tilifolia</i>	Dhaman	13	5	24
xxxvi)	<i>Lagerstroemia spp.</i>	Jarul	24.5	21.5	22.5

Table 2 (Continued)

Sl No.	Species of Wood		For Permanent Construction Strength per Nail		For Temporary Structures Strength per Nail (for Both Lengthening Joints and Node Joints)
	Botanical Name	Trade Name	Lengthening Joints	Node Joints	
			$N \times 10^2$	$N \times 10^2$	
(1)	(2)	(3)	(4)	(5)	(6)
xxxvii)	<i>Holoptelea integrifolia</i>	Kanju	12.5	9.5	20.5
xxxviii)	<i>Hopea parviflora</i>	Hopea	31.5	13	28.5
xxxix)	<i>Lagerstroemia spp.</i>	Lendi	19	5	26
xl)	<i>Madhuka Logifera</i> (Syn. <i>Barfia Latifolia</i>)	Mahua	23	7.5	26
xli)	<i>Mangifera indica</i>	Mango	11	9	16
xlii)	<i>Maniltoa polyandra</i>	Ping	26	23.5	32
xliii)	<i>Mesua ferrea</i>	Mesua	26	8	41
xliv)	<i>Michelia spp.</i>	Champ	13	9	20
xl v)	<i>Millingtonia spp.</i>	Phulsopa	10.5	6	17
xlvi)	<i>Morus alba</i>	Mulberry	13	10.5	22.5
xl vii)	<i>Melia azedarach</i>	Persian lilac (bakain)	10.5	2.5	10.5
xl viii)	<i>Morus alba</i>	Mulberry	13	10.5	22.5
xlix)	<i>Ougeinia oojeinensis</i>	Sandan	17	11	18
l)	<i>Phoebe spp.</i>	Bonsum	12	6	13
li)	<i>Pinus roxburghii</i>	Chir	11	10	16
lii)	<i>Pinus wallichiana</i>	Kail	7	3	9
liii)	<i>Planchonia andamanica</i>	Red bombwe	14	13	29
liv)	<i>Planchonia valida</i> (Syn. <i>P. andamonica</i>)	Red bombwe	14.5	13	29.5
lv)	<i>Poeciloneuron indicum</i>	Ballagi	16	7	15.5
lvi)	<i>Populus deltoides</i>	Poplar	16	11.5	17.5
lvii)	<i>Protium serratum</i> (Syn. <i>Burserra Serrata</i>) A. P.	–	25.5	7.0	22.5
lviii)	<i>Pterocarpus marsupium</i>	Bijasal	15	12	27
lix)	<i>Pterocarpus dalbergiodes</i>	Pauduak	19	14	23
lx)	<i>Quercus spp.</i>	Oak (Assam)	21	10	37.5

Table 2 (Concluded)

Sl No.	Species of Wood		For Permanent Construction Strength per Nail		For Temporary Structures Strength per Nail (for Both Lengthening Joints and Node Joints)
	Botanical Name	Trade Name	Lengthening Joints	Node Joints	
			$N \times 10^2$	$N \times 10^2$	
(1)	(2)	(3)	(4)	(5)	(6)
lxi)	<i>Quercus spp.</i>	Oak (Uttar Pradesh)	11	10.5	26.5
lxii)	<i>Scheichera cleosa</i>	Kusum	23	16	39.5
lxiii)	<i>Schima wallachi</i>	Chillauni	19.5	9.5	25.5
lxiv)	<i>Shorea assamica</i>	Makai	12.5	7.5	15.0
lxv)	<i>Shorea robusta</i>	Sal (Madhya Pradesh)	23	15.5	19.5
lxvi)	<i>Shorea robusta</i>	Sal (Uttar Pradesh)	10	5	19
lxvii)	<i>Stereospennum chelonoides</i>	Padriwood	16	8	19.5
lxviii)	<i>Syzygium spp.</i>	Jamum	15	12	25
lxix)	<i>Tectona grandis</i>	Teak	14	8	13
lxx)	<i>Terminalia alata</i>	Sain	16	16	29
lxxi)	<i>Terminalia arjuna</i>	Arjun	8.5	7	16.5
lxxii)	<i>Terminalia bellirica</i>	Bahera	10	10	14
lxxiii)	<i>Terminalia biolata</i>	White chuglam	18	9	21
lxxiv)	<i>Terminalia chebula</i>	Myrobalan	16.5	5.5	22.5
lxxv)	<i>Terminalia procera</i>	Badam	18	10.5	20
lxxvi)	<i>Terminalia manii</i>	Black chuglam	23	10	33
lxxvii)	<i>Terminalia myriocarpa</i>	Hollock	13	10	19.5
lxxviii)	<i>Toona spp.</i>	Toona	10	8	21
lxxix)	<i>Xylia xylacarpa</i>	Irul	23	6	33
lxxx)	<i>Toona ciliata</i>	Toon	16	9	21

NOTES

1 Nails of 3.55 mm (9 SWG) diameter are most commonly used. The above values can also be used for 4 mm (8 SWG) diameter 100 mm long nails.

2 The values in Newton (N) are approximate converted values from kgf. For exact conversion the value is 1 kgf = 9.806 65 N.

3 Preboring of nail-jointed timber construction is desirable and shall be done in accordance with good practice as in, 6.2.2 and Table 3 of [C-3A(7)].

**Table 3 Permissible Lateral Strengths (In Double Shear) of Nails 5.00 mm Dia (6 SWG),
125 mm and 150 mm Long
(Clauses 4.3, 7.8.2 and 8.3.2)**

Sl No.	Species of Wood		For Permanent Construction Strength per Nail		For Temporary Structures Strength per Nail (for Both Lengthening Joints and Node Joints)
	Botanical Name	Trade Name	Lengthening Joints	Node Joints	$N \times 10^2$
			$N \times 10^2$	$N \times 10^2$	
(1)	(2)	(3)	(4)	(5)	(6)
i)	<i>Abies pindrow</i>	Fir	16.5	4.5	21
ii)	<i>Acacia catechu</i>	Khair	42	25	71.5
iii)	<i>Acacia nilotica</i>	Babul	27	13.5	53
iv)	<i>Alibizia procera</i>	Safed siris	35	18	-
v)	<i>Alibizia odoratissima</i>	Kala siris	27.5	17.5	45
vi)	<i>Alstonia scholaris</i>	Chatian	9.5	5.5	27
vii)	<i>Anogeissus latifolia</i>	Axlewood	22.5	13	46.5
viii)	<i>Azadirachata indica</i>	Neem	23.5	16	47
ix)	<i>Cupressus torulosa</i>	Cypress	20	7	27
x)	<i>Cullenia rosayroana</i>	Karani	11	9.5	30
xi)	<i>Dalbergia sissoo</i>	Sissoo	17	15	43
xii)	<i>Diptocarous spp.</i>	Gurjan	19.5	9.5	33
xiii)	<i>Eucalyptus Prob. E. Tereticornis</i>	—	37.5	28.5	47
xiv)	<i>Hardwickia binata</i>	Anjan	32	19	59
xv)	<i>Hopea perviflora</i>	Hopea	60.5	25	61.5
xvi)	<i>Holoptelea integrifolia</i>	kanju	18	12.5	37.5
xvii)	<i>Mangifera indica</i>	Mango	22.5	15	32
xviii)	<i>Mesua ferrea</i>	Mesua	24	15.5	57.5
xix)	<i>Michelia champaca</i>	Champ	26	12.5	39
xx)	<i>Pterocarpus marsupium</i>	Bijasal	20.5	15	43
xxi)	<i>Pinus roxburghii</i>	Chir	9	6	24
xxii)	<i>Shorea robusta</i> (Uttar Pradesh)	Sal	19.5	17	37
xxiii)	<i>Shorea robusta</i>	Sal	30.5	20	41
xxiv)	<i>Schleichera cleosa</i>	Kusum	15	14	55
xxv)	<i>Stereospermum personatum</i>	Padriwood	22	8	34
xxvi)	<i>Syzygium cumini</i>	Jamum	18	14.5	38.5
xxvii)	<i>Terminalia myriocarpa</i>	Hollock	27.5	9	41
xxviii)	<i>Tectona grandis</i>	Teak	28	13	30
xxix)	<i>Toona ciliata</i>	Toon	19	7	32.5
xxx)	<i>Hopea utilis</i>	Karung kangoo	31	10	58
xxxi)	<i>Phoebe spp.</i>	Bonsum	20	7.5	30
xxxii)	<i>Pterocarpus dalbergiodes</i>	Padauk	27.4	13.4	37.7
xxxiii)	<i>Populus deltoides</i>	Poplar	21	15	38.5
NOTES 1 Nails of 5.00 mm diameter are most commonly used. 2 The values in Newton (N) are approximate converted values from kgf. For exact conversion the value is 1 kgf = 9.806 65 N. 3 Preboring of nail-jointed timber construction is desirable and shall be done in accordance with good practice as in, 6.2.2 and Table 3 of [C-3A(7)].					

Table 4 Permissible Percentage Moisture Content Values

(Clause 4.4)

Sl No.	Use	Zones (see Note)			
		I	II	III	IV
(1)	(2)	(3)	(4)	(5)	(6)
i)	Structural elements	12	14	17	20
ii)	Doors and windows:				
	a) 50 mm and above in thickness	10	12	14	16
	b) Thinner than 50 mm	8	10	12	14
iii)	Flooring strips for general purposes	8	10	10	12
iv)	Flooring strips for tea gardens	12	12	14	16

NOTE — The country has been broadly divided into the following four zones based on the humidity variations in the country:

- a) Zone I — Average annual relative humidity less than 40 percent;
- b) Zone II — Average annual relative humidity 40 percent to 50 percent;
- c) Zone III — Average annual relative humidity 50 percent to 67 percent; and
- d) Zone IV — Average annual relative humidity more than 67 percent.

For detailed zonal classification in the country, tolerances, etc, reference may be made to good practice [C-3A(4)].

4.5 Sawn Timber

4.5.1 Sizes

Preferred cut sizes of timber for use in structural components shall be as given in [Table 5](#), [Table 6](#) and [Table 7](#).

Table 5 Preferred Cut Sizes of Structural Timber for Roof Trusses (Span from 3 m to 20 m)

(Clause 4.5.1)

Sl No.	Thickness mm	Width mm							
		(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	20	40	50	60	80	100	—	—	—
ii)	25	40	50	60	80	100	120	140	160
iii)	30	40	50	60	80	100	120	140	160
iv)	35	—	—	60	80	100	120	140	160
v)	40	—	—	60	80	100	120	140	160
vi)	50	—	—	60	80	100	120	140	160
vii)	60	—	—	—	80	100	120	140	160
viii)	80	—	—	—	—	100	120	140	160

NOTES

1 For truss spans marginally above 20 m, preferred cut sizes of structural timber may be allowed.

2 Preferred lengths of timber — 1 m, 1.5 m, 2 m, 2.5 m and 3 m.

Table 6 Preferred Cut Sizes of Structural Timber for Roof Purlins, Rafters, Floor Beams, etc
(Clause 4.5.1)

SI No.	Thickness mm	Width mm						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	50	80	100	120	140	—	—	—
ii)	60	80	100	120	140	160	—	—
iii)	80	—	100	120	140	160	—	—
iv)	100	—	—	—	140	160	180	200

NOTE — Preferred lengths of timber — 1.5 m, 2 m, 2.5 m and 3 m.

**Table 7 Preferred Cut Sizes of Structural Timber for Partition Framing and
Covering and for Centering**
(Clause 4.5.1)

SI No.	Thickness mm	Width mm								
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	10	40	50	60	80	—	—	—	—	—
ii)	15	40	50	60	80	100	—	—	—	—
iii)	20	40	50	60	80	100	120	160	200	—
iv)	25	40	50	60	80	100	120	160	200	240
v)	30	40	50	60	80	100	120	160	200	240
vi)	40	40	—	60	80	100	120	160	200	240
vii)	50	—	50	—	80	100	120	160	200	240
viii)	60	—	—	60	80	100	120	160	200	240
ix)	80	—	—	—	80	100	120	160	200	240

NOTE — Preferred lengths of timber: 0.5 m, 1 m, 1.5 m, 2 m, 2.5 m and 3 m.

4.5.2 Tolerances

Permissible tolerances in measurements of cut sizes of structural timber shall be as follows:

a) For width and thickness:

- 1) Up to and including 100 mm : + 3 – 0 mm
- 2) Above 100 mm : + 6 – 3 mm

b) For length : + 10 – 0 mm

4.6 Grading of Structural Timber

4.6.1 Cut sizes of structural timber shall be graded, after seasoning, into three grades based on permissible defects given in Table 8:

- a) Select grade;
- b) Grade I; and
- c) Grade II.

Table 8 Permissible Defects for Cut Sizes of Timber for Structural Use

(Clauses [4.6.1](#) and [4.6.2.2](#))

All dimensions in millimetres.

Sl No.	Defects	Select Grade	Grade I	Grade II
(1)	(2)	(3)	(4)	(5)
i)	Wane	Shall be permissible at its deepest portion up to a limit of 1/8 of the width of the surface on which it occurs	Shall be permissible at its deepest portion up to a limit of 1/6 of the width of the surface on which it occurs	Shall be permissible at its deepest portion up to a limit of 1/4 of the width of the surface on which it occurs
ii)	Worm holes	Other than those due to powder post beetles are permissible	Other than those due to powder post beetles are permissible	Other than those due to powder post beetles are permissible
iii)	Slope of grain	Shall not be more than 1 in 20	Shall not be more than 1 in 15	Shall not be more than 1 in 12
iv)	Live knots:			

Sl No.	Width of Wide Faces of Cut Sizes of Timber, Max	Permissible Maximum Size of Live Knot on		Permissible Maximum Size of Live Knot on		Permissible Maximum Size of Live Knot on	
		Narrow Faces and ¼ of the Width Face Close to Edges of Cut Size of Timber	Remaining Central Half of the Width of the Wide Faces	Narrow Faces and ¼ of the Width Face Close to Edges of Cut Size of Timber	The Remaining Central Half of the Width of the Wide Faces	Narrow Faces and ¼ of the Width Face Close to Edges of Cut Size of Timber	Remaining Central Half of the Width of the Wide Faces
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	75	10	10	19	19	29	30
ii)	100	13	13	25	25	38	39
iii)	150	19	19	38	38	57	57
iv)	200	22	25	44	50	66	75
v)	250	25	29	50	57	66	87
vi)	300	27	38	54	75	81	114
vii)	350	29	41	57	81	87	123
viii)	400	32	44	63	87	96	132
ix)	450	33	47	66	93	99	141
x)	500	35	50	69	100	105	150
xi)	550	36	52	72	103	108	156
xii)	600	38	53	75	106	114	159

Table 8 (Concluded)

v) Checks and shakes:

<i>Sl No.</i>	<i>Width of the Face of the Timber Max</i>	<i>Permissible Depth Max</i>	<i>Permissible Depth Max</i>	<i>Permissible Depth Max</i>
(1)	(2)	(3)	(4)	(5)
i)	75	12	25	36
ii)	100	18	35	54
iii)	150	25	50	75
iv)	200	33	65	99
v)	250	40	81	120
vi)	300	50	100	150
vii)	350	57	115	171
viii)	400	66	131	198
ix)	450	76	150	225
x)	500	83	165	249
xi)	550	90	181	270
xii)	600	100	200	300

4.6.2 The prohibited defects given in [4.6.2.1](#) and permissible defects given in [4.6.2.2](#) shall apply to structural timber.

4.6.2.1 Prohibited defects

Loose grains, splits, compression wood in coniferous species, heartwood rot, sap rot, crookedness, worm holes made by powder post beetles and pitch pockets shall not be permitted in all the three grades.

4.6.2.2 Permissible defects

Defects to the extent specified in [Table 8](#) shall be permissible.

NOTE — Wanes are permitted provided they are not combined with knots and the reduction in strength on account of the wanes is not more than the reduction with maximum allowable knots.

4.6.3 Location of Defects

The influence of defects in timber is different for different locations in the structural element. Therefore, these should be placed during construction in such a way so that they do not have any adverse effect on the members, in accordance with the accepted standard [C-3A(5)].

4.7 Suitability

4.7.1 Suitability in Respect of Durability and Treatability for Permanent Structures

There are two choices as given in [4.7.1.1](#) and [4.7.1.2](#).

4.7.1.1 First choice

The species shall be any one of the following:

- Untreated heartwood of high durability. Heartwood if containing more than 15 percent sap wood, may need chemical treatment for protection;
- Treated heartwood of moderate and low durability and Class 'a' and Class 'b' treatability;
- Heartwood of moderate durability and Class 'c' treatability after pressure impregnation; and
- Sapwood of all classes of durability after thorough treatment with preservative.

4.7.1.2 Second choice

The species of timber shall be heartwood of moderate durability and Class 'd' treatability.

4.7.2 Choice of Species for Load-Bearing Temporary Structures or Semi-Structural Components at Construction Site

The species shall be any one of the following:

- a) Heartwood of low durability and Class 'e' treatability; or
- b) The species whose durability and/or treatability is yet to be established, as listed in [Table 1](#).

4.8 Fastenings

All structural members shall be framed, anchored, tied and braced to develop the strength and rigidity necessary for the purposes for which they are used. Allowable stresses or loads on joints and fasteners shall be determined in accordance with recognized principles. Common mechanical fastenings are of bar type such as nails and spikes, wood screws and bolts, and timber connectors including metallic rings or wooden disc-dowels. Chemical fastenings include synthetic adhesives for structural applications.

NOTE — Synthetic adhesives are able to achieve for timber what welding has achieved for steel.

4.9 Requirements of Structural Timber

The various other requirements of structural timber for use in building shall be in accordance with the accepted standard [C-3A(5)].

5 PERMISSIBLE STRESSES

5.1 Fundamental stress values of different groups of timber are determined on small clear specimen according to good practices [C-3A(2)]. These values are then divided by the appropriate factors of safety to obtain the permissible stresses. Values of these factors of safety for different strength characteristics are as per the relevant table of the accepted standard [C-3A(5)].

5.2 The permissible stresses for Groups A, B and C for different locations applicable to Grade I structural timber shall be as given in [Table 9](#) provided that the following conditions are satisfied:

- a) The timbers should be of high or moderate durability and be given the suitable treatment where necessary;
- b) Timber of low durability shall be used after proper preservative treatment to good practice [C-3A(6)]; and
- c) The loads should be continuous and permanent and not of impact type.

5.3 The permissible stresses (excepting E) given in [Table 9](#) shall be multiplied by the following factors to obtain the permissible stresses for other grades provided that the conditions laid down in [5.2](#) are satisfied:

- | | | |
|----------------------------|---|------|
| a) For select grade timber | : | 1.16 |
| b) For Grade II timber | : | 0.84 |

5.3.1 When low durability timbers are to be used [see [5.2\(b\)](#)] on outside locations, the permissible stresses for all grades of timber, arrived at by [5.2](#) and [5.3](#) shall be multiplied by 0.80.

5.4 Modification Factors for Permissible Stresses

5.4.1 Due to Change in Slope of Grain

When the timber has not been graded and has major defects like slope of grain, knots and checks or shakes but not beyond permissible value, the permissible stress given in [Table 1](#) shall be multiplied by modification factor K_1 for different slopes of grain as given in [Table 10](#).

**Table 9 Minimum Permissible Stress Limits (in N/mm²) in Three Groups of Structural Timbers
(For Grade I Material)**

(Clauses 5.2 and 5.3)

Sl No.	Strength Character	Location of Use	Group A	Group B	Group C
(1)	(2)	(3)	(4)	(5)	(6)
i)	Bending and tension along grain	Inside ¹⁾	18.0	12.0	8.5
ii)	Shear ²⁾ :				
	a) Horizontal	All locations	1.05	0.64	0.49
	b) Along grain	All locations	1.5	0.91	0.70
iii)	Compression parallel to grain	Inside ¹⁾	11.7	7.8	4.9
iv)	Compression perpendicular to grain	Inside ¹⁾	4.0	2.5	1.1
v)	Modulus of elasticity ($\times 10^3$ N/mm ²)	All locations and grade	12.6	9.8	5.6

Table 10 Modifications Factor K_1 to Allow for Change in Slope of Grain

(Clause 5.4.1)

Sl No.	Slope	Modification Factor K_1	
		Strength of Beams, Joists and Ties	Strength of Posts or Columns
(1)	(2)	(3)	(4)
i)	1 in 10	0.80	0.74
ii)	1 in 12	0.90	0.82
iii)	1 in 14	0.98	0.87
iv)	1 in 15 and flatter	1.00	1.00

NOTE — For intermediary slopes of grains, values of modification factor may be obtained by interpolation.

¹⁾ For working stresses for other locations of use, that is, outside and wet, generally factors of 5/6 and 2/3 are applied.

²⁾ The values of horizontal shear to be used only for beams. In all other cases shear along grain to be used.

5.4.2 Due to Duration of Load

For different durations of design load, the permissible stresses given in [Table 1](#) shall be multiplied by the modification factor K_2 given in [Table 11](#).

NOTE — The strength properties of timber under load are time-dependent.

Table 11 Modification Factor K_2 , for Change in Duration of Loading		
(Clause 5.4.2)		
Sl No.	Duration of Loading	Modification Factor K_2
(1)	(2)	(3)
i)	Continuous (normal)	1.0
ii)	Two months	1.15
iii)	Seven days	1.25
iv)	Wind and earthquake	1.33
v)	Instantaneous or impact	2.00

5.4.2.1 The factor K_2 is applicable to modulus of elasticity when used to design timber columns, otherwise they do not apply thereto.

5.4.2.2 If there are several duration of loads (in addition to the continuous) to be considered, the modification factor shall be based on the shortest duration load in the combination, that is, the one yielding the largest increase in the permissible stresses, provided the designed section is found adequate for a combination of other larger duration loads.

NOTE — In any structural timber design for dead loads, snow loads and wind or earthquake loads, members may be designed on the basis of total of stresses due to dead, snow and wind loads using modification factor $K_2 = 1.33$ for the permissible stresses (see [Table 1](#)) to accommodate the wind load, that is, the shortest of duration and giving the largest increase in the permissible stresses. The section thus found is checked to meet the requirements based on dead loads alone with modification factor $K_2 = 1.00$.

5.4.2.3 Modification factor K_2 shall also be applied to allowable loads for mechanical fasteners in design of joints, when the wood and not the strength of metal determines the load capacity.

6 DESIGN CONSIDERATIONS

6.1 All structural members, assemblies or framework in a building, in combination with the floors, walls and other structural parts of the building shall be capable of sustaining, with due stability and stiffness the whole dead and imposed loadings as per Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS, without exceeding the limits of relevant stresses specified in this Subsection.

6.2 Buildings shall be designed for all dead and imposed loads or forces assumed to come upon them during construction or use, including uplifts or horizontal forces from wind and forces from earthquakes or other loadings. Structural members and their connections shall be proportioned to provide a sound and stable structure with adequate strength and stiffness. Wooden components in construction generally include panels for sheathing and diaphragms, siding, beams, girder, columns, light framings, masonry wall and joist construction, heavy-frames, glued laminated structural members, structural sandwiches, prefabricated panels, lamella arches, portal frames and other auxiliary constructions.

6.3 Net Section

6.3.1 The net section is obtained by deducting from the gross sectional area of timber the projected area of all material removed by boring, grooving or other means at critical plane. In case of nailing, the area of the prebored hole shall not be taken into account for this purpose.

6.3.2 The net section used in calculating load carrying capacity of a member shall be at least net section determined as above by passing an imaginary plane or a series of connected planes transversely through the members.

6.3.3 Notches shall be in no case remove more than one quarter of the section.

6.3.4 In the design of an intermediate or a long column, gross section shall be used in calculating load carrying capacity of the column.

6.4 Loads

6.4.1 The loads shall conform to those given in Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS.

6.4.2 The worst combination and location of loads shall be considered for design. Wind and seismic forces shall not be considered to act simultaneously.

6.5 Flexural Members

6.5.1 Such structural members shall be investigated for the following:

- a) Bending strength;
- b) Maximum horizontal shear;
- c) Stress at the bearings; and
- d) Deflection.

6.5.2 Effective Span

The effective span of beams and other flexural members shall be taken as the distance from face of supports plus one-half of the required length of bearing at each end except that for continuous beams and joists the span may be measured from centre of bearing at those supports over which the beam is continuous.

6.5.3 Usual formula for flexural strength shall apply in design:

$$f_{ab} = \frac{M}{Z} \leq f_b$$

6.5.4 Form Factors for Flexural Members

The following form factors shall be applied to the bending stress:

- a) *Rectangular section* — For rectangular sections, for different depths of beams, the form factor K_3 shall be taken as:

$$K_3 = 0.81 \left(\frac{D^2 + 89\,400}{D^2 + 55\,000} \right)$$

NOTE — Form factor (K_3) shall not be applied for beams having depth less than or equal to 300 mm.

- b) *Box beams and I-beams* — For box beams and I-beams the form factor K_4 shall be obtained by using the formula:

$$K_4 = 0.8 + 0.8y \left(\frac{D^2 + 89\,400}{D^2 + 55\,000} - 1 \right)$$

where

$$y = p_1^2 (6 - 8p_1 + 3p_1^2)(1 - q_1) + q_1$$

- c) *Solid circular cross sections* — For solid circular cross sections the form factor K_5 shall be taken as 1.18; and
- d) *Square cross sections* — For square cross sections where the load is in the direction of diagonal, the form factor K_6 shall be taken as 1.414.

6.5.5 Width

The minimum width of the beam or any flexural member shall not be less than 50 mm or 1/50 of the span, whichever is greater.

6.5.6 Depth

The depth of beam or any flexural member shall not be taken more than three times of its width without lateral stiffening.

6.5.6.1 Stiffening

All flexural members having a depth exceeding three times its width or a span exceeding 50 times its width or both shall be laterally restrained from twisting or buckling and the distance between such restraints shall not exceed 50 times its width.

6.5.7 Shear

6.5.7.1 The following formulae shall apply:

- The maximum horizontal shear, when the load on a beam moves from the support towards the centre of the span, and the load is at a distance of three to four times the depth of the beam from the support, shall be calculated from the following general formula:

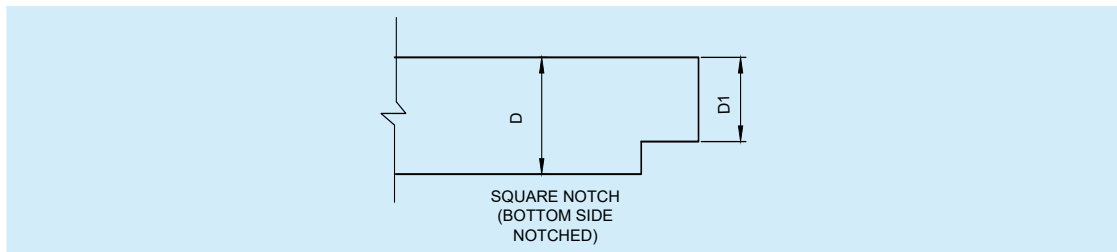
$$H = \frac{VQ}{Ib}$$

- For rectangular beams:

$$H = \frac{3V}{2bD}$$

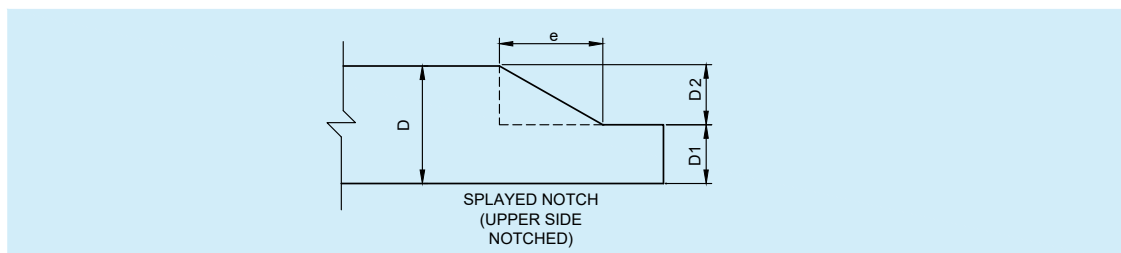
- For notched beams, with tension notch at supports (see [6.5.7.3](#)):

$$H = \frac{3VD}{2bD_1^2}$$



- For notched at upper (compression) face, where $e > D$:

$$H = \frac{3VD}{2bD_1}$$



- e) For notched at upper (compression) face, where $e < D$:

$$H = \frac{3V}{2b \left[D - \left(\frac{D_2}{D} \right) e \right]}$$

6.5.7.2 For concentrated loads:

$$V = \frac{10C(l-x)(x/D)^2}{9l[2 + (x/D)^2]} \text{ and}$$

For uniformly distributed loads:

$$V = \frac{W}{2} \left(1 - \frac{2D}{l} \right)$$

After arriving at the value of V, its value shall be substituted in the formula:

$$H = \frac{VQ}{Ib}$$

6.5.7.3 In determining the vertical reaction V, the following deductions in loads may be made:

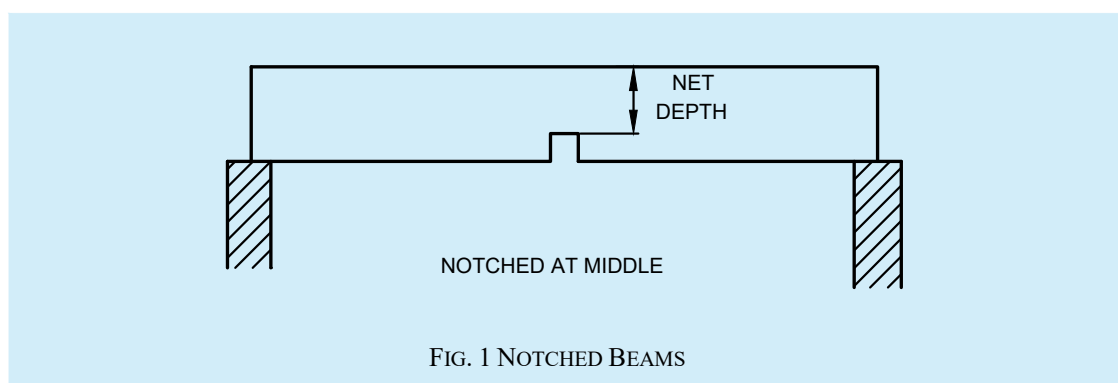
- Consideration shall be given to the possible distribution of load to adjacent parallel beams, if any;
- All uniformly distributed loads within a distance equal to the depth of the beam from the edge of the nearest support may be neglected except in case of beam hanging downwards from a particular support; and
- All concentrated loads in the vicinity of the supports may be reduced by the reduction factor applicable according to [Table 12](#).

Table 12 Reduction Factor for Concentrated Loads in the Vicinity of Supports [Clause 6.5.7.3(c)]					
SI No.	Distance of Load from the Nearest Support	1.5 D or Less	2 D	2.5 D	3 D or More
(1)	(2)	(3)	(4)	(5)	(6)
i)	Reduction factor	0.60	0.40	0.20	No reduction
NOTE — For intermediate distances, factor may be obtained by linear interpolation.					

6.5.7.4 Unless the local stress is calculated and found to be within the permissible stress, flexural member shall not be cut, notched or bored except as follows:

- Notches may be cut in the top or bottom neither deeper than one-fifth of the depth of the beam nor farther from the edge of the support than one-sixth of the span;
- Holes not larger in diameter than one quarter of the depth may be bored in the middle third of the depth and length; and

- c) If holes or notches occur at a distance greater than three times the depth of the member from the edge of the nearest support, the net remaining depth shall be used in determining the bending strength (see [Fig. 1](#)):



6.5.8 Bearing

6.5.8.1 The ends of flexural members shall be supported in recesses which provide adequate ventilation to prevent dry rot and shall not be enclosed. Flexural members except roof timbers which are supported directly on masonry or concrete shall have a length of bearing of not less than 75 mm. Members supported on corbels, offsets and roof timbers on a wall shall bear immediately on and be fixed to wall-plate not less than 75 mm × 40 mm.

6.5.8.2 Timber joists or floor planks shall not be supported on the top flange of steel beams unless the bearing stress, calculated on the net bearing as shaped to fit the beam, is less than the permissible compressive stress perpendicular to the grain.

6.5.8.3 Bearing stress

6.5.8.3.1 Length and position of bearing

- At any bearing on the side grain of timber, the permissible stress in compression perpendicular to the grain, f_{cn} , is dependent on the length and position of the bearing.
- The permissible stresses given in [Table 1](#) for compression perpendicular to the grain are also the permissible stresses for any length at the ends of a member and for bearings 150 mm or more in length at any other position.
- For bearings less than 150 mm in length located 75 mm or more from the end of a member as shown in [Fig. 2](#), the permissible stress may be multiplied by the modification factor K_7 given in [Table 13](#).
- No allowance need be made for the difference in intensity of the bearing stress due to bending of a beam.
- The bearing area should be calculated as the net area after allowance for the amount of wane.
- For bearings stress under a washer or a small plate, the same coefficient specified in [Table 13](#) may be taken for a bearing with a length equal to the diameter of the washer or the width of the small plate.
- When the direction of stress is at angle to the direction of the grain in any structural member, then the permissible bearing stress in that member shall be calculated by the following formula:

$$f_{c\theta} = \frac{f_{cp} \times f_{cn}}{f_{cp}\theta + f_{cn}\theta}$$

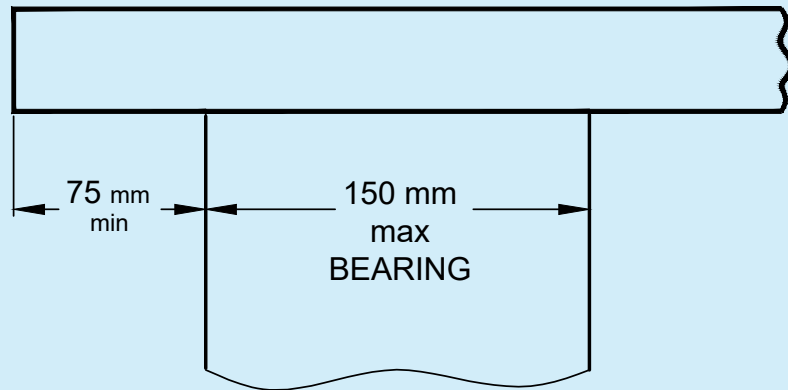


FIG 2. POSITION OF END BEARINGS

Table 13 Modification Factor K_7 for Bearing Stresses

[Clauses 6.5.8.3.1(c) and (f)]

Sl No.	Length of Bearing	15	25	40	50	75	100	150 or More
	mm							
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	Modification factor, K_7	1.67	1.40	1.25	1.20	1.13	1.10	1.00

6.5.9 Purlin is a structural member, essentially a beam subjected to transverse loads and rests on the top chord of roof truss. Being under bi-axial bending, vertical loads are resolved into two components, that is, perpendicular to and parallel to roof slopes (see Fig. 3). The effect of normal and tangential components of the vertical load is worked out separately and the in fibre stress in bending, shall be the algebraic sum of both.

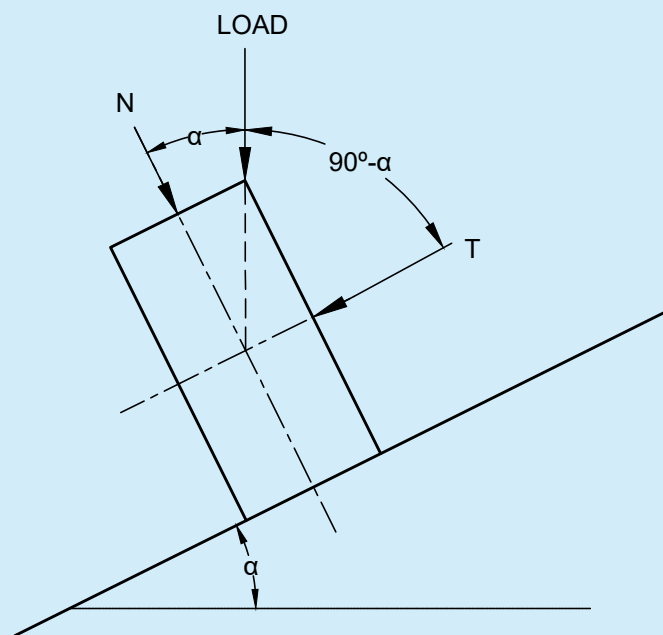


FIG. 3 PURLIN

6.5.10 Deflection

The deflection in the case of all flexural members supporting brittle materials like gypsum ceilings, slates, tiles and asbestos sheets shall not exceed 1/360 of the span. The deflection in the case of other flexural members shall not exceed 1/240 of the span and 1/150 of the freely hanging length in the case of cantilevers.

6.5.10.1 Usual formula for deflection shall apply:

$$\delta = \frac{KWL^3}{EI} \quad (\text{ignoring deflection due to shear strain})$$

K-value = 1/3 for cantilevers with load at free end;
= 1/8 for cantilevers with uniformly distributed load;
= 1/48 for beams supported at both ends with point load at centre; and
= 5/384 for beams supported at both ends with uniformly distributed load.

6.5.10.2 In order to allow the effect of long duration loading on *E*, for checking deflection in case of beams and joists the effective loads shall be twice the dead load if timber is initially dry.

NOTE — The tendency of flexural members to acquire larger permanent deflections under long continued loadings has to be taken care of either by considering double the dead load in calculations of deflection or by giving a factor of safety to MOE.

6.5.10.3 Self weight of beam shall be considered in design.

6.6 Columns

NOTE — The formulae given are for columns with pin end conditions and the length shall be modified suitably with other end conditions.

6.6.1 Solid Columns

Solid columns shall be classified into short, intermediate and long columns depending upon their slenderness ratio (*S/d*) as follows:

- Short columns* — where *S/d* does not exceed 11;
- Intermediate columns* — where *S/d* is between 11 and *K₈*; and
- Long columns* — where *S/d* is greater than *K₈*.

6.6.1.1 For short columns, the permissible compressive stress shall be calculated as follows:

$$f_c = f_{cp}$$

6.6.1.2 For intermediate columns, the permissible compressive stress is calculated by using the following formula:

$$f_c = f_{cp} \left[1 - \frac{1}{3} \left(\frac{S}{K_8 d} \right)^4 \right]$$

6.6.1.3 For long columns, the permissible compressive stress shall be calculated by using the following formula:

$$f_c = \frac{0.329E}{\left(\frac{S}{d} \right)^2}$$

6.6.1.4 In case of solid columns of timber, *S/d* ratio shall not exceed 50.

6.6.1.5 The permissible load on a column of circular cross section shall not exceed that permitted for a square column of an equivalent cross sectional area, where side of square is equal to 0.886 times the diameter.

6.6.1.6 For determining S/d ratio of a tapered column, such as for wooden poles, its least dimension shall be taken as the sum of the corresponding least dimensions at the small end of the column and one-third of the difference between this least dimension at the small end and the corresponding least dimension at the large end, but in no case shall the least dimension for the column be taken as more than one and a half times the least dimension at the small end. The induced stress at the small end of the tapered column shall not exceed the permissible compressive stress in the direction of grain.

6.6.2 Built-up Columns

6.6.2.1 Box column

Box columns shall be classified into short, intermediate and long columns as follows:

- Short columns* — where $\frac{S}{\sqrt{d_1^2 + d_2^2}}$ is less than 8;
- Intermediate columns* — where $\frac{S}{\sqrt{d_1^2 + d_2^2}}$ is between 8 and K_9 ; and
- Long columns* — where $\frac{S}{\sqrt{d_1^2 + d_2^2}}$ is greater than K_9 ;

6.6.2.2 For short columns, the permissible compressive stress shall be calculated as follows:

$$f_c = qf_{cp}$$

6.6.2.3 For intermediate columns, the permissible compressive stress shall be obtained using the following formula:

$$f_c = qf_{cp} \left[1 - \frac{1}{3} \left(\frac{S}{K_9 \sqrt{d_1^2 + d_2^2}} \right)^4 \right]$$

6.6.2.4 For long columns, the permissible compressive stress shall be calculated by using the following formula:

$$f_c = \frac{0.329UE}{\left(\frac{S}{\sqrt{d_1^2 + d_2^2}} \right)^2}$$

6.6.2.5 The following values of U and q , depending upon plank thickness (t) in [6.6.2.3](#) and [6.6.2.4](#), shall be used for width of plank not greater than 5 times the plank thickness:

<i>Sl No.</i>	<i>t</i> mm	<i>U</i>	<i>q</i>
(1)	(2)	(3)	(4)
i)	25	0.80	1.00
ii)	50	0.60	1.00

6.6.3 Spaced Columns

6.6.3.1 The formulae for solid columns as specified in [6.6.1](#) are applicable to spaced columns with a restraint factor of 2.5 or 3, depending upon distances of end connectors in the column.

NOTE — A restrained factor of 2.5 for location of centroid group of fasteners at $S/20$ from end and 3 for location at $S/10$ to $S/20$ from end shall be taken.

6.6.3.2 For intermediate spaced column, the permissible compressive stress shall be:

$$f_c = f_{cp} \left[1 - \frac{1}{3} \left(\frac{S}{K_{10}d} \right)^4 \right]$$

6.6.3.3 For long spaced columns, the permissible compressive stress shall be:

$$f_c = \frac{0.329E \times 2.5}{(S/d)^2}$$

6.6.3.4 For individual members of spaced columns, S/d ratio shall not exceed 80.

6.6.4 Compression members shall not be notched. When it is necessary to pass services through such a member, this shall be effected by means of a bored hole provided that the local stress is calculated and found to be within the permissible stress specified. The distance from the edge of the hole to the edge of the member shall not be less than one quarter of width of the face.

6.7 Tension Member (Ties)

The stress is axial tension parallel to grain of wood shall be calculated on the basis of net-section area which shall not exceed the safe permissible value in tension parallel to grain:

$$f_{at} = \frac{\text{Maximum force prevailing}}{\text{Area of cross section}} \leq f_t$$

6.8 Structural Members Subject to Bending and Axial Stresses

6.8.1 Structural members subjected both to bending and axial compression shall be designed to comply with the following:

$$\frac{f_{ac}}{f_c} + \frac{f_{ab}}{f_b} \text{ is not greater than } 1$$

6.8.2 Structural members subjected both to bending and axial tension shall be designed to comply with the following:

$$\frac{f_{at}}{f_t} + \frac{f_{ab}}{f_b} \text{ is not greater than } 1$$

6.9 Timber Roof Truss

Trusses are framed structures in which separate straight members are so arranged and connected at their ends that members form triangles. The external loads cause direct stresses in the members. Essentially a plane structure which is very stiff in the plane of the members but very flexible in every other direction. For members subjected to reversal of stresses, design has to be for one stress and tested for the other applied at the panel points. Eccentricity of meeting members at joint may result in the increased stresses in design.

7 DESIGN OF COMMON STEEL WIRE NAIL JOINTS

7.1 General

Nail jointed timber construction is suitable for light and medium timber framings (trusses, etc) up to 15 m spans. With the facilities of readily available materials and simpler workmanship in mono-chord and split-chord constructions, this type of fabrication has a large scope. Preboring of nail-jointed timber construction is desirable and shall be done in accordance with good practice as in, 6.2.2 and Table 3 of good practice [C-3A(7)].

7.2 Dimensions of Members

The dimension of an individual piece of timber (that is any single member) shall be within the range given below:

- The minimum thickness of the main members in mono-chord construction shall be 30 mm;
- The minimum thickness of an individual piece of members in split-chord construction shall be 20 mm for web members and 25 mm for chord members; and
- The space between two adjacent pieces of timber shall be restricted to a maximum of 3 times the thickness of the individual piece of timber of the chord member. In case of web members, it may be greater for joining facilities.

7.3 No lengthening joint shall preferably be located at a panel point. Generally, not more than two, but preferably one, lengthening joint shall be permitted between the two panel points of the members.

7.4 Specification and Diameter of Nails

7.4.1 The nails used for timber joints shall conform to Part B ‘Building Materials’ of this NBCS. The nails shall be diamond pointed.

7.4.2 The diameter of nail shall be within the limits of one-eleventh to one-sixth of the least thickness of members being connected.

7.4.3 Where the nails are exposed to be saline conditions, common wire nails shall be galvanized.

7.5 Arrangement of Nails in the Joints

The end distances, edge distances and spacing of nails in a nailed joint should be such as to avoid undue splitting of the wood and shall not be less than those given in [7.5.1](#) and [7.5.2](#).

7.5.1 Lengthening Joints

The requirement of spacing of nails in a lengthening joint shall be as given in [Table 14](#) (see also [Fig. 4](#)):

Table 14 Spacing of Nails (Clause 7.5.1)			
Sl No.	Spacing of Nails	Type of Stress in the Joint	Requirement <i>Min</i>
(1)	(2)	(3)	(4)
i)	End distance	Tension compression	12 <i>n</i> 10 <i>n</i>
ii)	In direction of grain	Tension compression	10 <i>n</i> 5 <i>n</i>
iii)	Edge distance	—	5 <i>n</i>
iv)	Between row of nails perpendicular to the grain	—	5 <i>n</i>
NOTES			
1 <i>n</i> is shank diameter of nails.			
2 The 5 <i>n</i> distance between rows perpendicular to the grain may be increased subject to the availability of width of the member keeping edge distance constant.			

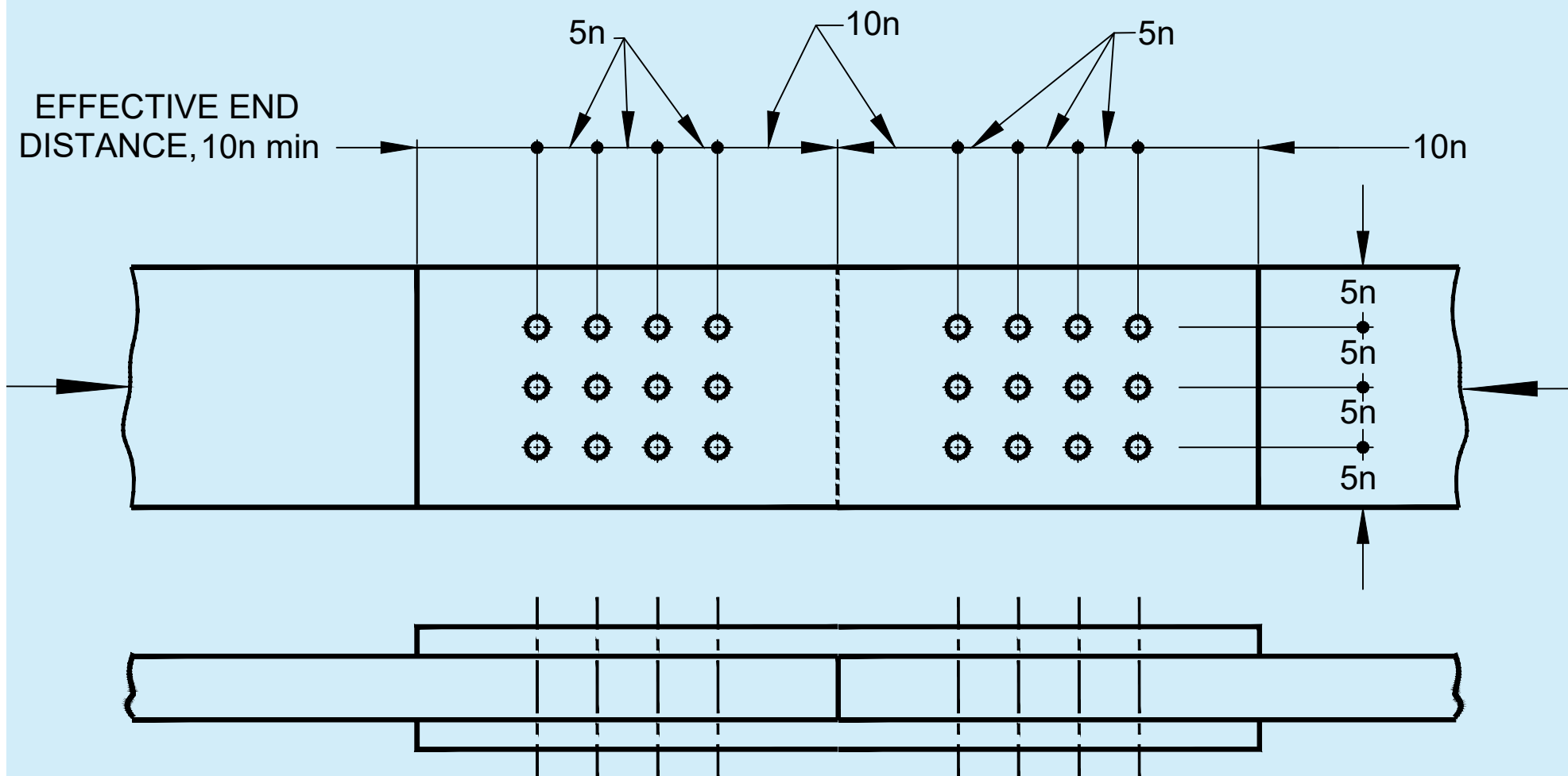


FIG. 4A MONOCHORD TYPE BUTT JOINT SUBJECT TO COMPRESSION

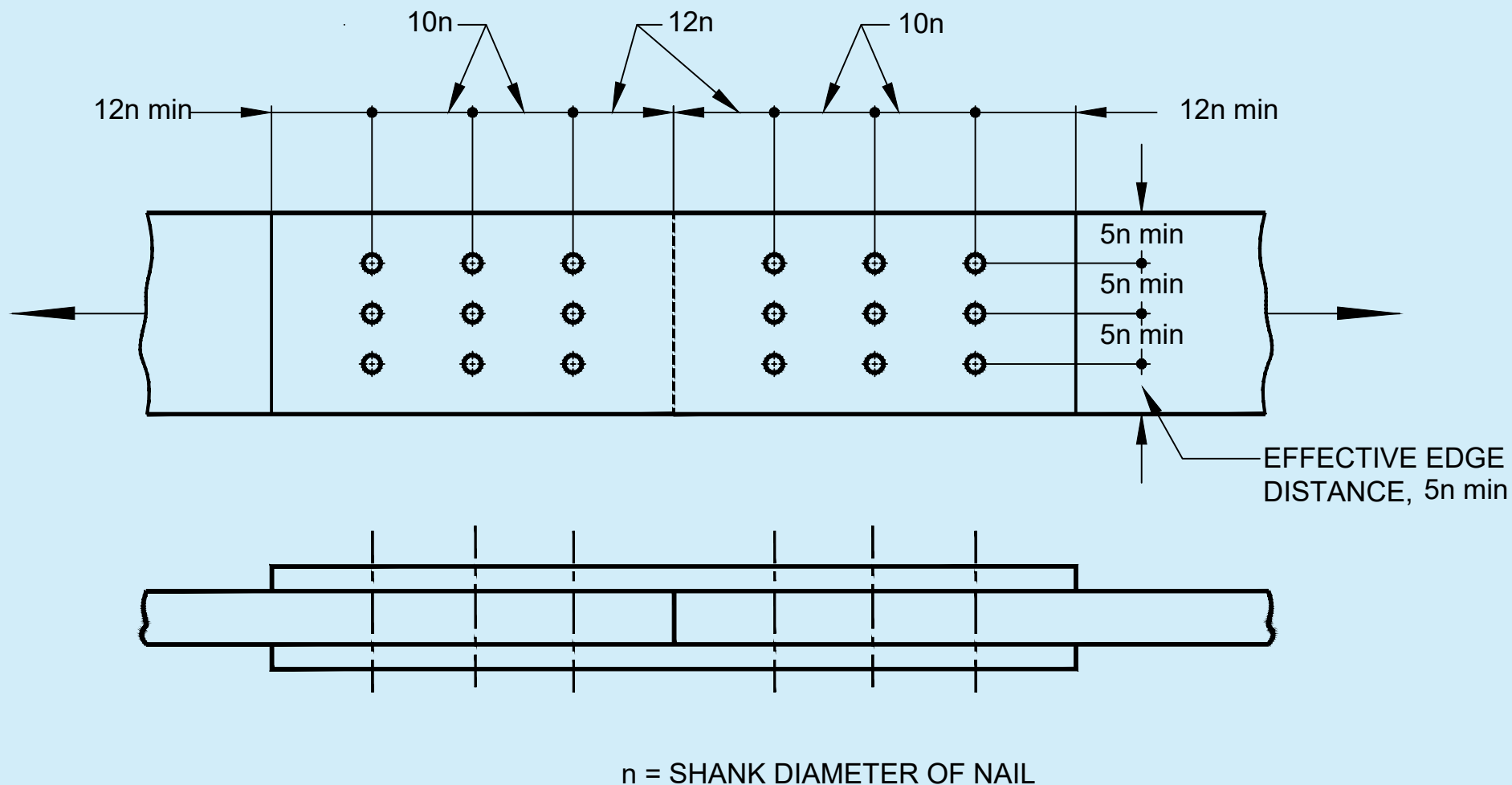


FIG. 4B MONOCHORD TYPE BUTT JOINT SUBJECT TO TENSION

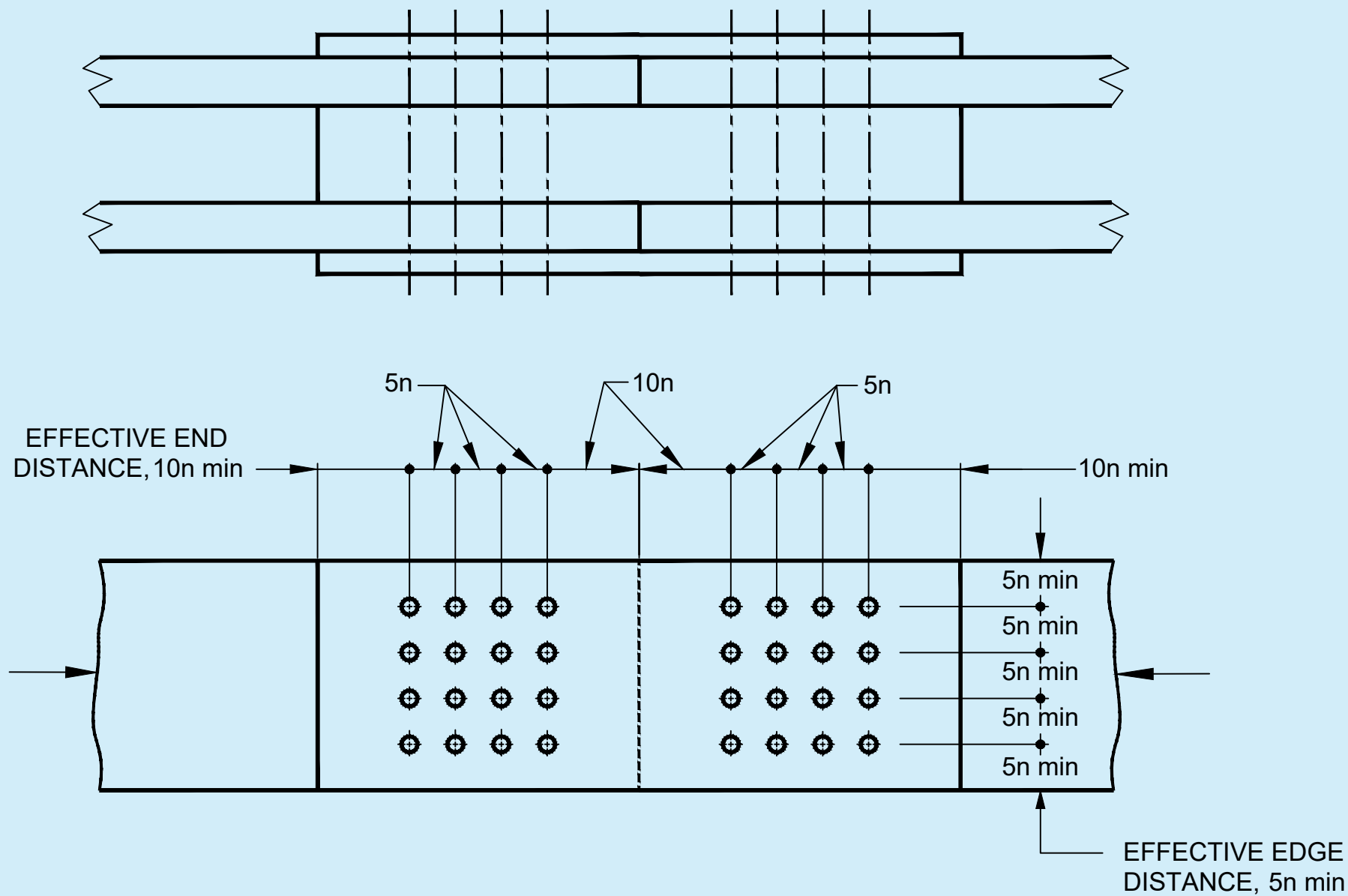
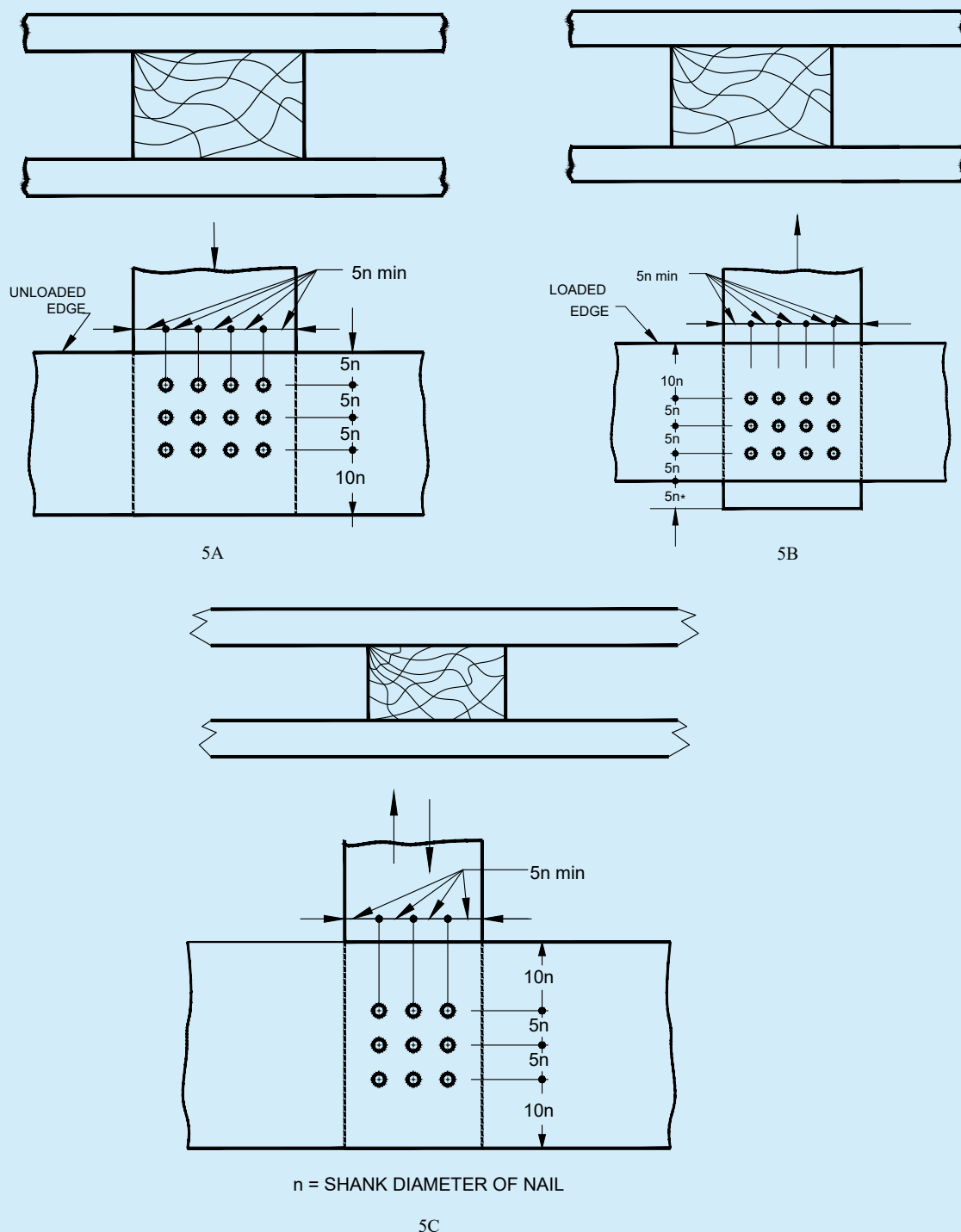


FIG. 4C SPLIT - CHORD TYPE BUTT JOINT SUBJECT TO COMPRESSION



7.5.2 Node Joints

The requirement for spacing of nails in node joints shall be as specified in [Fig. 5](#) where the members are at right angle and as in [Fig. 6](#) where the members are inclined to one another at angles other than 90° and subjected to either pure compression or pure tension.



* $5n$ MAY BE INCREASED TO $10n$, IF THE DESIGNED WIDTH OF CHORD MEMBER PERMITS. OTHERWISE THE END OF THE LOADED WEB MEMBER MAY BE EXTENDED BY $5n$ min

FIG. 5 SPACING OF NAILS WHERE MEMBERS ARE AT RIGHT ANGLES TO ONE ANOTHER

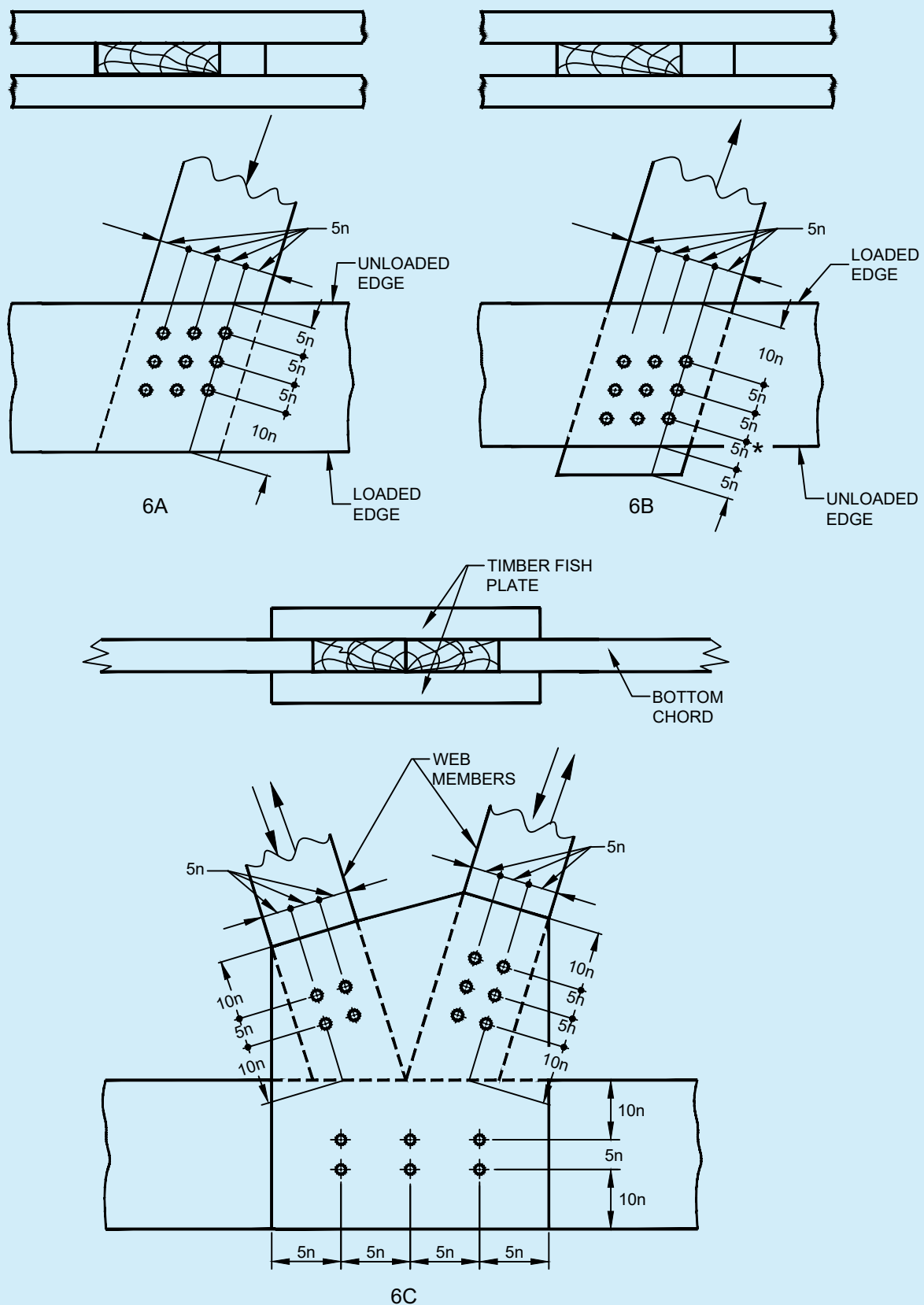


FIG. 6 SPACING OF NAILS AT NODE JOINTS WHERE MEMBERS ARE INCLINED TO ONE ANOTHER

7.6 Penetration of Nails

7.6.1 For a lap joint when the nails are driven from the side of the thinner member, the length of penetration of nails in the thicker member shall be one and a half times the thickness of the thinner member subject to maximum of the thickness of the thicker member.

7.6.2 For butt joints the nails shall be driven through the entire thickness of the joint.

7.7 Design Considerations

7.7.1 Where a number of nails are used in a joint, the allowable load in lateral resistance shall be the sum of the allowable loads for the individual nails, provided that the centroid of the group of these nails lies on the axis of the member and the spacings conform to [7.5](#). Where a large number of nails are to be provided at a joint, they should be so arranged that there are more of rows rather than more number of nails in a row.

7.7.2 Nails shall, as far as practicable, be arranged so that the line of force in a member passes through the centroid of the group of nails. Where this is not practicable, allowance shall be made for any eccentricity in computing the maximum load on the fixing nails as well as the loads and bending moment in the member.

7.7.3 Adjacent nails shall preferably be driven from opposite faces, that is, the nails are driven alternatively from either face of joint.

7.7.4 For a rigid joint, a minimum of 2 nails for nodal joints and 4 nails for lengthening joint shall be driven.

7.7.5 Two nails in a horizontal row are better than using the same number of nails in a vertical row.

7.7.6 The permissible lateral strength (in double shear) of mild steel common wire for different species of timber shall be as per [4.3](#).

7.8 Special Consideration in Nail-Jointed Truss Construction

7.8.1 The initial upward camber provided at the centre of the lower chord of nail-jointed timber trusses shall be not less than 1/200 of the effective span for timber structures using seasoned wood and 1/100 for unseasoned or partially seasoned wood.

7.8.2 The total combined thickness of the gusset or splice plates on either side of the joint in a mono-chord type construction shall not be less than one and a half times the thickness of the main members subject to a minimum thickness of 25 mm of individual gusset plate.

NOTES

1 The allowable load or lateral strength values of nails shall be those as given in [Table 2](#) and [Table 3](#).

2 The strength data for joints given in the Subsection apply to gusset or splice or fish plates of solid wood; however, materials other than solid wood may be used for gusset when field tests are made and their strength requirements have been established.

7.8.3 The total combined thickness of all spacer blocks or plates or both including outer splice plates, at any joint in a split-chord type construction shall not be less than one and a half times the total thickness of all the main members at that joint.

7.9 Fabrication

The fabrication of nail-jointed timber construction shall be done in accordance with good practice [C-3A(7)].

8 DESIGN OF NAIL LAMINATED TIMBER BEAMS

8.1 Method of Arrangement

8.1.1 The beam shall be made up of 20 mm to 30 mm thick planks placed vertically with joints staggered in the adjoining planks with a minimum distance of 300 mm. The planks are laminated with the help of wire nails at regular intervals to take up horizontal shear developed in the beam besides keeping the planks in position (see [Fig. 7](#)).

8.1.2 The advantage in laminations lies in dimensional stability, possible dispersal of defects and better structural performance.

8.2 Sizes of Planks and Beams

8.2.1 The plank thickness for fabrication of nailed laminated beams recommended are 20 mm, 25 mm and 30 mm.

8.2.2 In case of nailed laminated timber beam the maximum depth and length of planks shall be limited to 250 mm and 2 000 mm, respectively.

8.2.3 In order to obtain the overall width of the beam, the number and thickness of planks to form vertical nailed laminated beams, and also type and size of wire nail shall be as mentioned in [Table 15](#). The protruding portion of the nail shall be cut off or clenched across the grains.

8.3 Design Considerations

8.3.1 Nail laminated beams shall be designed in accordance with [6](#) and provisions in the good practice [C-3A(8)].

8.3.1.1 The deflection in the case of nailed laminated timber beams, joists, purlins, battens and other flexural members supporting brittle materials like gypsum, ceiling slates, tiles and asbestos sheets shall not exceed 1/480 of the span. The deflection in case of other flexural members shall not exceed 1/360 of the span in the case of beams and joists and 1/225 of the freely hanging length in case of cantilevers.

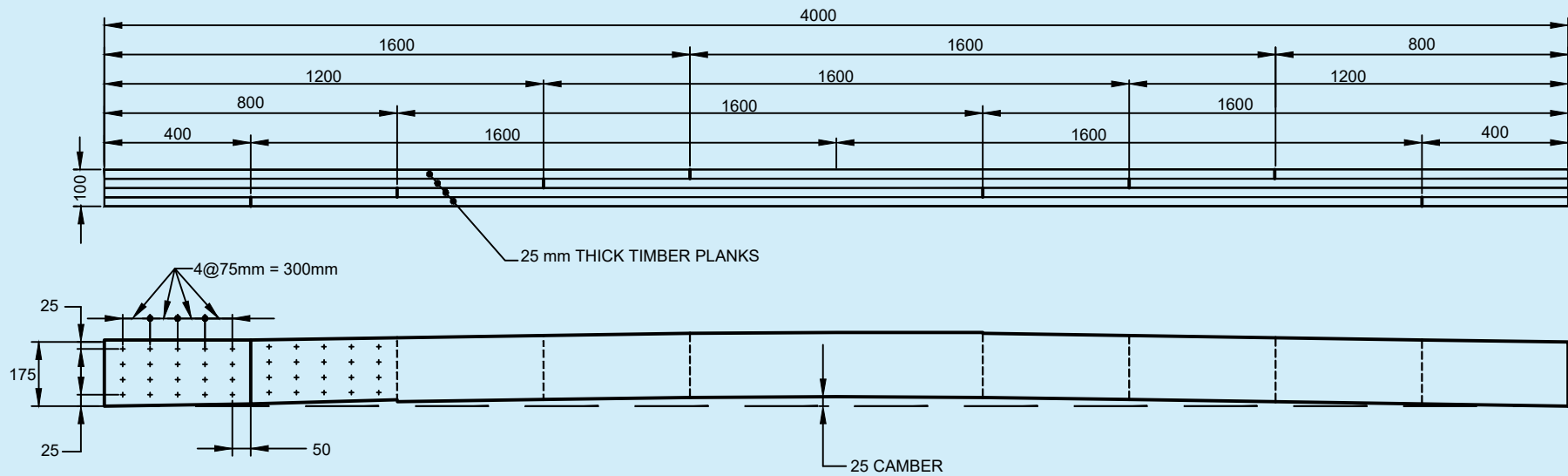
8.3.2 Permissible lateral strength of diamond pointed mild steel wire nails shall be as given in [Table 2](#) and [Table 3](#) for Indian species of timber, which shall apply to nails that have their points cut flush with the faces. For nails clenched across the grains the strength may be increased by 20 percent over the values for nails with points cut flush.

Table 15 Number and Size of Planks and Nails for Nailed Laminated Beams

(Clause [8.2.3](#))

Sl No.	Overall Width of Beam mm	No. of Planks	Thickness of Each Plank mm	Type and Size of Nail to be Used mm
(1)	(2)	(3)	(4)	(5)
i)	50	2	25	80 long 3.55 dia
ii)	60	3	20	- do -
iii)	70	3	(2 × 25) (1 × 20)	- do -
iv)	80	4	20	100 long 4.0 dia
v)	90	3	30	- do -
vi)	100	4	25	125 long 5.0 dia
vii)	110	4	(3 × 30) (1 × 20)	- do -
viii)	120	4	30	- do -
ix)	150	5	30	150 long 5.0 dia

NOTE — A number of combinations of the different thickness of planks may be adopted as long as the minimum and maximum thickness of the planks are adhered to.



All dimensions in millimetres.

FIG. 7 PLAN AND ELEVATION OF A TYPICAL NAILED LAMINATED TIMBER BEAM

8.3.3 Arrangement of Nails

8.3.3.1 A minimum number of four nails in a vertical row at regular interval not exceeding 75 mm to take up horizontal shear as well as to keep the planks in position shall be used. Near the lengthening joints of the planks this distance may, however, be limited to 50 mm instead of 75 mm.

8.3.3.2 Shear shall be calculated at various points of the beam and the number of nails required shall be accommodated within the distance equal to the depth of the beam, with a minimum of 4 nails in a row at a standard spacing as shown in [Fig. 8](#).

8.3.3.3 If the depth of the beam is more, then the vertical intermediate spacing of nails may be increased proportionately.

8.3.3.4 If the nails required at a point are more than that can be accommodated in a row, then these shall be provided lengthwise of the beam within the distance equal to the depth of the beam at standard lengthwise spacing.

8.3.3.5 For nailed laminated beam minimum depth of 100 mm for 3.55 mm and 4 mm diameter nails and 125 mm for 5 mm diameter nails shall be provided.

8.4 Fabrication

The fabrication of nailed laminated timber beams shall be done in accordance with good practice [C-3A(8)].

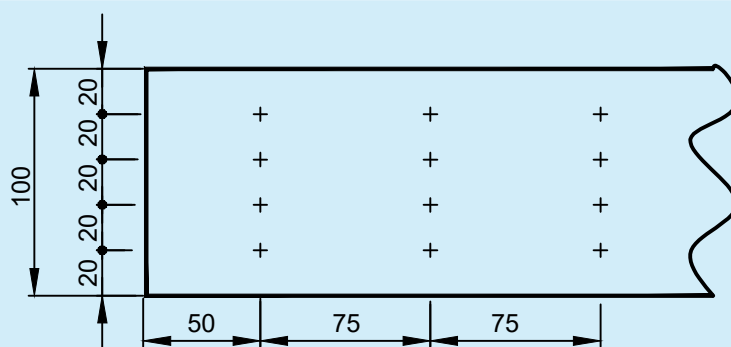


FIG. 8A FOR 3.55 mm AND 4 mm DIAMETER NAILS

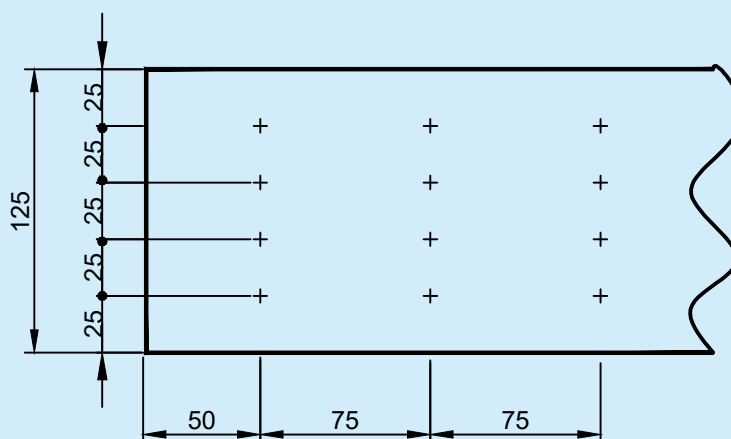


FIG. 8B FOR 5 mm DIAMETER NAILS

All dimensions in millimetres.

FIG. 8 STANDARD LENGTHWISE SPACING IN NAILED LAMINATED BEAM

9 DESIGN OF BOLTED CONSTRUCTION JOINTS

9.1 General

Bolted joints suit the requirements of prefabrication in small and medium span timber structures for speed and economy in construction. Bolt jointed construction units offer better facilities as regards to workshop ease, mass production of components, transport convenience and re-assembly at site of work particularly in defence sector for high altitudes and far off situations. Designing is mainly influenced by the species, size of bolts, moisture conditions and the inclination of loadings to the grains. In principle, bolted joints follow the pattern of riveted joints in steel structures.

9.2 Design Considerations

9.2.1 Bolted timber construction shall be designed in accordance with [6](#). The concept of critical section, that is, the net section obtained by deducting the projected area of bolt-holes from the cross-sectional area of member is very important for the successful design and economy in timber.

9.2.2 Bolt Bearing Strength of Wood

The allowable load for a bolt in a joint consisting of two members (single shear) shall be taken as one half the allowable loads calculated for a three member joint (double shear) for the same t'/d_3 ratio. The percentage of safe working compressive stress of timber on bolted joints for different t'/d_3 ratios shall be as given in [Table 16](#).

Table 16 Percentage of Safe Working Compressive Stress of Timber for Bolted Joints in Double Shear (Clause 9.2.2)			
Sl No.	t'/d_3 Ratio	Stress Percentage	
		Parallel to Grain λ_1	Perpendicular to Grain λ_2
(1)	(2)	(3)	(4)
i)	1.0	100	100
ii)	1.5	100	96
iii)	2.0	100	88
iv)	2.5	100	80
v)	3.0	100	72
vi)	3.5	100	66
vii)	4.0	96	60
viii)	4.5	90	56
ix)	5.0	80	52
x)	5.5	72	49
xi)	6.0	65	46
xii)	6.5	58	43
xiii)	7.0	52	40
xiv)	7.5	46	39
xv)	8.0	40	38
xvi)	8.5	36	36
xvii)	9.0	34	34
xviii)	9.5	32	33

Table 16 (Concluded)			
Sl No.	t'/d_3 Ratio	Stress Percentage	
		Parallel to Grain λ_1	Perpendicular to Grain λ_2
(1)	(2)	(3)	(4)
xix)	10.0	30	31
xx)	10.5	—	31
xxi)	11.0	—	30
xii)	11.5	—	30
xiii)	12.0	—	28

9.2.2.1 Where a number of bolts are used in a joint, the allowable loads shall be the sum of the allowable loads for the individual bolts.

9.2.2.2 The factors for different bolt diameter used in calculating safe bearing stress perpendicular to grain in the joint shall be as given in [Table 17](#).

9.2.3 Dimensions of Members

- The minimum thickness of the main member in mono-chord construction shall be 40 mm;
- The minimum thickness of side members shall be 20 mm and shall be half the thickness of main members; and
- The minimum individual thickness of spaced member in split-chord construction shall be 20 mm and 25 mm for webs and chord members, respectively.

9.2.4 Bolts and Bolting

- The diameter of bolt in the main member shall be so chosen to give larger slenderness (t'/d_3) ratio of bolt;
- There shall be more number of small diameter bolts rather than small number of large diameter bolts in a joint;
- A minimum of two bolts for nodal joints and four bolts for lengthening joints shall be provided;
- There shall be more number of rows rather than more bolts in a row;
- The bolt holes shall be of such diameter that the bolt can be driven easily; and
- Washers shall be used between the head of bolt and wood surface as also between the nut and wood.

Table 17 Bolt Diameter Factor		
(Clause 9.2.2.2)		
Sl No.	Diameter of Bolt mm	Diameter Factor (d_f)
(1)	(2)	(3)
i)	6	5.70
ii)	10	3.60
iii)	12	3.35
iv)	16	3.15
v)	20	3.05
vi)	22	3.00
vii)	25	2.90

9.3 Arrangement of Bolts

9.3.1 The following spacings in bolted joints shall be followed (see [Fig. 9](#)):

- a) *Spacing of bolts in a row* — For parallel and perpendicular to grain loading = $4 d_3$;
- b) *Spacing between rows of bolts*:
 - 1) *For perpendicular to grain loading* — $2.5 d_3$ to $5 d_3$ ($2.5 d_3$ for t'/d_3 ratio of 2 and $5 d_3$ for t'/d_3 ratio of 6 or more. For ratios between 2 and 6 the spacing shall be obtained by interpolation.
 - 2) *For parallel to grain loading* — At least $(N - 4) d_3$ with a minimum of $2.5 d_3$. Also governed by net area at critical section which should be 80 percent of the total area in bearing under all bolts.
- c) *End distance* — $7d_3$ for softwoods in tension, $5 d_3$ for hardwoods in tension and $4 d_3$ for all species in compression; and
- d) *Edge distance*:
 - 1) For parallel to grain loading $1.5 d_3$ or half the distance between rows of bolts, whichever is greater.
 - 2) For perpendicular to grain loading, the loaded edge distance shall be at least $4 d_3$.

9.3.2 For inclined members, the spacing given above for perpendicular and parallel to grain of wood may be used as a guide and bolts arranged at the joint with respect to loading direction.

9.3.3 The bolts shall be arranged in such a manner so as to pass the centre of resistance of bolts through the intersection of the gravity axis of the members.

9.3.4 Staggering of bolts shall be avoided as far as possible in case of members loaded parallel to grain of wood. For loads acting perpendicular to grain of wood, staggering is preferable to avoid splitting due to weather effects.

9.3.5 Bolting

The bolt holes shall be bored or drilled perpendicular to the surface involved. Forcible driving of the bolts shall be avoided which may cause cracking or splitting of members. A bolt hole of 1.0 mm oversize may be used as a guide for preboring.

9.3.5.1 Bolts shall be tightened after one year of completion of structure and subsequently at an interval of two to three years.

9.4 Outline for Design of Bolted Joints

Allowable load on one bolt (unit bearing stress) in a joint with wooden splice plates shall not be greater than value of P , R , F as determined by one of the following equations:

- a) *For loads parallel to grain*:

$$P = f_{cp} a \lambda_1$$

- b) *For loads perpendicular to grain*:

$$R = f_{cn} a \lambda_2 d_f$$

- c) *For loads at an angle to grain*:

$$F = \frac{PR}{P \sin^2 \theta + R \cos^2 \theta}$$

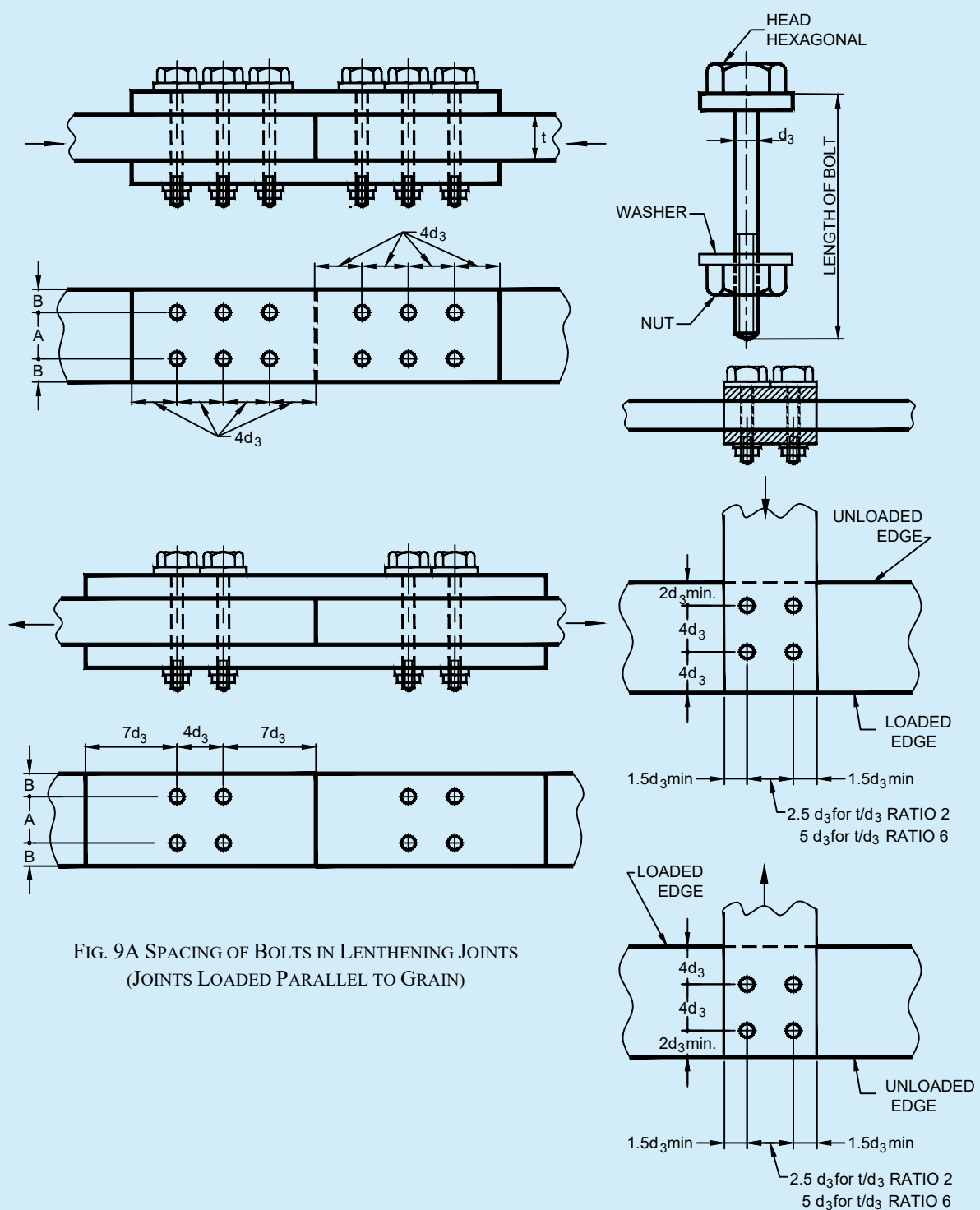


FIG. 9A SPACING OF BOLTS IN LENTHENING JOINTS
(JOINTS LOADED PARALLEL TO GRAIN)

FIG. 9B SPACING OF BOLTS AT NODE JOINTS

FIG. 9 TYPICAL SPACING OF BOLTS IN STRUCTURAL JOINTS

9.5 Fabrication

The fabrication of bolt jointed construction shall be in accordance with good practice [C-3A(9)].

To meet the specific requirements of 'all-timber construction' for structures where chemical corrosion of steel elements is a problem, and in order to erect anti-magnetic laboratories, sheds, etc, design should be made on the basis of mild steel bolt joints with the following reductions for timber bolts and bamboo pins:

Allowable safe working load per hardwood timber bolt	= 66 percent of the safe working load per mild steel bolt
Safe working load per solid bamboo pin	= 56 percent of the working load per mild bamboo per steel bolt

10 DESIGN OF TIMBER CONNECTOR JOINTS

10.1 In large span structures, the members have to transmit very heavy stresses requiring stronger jointing techniques with metallic rings or wooden disc-dowels. Improvised metallic ring connector is a split circular band of steel made from mild steel pipes. This is placed in the grooves cut into the contact faces of the timber members to be joined, the assembly being held together by means of a connecting bolt (see [Fig. 10](#)).

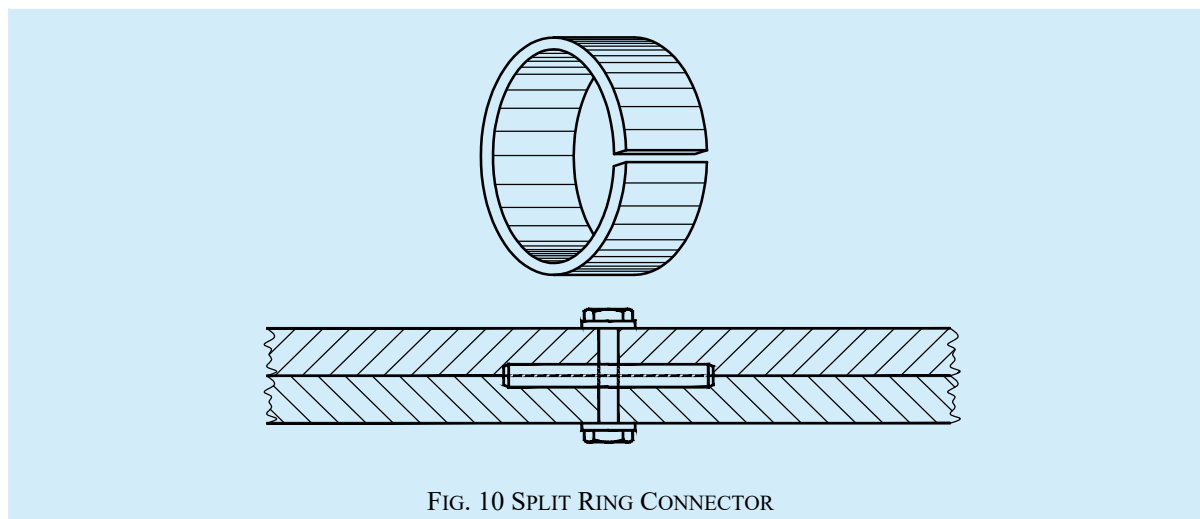


FIG. 10 SPLIT RING CONNECTOR

10.1.1 Dimensions of Members

Variation of thickness of central (main) and side members affect the load carrying capacity of the joint. The thickness of main member shall be at least 57 mm and that of side member 38 mm with length and width of members governed by placement of connector at joint.

The metallic connector shall be so placed that the loaded edge distance is not less than the diameter of the connector and the end distance not less than 1.75 times the diameter on the loaded side.

10.1.2 Design Considerations

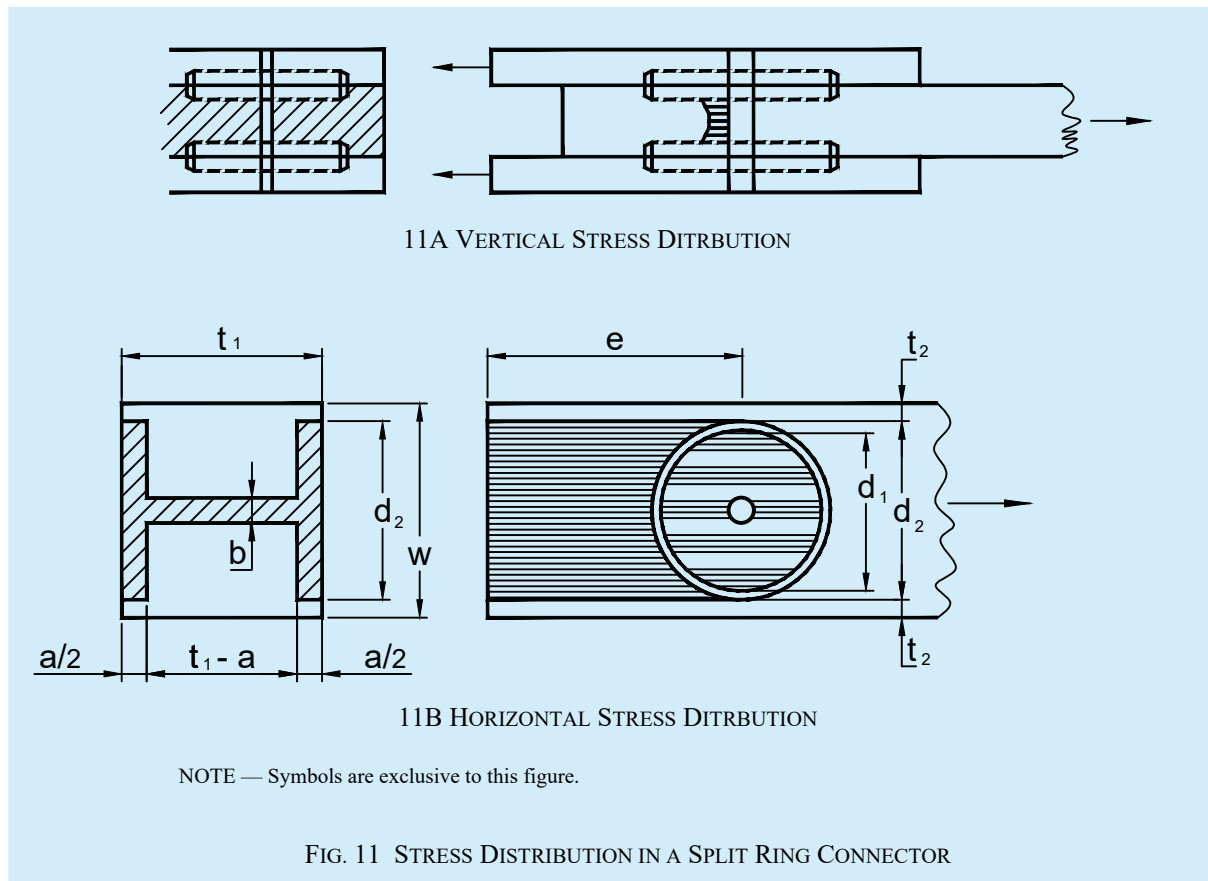
The primary stresses in a split ring connector joint under tension are illustrated in [Fig. 11](#). The shaded areas represent the part of wood in shear, compression and tension. Related formulae for the same are given in [10.1.2.1](#).

For fabrication of structural members, a hole of the required size of the bolt is drilled into the member and a groove is made on the contact faces of the joint.

Theoretical safe loads in design shall be corroborated with sample tests done in accordance with a [C-3A(10)].

NOTE — A pilot study on determination of strength of improvised split ring connector joint in a compression test for a specific design problem conducted using *Mesua ferrea* (Mesua), yielded the data which is given below, for guidance:

Sl No.	No. and Diameter of Ring Used in a Joint		No. and Size of Bolt Used in a Joint		Side Member		Central Member		Load Direction with respect to Grains of Wood	End Distance	Inter-mediate Distance	Load Per Pair of Connector
	No.	Size mm	No.	Size mm	Thickness mm	Width mm	Thickness mm	Width mm		mm	mm	kgf
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
i)	2	63	1	12 × 125	31	92	38	92	Parallel	63	–	3 930
ii)	2	63	1	12×125	31	92	38	92	Parallel	75	–	4 185
iii)	2	63	1	12 × 150	38	92	63	92	Parallel	75	–	4 010
iv)	2	63	1	12 × 150	38	117	63	117	Parallel	75	–	4 450
v)	2	63	1	12 × 125	31	138	38	92	Perpendicular	69	–	2 520
vi)	2	63	1	12 × 125	38	138	38	92	Perpendicular	69	–	3 515
vii)	2	100	1	19 × 175	38	138	66	138	Parallel	100	–	7 075
viii)	2	100	1	19 × 175	38	138	66	138	Parallel	125	–	7 370
ix)	2	100	1	19 × 175	41	138	75	138	Parallel	100	–	7 220
x)	2	100	1	19 × 175	41	138	75	138	Parallel	125	–	7 645
xi)	4	100	2	19 × 200	38	138	66	138	Parallel	100	150	5 655
xii)	4	100	2	19 × 200	41	138	75	138	Parallel	100	150	5 925
xiii)	4	100	2	19 × 200	41	138	75	138	Parallel	125	200	7 135



10.1.2.1 Strength of joint is controlled by one or another or combination of three properties:

a) Tension area (see [Fig. 11A](#)):

$$t_1 w - \left[2 \left(\frac{a d_2}{2} \right) + b(t_1 - a) \right]$$

b) Compression area (see [Fig. 11B](#)):

$$2 \left(\frac{a d_2}{2} \right) + b(t_1 - a)$$

c) Shear area (see [Fig. 11B](#)):

- 1) Within core : $2 \left(\frac{\pi d_1^2}{4} \right)$
- 2) Below core : $2 \left[d_2 e - \frac{1}{2} \left(\frac{\pi d_2^2}{4} \right) + 2 \left(\frac{a e}{2} \right) \right]$

where

- d_1 = inside diameter of connector;
- t_1 = thickness of member;
- w = width of member;
- a = depth of connector;
- d_2 = outside diameter of connector;
- b = diameter of bolt;
- e = end distance from centre of connector to end of member; and
- t_2 = thickness of metal.

10.2 Wooden Disc-Dowel

10.2.1 It is a circular hardwood disc generally tapered each way from the middle so as to form a double conical frustum. Such a disc is made to fit into preformed holes (recesses), half in one member and the other half in another, the assembly being held by one mild steel bolt through the centre of the disc to act as a coupling for keeping the jointed wooden members from spreading apart.

As timber always fails by shearing or crushing in front of bolts, the function of disc-dowel incorporated in the joint is to distribute the shearing stresses on a larger area of wood so as to transmit the full amount of stresses.

10.2.2 Dimensions of Members

The thickness of dowel may vary from 25 mm to 35 mm and diameter from 50 mm to 150 mm. The diameter of dowel shall be 3.25 to 3.50 times the thickness.

The edge clearance shall range from 12 mm to 20 mm as per the size of the dowel. The end clearance shall be at least equal to the diameter of dowel for joints subjected to tension and three-fourth the diameter for compression joints. Disc dowel shall be turned from radial (quarter sawn) planks of seasoned material.

10.2.3 Choice of Species

Wood used for making dowels shall be fairly straight grained, free from excessive liability to shrink and warp, and retain shape well after seasoning and having high shear strength. Species recommended include:

- a) Babul;
- b) Dhaman;
- c) Irul;
- d) Sissoo;
- e) Rose wood;
- f) Sandal;
- g) Axle-wood;
- h) Padauk; and
- j) Yon.

NOTE – Data on the above species are as per [Table 1](#).

10.2.4 Design Considerations

The forces on dowel in a lap joint and butt joint are illustrated in [Fig. 12](#). Dowel is subjected to shearing at the mid-section, and compression along the grain at the bearing surfaces. For equal strength in both the forces, formula equations are given in the [Fig. 12](#) to determine the size of dowel.

The making of wooden discs may present some problems in the field, but they may be made in small workshop to the specifications of the designer. This is also economically important. Once the wood fittings are shop tailored and made, the construction process in the field is greatly simplified.

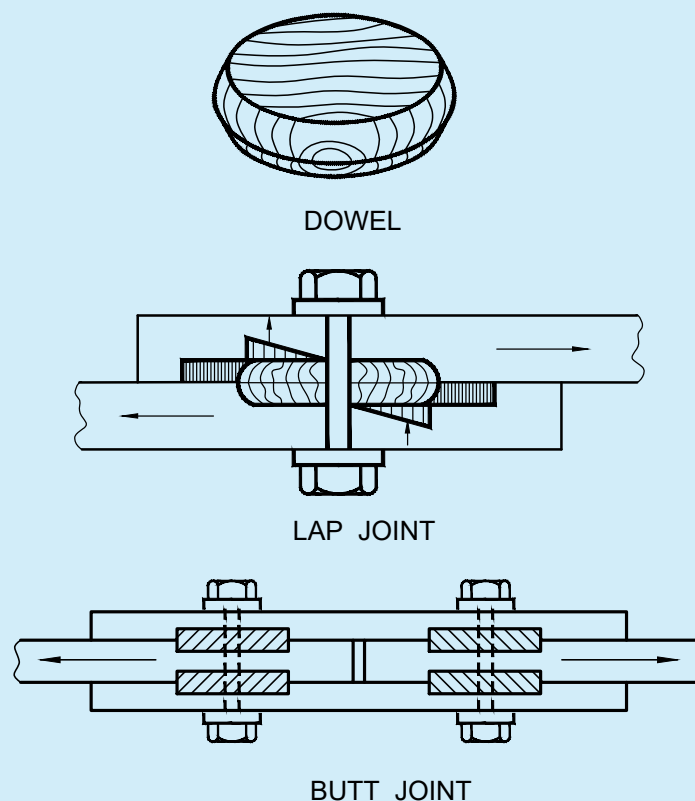
Theoretical safe loads in design shall be confirmed through sample tests.

NOTE — Some experimental studies have indicated the following safe loads, in kN, for dowels bearing parallel to the grain, which are given here for guidance only.

Sl No.	Species		Safe Loads for Dowels Bearing Parallel to the Grain, in kN					
	Members	Dowels	Size of Dowel (mm)					
			62 × 25	75 × 25	87 × 25	100 × 31	112 × 37	125 × 37
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	Sal	Babul	6.671	9.810	13.342	17.805	22.269	27.566
ii)	Sal	Sissoo	5.346	7.554	10.251	13.342	16.922	20.895

10.2.4.1 Rigid specifications regarding use of particular planks for fabrication of wooden disc-dowels, and the specific alignment of grains to be in line with grains of members forming the joint, can be relaxed. Application of appropriate factors in design of such joints whatever be the angle of load and members, etc, would eliminate risks involved by providing 60 percent more number of dowels at a joint. Thus, decrease the calculated load per dowel to 60 percent.

10.2.4.2 Analysis of variance for safe load in case of dowel joints made with MS bolt, timber bolt, bamboo pin as joiners showed no significant difference. Thus, the concept of 'all-timber construction' hold good for chemical industries where corrosion due to acid fumes is a recurring problem.



Lap joint — Bolt in simple tension due to clockwise turning moment on dowel.

Butt joint — No tilting moment in dowel due to balancing effect [dowels are in shear (no bending, shearing and tensile stress on bolts)]

Size of dowel for equal strength in both shearing and bearing.

$$\frac{\pi d^2}{4} \times S = d \times \frac{t}{2} \times c$$

where

d = mid diameter of the dowel;

t = thickness of dowel;

s = safe working stress in shear along grain; and

c = safe compressive stress along grain.

NOTE — Symbols are exclusive to this figure.

FIG. 12 DISTRIBUTION OF FORCES IN DOWEL JOINT

11 GLUED LAMINATED CONSTRUCTION

11.1 Developments in the field of synthetic adhesive have brought gluing techniques within the range of engineering practice. Timber members of larger cross-sections and long lengths can be fabricated from small sized planks by the process of glulam. The term glued laminated timber construction as applied to structural members refers to various laminations glued together, either in straight or curved form, having grain of all laminations essentially parallel to the length of the member.

11.1.1 Choice of Glue

The adhesive used for glued laminated assembly are 'gap filling' type. A 'filler' in powder form is introduced in the adhesive. Structural adhesives are supplied either in powder form to which water is added or in resin form to which a hardener or catalyst is added. For choice of glues, reference may be made to good practice [C-3A(11)]. However, it is important that only boiling water proof (BWP) grade adhesives shall be used for fabrication of glulam in tropical, high humid climates like India.

11.1.2 Manufacturing Schedule

In absence of a systematic flow-line in a factory, provisions of intermediate technology shall be created for manufacturing structural elements. The schedule involves the following steps:

- a) Drying of planks;
- b) Planning;
- c) End-jointing by scarfs or fingers;
- d) Machining of laminations;
- e) Setting up dry assembly of structural unit;
- f) Application of glue;
- g) Assembly and pressing the laminations;
- h) Curing the glue lines, as specified; and
- j) Finishing, protection and storage.

11.1.3 The information with regard to block shear test results on glued timber joints is given in [Table 18](#).

Table 18 Block Shear Test Results on Glued Timber Joints				
(Clause 11.1.3)				
Sl No.	Timber Species	Ultimate Shear Stress (Inside Location) N/mm ²		
		Solid Timber Along the Grain (Un-Jointed)	With Urea Formaldehyde	With Phenol Formaldehyde
(1)	(2)	(3)	(4)	(5)
i)	<i>Abies pindrow</i> (Fir)	5.984	7.652	9.614
ii)	<i>Acacia catechu</i> (Khair)	15.107	7.456	—
iii)	<i>Acacia milotica</i> (Babul)	14.126	5.199	3.237
iv)	<i>Albizia odoratissima</i> (Karasiris)	15.009	14.126	5.984
v)	<i>Azadirachta indica</i> (Neem)	12.557	7.259	8.240
vi)	<i>Castanopsis hystrix</i> (Chestnut)	8.044	1.668	—
vii)	<i>Cedrus deodar</i> (Deodar)	6.867	4.218	3.237

Table 18 (Concluded)

Sl No.	Timber Species	Ultimate Shear Stress (Inside Location) N/mm ²		
		Solid Timber Along the Grain (Un-Jointed)	With Urea Formaldehyde	With Phenol Formaldehyde
(1)	(2)	(3)	(4)	(5)
viii)	<i>Dalbergia sissoo</i> (Sissoo)	10.595	15.107	9.320
ix)	<i>Eucalyptus eugenioides</i> (Eucalyptus)	12.164	10.987	5.101
x)	<i>Grewia tiliefolia</i> (Dhawan)	8.927	10.889	6.475
xi)	<i>Hopea utilis</i> (Karunkangoo)	14.813	9.320	—
xii)	<i>Hopea glabra</i> & <i>H. parviflora</i> (Hopea)	15.009	13.832	0.491
xiii)	<i>Mangifera Indica</i> (Mango)	9.418	9.418	3.826
xiv)	<i>Madhuca longifolia</i> (Mahua)	9.810	0.883	7.456
xv)	<i>Mesua ferrea</i> (Mesua)	12.164	1.079	2.256
xvi)	<i>Morus alba</i> (Mulberry)	9.810	13.734	8.142
xvii)	<i>Pinus wallichiana</i> (Kail)	5.494	6.377	7.259
xviii)	<i>Pinus roxburghii</i> (Chir)	6.082	6.769	7.750
xix)	<i>Poecilone indicum</i> (Ballagi)	14.911	5.297	5.984
xx)	<i>Pterocarpus mersupium</i> (Bijasal)	9.221	5.003	6.671
xxi)	<i>Quercus spp.</i> (Oak)	8.142	10.791	5.396
xxii)	<i>Ougeinia oojeinensis</i> (Sanetan)	13.440	5.396	9.516
xxiii)	<i>Shorea robusta</i> (Sal) M.P	9.123	5.003	—
xxiv)	<i>Shorea robusta</i> (Sal) UP	12.361	5.396	1.472
xxv)	<i>Sterospermum chelonoides</i> (Padriwood)	11.674	9.221	8.829
xxvi)	<i>Syzygium cuzainii</i> (Jaman)	10.791	9.418	10.301
xxvii)	<i>Terminalia alata</i> (Sain)	9.320	13.538	8.142
xxviii)	<i>Tectona grandis</i> (Teak)	10.987	5.984	6.377
xxix)	<i>Terminalia myriocarpa</i> (Hollock)	8.339	9.221	9.810
xxx)	<i>Toona ciliata</i> (Toon)	6.867	4.905	6.180

NOTE — The information given in the above table is based on the block shear test results on glued timber joints conducted on the basis of BS 4169 : 1970 'Specification for manufacture of glued-laminated timber structural members' (since revised and subsequently superseded by BS EN 392 : 1995, BS EN 391 : 1995, BS EN 386 : 1995) and the failures observed with regard to glue line and the wood. Ultimate shear stresses for 29 species were studied keeping design glue shear strength at least one-third of ultimate. Glue mixing, spreading, pressing and curing was done as per the instructions of adhesive manufacturers under improvised techniques locally developed. Sustainable period before delamination of glue bond (urea formaldehyde) in weathering tests was recorded for over a year. Limited number of species survived with most of them failed within a period of six months.

11.2 Design of Glued Laminated Beams

11.2.1 General

Glued laminated structural members shall be fabricated only where there are adequate facilities for accurate sizing and surfacing of planks, uniform application of glue, prompt assembly, and application of adequate pressure and prescribed temperature for setting and curing of the glue. Design and fabrication shall be in accordance with established engineering principles and good practice. A glued laminated beam is a straight member made from a number of laminations assembled both ways either horizontally or vertically (see Fig. 13). While vertical laminations have limitations in restricting the cross section of a beam by width of the plank, horizontally laminated section offers wider scope to the designer in creating even the curved members. Simple straight beams and joists are used for many structures from small domestic rafters or ridges to the light industrial structures.

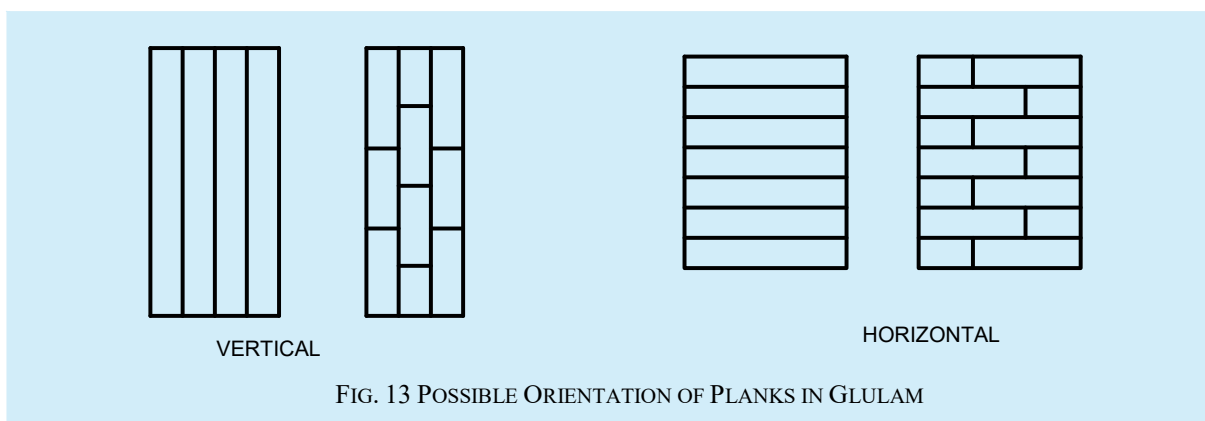


FIG. 13 POSSIBLE ORIENTATION OF PLANKS IN GLULAM

11.2.2 Design

The design of glue laminated wood elements shall be in accordance with good engineering practice and shall take into consideration the species and grade of timber used, presence of defects, location of end joints in laminations, depth of beams and moisture contents expected while in service. Beams of large spans shall be designed with a suitable camber to assist in achieving the most cost effective section where deflection governs the design. The strength and stiffness of laminated beams is often governed by the quality of outer laminations. Glued laminated beams can be tapered to follow specific roof slopes across a building and/or to commensurate with the varying bending moments. Glulam allows the engineers to mix grades within a beam. Obviously, the lower tensile portion should be of high grade material. However, it is possible to incorporate a considerable volume of lower grade material near the neutral axis taking care of the horizontal shear stress. Similarly, species can also be mixed depending upon the modulus of elasticity.

Straight beams that will be simply supported should be cambered during fabrication to allow for deflection due to $\frac{\text{span}}{400}$ dead loads over a period of time. The usual camber is $\frac{\text{span}}{400}$, unless otherwise specified.

11.2.2.1 Design outline for horizontally laminated beams

A design outline for horizontal laminated beams, specifying data required and a possible solution is given below.

a) Data required:

- 1) Span (L), loads including self weight of beam (W), species of timber, thickness of lamination proposed; and
- 2) Mechanical properties of timber species (permissible extreme fibre stress in bending-tension, permissible compressive stress perpendicular to grain, modulus of elasticity, permissible shear stress parallel to grain).

The permissible shear stress at the glue line is at least the same as the permissible shear stress parallel to grain of the species in design, unless otherwise tested.

b) *Solution:*

- 1) Let, b = width of beam, and d = depth of beam, for lateral stability, $b : d = 1 : 4$ (say).
- 2) Determine bending moment on the basis of span, loading and support conditions (say, $WL/8$ for beams with both ends fixed).
- 3) Also, bending moment (M) = Bending stress (f_b) $\times$ Section modulus (Z):

$$\text{Therefore, } M = f_b \times \frac{bd^2}{6} = \frac{WL}{8}$$

$$\text{or } bd^2 = \frac{WL}{8} \times \frac{6}{f_b}$$

- 4) From bd^2 obtained at (c) and $b : d$ assumed at (a), arrive at appropriate values of b and d .
- 5) Check for deflection of beam. Since dead load is for long term, double this for deflection check, that is:

$$W = (2 \times \text{Dead load}) + \text{Imposed load}$$

Deflection is to be limited to the specified requirement, say 0.003 of span.

Substituting values of b and d from (d) and $I (= bd^3/12)$ in the formula for deflection, perform check against the limit for deflection.

- 6) Decide the number and thickness of laminations for the required depth of section.

11.2.3 Material

Laminating boards shall not contain decay, knots or other strength reducing characteristics in excess of those sizes or amounts permitted by specifications. The moisture content shall approach that expected in service and shall in no case exceed 15 percent at the time of gluing. The moisture content of individual laminations in a structural member shall not differ by more than 3 percent at the time of gluing. Glue shall be of type suitable for the intended service of a structural member.

11.2.4 Fabrication/Manufacture

In order to assure a well-bonded and well-finished member of true shape and size, all equipment, end-jointing, glue spread, assembly, pressing, curing or any other operation in connection with the manufacture of glued structural members shall be in accordance with the available good practices and as per glue manufacturers' instructions as applicable.

11.2.4.1 End joints shall be scattered in adjacent laminations, which shall not be located in very highly stressed outer laminations.

11.2.5 Testing

Quality control, inspection and testing during production is essential to structural glulam. Special attention needs to be given to members exposed to weather.

For examining the quality of glue and its relative strength namely species of timber in glued laminated construction, it is necessary to conduct block shear and other related tests in accordance with accepted standard [C-3A(11)]. Structural loading tests on prototype sizes provide information on the strength properties, stiffness or rigidity against deflection of a beam. Information on different strength functions as obtained from such indigenous tests carried out on laminated timber beams are given in [Table 19](#), for guidance in design.

Table 19 Laminated Timber Beams — Structural Bending Test Under Three Point Loading

(Clause [11.2.5](#))

Species — *Pinus wallichiana* (Kail);

Effective span — 2.3 m, Cross section: 100 × 240 mm; and

Adhesive — Urea formaldehyde.

Sl No.	Strength Functions	Nail Laminated Vertical Laminates with Four Full Size Planks Using 8 SWG, 100 mm Long Nails	Glue Laminated		Solid (Control) Specimen
			Vertical Laminates Four Planks, Each 25 mm Thick	Horizontal Laminates Eight Planks, Each 30 mm Thick	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Modulus of rupture, N/mm ²	27.439	31.363	21.568	30.509
ii)	Modulus of elasticity (× 10 ³ N/mm ²)	8.167	8.903	8.210	7.554
iii)	Fibre stress at limit of proportionality, N/mm ²	18.136	14.166	13.259	18.276
iv)	Maximum horizontal shear stress, N/mm ²	1.723	1.967	1.319	1.906

12 LAMINATED VENEER LUMBER

12.1 Certain reconstituted lignocellulosic products with fibre oriented along a specific direction have been developed and are being adopted for load bearing applications. Laminated veneer lumber is one such product developed as a result of researches in plantation grown species of wood. Density of laminated veneer lumber ranges from 600 kg/m³ to 750 kg/m³ which is manufactured in accordance with accepted standard [C-3A(12)].

12.1.1 Dimensions

Sizes of laminated veneer lumber composite shall be inclusive of margin for dressing and finishing unless manufactured to order. The margin for dressing and finishing shall not exceed 3 mm in the width and thickness and 12 mm in the length.

12.1.2 Permissible Defects

- | | | |
|-------------------|---|---|
| a) Jointing gaps | : | Not more than 3 mm wide, provided they are well staggered in their spacing and position between the successive plies; |
| b) Slope of grain | : | Not exceeding 1 in 10 in the face layers; |
| c) Tight knot | : | Three numbers up to 25 mm diameter in 1 m ² provided they are spaced 300 mm or more apart; and |
| d) Warp | : | Not exceeding 1.5 mm per metre length. |

12.1.3 Strength Requirements

The strength requirements for laminated veneer lumber shall be as per [Table 20](#).

Table 20 Requirements of Laminated Veneer Lumber

(Clause [12.1.3](#))

Sl No.	Properties	Requirement
(1)	(2)	(3)
i)	Modulus of rupture, N/mm^2 , <i>Min</i>	50
ii)	Modulus of elasticity, N/mm^2 , <i>Min</i>	7 500
iii)	Compressive strength parallel to grain, N/mm^2 , <i>Min</i>	35
iv)	Horizontal shear:	
	a) Parallel to laminae, N/mm^2 , <i>Min</i>	6
	b) Perpendicular to laminae, N/mm^2 , <i>Min</i>	8
v)	Tensile strength parallel to grain, N/mm^2 , <i>Min</i>	55
vi)	Screw holding power:	
	a) Edge (N), <i>Min</i>	2 300
	b) Face (N), <i>Min</i>	2 700
vii)	Thickness swelling in 2 h water soaking (percent), <i>Max</i>	3

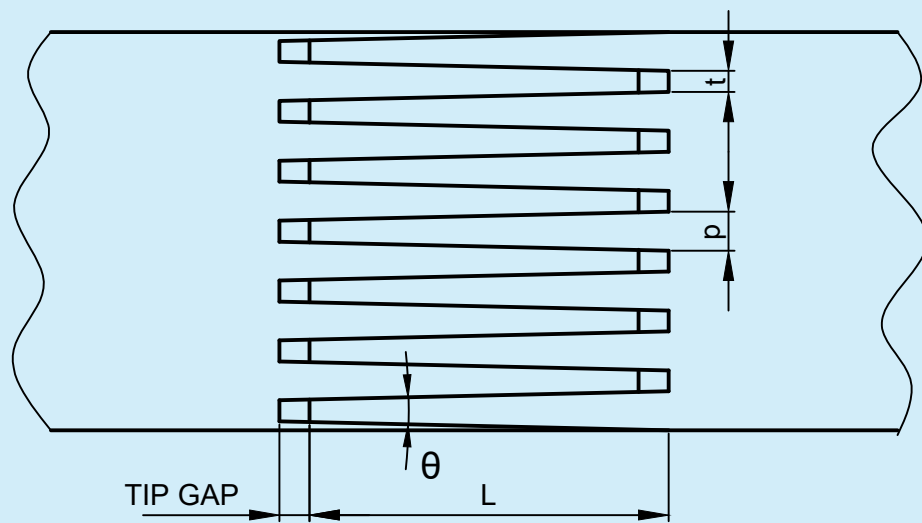
13 GLUED FINGER JOINTS

13.1 Finger joints are glued joints connecting timber members end-to-end (see [Fig. 14](#)). Such joints shall be produced by cutting profiles (tapered projections) in the form of v-shaped grooves to the ends of timber planks or scantling to be joined, gluing the interfaces and then mating the two ends together under pressure. Finger joints provide long lengths of timber, ideal for upgrading timber by permitting removal of defects, minimizing warping and reducing wastage by avoiding short off-cuts. To improve the long-term durability and performance of glued finger joints, the timber used should be well-seasoned to maintain dimensional stability. Additional protective treatments may be required for finger-jointed timber used in outdoor or high-moisture environments to prevent decay and degradation of the adhesive bonds.

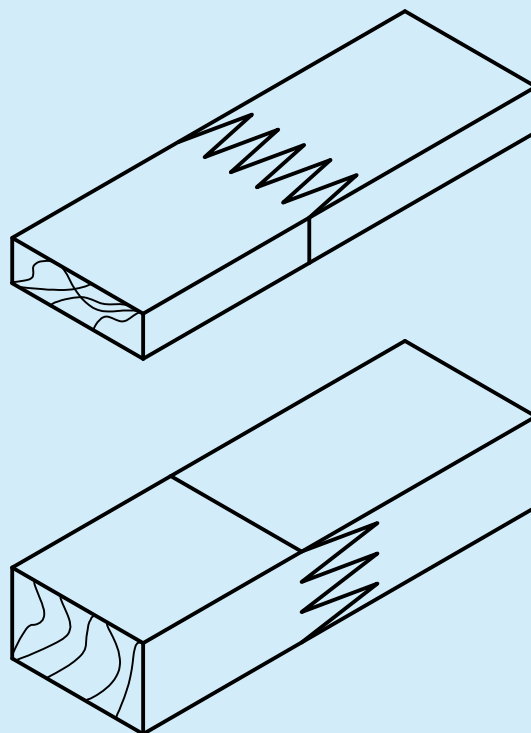
13.1.1 In finger joints the glued surfaces are on the side grain rather than on the end grain and the glue line is stressed in shear rather in tension.

13.1.1.1 The fingers can be cut from edge-to-edge or from face-to-face. The difference is mainly in appearance, although bending strength increases if several fingers share the load. A joist is slightly stronger with edge-to-edge finger joints and a plank is stronger with face-to-face finger joint. The choice of adhesive should consider the wood species being used, as different timber species may exhibit varying adhesive compatibility. For example, denser species may require stronger bonding adhesives, while lower-density woods may work well with more flexible adhesives. Testing of specific adhesives with the wood species used is recommended to ensure optimal joint strength and performance.

13.1.1.2 For structural finger jointed members for interior dry locations, adhesives based on melamine formaldehyde cross linked polyvinyl acetate (PVA) are suited. For high humid and exterior conditions, phenol formaldehyde and resorcinol formaldehyde type adhesives are recommended. Proper adhesives should be selected in consultation with the designer and adhesive manufacturers and assessed in accordance with accepted standard [C-3A(11)]. The environmental impact of adhesives used in finger-jointed construction should also be considered. Where possible, the use of low- volatile organic compounds (VOC) or environmentally friendly adhesives should be promoted, particularly in indoor applications, to reduce emissions and promote healthier indoor air quality.



L = FINGER LENGTH
 p = PITCH
 t = TIP THICKNESS
 $\theta = 6^\circ$



ORIENTATION OF FINGER JOINTS

FIG. 14 TYPICAL FINGER JOINT GEOMETRY

13.1.2 Manufacturing Process

In the absence of sophisticated machinery, the finger joints shall be manufactured through intermediate technology with the following steps:

- a) Drying of wood;
- b) Removal of knots and other defects;
- c) Squaring the ends of the laminating planks;
- d) Cutting the profile of finger joint in the end grain;
- e) Applying adhesives on the finger interfaces;
- f) Pressing the joint together at specified pressure;
- g) Curing of adhesive line at specified temperature; and
- h) Planing of finger-jointed planks for smooth surface.

For higher precision and productivity, modern manufacturing facilities could implement computer numerical control (CNC) machines for cutting finger joints. Automation helps achieve consistent profile geometry, improves the quality of adhesive application, and reduces the margin for human error during production.

13.1.3 Strength

Strength of finger joints depends upon the geometry of the profile for structural purpose; this is generally 50 mm long, 12 mm pitch.

13.1.3.1 The strength properties of glued finger joints are given in [Table 21](#) for information and guidance in design.

13.1.3.2 Tip thickness shall be as small as practically possible.

14 STRUCTURAL USE OF PLYWOOD

Unlike sawn timber, plywood is a layered panel product comprising veneers of wood bonded together with adjacent layers usually at right angles. As wood is strongest when stressed parallel to grain, and weak perpendicular to grain, the lay up or arrangement of veneers in the panel determines its properties. When the face grain of the plywood is parallel to the direction of stress, veneers parallel to the face grain carry almost all the load. Some information/guidelines for structural use of plywood which should be manufactured in accordance with accepted standards [C-3A(13)] are given in [14.1](#) to [14.3](#).

14.1 The plywood has a high strength to weight ratio and is dimensionally stable material available in sheets of a number of thicknesses and construction. Plywood can be sawn, drilled and nailed with ordinary wood working tools. The glues used to bond these veneers together are derived from synthetic resins which are set and cured by heating. The properties of adhesives can determine the durability of plywood.

14.2 In glued plywood construction, structural plywood is glued to timber resulting in highly efficient and light structural components like web beams (I and box sections) (see [Fig. 15](#) and [Fig. 16](#)) stressed skin panels (see [Fig. 17](#)) used for flooring and walling and pre-fabricated houses, cabins, etc. Gluing can be carried out by nail gluing techniques with special clamps. High shear strength of plywood in combination with high flexural strength and stiffness of wood result in structures characterized by high stiffness for even medium spans. Plywood can act as web transmitting shear stress in web bearing or stressed skin or sandwich construction. The effective moment of inertia of web beam and stressed skin construction depends on modular ratio that is, E of wood to E of plywood.

Table 21 Strength Characteristics of Glued Finger Joints

(Clause [13.1.3.1](#))

Timber species – *Pinus wallichiana* (Kail)

(Averaged test results)

Sl No.	Character →	Compressive Parallel to Grain Tests			Tension Parallel to Grain Tests			Static Bending Tests											
	Treatment ↓							Central Point Loading						Two Point Loading					
								Maximum Compressive Stress (N/mm ²)	Fibre Stress at EL (N/mm ²)	MOE (×10 ³ N/mm ²)	Maximum Tensile Stress (N/mm ²)	Tensile Stress at EL (N/mm ²)	MOE (×10 ³ N/mm ²)	Edge-wise			Flat-wise		
		Fibre Stress at EL (N/mm ²)	MOR (N/mm ²)	MOE (×10 ³ N/mm ²)	Fibre Stress at EL (N/mm ²)	MOR (N/mm ²)	MOE (×10 ³ N/mm ²)							Fibre Stress at EL (N/mm ²)	MOR (N/mm ²)	MOE (×10 ³ N/mm ²)	Fibre Stress at EL (N/mm ²)	MOR (N/mm ²)	MOE (×10 ³ N/mm ²)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
i)	UL ₁	31.196	16.971	10.202	30.019	23.446	10.202	20.993	29.136	6.769	39.142	46.401	8.142	25.114	28.743	8.044	32.471	38.553	9.516
ii)	UL ₂	37.474	17.462	11.282	30.607	23.642	10.301	20.209	31.686	6.671	35.120	46.205	7.161	23.446	42.085	7.652	41.202	51.306	9.712
iii)	UL ₃	35.905	19.914	10.595	32.667	23.838	10.006	22.171	35.022	6.867	38.750	47.480	8.437	25.898	29.234	7.848	29.626	38.357	8.633
iv)	PL ₁	28.743	21.484	9.908	20.895	17.560	10.497	20.699	23.446	8.142	22.955	27.076	9.123	18.443	21.876	7.848	25.506	27.174	8.927
v)	PL ₂	28.743	19.424	10.006	18.149	17.069	9.614	16.285	18.639	5.984	24.133	26.095	8.731	16.579	17.854	7.259	26.389	28.547	8.535
vi)	PL ₃	27.860	20.111	9.908	24.721	19.326	10.497	15.009	17.069	7.652	21.680	26.193	8.829	10.301	11.183	6.867	26.487	30.313	11.183
vii)	C	35.512	27.959	10.301	31.490	23.152	10.104	25.506	45.518	7.063	36.788	57.192	7.063	31.883	44.537	7.946	38.161	52.582	9.221

MOE – Modulus of elasticity (values rounded off for decimals)

MOR – Modulus of rupture

EL – Elastic limit

U – Urea formaldehyde with liquid hardener

P – Cold setting phenolic glue with hardener

L₁ – Finger length 38 mm

L₂ – Finger length 50 mm

L₃ – Finger length 62 mm

C – Control (unjointed specimen)

NOTE — The information given in the above table is based on the tests conducted on glued finger joints of varying lengths/pitch with two types of adhesives.

14.3 Structural plywood is also very efficient as cladding material in wood frame construction such as houses. This type of sheathing is capable of resisting racking due to wind and earthquake forces. Structural plywood has been widely used as diaphragm (horizontal) as in roofing and flooring in timber frame construction. It has been established that 6 mm thick plywood can be used for sheathing and even for web and stressed skin construction, 9 mm to 12 mm thick plywood is suitable for beams, flooring diaphragms, etc. Phenol formaldehyde (PF) and PRF adhesive are suitable for fabrication of glued plywood components. 6 mm to 12 mm thick structural plywood can be very well used as nailed or bolted gussets in fixing members of trusses or lattice girders or trussed rafters.

Normally, scarf joints are used for fixing plywood to required length and timber can be joined by using either finger or scarf joints. Arch panels, folded plates, shelves are other possibilities with this technique.

15 TRUSSED RAFTER

15.1 General

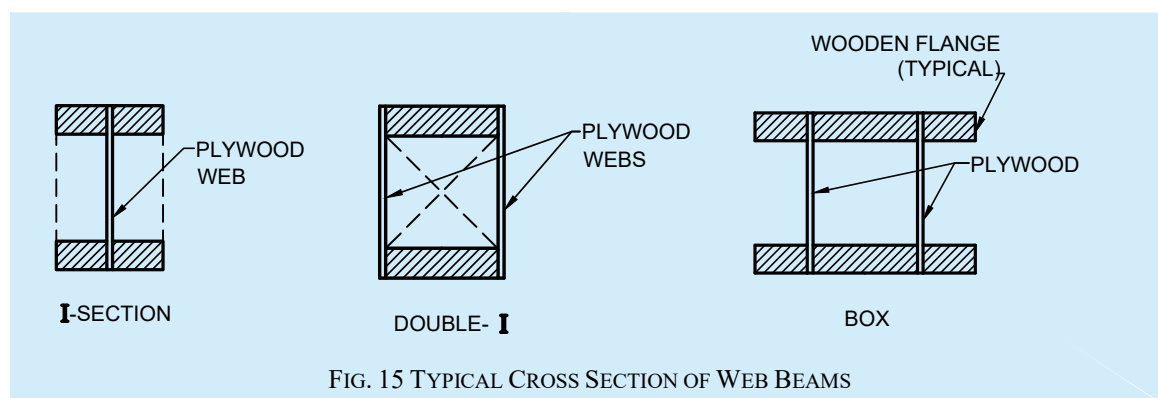
A roof truss is essentially a plane structure which is very stiff in the plane of the members, that is the plane in which it is expected to carry loads, but very flexible in every other direction. Thus, it can virtually be seen as a deep, narrow girder liable to buckling and twisting under loads. In order, therefore, to reduce this effect, eccentricity of loading and promote prefabrication for economy, low-pitched trussed rafters are designed with bolt ply/nail ply joints. Plywood as gussets, besides being simple have inherent constructional advantage of grain over solid wood for joints and a better balance is achievable between the joint strength and the member strength.

Trussed rafters are light weight truss units spaced at close centres for limited spans to carry different types of roof loads. They are made from timber members of uniform thickness fastened together in one plane. The plywood gussets may be nailed or glued to the timber to form the joints. Conceptually a trussed rafter is a triangular pin jointed system, traditionally meant to carry the combined roof weight, cladding services and wind loads. There is considerable scope for saving timber by minimizing the sections through proper design without affecting structural and functional requirements.

Trussed rafters are required to be supported only at their ends so that there is no need to provide load bearing internal walls, purlins, etc. In comparison with traditional methods of construction they use less timber and considerably reduce on-site labour and mass production of reliable units can be carried out under workshop controls.

Additional consideration may include:

- Climatic Suitability** — Trussed rafters should be designed with consideration for India's diverse climatic zones. Timber species suitable for humid, dry and coastal regions should be preferred to prevent issues related to swelling, shrinkage, or decay due to moisture variations;
- Use of Local Timber Species** — Indian species such as Teak (*Tectona grandis*), Sal (*Shorea robusta*), Neem (*Azadirachta indica*), and other durable hardwoods should be prioritized. The use of locally available plantation-grown timber helps reduce transportation costs and supports sustainability; and
- Fire and Termite Resistance** — Due to the prevalence of termites in many regions of India, timber rafters should be treated with appropriate termite-resistant chemicals. Similarly, the risk of fire in construction should be mitigated through the application of fire-retardant coatings.



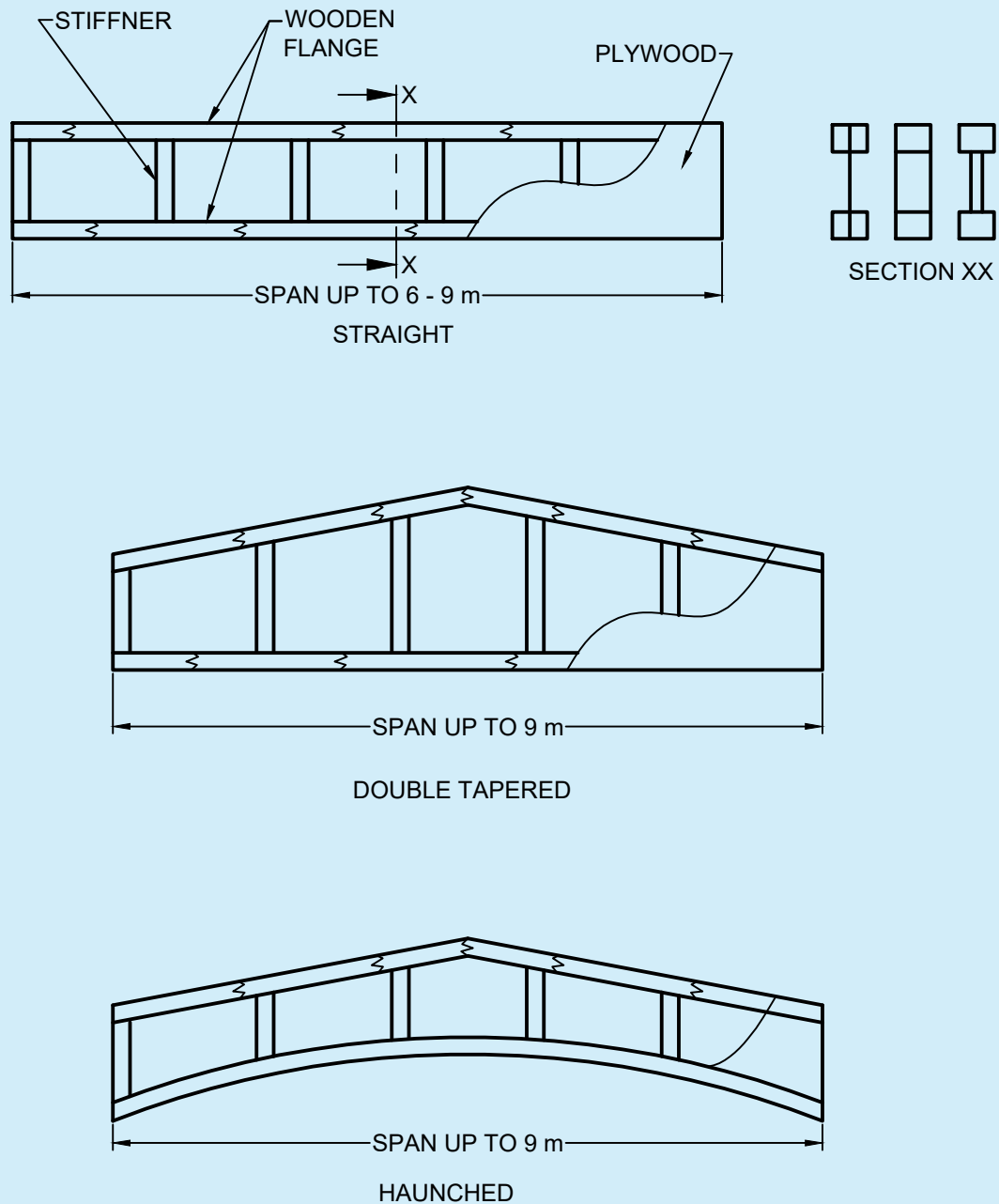


FIG. 16 WEB BEAM CONFIGURATIONS

FIG. 16 WEB BEAM CONFIGURATIONS

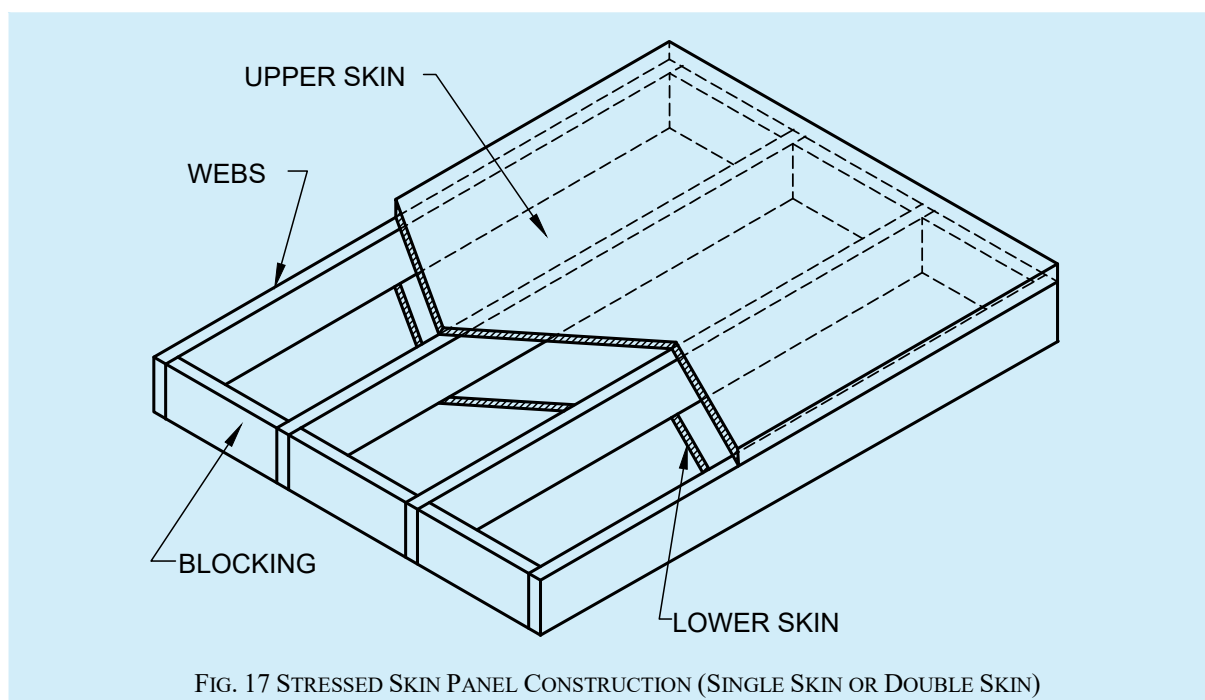


FIG. 17 STRESSED SKIN PANEL CONSTRUCTION (SINGLE SKIN OR DOUBLE SKIN)

15.2 Design

15.2.1 The design may be based on the use of Group ‘C’ species of standard grade timber in accordance with good practice [C-3A(7)] for nail jointed timber construction. As the strength and stiffness of any framing depends upon efficiency of joints, design calls for proper size and number of fasteners, their spacings and thickness of gusset plates for transfer of stress from one member to the other.

Trussed rafter shall be designed to sustain the dead and imposed loads specified in Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS and the combinations expected to occur. Extra stresses/ deflections during handling, transportation and erection shall be taken care of. Structural analysis, use of load-slip and moment, rotation characteristics of the individual joints may be used if feasible. Alternatively, the maximum direct force in a member may be assessed to be given by an idealized pin-jointed framework, fully loaded with maximum dead and imposed load in the combination in which they may reasonably be expected to occur.

NOTE — Some trussed rafters with plywood gusseted joint, span 6 m, designed in *Pinus roxburghii* (Chir) were tested in prototypes at Forest Research Institute, Dehradun for structural adequacy as under, in accordance with accepted standards [C-3A(14)]:

- a) Destructive test (short-term); and
- b) Proof test (long-term).

Results demonstrated an apparent factor at safety of 4 and actual factor of safety of 3. Deflection was well within the allowable deflection at design load. A pre-camber of 1/240 of span at the bottom chord was recommended.

15.3 Timber

The species of timber including plantation grown species which can be used for trussed rafter construction and permissible stresses thereof shall be in accordance with [Table 1](#). Moisture contents shall be as per zonal requirements in accordance with [4.4](#).

15.4 Plywood

Boiling water proof (BWP) grade preservative treated plywood shall be used in accordance with accepted standards [C-3A(13)]. Introduction of a plywood gusset simplifies the jointing and in addition provides rigidity to the joint. Preservation of plywood and other panel products shall be done in accordance with good practice [C-3A(15)].

15.5 Some design data on the lateral load bearing strength of common wire nails in plywood-to-wood joints are presented for timber species, Sal and Chir in [Table 22](#), for guidance in design.

16 STRUCTURAL SANDWICHES

16.1 General

Sandwich constructions are composites of different materials including wood based materials formed by bonding two thin facings of high strength material to a light weight core which provides a combination of desirable properties that are not attainable with the individual constituent materials (see Fig. 18). The thin facings are usually of strong dense material since that are the principal load carrying members of the construction. The core should be stiff enough to ensure the faces remain at the correct distance apart. The sandwiches used as structural elements in building construction shall be adequately designed for their intended services and shall be fabricated only where there are adequate facilities for gluing or otherwise bonding cores to facings to ensure a strong and durable product. The entire assembly provides a structural element of high strength and stiffness in proportion to its mass.

Non-structural advantages can also be derived by proper selection of facing and core material, for example, an impermeable facings can be used to serve as a moisture barrier for walls and roof panels and core may also be selected to provide thermal and/or acoustic insulation, fire resistance, etc, besides the dimensional stability.

16.2 Cores

Sandwich cores shall be of such characteristics as to give to the required lateral support to the stressed facings to sustain or transmit the assumed loads or stresses. Core generally carries shearing loads and to support the thin facings due to compressive loads. Core shall maintain the strength and durability under the conditions of service for which their use is recommended. A material with low E and small shear modulus may be suitable.

16.3 Facings

Facings shall have sufficient strength and rigidity to resist stresses that may come upon them when fabricated into a sandwich construction. They shall be thick enough to carry compressive and tensile stresses and to resist puncture or denting that may be expected in normal usages.

16.4 Designing

Structural designing may be comparable to the design of I-beams, the facings of the sandwich represent the flanges of the I-beam and the sandwich core I-beam web.

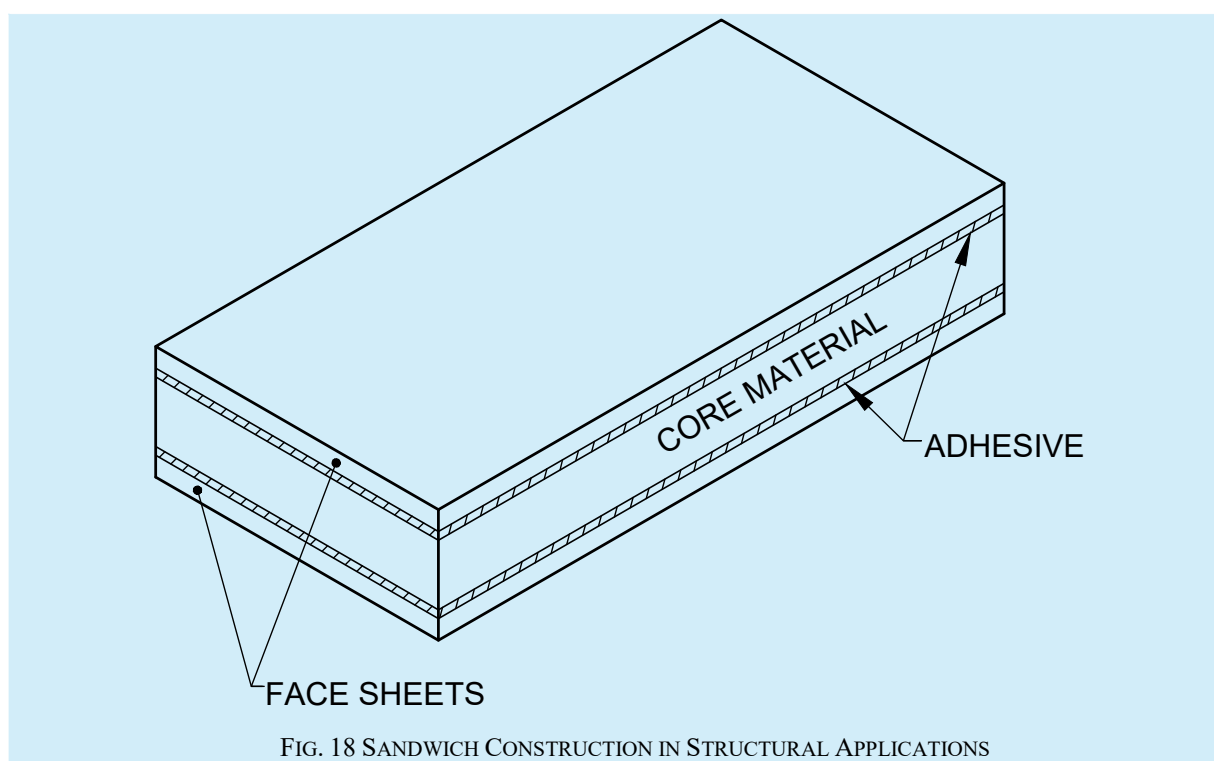


Table 22 Lateral Load Bearing Strength of Wire Nails in Plywood-to-Wood Composite Joints Permissible Lateral Strength of 9 SWG (3.55 mm diameter) Nail in Double Shear (kg) Safe Load per Nail
(Clause [15.5](#))

Sl No.	Thickness of Members mm		Facing Veneer of Plywood Along the Loading Direction								Across Loading Direction	
			(Loading Direction in the Main Member of the Joint)								(Loading Direction in the Main Member of the Joint)	
	Main Member Timber	Side Members Plywood	Parallel to Grain		Perpendicular to Grain		At 60° Inclination to Grain		At 30° Inclination to Grain		Loaded Parallel to Grain	
Permanent Structures			Temporary Structures	Permanent Structures	Temporary Structures	Permanent Structures	Temporary Structures	Permanent Structures	Temporary Structures	Permanent Structures	Temporary Structures	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
i)	Species — <i>Shorea robusta</i> (Sal):											
	a) 30	9	109	121	135	156	151	168	140	164	142	151
	b) 40		123	124	121	166	142	181	145	158		
	c) 50		89	123	157	167	162	170	99	139		
	d) 30	12	157	177	150	154	164	179	111	171	135	180
	e) 40		179	191	145	193	154	194	132	192		
	f) 50		151	183	149	191	155	202	128	182		
	g) 30	19	245	283	Larger thickness of plywood is generally not recommended to be used in structures unless heavily stressed						150	209
	h) 40		259	264								
	j) 50		231	316								
ii)	Species — <i>Pinus roxburghii</i> (Chir):											
	a) 30	9	102	127	112	141	60	125	92	124	107	148
	b) 40		103	148	100	153	98	142	101	141		
	c) 50		87	133	119	150	50	168	50	114		
	d) 30	12	133	163	116	111	80	132	127	150	118	174
	e) 40		114	173	112	136	64	157	137	174		
	f) 50		99	194	117	138	90	170	119	161		

Table 22 (Concluded)

Sl No.	Thickness of Members mm		Facing Veneer of Plywood Along the Loading Direction								Across Loading Direction	
			(Loading Direction in the Main Member of the Joint)								(Loading Direction in the Main Member of the Joint)	
	Main Member Timber	Side Members Plywood	Parallel to Grain		Perpendicular to Grain		At 60° Inclination to Grain		At 30° Inclination to Grain		Loaded Parallel to Grain	
Permanent Structures			Temporary Structures	Permanent Structures	Temporary Structures	Permanent Structures	Temporary Structures	Permanent Structures	Temporary Structures	Permanent Structures	Temporary Structures	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
	g) 30	19	219	309	Larger thickness of plywood generally not used in structures unless heavily stressed						158	249
	h) 40		238	306								
	j) 50		236	333								
NOTES												
1 Plywood shall be of shuttering grade as per the accepted standard [C-3A(16)].												
2 Subject to satisfying all the design requirements in structures:												
a) For denser species, timber thickness of 30 mm may be used by means of a plywood of 9 mm thickness for all practical purposes. For species like Chir, plywood thickness of 12 mm may be used with 30 mm or 40 mm timber thickness.												
b) Regarding aligning the veneers of plywood plate; nail strength data may be used as design function irrespective of alignment of face veneer to overcome practical difficulties.												
c) While prebore is essential to drive nail in structural timber as per the good pratice [C-3A(7)], plywood plates need not be prebored for nailing.												

16.5 Tests

Panels of sandwich construction shall be subject to testing in accordance with accepted standards [C-3A(17)]. Tests shall include, as applicable, one or more of the following:

- a) Flexural strength/stiffness;
- b) Edge-wise compressions;
- c) Flat-wise compression;
- d) Shear in flat-wise plane;
- e) Flat-wise tensions;
- f) Flexural creep (creep behaviour of adhesive);
- g) Cantilever vibrations (dynamic property); and
- h) Weathering for dimensional stability.

17 LAMELLA ROOFING

17.1 General

The lamella roofing offers an excellent architectural edifice in timber, amenable to prefabrication, light weight structure with high central clearance. It is essentially an arched structure formed by a system of intersecting skewed arches built-up of relatively short timber planks of uniform length and cross section. Roof is designed as a two hinged arch with a depth equal to the depth of an individual lamella and width equal to the span of the building. The curved lamellas (planks) are bevelled and bored at the ends and bolted together at an angle, forming a network (grid) pattern of mutually braced and stiffened members (see [Fig. 19](#) and [Fig. 20](#)).

The design shall be based on the balanced or unbalanced assumed load distribution used for roof arches. Effect of deformation or slip of joints under load on the induced stresses shall be considered in design. Thrust components in both transverse and longitudinal directions of the building due to skewness of the lamella arch shall be adequately resisted. Thrust at lamella joints shall be resisted by the moment of inertia in the continuous lamella and roof sheathing (decking) of lamella roofing. The interaction of arches in two directions adds to the strength and stability against horizontal forces. For design calculations several assumption tested and observed derivations, long-duration loading factors, seasoning advantages and effects of defects should be taken into account.

17.2 Lamellas

Planking shall be of a grade of timber that is adequate in strength and stiffness to sustain the assumed loads, forces, thrust and bending moments generated in lamella roofing. Lamella planks shall be seasoned to a moisture content approximating that they will attain in service. Lamella joints shall be proportioned so that allowable stresses at bearings of the non-continuous lamellas on the continuous lamellas or bearings under the head or washer of bolts are not exceeded.

17.2.1 Design Sequence and Methods

A lamella roof is essentially an arch (segmental, parabolic or elliptical) formed by a system of intersecting skewed arches, built up of relatively short members of small dimensional planks called 'lamellas'. The arch is designed as a two hinged arch with a depth equal to the depth of the individual lamella and a width equal to the span of the building of certain length. For design purposes, a unit length of the given span forming a single arch is considered loaded with the given loads. For the particular arch, curvature offers resultant outward suction on the roof to be less than its self weight to facilitate design and construction for economy (this is possible only when the semi-angle of arch is less than 42°).

Some design assumptions that should be considered include:

- a) Self weight of timber components (in particular species), namely lamella planks, roof boarding, wall plates, etc is assumed for the particular timber section;

- b) Imposed loads as per Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS for curved roofs where slope at springing point is greater than 10°. Take the reduced worked out values of imposed loads for actual design for timber components as per this Subsection;
- c) Wind load is taken as per Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS for structures with large openings, that is, openings larger than 20 percent of wall area (or as the case may be);
- d) Stresses developed in curved lamella planks should not exceed the allowable safe stresses of timber species used;
- e) Range of various sizes of timber planks for lamellas shall be as under:
 - 1) *Length* — 1.0 m to 2.5 m;
 - 2) *Section*:
 - i) *Thickness* — 25 mm to 50 mm; and
 - ii) *Width of uncurved plank* — 100 mm to 250 mm.
- f) Limitations to be observed for spacing of lamellas over wall plate (or bressummer):
 - 1)
 - i) For skew angle of 21° (diamond angle 42°);
 - ii) For 2.5 m length of plank, spacing shall be 870 mm maximum; and
 - iii) For 1.0 m length, spacing shall be 360 mm maximum.
 - 2)
 - i) For skew angle of 19° (diamond angle 38°);
 - ii) For 2.5m length of plank, spacing shall be 800 mm maximum; and
 - iii) For 1.0 m length of plank, spacing shall be 320 mm maximum.
- g) Value of horizontal thrust, resultant thrust and maximum bending moment, are taken on per metre length of structure in case of transverse direction of arch;
- h) Bending moment may be suitably reduced, if so desired by the designer, by applying stability factor if the ratio of the length of structure to the curved length of arch so warrants (as this ratio increases, the stability of the structure decreases);
- j) The thrust in the skewed direction of arch, within one spacing of lamellas is shared by two lamella-lines in skewed direction; and
- k) More than one bolt (MS) shall be used for nodal joint. For design check of bolt against tension, half the value of thrust in skewed direction will be considered for calculations purposes in light of (j) {in case of stiff roof covering material [that is, corrugated galvanized iron (CGI) sheet, close boarding, etc] all lamella planks will be held jointly behaving as a monolithic unit to a large extent and the actual stresses developed in bolt will be reduced resulting in lesser actual tension in bolts than the calculated design value}.

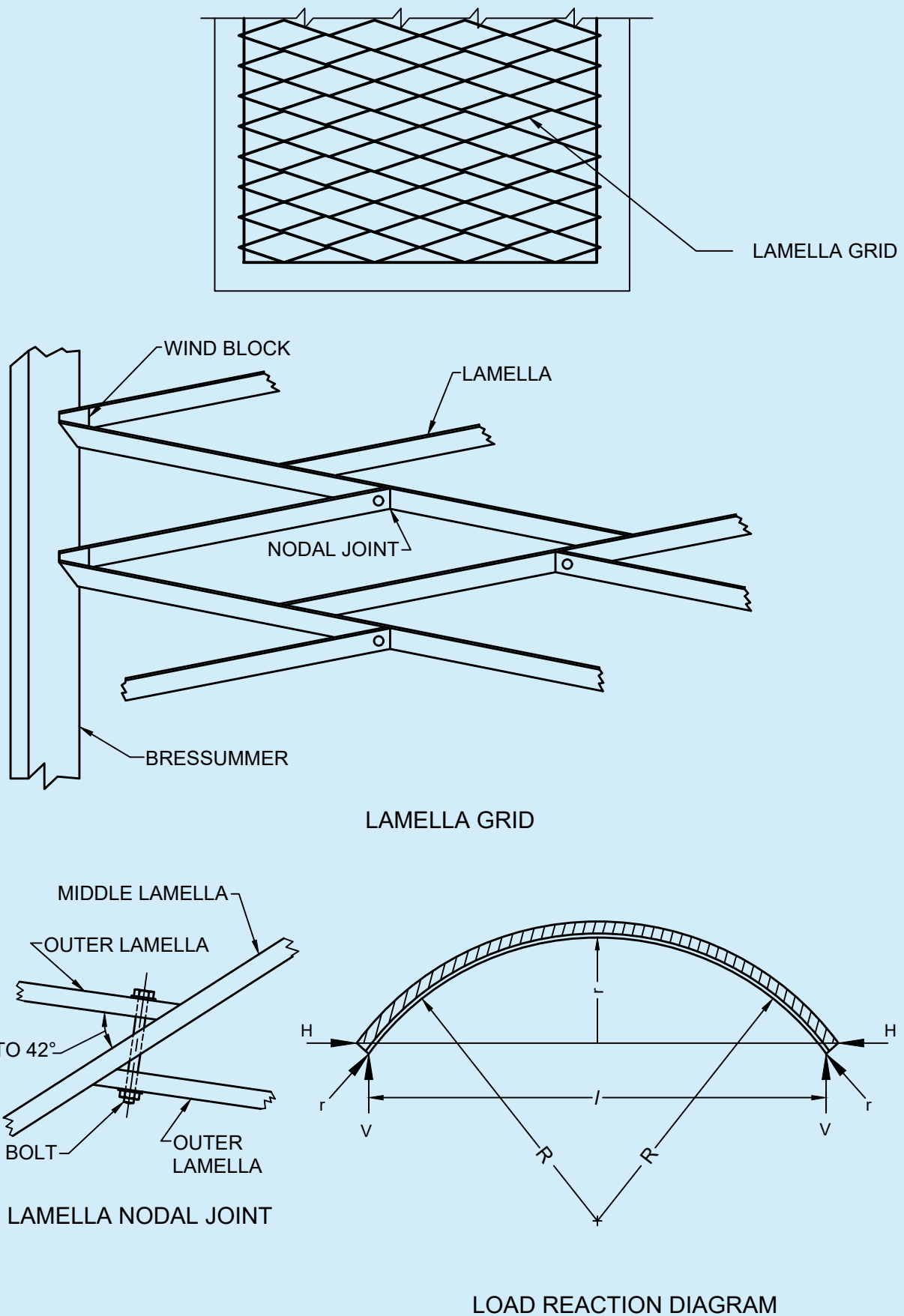
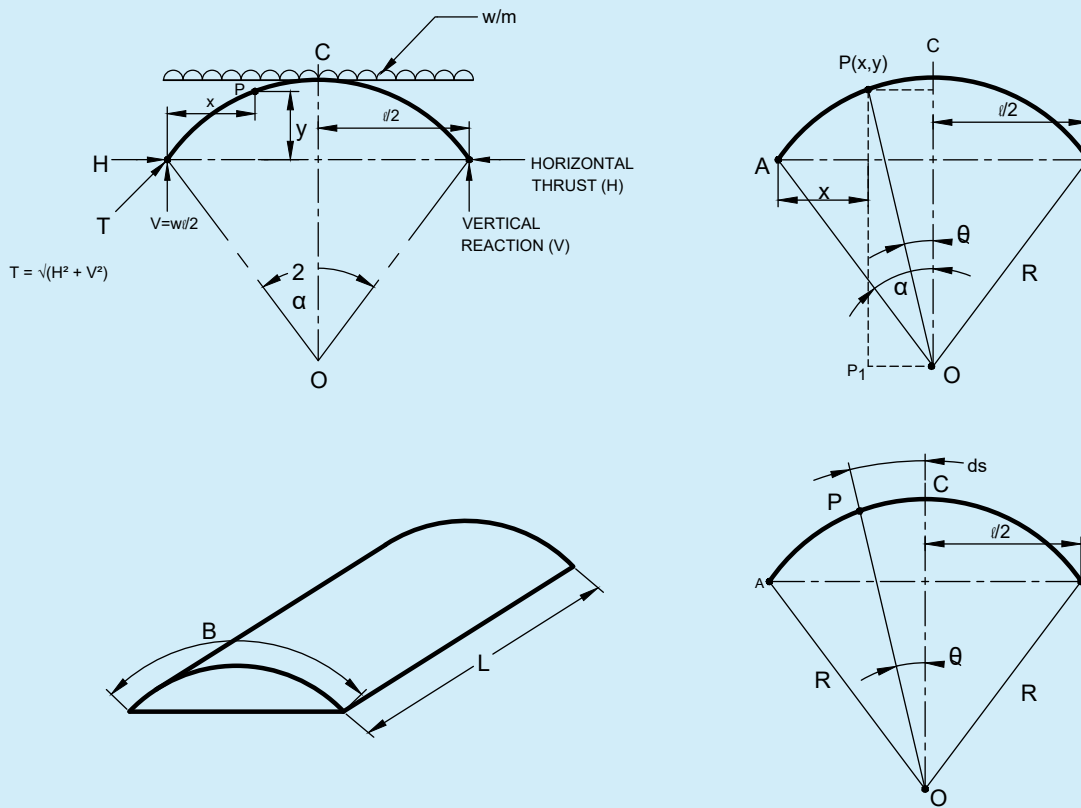


FIG. 19 TYPICAL ARRANGEMENT OF LAMELLA ROOFING



NOTES

1 When the ratio of length of the structure (L) to the circumference of the arch (B) increases, stability of the structure decreases. The roof should be designed as:

- short roof (say, $L/B = 1$), or
- long roof (say, $L/B = 1.5$).

Shorter roofs have an advantage over long roofs for economical design. However, there is no such advantage beyond L/B of 4.

2 Due to stiff roof covering material, stiffness is induced in an arch, which results in safer design.

FIG. 20 STABILITY OF LAMELLA ARCH ROOF

17.2.1.1 Horizontal thrust

Any of the known methods for solving a statically indeterminate structure is used to evaluate horizontal thrust, bending moment, etc, to which a segmental arch will be subjected under the given condition of loadings. For 60° segmental arch, the radius of an arch will be equal to the span of the arch. Horizontal thrust can be calculated by the following two methods:

a) Method 1

Horizontal thrust (H_B) per length of one bay (that is, distance between 'tie-rod spacing' or 'column centres'):

$$H_B = (DL + IL) \times \left[\frac{l^2 + s_t^2}{8r} \right]$$

where

H_B = horizontal thrust, in N;

- DL = dead load, in N;
 IL = imposed load, in N;
 l = span of arch, in m;
 s_t = tie-rod spacing, in m; and
 r = rise of roof arch, in m.

b) *Method 2*

Horizontal thrust (H_1) per metre length of structure (in N):

$$H_1 = C \times w \times R$$

where

C = coefficient varying with the angle of arch and radius of arch;

w = equivalent distributed load per metre of span (in N/m); and

R = radius of circular arch (in m).

Horizontal thrust (H_1) is evaluated as:

$$H_1 = \frac{\int \mu y ds}{\int y^2 ds}$$

where

H_1 = horizontal thrust (per metre length of structure), in N;

μ = bending moment at any point 'P' distance 'x' metre from one end of springing point of arch and for vertical loads only (consider arch as a beam simply supported carrying the same load);

y = height of point 'P' above springing line of arch;

θ = angle which the point 'P' makes at the centre of arch with the vertical; and

ds = $Rd\theta$.

Now solution for numerator:

$$\begin{aligned}
 \mu &= \frac{wl}{2}x - wx \frac{x}{2} \\
 &= \frac{wx}{2} (l - x)
 \end{aligned}$$

where

$$\begin{aligned}
 x &= \frac{l}{2} - R \sin \theta \\
 x &= \frac{R}{2} - R \sin \theta \\
 x &= R \left(\frac{1}{2} - \sin \theta \right)
 \end{aligned}$$

and $y = R \cos \theta - R \cos \alpha$

$$y = R(\cos \theta - \cos \alpha)$$

and $ds = Rd\theta$

Substitute the values in expression:

$$\int \mu y ds = \int \frac{wx(l-x)}{2} R(\cos \theta - \cos \alpha) R d\theta$$

$$\text{Solve for limits of } \theta \text{ from } -\alpha \text{ to } +\alpha = \int_0^\alpha wR^4 \left(\frac{1}{4} - \sin^2 \theta\right) (\cos \theta - \cos \alpha) d\theta$$

Again for denominator;

$$\begin{aligned} \int y^2 ds &= \int_{-\alpha}^{+\alpha} R^2 (\cos \theta - \cos \alpha)^2 R d\theta \\ &= 2R^3 \int_0^\alpha (\cos \theta - \cos \alpha)^2 d\theta \end{aligned}$$

$$\text{Therefore, } H_1 = \frac{\int \mu y ds}{\int y^2 ds}$$

$$= \frac{wR^4 \int_0^\alpha \left[\frac{1}{4} - \sin^2 \theta\right] (\cos \theta - \cos \alpha) d\theta}{2R^3 \int_0^\alpha (\cos \theta - \cos \alpha)^2 d\theta}$$

On simplification of the above equation for a particular arch, $\alpha = 30^\circ$, $R = 1$ (α in radian = 0.5 approx),

$$\begin{aligned} \sin \alpha &= 1/2 \\ \cos \alpha &= \sqrt{3}/2 \end{aligned}$$

$$H_1 = 0.923 wR$$

Vertical reaction, $V_1 = \frac{\text{Total load} \times \text{span}}{2}$ at each end of arch due to dead and live loads per metre length (curved strip) of arch roofing.

17.2.1.2 Resultant thrust, T_1

Due to horizontal thrust and the vertical reaction, resultant transverse thrust, in kg per metre length on the arch can be worked out:

$$T_1 = \sqrt{H_1^2 + V_1^2}$$

17.2.1.3 Maximum bending moment

Bending moment at any point 'P' on arch making an angle of θ with the vertical line OC is, $BM = \mu - (H \times y)$

$$= \frac{wR^2}{2} \left(\frac{1}{4} - \sin^2 \theta\right) - 0.923wR^2 \left(\cos \theta - \frac{\sqrt{3}}{2}\right)$$

Differentiating the above expression for BM, we get

$$\frac{dM}{d\theta} = (-\sin \theta \cos \theta + 0.923 \sin \theta) wR^2$$

It is well known that $\frac{dM}{d\theta} = \text{shear force}$

This shear is zero when BM is maximum, that is:

either $\theta = \text{zero}$

or $\cos \theta = 0.923$

Therefore, $\theta = \cos^{-1} 0.923 = 22.6^\circ$

Substituting these values in the above equation of bending moment, we get maximum BM, that is, the maximum BM occurs at:

- a) crown of arch; and
- b) at 22.6° from the central line OC.

Thus when $\theta = 0$

$$\begin{aligned} BM &= \frac{wR^2}{2} \times \frac{1}{4} - 0.923wR^2 \left(1 - \frac{\sqrt{3}}{2}\right) \\ &= wR^2(0.125 - 0.12368) \\ &= 0.00132wR^2 \end{aligned}$$

When $\theta = 22.6^\circ$, $\sin \theta = 0.384$

$$\begin{aligned} BM &= \frac{wR^2}{2} \left(\frac{1}{4} - 0.384\right)^2 - 0.923wR^2(0.923 - 0.866) \\ &= wR^2(0.125 - 0.073728) - 0.923 \times 0.057wR^2 \\ &= -0.001339wR^2 \text{ or say } -0.00134wR^2 \end{aligned}$$

The arch has to be designed for the greater of the two values (subject to the longitudinal stiffness coefficient on the basis of ratio L/B of length of the structure to the circumference of the arch) for economical design.

NOTE — Value of horizontal thrust, H_1 , resultant thrust, T_1 and bending moment, M_1 shall be for 1 m length of building.

17.2.1.4 Spacing of lamellas

Lengthwise spacing of lamellas depends upon the angle of skew lamellas and the length of the planks used for lamellas (As a general guidance it may be stated that the angle enclosed between the intersecting skewed arches should not exceed 45° . An angle between 38° and 40° is always preferred).

17.2.1.5 Design of lamella sizes

Net section of each of lamellas is assumed in curved profile conforming to the curvature of the roof surface.

Maximum BM = M_s over 600 mm spacing of lamella in the direction of skew angle, for example:

$$= \frac{0.6M_1 \times 10}{\cos 19^\circ} \text{ Nm}$$

where

M_1 = maximum bending moment per metre length of structure in transverse direction.

And maximum thrust N (in Newton) on this strip of 600 mm in skewed direction is:

$$N = \frac{0.6T_1}{\cos 19^\circ} \times \frac{1}{2}$$

(as the force T_1 is shared by two lines of lamellas)

where

N = maximum thrust on 600 mm strip in skewed direction; and

T_1 = resultant thrust in transverse direction (in N) per metre length of structure.

Now total compressive and flexural stresses should not exceed the allowable safe stress of timber species used for lamella plank. For fulfilling this condition the following equation should be checked:

$$\frac{N/A}{f_{c \parallel}} + \frac{M_s/Z}{f_t} \leq 1$$

where

- N = maximum thrust in skewed direction on strip of 600 mm (in N) corresponding to the maximum bending moment;
 A = cross-sectional area of lamella plank, in mm²;
 Z = section modulus of lamella plank, in mm³;
 $f_{c \parallel}$ = safe compression strength of timber fibre parallel to grain, N/mm²;
 M_s = bending moment over 600 mm strip of arch in skewed direction, Nmm;
 f_t = safe tensile strength of timber parallel to grain, N/mm²; and

NOTE — In case the lamella plank is under stressed, the minimum section for practical reasons shall be 25 mm × 100 mm subject to satisfying minimum edge and intermediate distances for accommodating bolts at nodal joints.

17.2.1.6 Design of bolts for nodal joint

To avoid any possibility of hinge action with the use of bolts, it is imperative to adopt two or more bolts of smaller diameter and adequately spaced in accordance with the good practice [6-3A(9)].

Assume diameter of bolt 'd' in mm, and length of bolt in one plank 't' mm, that is, thickness of middle lamella plank.

Area of contact of bolt with timber plank = $d \times t$ mm²

- a) *Check against crushing of timber at contact point of bolt* — For the bolt to transmit load without crushing the timber fibres, the following condition shall be satisfied:

$$t \times d \times \text{compressive stress perpendicular to grain for the timber species} > P$$

where

P = total load to be transmitted through bolt, in N.

- b) *Check against tension in bolt* — Arrangement of bolts at the junction of lamellas result in pure tension in bolt as bolt line is at right angle to the outer two lamellas:

Therefore, $Z_t = N \times \cot 2\phi$

where

Z_t = tension in bolt, in N;

N = normal thrust corresponding to maximum moment M_s (in N); and

$$= \frac{T_1}{\cos} \times \frac{1}{2}$$

T_1 = resultant thrust in transverse direction (in N) per metre length of structure.

Therefore;

Area of bolt (A_t) × tensile stress of steel (F_t) ≥ Z_t

Area of bolt required for safe design = Z_t/F_t

Select the diameter to engage at least two bolts.

(Required area of bolt is the effective area after considering reduction of area at the threaded end, which is generally 17 percent of the actual bolt diameter).

17.2.1.7 Design of steel washer at joints

Area of steel plate = A_t mm²

Stress perpendicular to grain of wood at the point of contact with steel plate washer = Z_t/A_t N/mm²

If $Z_t/A_t \leq$ compressive stress perpendicular to grain of the species of timber used, the assumed section is safe.

17.3 Construction

Design and construction of lamella roofs in India assumes the roof surfaces to be cylindrical with every individual lamella an elliptic segment of an elliptical arch of constant curved length but of different curvature. Lamella construction is thus more of an art than science as there is no analytical method available for true generation of schedule of cutting lengths and curvature of curved members forming the lamella grid. Dependence of an engineer on the practical ingenuity of master carpenter is almost final. All the lamella joints shall be accurately cut and fitted to give full bearing without excessive deformation or slip. Bolts at lamella splices shall be adequate to hold the members in their proper position and shall not be over tightened to cause bending of the lamellas or mashing of wood under the bolt heads. Connection of lamellas to the end arches shall be adequate to transmit the thrust or any other force. Sufficient false work or sliding jig shall be provided for the support of lamella roof during actual construction/erection. To avoid excessive deformation in a lamella system, all joints should be carefully fitted and well tightened.

18 NAIL AND SCREW HOLDING POWER OF TIMBER

18.1 General

One of the most common ways of joining timber pieces to one another is by means of common wire nails and wood screws. Timber is used for structural and non-structural purposes in form of scantlings, rafters, joists, boarding, crating and packing cases, etc, needing suitable methods of joining them. Nevertheless, it is the timber which holds the nails or screws and as such pulling of the nails/screws is the chief factor which come into play predominantly. In structural nailed joints, nails are essentially loaded laterally, the design data for which is already available in good practice [C-3A(7)]. Data on holding power of nails/screws in different species is also useful for common commercial purposes. The resistance of mechanical fastenings is a function of the specific gravity of wood, direction of penetration with respect to the grain direction, depth of penetration and the diameter of fastener assuming that the spacing of fasteners should be adequate to preclude splitting of wood.

18.2 Nails

Nails are probably the most common and familiar fastener. They are of many types and sizes in accordance with the accepted standard [C-3A(18)]. In general nails give stronger joints when driven into the side grain of wood than into the end grain. Nails perform best when loaded laterally as compared to axial withdrawal so the nailed joints should be designed for lateral nail bearing in structural design. Information on withdrawal resistance of nails is available and joints may be designed for that kind of loading as and when necessary.

18.3 Screw

Next to the hammer driven nails, the wood screw may be the most commonly used fastener. Wood screws are seldom used in structural work because of their primary advantage is in withdrawal resistance for example for fixing of ceiling boards to joists, purlin cleats, besides the door hinges, etc. They are of considerable structural importance in fixture design and manufacture. Wood screws are generally finished in a variety of head shapes and manufactured in various lengths for different screw diameters or gauges in accordance with the accepted standards [C-3A(19)].

The withdrawal resistance of wood screws is a function of screw diameter, length of engagement of the threaded portion into the member, and the specific gravity of the species of wood. Withdrawal load capacity of wood

screws are available for some species and joints may be designed accordingly. End grain load on wood screws are unreliable and wood screws shall not be used for that purpose.

18.4 Nail and screw holding power is one of the very important properties of timber which governs its utilization in diverse industrial and engineering fields. This has been evaluated by experimental method followed for evaluation of such properties as per normal procedure given in the accepted standards [C-3A(20)], using:

- Specimen* — of size 150 mm × 50 mm × 50 mm;
- Nail* — diamond pointed wire nail of 50 mm length and 2.50 mm shank diameter;
- Screw* — galvanized gimlet pointed, No. 8; and
- Conditions* — penetration 25 mm.

While nails were driven by a hand hammer, screws were driven by a screw driver in a 2.5 mm diameter prebore on radial, tangential and end faces.

Composite nail and screw holding power of some Indian timbers, evaluated as above, is presented in [Table 23](#) for guidance in design.

Table 23 Composite Nail and Screw Holding Power of Some Indian Timbers					
(Clause 18.4)					
Sl No.	Timber Species	Locality	Standard Specific Gravity (Weight Oven Dry)/Volume Green	Composite Holding Power	
				N	
				Nail	Screw
(1)	(2)	(3)	(4)	(5)	(6)
i)	<i>Acacia catechu</i> (Khair)	Uttar Pradesh	0.888	2 119	5 454
ii)	<i>Acacia ferruginea</i>	Maharashtra	0.876	1 893	5 700
iii)	<i>Acacia leucophloea</i>	Madhya Pradesh	0.660	1 776	4 503
iv)	<i>A. nilotica</i> (Babul)	Uttar Pradesh	0.670	1 815	5 219
v)	<i>Acrocarpus fraxinifolius</i> (Mundani)	Uttar Pradesh	0.662	1 020	2 865
vi)	<i>Adina oligocephala</i>	Arunachal Pradesh	0.612	1 795	3 453
vii)	<i>Aesculus indica</i> (Horse chestnut)	Himachal Pradesh	0.428	755	2 119
viii)	<i>Albizia lucida</i>	Arunachal Pradesh	0.501	1 462	3 188
ix)	<i>Anogeissus sericea</i>	Madhya Pradesh	0.738	1 570	4 944
x)	<i>Aphanamixis polgstachya</i> (Pitraj)	West Bengal	0.557	1 648	3 110
xi)	<i>Araucaria cunning hanvil</i>	Uttar Pradesh	0.486	1 128	2 943
xii)	<i>Aritocarpus lakoocha</i> (Lakooch)	Uttar Pradesh	0.530	1 050	2 982
xiii)	<i>Azadirachta indica</i> (Neem)	Uttar Pradesh	0.693	2 001	5 219
xiv)	<i>Betula alnoides</i> (Birch)	West Bengal	0.569	1 648	3 237
xv)	<i>Broussonetia papyrifera</i>	Uttar Pradesh	0.394	746	1 383
xvi)	<i>Canarium strictum</i> (White dhup)	Assam	0.486	1 207	2 619
xvii)	<i>Cassia fistula</i> (Amaltos)	Uttar Pradesh	0.746	2 109	4 748
xviii)	<i>Cassia siamea</i>	Madhya Pradesh	0.697	1 315	3 649

Table 23 (Continued)

Sl No.	Timber Species	Locality	Standard Specific Gravity (Weight Oven Dry)/Volume Green	Composite Holding Power N	
				Nail	Screw
(1)	(2)	(3)	(4)	(5)	(6)
xix)	<i>Castanopsis indica</i> (Indian chestnut)	Meghalaya	0.591	1 697	3 090
xx)	<i>Cupressus cashmeriana</i>	Uttar Pradesh	0.409	903	2 325
xxi)	<i>Dipterocarpus bourdillonii</i> (Gurjan)	Kerala	0.577	912	2 766
xxii)	<i>Dipterocarpus indicus</i> (Gurjan)	Karnataka	0.618	1 109	2 904
xxiii)	<i>Eucalyptus hybrid</i>	Uttar Pradesh	0.632	1 736	2 453
xxiv)	<i>Exbucklandia populnea</i> (Pipli)	West Bengal	0.558	1 403	3 443
xxv)	<i>Fraxinus udheri</i>	Uttar Pradesh	0.539	1 354	3 483
xxvi)	<i>Gluta travancorica</i> (Gluta)	Tamil Nadu	0.621	1 324	2 923
xxvii)	<i>Grewia tiliifolia</i> (Dhaman)	Karnataka	0.679	1 746	2 884
xxviii)	<i>Holarhena antidysenterica</i> (Kurchi)	Uttar Pradesh	0.445	1 020	2 698
xxix)	<i>Lagerstroemia speciosa</i> (Jarul)	Kerala	0.531	1 697	2 717
xxx)	<i>Lannea coromandelica</i> (Jhingan)	Rajasthan	0.497	1 216	2 894
xxxi)	<i>Magnolia hookeri</i>	Arunachal Pradesh	0.403	1 128	1 746
xxxii)	<i>Mangifera indica</i> (Mango)	Uttar Pradesh	0.411	1 383	2 433
xxxiii)	<i>Manglietia spp.</i>	Assam	0.409	912	2 031
xxxiv)	<i>Mesna assamica</i> (Sianhor)	Assam	0.687	1 795	3 502
xxxv)	<i>Michelia champaca</i> (Champ)	Assam	0.488	1 128	2 737
xxxvi)	<i>Mimusops elengi</i>	Karnataka	0.762	2 158	5 101
xxxvii)	<i>Morus alba</i> (Mulberry)	Uttar Pradesh	0.557	1 030	4 091
xxxviii)	<i>Morus laevigata</i> (Bola)	Assam	0.567	932	2 600
xxxix)	<i>Morus spp.</i>	Uttar Pradesh	0.594	1 589	3 129
xl)	<i>Nephelium litchi</i>	Uttar Pradesh	0.762	1 825	4 238
xli)	<i>Oroxylum indicum</i>	Andaman	0.370	961	1 736
xl ii)	<i>Ougenia oogeinensis</i> (Sandan)	Uttar Pradesh	0.701	1 570	5 317
xl iii)	<i>Pinus kesiya</i>	Nagaland	0.489	1 089	2 354
xl iv)	<i>Pinus kesiya</i>	Orissa	0.453	1 118	2 325
xl v)	<i>Pinus roxburghii</i> (Chir)	Uttar Pradesh	0.453	1 128	2 482
xl vi)	<i>Pinus wallichiana</i> (Kail)	Himachal Pradesh	0.468	873	1 521
xl vii)	<i>Pistacia integerrima</i>	Jammu and Kashmir	0.733	1 324	4 120
xl viii)	<i>Polyalthia spp.</i> (Debdara)	Madhya Pradesh	0.586	1 452	2 953
xl ix)	<i>Pometa pinnata</i>	Andaman	0.636	1 579	3 532

Table 23 (Concluded)

Sl No.	Timber Species	Locality	Standard Specific Gravity (Weight Oven Dry)/Volume Green	Composite Holding Power N	
				Nail	Screw
(1)	(2)	(3)	(4)	(5)	(6)
l)	<i>Prunus nepalensis</i> (Arupati)	West Bengal	0.482	1 099	2 531
li)	<i>Pterocarpus marsupium</i>	Bihar	0.587	1 305	2 982
lii)	<i>Pterygota alata</i> (Narikel)	Assam	0.475	1 403	3 463
liii)	<i>Pterygota alata</i> (Narikel)	Kerala		1 089	2 639
liv)	<i>Robinia spp.</i>	Jammu and Kashmir	0.629	2 227	3 777
lv)	<i>Sageraea elliptica</i> (Chooi)	Andaman	0.682	1 805	4 522
lvi)	<i>Schleichera oleosa</i> (Kusum)	Bihar	0.841	1 884	4 277
lvii)	<i>Shorea robusta</i> (Sal)	Uttar Pradesh	0.774	1 216	3 620
lviii)	<i>Soyimida ferifuga</i>	Tamil Nadu	0.965	1 687	6 033
lix)	<i>Tectona grandis</i> (Teak)	Uttar Pradesh	0.574	1 020	3 404
lx)	<i>Tectona grandis</i> (Teak)	Tamil Nadu	0.633	1 364	3 483
lxi)	<i>Tectona grandis</i> (Teak)	Maharashtra	0.526	971	3 728
lxii)	<i>Terminalia alata</i> (Laurel)	Uttar Pradesh	0.806	2 521	3 806
lxiii)	<i>Terminalia citrina</i>	Assam	0.629	1 540	3 855
lxiv)	<i>Terminalia paniculata</i> (Kindal)	Kerala	0.630	1 982	3 257
lxv)	<i>Tetrameles nudiflora</i> (Maina)	Assam	0.289	667	1 413
lxvi)	<i>Thespesia populnea</i> (Bhendi)	Maharashtra	0.647	1 923	5 425
lxvii)	<i>Tsuga brunonsiana</i>	West Bengal	0.380	540	2 050
lxviii)	<i>Ulmus coallichiana</i> (Elm)	Himachal Pradesh	0.455	834	1 766
lxix)	<i>Zanthoxylum spp.</i>	West Bengal	0.496	1 138	3 071

19 PROTECTION AGAINST TERMITE ATTACK IN BUILDINGS

19.1 Two groups of organisms which affect the mechanical and aesthetic properties of wood in houses are fungi and insects. The most important wood destroying insects belong to termites and beetles. Of about 250 species of wood destroying termites recorded in India, not more than a dozen species attack building causing about 90 percent of the damage to timber and other cellulosic materials. Subterranean termites are the most destructive of the insects that infest wood in houses justifying prevention measures to be incorporated in the design and construction of buildings.

19.1.1 Control measures consist in isolating or sealing off the building from termites by chemical and non-chemical construction techniques. It is recognized that 95 percent damage is due to internal travel of the termites from ground upwards rather than external entry through entrance thus calling upon for appropriate control measures in accordance with good practices [C-3A(21)].

19.2 Chemical Methods

Termites live in soil in large colonies and damage the wooden structure in the buildings by eating up the wood or building nests in the wood. Poisoning the soil under and around the building is a normal recommended practice. Spraying of chemical solution in the trenches of foundations in and around walls, areas under floors before and

after filling of earth, etc. In already constructed building the treatment can be given by digging trenches all around the building and then giving a liberal dose of chemicals and then closing the trenches.

19.3 Wood Preservatives

Natural resistance against organisms of quite a few wood species provides durability of timber without special protection measure. It is a property of heartwood while sapwood is normally always susceptible to attack by organisms. Preservatives should be well applied with sufficient penetration into timber. For engineers, architects and builders, the following are prime considerations for choice of preservatives:

- a) Inflammability of treated timber is not increased and mechanical properties are not decreased;
- b) Compatibility with the glue in laminated wood, plywood and board material;
- c) Water repellent effect is preferred;
- d) Possible suitability for priming coat;
- e) Possibility of painting and other finishes;
- f) Non-corrosive nature in case of metal fasteners; and
- g) Influence on plastics, rubber, tiles and concrete.

19.4 Constructional Method

Protection against potential problem of termite attack can simply be carried out by ordinary good construction which prevents a colony from gaining access by:

- a) periodic visual observations on termite galleries to be broken off;
- b) specially formed and properly installed metal shield at plinth level; and
- c) continuous floor slabs, apron floors and termite grooves on periphery of buildings.

20 CROSS-LAMINATED TIMBER (CLT)

Cross-laminated timber (CLT) is an engineered wood product used as a structural component in mass timber construction. CLT panels shall consist of not less than three layers of solid-sawn lumber or structural composite lumber, with adjacent layers cross-oriented and bonded with structural adhesives to form a solid wood element.

20.1 General Requirements

20.1.1 Material Specifications

CLT panels shall be manufactured to meet stringent performance criteria for mechanical properties, including load-bearing capacity, dimensional tolerances, and adhesive bond durability to ensure structural reliability. Panels should also demonstrate resistance to delamination and other essential physical properties through rigorous testing to validate long-term performance under various environmental conditions.

The timber used for CLT production shall be kiln-dried to a moisture content of 12 percent $\pm$ 2 percent, ensuring stability and resistance to deformation.

20.1.2 Dimensions

CLT panels shall have a minimum thickness of 50 mm, with thicknesses ranging up to 500 mm for structural applications. The typical panel dimensions for commercial use may be up to 18 000 mm in length and 3 000 mm in width.

The number of layers shall be odd (for example, 3, 5, 7), with layer thicknesses adjusted according to structural requirements.

20.2 Physical Properties

20.2.1 Delamination Resistance

Delamination testing shall be conducted to ensure that total delamination ($Delam_{tot}$) shall not exceed 10 percent, and maximum delamination ($Delam_{max}$) shall not exceed 40 percent, ensuring durability and structural integrity.

Results from studies indicate that CLT made from plantation-grown hardwoods such as Rubberwood, *Melia dubia* (Malabar Neem), and Eucalyptus meets these requirements, with delamination values comparable to Radiata pine, a commonly used species in CLT production.

20.2.2 Water Absorption and Swelling Properties

CLT panels shall be evaluated for water absorption (WA), thickness swelling (TS), and volumetric swelling (VS) using the water soaking method. These parameters shall be assessed to ensure that the panels meet acceptable limits for moisture-induced dimensional changes, maintaining stability under varying environmental conditions.

Reference to results from studies ([Table 24](#) and [Table 25](#)) shows that hardwood species demonstrate comparable performance to Radiata pine, indicating their suitability for structural applications.

For detailed information, a reference may be made to *A Hybrid Approach to Green Building Construction Using Mass Timber* February 2024, DOI: 10.58532/V3BICE3P8CH2 by Ram Kumar V R and Amitava Sil.

Table 24 Physical Properties of Polyurethane (PUR) Bonded Homogenous CLT in Comparison to Radiata Pine CLT
(Clause [20.2.2](#))

Sl No.	Type of CLT	$Delam_{tot}$ Percent	$Delam_{max}$ Percent	WA Percent		TS Percent		VS Percent	
		CLT	CLT	CLT	Solid wood	CLT	Solid wood	CLT	Solid Wood
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	Rubberwood	6.49 (0.98)	29.01 (2.90)	31.23 (4.19)	52.91 (5.47)	3.26 (0.93)	9.93 (1.82)	3.26 (0.93)	9.93 (1.82)
ii)	<i>Melia dubia</i>	6.77 (0.78)	26.65 (3.86)	23.51 (3.48)	39.98 (9.94)	4.38 (2.28)	9.15 (3.29)	7.86 (1.89)	13.58 (6.85)
iii)	Silver oak	8.08 (1.21)	21.3 (4.13)	14.10 (3.66)	35.88 (12.28)	5.05 (1.02)	13.19 (1.89)	8.37 (1.43)	20.83 (2.62)
iv)	Eucalyptus	6.61 (2.29)	34.10 (3.28)	25.68 (7.57)	38.97 (7.57)	4.06 (1.24)	15.08 (7.76)	7.39 (1.70)	17.92 (7.74)
v)	Radiata pine	6.63 (0.86)	26.70 (3.63)	29.35 (5.46)	40.03 (7.20)	4.29 (0.29)	12.47 (3.01)	14.09 (0.66)	20.62 (2.82)
NOTE — Values in the parentheses indicate standard deviation.									

Table 25 Physical Properties of Hybrid CLT in Comparison to Radiata Pine CLT

(Clause 20.2.2)

Sl No.	Type of CLT	<i>Delam</i> _{tot} Percent	<i>Delam</i> _{max} Percent	WA Percent	TS Percent	VS Percent
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Hybrid with <i>Eucalyptus</i> (E) and <i>Melia dubia</i> (M) (EME)	7.68 (0.57)	31.62 (4.37)	26.43 (3.66)	6.33 (1.77)	15.86 (1.47)
ii)	Hybrid with <i>Melia dubia</i> (M) and <i>Eucalyptus</i> (E) (MEM)	6.68 (0.53)	28.65 (3.82)	24.35 (6.86)	5.59 (1.38)	8.85 (1.93)
iii)	Hybrid with <i>Melia dubia</i> (M) and Silver oak (S) (MSM)	7.06 (1.22)	28.96 (1.12)	21.86 (5.09)	5.43 (1.84)	7.04 (3.94)
iv)	Hybrid with silver oak (S) and <i>Melia dubia</i> (M) (SMS)	7.65 (0.23)	26.62 (1.45)	15.51 (5.6)	8.43 (2.30)	8.24 (2.96)
v)	Radiata pine	6.63 (0.86)	26.70 (3.63)	29.35 (5.46)	4.29 (0.29)	14.09 (0.66)

NOTE — Values in the parentheses indicate standard deviation.

20.2.3 Density

The density of CLT panels shall be determined according to the timber species and adhesive used. The density of CLT panels shall range from 5 500 kg/m³ to 8 500 kg/m³, depending on the species (for example, Rubberwood, Eucalyptus). Table 26 provides the density values for different species used in CLT production.

Table 26 Density of CLT Panels

(Clause 20.2.3)

Sl No.	Species	Density of Wood (at 12 Percent Moisture Content) kg/m ³	Density of CLT kg/m ³
(1)	(2)	(3)	(4)
i)	Rubberwood	6 200 (0.05)	6 500 (0.08)
ii)	<i>Melia dubia</i>	5 100 (0.04)	5 500 (0.01)
iii)	Eucalyptus	8 000 (0.05)	8 500 (0.03)
iv)	Silver oak	6 200 (0.02)	6 800 (0.02)
v)	Radiata pine	4 200 (0.03)	4 500 (0.03)
vi)	EME	—	7 700 (0.02)
vii)	MEM	—	6 400 (0.03)
viii)	MSM	—	5 500 (0.02)
ix)	SMS	—	6 700 (0.01)

NOTE — Values in the parentheses indicate standard deviation.

20.3 Mechanical Properties

20.3.1 Bending, Compression, and Shear Strength

Mechanical testing shall include four-point bending tests, compression parallel to grain tests and block shear tests. The results shall comply with the minimum strength requirements for structural applications as specified in Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS.

[Table 27](#) and [Table 28](#) present the mechanical properties of homogeneous and hybrid CLT panels, showing adequate performance for structural uses. Rubberwood CLT exhibited higher mechanical properties than hybrid configurations in some tests.

20.3.2 Impact of Adhesive Type on Mechanical Performance

The choice of adhesive (for example, phenol resorcinol formaldehyde, PUR) shall be considered, as it affects the bonding strength and durability. Mechanical properties of CLT bonded with different adhesives, indicating that PRF-bonded panels have superior performance.

Table 27 Mechanical Properties of Homogeneous CLT Panels					
(Clause 20.3.1)					
Sl No.	CLT Made from	MoE MPa	MoR MPa	Compression Parallel to Grain MPa	Block Shear Strength MPa
(1)	(2)	(3)	(4)	(5)	(6)
i)	Rubberwood	8 600 (328.3)	48.7 (13.8)	25.0 (2.3)	6.7 (1.4)
ii)	<i>Melia dubia</i>	6 758 (598.6)	44.8 (5.7)	23.0 (1.8)	6.3 (0.7)
iii)	Silver oak	8 835 (739.5)	64.2 (6.9)	26.34 (2.0)	6.7 (0.5)
iv)	<i>Eucalyptus</i>	13 811 (1468.7)	81.6 (8.3)	35.40 (6.7)	6.65 (0.6)
v)	Radiata pine	4 542 (1 024.3)	49.6 (10.3)	16.4 (2.3)	3.1 (0.5)
NOTE — Values in the parentheses indicate standard deviation.					

Table 28 Mechanical Properties of Hybrid CLT Panels					
(Clause 20.3.1)					
Sl No.	CLT Made from	MoE MPa	MoR MPa	Compression Parallel to Grain MPa	Block Shear Strength MPa
(1)	(2)	(3)	(4)	(5)	(6)
i)	EME	10 943 (1 074.8)	70.0 (7.2)	30.5 (3.7)	5.8 (0.6)
ii)	MEM	7 770 (938.5)	47.0 (4.5)	24.6 (2.2)	5.8 (0.6)
iii)	SMS	7 701 (600.5)	61.6 (7.5)	33.5 (4.0)	5.6 (0.5)
iv)	MSM	6 328 (652.8)	45.5 (4.3)	24.2 (3.1)	5.3 (0.5)
NOTE — Values in the parentheses indicate standard deviation.					

20.4 Fire Safety Requirements

20.4.1 Fire Resistance Ratings

CLT panels should have a fire-resistance rating in accordance with Part F 'Fire and Life Safety' of this NBCS. This shall be achieved through inherent fire resistance, encapsulation with materials like gypsum board (minimum thickness 13 mm), or a combination of methods.

The charring rate of CLT shall be considered in fire design calculations, with a recommended charring rate of 0.65 mm/min for calculating the load-bearing capacity during fire exposure.

20.4.2 Protection of Connections

Steel connectors and other metal components shall be protected with fire-resistant coatings or encased in non-combustible materials to ensure structural integrity during a fire.

20.5 Durability and Biological Resistance

20.5.1 Preservative Treatments

CLT panels shall be treated with preservatives like bifenthrin or copper chrome borate to enhance resistance to biological degradation. Testing has shown that preservative-treated CLT has improved resistance to termite and fungal attacks compared to untreated samples (see [Table 29](#) for termite degradation results).

Table 29 Degradation of Samples due to Termite Attack			
(Clause 20.5.1)			
Sl No.	Sample	Average Degradation percent	Duration of Exposure months
(1)	(2)	(3)	(4)
i)	Control solid wood	100	6
ii)	Control CLT	100	9
iii)	Calcium channel blockers (CCB) treated CLT	60	36
iv)	Bifenthrin treated CLT	0	36

20.5.2 Performance Under Biological Exposure

Durability testing shall include assessments for resistance to termites, white rot and brown rot fungi. Untreated rubberwood is highly susceptible to biological degradation, while treated samples showed minimal degradation after prolonged exposure.

20.5.3 Effect of Preservatives on Mechanical Properties

The use of preservatives should not adversely affect the mechanical properties of CLT. Block shear strength tests indicate that treated panels retain comparable mechanical performance to untreated panels.

20.6 Handling, Storage and Installation

20.6.1 Moisture Protection

During construction, CLT panels shall be protected from moisture exposure. Panels should be stored in a dry environment and water infiltration during transportation and installation shall be minimized.

21 MASS TIMBER

Structural elements primarily composed of solid, built-up, panelized, or engineered wood products are categorized as products which include, but are not limited to, cross-laminated timber (CLT), glued-laminated timber (glulam), and laminated veneer lumber (LVL). It consists of multiple layers of wood, mechanically or adhesively bonded together to create large, strong and stable components suitable for load-bearing and non-load-bearing uses in construction. Mass timber elements can be used for structural framing, walls, floors, roofs and columns. It applies to the construction of residential, commercial, and public buildings where structural elements, including floors, walls, roofs and columns, are made from engineered wood products like cross-laminated timber (CLT), glued-laminated timber (glulam) and other solid or panellised wood products.

These load-bearing and non-load-bearing products include:

- a) Cross-laminated timber (CLT);
- b) Glued-laminated timber (Glulam);
- c) Nail-laminated timber (NLT); and
- d) Laminated veneer lumber (LVL).

21.1 Design and Structural Integrity

21.1.1 Structural Design Requirements

- a) *Load-bearing capacity* — Mass timber elements shall be designed to carry the vertical and lateral loads imposed on the building in compliance with Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS. The design shall account for dead loads, live loads, wind loads, seismic loads and other imposed loads;
- b) *Cross-sectional dimensions* — The minimum cross-sectional dimensions for mass timber elements (for example, columns, beams, floor slabs) shall be specified based on building height, function and occupancy. For example, CLT panels used for floor slabs should meet a minimum thickness of 100 mm, depending on the structural requirements; and
- c) *Deflection limits* — The design of mass timber structures shall ensure that the deflection under service load conditions remains within acceptable limits as defined in this Part of this NBCS.

21.1.2 Hybrid Systems

Where mass timber is used in conjunction with steel or concrete, the design should ensure that load paths are continuous and that materials are compatible. Load transfer mechanisms shall be adequately designed to prevent failure at the connection points between materials.

21.1.3 Seismic Performance

Mass timber buildings located in seismic zones should be designed to accommodate lateral loads and displacements as per Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS. Timber's flexibility can offer good seismic performance, but connections and structural integrity should be robustly designed to prevent failure.

21.2 Fire and Life Safety

It should be ensured in accordance with Part F ‘Fire and Life Safety’ of this NBCS.

21.2.1 Protection of Connections

All steel connectors, bolts and fasteners used in mass timber construction shall be protected with fire-resistant coatings or enclosed within non-combustible materials to prevent failure during fire exposure.

21.3 Minimum Dimensions and Thickness of Mass Timber Elements

21.3.1 Columns and Beams

Mass timber columns and beams shall be designed to meet the structural load requirements with a minimum cross-sectional dimension of 200 mm for columns and 150 mm for beams, depending on the span and load requirements.

21.3.2 Floor and Roof Panels

CLT panels used for floor and roof assemblies should meet a minimum thickness of 100 mm for floor systems and 85 mm for roof systems to achieve the required fire and structural performance.

21.3.3 Wall Panels

CLT or NLT wall panels shall meet a minimum thickness of 125 mm to ensure fire resistance and structural stability under lateral and vertical loads.

21.4 Construction Practices

21.4.1 Handling and Installation of Mass Timber Elements

During construction, care should be taken to protect mass timber elements from moisture. Elements shall be stored in dry conditions and protected from water exposure during construction to prevent damage and degradation.

21.4.2 Assembly and Connections

Connections between mass timber elements (for example, floor-to-wall, beam-to-column) should be engineered to handle both static and dynamic loads. These connections shall be made using metal connectors, dowels, or bolts designed to prevent movement and ensure continuous load transfer. Special attention should be paid to fire protection at these connection points.

21.4.3 Fire Sealing of Penetrations

Any penetrations for services (for example, plumbing, electrical) through mass timber elements shall be sealed with approved fire-rated materials to maintain the integrity of the fire-rated assembly.

21.5 Sustainability and Environmental Benefits

21.5.1 Environmental Performance

Mass timber is recognized for its carbon sequestration properties, where wood captures and stores carbon dioxide. Buildings made from mass timber reduce the overall carbon footprint of the construction process compared to traditional materials like steel and concrete.

21.5.2 Energy Efficiency

Mass timber inherent thermal insulation properties help to regulate indoor temperatures, thereby reducing energy consumption for heating and cooling. The thermal resistance of wood should be considered in the building envelope design, allowing for thinner wall assemblies with better insulation compared to conventional materials.

22 LIFE CYCLE ASSESSMENT (LCA) OF TIMBER FOR USE IN BUILDINGS

22.1 Timber used in building structures and elements shall undergo a comprehensive life cycle assessment (LCA) to evaluate its environmental impacts across its entire lifecycle, from raw material extraction to disposal or recycling. LCA provides valuable insights into the sustainability of timber and its environmental footprint in building applications. The assessment shall cover the following stages:

- a) *Production and Construction* — This stage evaluates the energy and resources required for the extraction, transportation and processing of raw timber into building materials. It also includes the energy

consumption and waste generated during the construction process, as well as the transportation of materials to the construction site.

- b) *Use Stage* — The use stage assesses the environmental impacts of timber during the building's operational lifespan. This includes maintenance, repairs and potential replacement of timber components, considering the durability, performance and long-term sustainability of the material.
- c) *End-of-Life* — At the end of the building's useful life, the timber structure may undergo demolition, with the materials being processed for waste management, recycling, or reuse. This stage assesses the environmental effects of these processes and the potential for timber materials to be repurposed.
- d) *Externalized Impacts beyond the System Boundary* — The incidental effects of recycling, reusing and recovering energy, water, or materials beyond the system boundaries of the LCA study are included in this stage. These externalized impacts consider how recycled timber or materials might contribute to environmental benefits outside the building's lifecycle.

22.2 Environmental Impact Categories

The LCA may be used to assess the environmental impact of timber buildings in several categories, which encompass emissions and pollutants generated throughout the lifecycle. These categories include:

- a) *Global Warming Potential* — The contribution of greenhouse gas emissions to climate change;
- b) *Acidification Potential* — The potential for acidic compounds to form in the environment, affecting ecosystems and infrastructure;
- c) *Eutrophication Potential* — The impact of nutrient loading on water bodies, potentially causing excessive algal growth;
- d) *Ozone Depletion Potential* — The impact of substances that contribute to the depletion of the ozone layer; and
- e) *Smog Formation Potential* — The potential for air pollution to contribute to the formation of ground-level ozone and smog.

All relevant material and process quantities shall be included in the LCA inventory and their impacts shall be calculated using recognized tools and methodologies.

22.3 Requirements for Life Cycle Assessment

In the context of building with timber, LCA may be required for the following purposes:

- a) *Decision-Making Support* — Providing building owners and stakeholders with data on the environmental and sustainability aspects of timber, aiding informed decision-making;
- b) *Design Evaluation* — Offering insights into material selection and environmental performance across different design possibilities;
- c) *Product Comparison* — Demonstrating the environmental advantages of timber in comparison to alternative building materials;
- d) *Environmental Benefit Evaluation* — Helping assess the ecological benefits of innovative timber products or policies; and
- e) *Performance Benchmarking* — Comparing the environmental performance of timber buildings against established benchmarks and sustainability goals.

22.4 Methodology for Conducting LCA

The LCA of timber buildings may be conducted iteratively, with increasing detail as the design or construction phases progress. The process should follow these steps:

- a) Establish the purpose and scope of the LCA, focusing on the objectives of evaluating timber's environmental impact;
- b) Gather data on the timber materials, including their types, quantities, lifespans and lifecycle stages. Special attention should be given to materials used in the production and use stages;

- c) Using LCA tools (for example, SimaPro, MATLAB), assess the environmental impacts of timber buildings. The evaluation shall include biogenic carbon considerations, as timber stores carbon from the atmosphere, contributing to carbon sequestration during its lifecycle;
- d) Interpret the LCA results by categorizing environmental impacts by building component, material type, and lifecycle stage. Identify key contributors to the building's overall environmental footprint and visualize the outcomes; and
- e) Document all findings, including assumptions, methodology and conclusions and present the results in reports for stakeholders. The results shall guide sustainable construction practices and decision-making in timber building projects.

22.5 Reporting

The results of the LCA should be documented, including all assumptions, methodologies and conclusions. Reports should be prepared in accordance with relevant standards and may be used to guide sustainable building practices and decision-making in construction.

(Clause 4.1)

A-1 The [Table 30](#) below specifies the mechanical properties of selected foreign timber species, indicating their density, modulus of elasticity, permissible stresses, and preservative characteristics. This information is intended to serve as a reference for designers and engineers, particularly in cases where such species are imported for structural or allied applications, while ensuring that due consideration is given to relevant standards and treatment requirements.

(Clause A-1)

Sl No.	Species		Locality from where Tested	Average Density at 12 Percent Moisture Content	Modulus of Elasticity (All Grades and All Locations) × 10 ³	Permissible Stress			
	Botanical Name	Trade Name				Bending-Tension Along Grain, Extreme Fibre Stress	Shear all Locations Horizontal	Compression Parallel to Grain	Compression Perpendicular to Grain
						N/mm ²	N/mm ²	N/mm ²	N/mm ²
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(8)	(8)
i)	<i>Pseudotsuga menziesii</i>	Douglas fir (DF)	Pacific NW, USA	513	13.4	Inside: 11.0 Outside: 9.4 Wet: 9.4	1.24 (Inside) 1.05 (Outside)	Inside: 11.7 Outside: 9.9 Wet: 9.9	2.9
ii)	<i>Tsuga heterophylla</i>	Western hemlock (WH)	Pacific NW, USA	500	11.2	Inside: 9.3 Outside: 7.9 Wet: 7.9	1.03 (Inside) 0.88 (Outside)	Inside: 9.3 Outside: 7.9 Wet: 7.9	2.4

NOTE — The above data on foreign timber species is reproduced from the *USDA Wood Handbook (USA)*. This information is for reference only. At present, no indigenous test data under Indian climatic and service conditions is available for these species. Therefore, before adopting them in structural applications in India, the properties should be verified through appropriate laboratory testing as per accepted standards.

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of the enforcement of this NBCS. The standards listed may be used for conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parenthesis indicates the number of the reference in this Subsection.

	<i>IS No.</i>	<i>Title</i>
(1)	IS 707 : 2011	Timber technology and utilization of wood, bamboo and cane — Glossary of terms (<i>third revision</i>)
(2)	IS 1708 (Parts 1 to 18) : 1986	Methods of testing small clear specimens of timber (<i>second revision</i>)
	IS 2408 : 1963	Methods of static tests of timbers in structural sizes
	IS 2455 : 1990	Methods of sampling of model trees and logs and their conversion for timer testing (<i>second revision</i>)
(3)	IS 4970 : 1973	Key for identification of commercial timbers (<i>first revision</i>)
(4)	IS 287 : 1993	Permissible moisture content of timber used for different purposes — Recommendations (<i>third revision</i>)
(5)	IS 3629 : 1986	Specification for structural timber in building (<i>first revision</i>)
(6)	IS 401 : 2001	Preservation of timber — Code of practice (<i>fourth revision</i>)
(7)	IS 2366 : 1983	Nail-jointed timber construction — Code of practice (<i>first revision</i>)
(8)	IS 4983 : 1968	Code of practice for design and construction of nailed laminated timber beams
(9)	IS 11096 : 1984	Code of practice for design and construction of bolt jointed timber construction
(10)	IS 4907 : 2004	Methods of testing timber connector joints (<i>first revision</i>)
(11)	IS 9188 : 1979	Performance requirements for adhesive for structural laminated wood products for use under exterior exposure conditions
(12)	IS 14616 : 1999	Laminated veneer lumber — Specification
(13)	IS 10701 : 2012	Structural plywood — Specification (<i>first revision</i>)
	IS 4990 : 2024	Plywood for concrete shuttering works — Specification (<i>fourth revision</i>)
(14)	IS 4924	Method of test for nail-jointed timber trusses:
	(Part 1) : 1968	Destructive test
	(Part 2) : 1968	Proof test
(15)	IS 12120 : 1987	Code of practice for preservation of plywood and other panel products
(16)	IS 4990 : 2024	Plywood for concrete shuttering works — Specification (<i>fourth revision</i>)
(17)	IS 9307 (Parts 1 to 8) : 1979	Methods of test for wood-based structural sandwich construction
(18)	IS 723 : 2023	Steel countersunk head wire nails — Specification (<i>second revision</i>)
(19)	IS 451 : 1999	Technical supply conditions for wood screws (<i>third revision</i>)
	IS 6736 : 1972	Specification for slotted raised countersunk head wood screw

	<i>IS No.</i>	<i>Title</i>
	IS 6739 : 1972	Specification for slotted round head wood screws
	IS 6760 : 1972	Specification for slotted countersunk head wood screws
(20)	IS 1708 (Parts 1 to 18) : 1986	Methods of testing small clear specimens of timber (<i>second revision</i>)
(21)	IS 6313 (Part 1) : 1981	Code of practice for anti-termite measures in buildings: Part 1 Constructional measures (<i>first revision</i>)
	IS 6313	Anti-termite measures in buildings — Code of practice:
	(Part 2) : 2022	Pre-constructional chemical treatment measures (<i>fourth revision</i>)
	(Part 3) : 2022	Treatment for existing buildings (<i>fourth revision</i>)

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 3 TIMBER AND BAMBOO: 3B BAMBOO

BUREAU OF INDIAN STANDARDS

C O N T E N T S

FOREWORD	...	3
1 SCOPE	...	7
2 TERMINOLOGY	...	7
3 SYMBOLS	...	10
4 MATERIALS	...	11
5 PERMISSIBLE STRESSES	...	14
6 DESIGN CONSIDERATIONS	...	18
7 DESIGN, TECHNIQUES OF JOINTS AND CONSTRUCTION PRACTICES	...	22
8 BAMBOO AS REINFORCEMENT IN CONCRETE	...	31
9 WALLING, FLOORING, ROOFING, AND IN-FILL PANELS FOR WALL AND COMMON ROOF COVERING/CLADDING MATERIALS		33
10 LIFE CYCLE ASSESSMENT OF BAMBOO FOR USE IN BUILDINGS	...	40
11 TYPICAL DETAILS OF CONVENTIONAL BAMBOO BUILDING CONSTRUCTION	...	41
ANNEX A SOURCE AND LOCAL NAMES OF SOME OF THE SPECIES OF BAMBOO	...	42
ANNEX B RECOMMENDATIONS FOR BAMBOO STRUCTURES IN HIGH WIND/CYCLONE PRONE AREAS	...	45
ANNEX C REPRESENTATIVE DESIGNS AND CONSTRUCTION DETAILS FOR BAMBOO STRUCTURES	...	46
LIST OF STANDARDS	...	51

National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part C/Subsection 3B) covers the general principles involved in the design of structural bamboo in buildings with regard to mechanical resistance and durability of structures. Design of both bamboo (round bamboo, split bamboo, glued laminated bamboo) and bamboo-based panels joined together with adhesives or mechanical fasteners are covered in this Subsection. It also covers minimum strength data, dimensional stability, grading requirements and traditional bamboo joints for quality assurance. Constructional aspects using bamboo, such as, work on site, fabrication of components off-site and their erection on site is also covered to the extent necessary to indicate and ensure the quality of material and standard of workmanship to comply with the assumptions of the design rules and the limitations.

Bamboo is a versatile resource possessing high strength-to-weight ratio and cost ratio and offers considerable ease in working with simpler tools. It has a long and well established tradition as a building material. The application of bamboo as a constructional material is largely based on established traditions and intuitions of forefathers throughout the tropical and sub-tropical regions.

Resilience coupled with light weight makes bamboo an ideal material for housing in disaster prone/earthquake prone areas. It has the capacity to withstand repetitive loadings and absorb more energy and show larger deflections before collapse and as such is safer under earth tremors. As the design guidelines were inadequate, the application of bamboo as an engineering material was largely based on practical and engineering experience. A need was therefore felt to develop a Code for design and construction with bamboo so as to cater to a number of social and trade advantages, engineering recognition and the improved status of bamboo as an engineering material. This was achieved by adding a new chapter in the 2005 revision of this code, covering structural design with bamboo. Forest Research Institute, Dehradun and some other organizations were engaged in bamboo research to establish its silviculture, botanical, entomological and pathological aspects besides creating a utilization base. All these works were utilized in the preparation of the 2005 version of this Subsection of this

Some of the suitable species grown in India and neighbouring countries are enlisted in [Annex A](#) along with their local names and source, for general information.

The bamboo culm has a tubular structure consisting essentially of nodes and inter-nodes. In the inter-nodes the cells are axially oriented while the nodes provide the transverse inter-connection. The disposition of the nodes and the wall thickness are significant in imparting strength to bamboo against bending and crushing. In a circular cross-section, bamboo is generally hollow and for structural purposes this form is quite effective and advantageous. Each of the species of bamboo has widely different characteristics affecting its usefulness as constructional material. The strength of bamboo culms, their straightness, lightness combined with hardness, range and size of hollowness make them potentially suitable for a variety of applications both structural and non-structural. With good physical and mechanical properties, low shrinkage and good average density, bamboo is well suited to replace wood in several applications, especially in slats and panel form. Mass production of bamboo components, if required, may be integrated with treatment facilities near to the site of plantation or site of construction, for speed and economy in construction.

The Section on structural design with bamboo was first published in 2005 as a part of National Building Code of India under Subsection 3B of Part 6. In the 1983 version of this code, timber was covered under Section 3 of Part 6 'Structural Design' under the title 'Wood', which did not cover bamboo. In the 2005 version of this code, the scope of Section 3 was enlarged as Section 3 'Timber and bamboo', which was subdivided into Subsection 3A 'Timber' and Subsection 3B 'Bamboo'. This Subsection pertains to 3B 'Bamboo'.

Key changes to the 2016 version included: inclusion of provisions on construction practices using bamboo that was addressed in erstwhile Part 7 'Construction Management, Practices and Safety'; updated provisions for bamboo trusses, bamboo foundations; bamboo as reinforcement in concrete; introduction of the use of bamboo-based materials for walling, flooring, roofing, and in-fill panels, along with provisions for reconstituted wood from bamboo; and updates to the terms and definitions related to bamboo-based panel products.

With the standardization on design and construction aspects of bamboo in this Subsection of this code, it was expected to ensure durable structures, however, some specific guidelines were required in the event of tropical cyclones in the sub-continent and the traditional non-engineered structures generally constructed. Keeping this in view, some recommendations are given in [Annex B](#) which may be followed in addition to provisions given in the main text of this Subsection.

As a result of experience gained in implementation of 2016 version of this code and further research work done, feedback received as well as revision/updation of some of the existing standards, a need to revise this Subsection was felt. This revision has, therefore, been brought out to take care of these aspects.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

The significant changes incorporated in this revision include the following:

- a) Bamboo grading, in [4.4](#), has been expanded to include special grades based on mean outer diameter;
- b) Bamboo mat corrugated sheets (BMCS) are included, in [6.9.9](#), as a sustainable roofing material with high strength, resilience, and thermal resistance for diverse climates;
- c) Comprehensive life cycle assessment to evaluate environmental impacts from production to end-of-life of bamboo structures;
- d) [Annex C](#) has been introduced illustrating conventional bamboo construction details such as trusses, roofing, walls, and flooring; and
- e) References to all the concerned Indian Standards have been updated.

Reference may be made to Part F ‘Fire and Life Safety’ of this NBCS for fire protection of the bamboo buildings.

The information contained in this Subsection is largely based on the works carried out at Forest Research Institute, Dehradun; Institute of Wood Science and Technology, Bangalore; International Network for Bamboo and Rattan (INBAR) documents and Indian Standards given below. Assistance has also been derived from ISO 22156 : 2021 ‘Bamboo structures- Bamboo culms- Structural design’, apart from the following:

IS 9096 : 2006	Preservation of bamboo for structural purpose — Code of practice
IS 15912 : 2018	Structural design using bamboo — Code of practice

All standards, whether given herein above or cross-referred to in the main text of this Subsection, are subject to revision. The parties to agreement based on this Subsection are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Subsection is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 ‘Rules for rounding off numerical values (*second revision*)’. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Subsection.

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Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to “**accepted standard**” in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

- a) Good practices [C-3B(1)] refers to the Indian Standard(s) give at serial number (1) of the list of standards given at the end of this Subsection, that is, IS 9096 : 2006, ‘Code of practice for preservation of bamboo for structural purposes (*first revision*)’

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 3 TIMBER AND BAMBOO: 3B BAMBOO

1 SCOPE

1.1 This NBCS (Part C/Subsection 3B) covers the general principles involved in the design of structural bamboo in buildings with regard to mechanical resistance and durability of structures. Design of both bamboo (round bamboo, split bamboo, glued laminated bamboo) and bamboo-based panels joined together with adhesives or mechanical fasteners are covered in this Subsection. It also covers minimum strength data, dimensional stability, grading requirements and traditional bamboo joints for quality assurance. Constructional aspects using bamboo, such as, work on site, fabrication of components off-site and their erection on site are also covered to the extent necessary to indicate and ensure the quality of material and standard of workmanship to comply with the assumptions of the design rules and the limitations.

1.2 The following aspects are not covered in this Subsection:

- a) Limit state design and performance of structures; and
- b) Scientific designing of bamboo joints and their fastenings.

2 TERMINOLOGY

For the purpose of this Subsection, the following definitions shall apply.

2.1 Anatomical Purpose Definitions

2.1.1 *Bamboo* — Tall perennial grasses found in tropical and sub-tropical regions. They belong to the family *Poaceae* and sub-family *Bambusoideae*.

2.1.2 *Bamboo Culm* — A single shoot of bamboo usually hollow except at nodes which are often swollen.

2.1.3 *Bamboo Clump* — A cluster of bamboo culms emanating from two or more rhizomes in the same place.

2.1.4 *Cellulose* — A carbohydrate, forming the fundamental material of all plants and a main source of the mechanical properties of biological materials.

2.1.5 *Cell* — A fundamental structural unit of plant and animal life, consisting of cytoplasm and usually enclosing a central nucleus and being surrounded by a membrane (animal) or a rigid cell wall (plant).

2.1.6 *Cross Wall* — A wall at the node closing the whole inside circumference and completely separating the hollow cavity below from that above (*Syn.* Diaphragm, that is, horizontal partition in a standing culm) (*see also* [2.2.19](#) and [2.2.32](#)).

2.1.7 *Hemi Cellulose* — The polysaccharides consisting of only 150 to 200 sugar molecules, also much less than the 10 000 of cellulose.

2.1.8 *Lignin* — A polymer of phenyl propane units, in its simple form ($C_6H_5CH_2CH_2CH_3$).

2.1.9 *Sliver* — Thin strips of bamboo processed from bamboo culm.

2.1.10 *Tissue* — Group of cells, which in higher plants consist of (a) Parenchyma — a soft cell of higher plants as found in stem pith or fruit pulp, (b) Epidermis — the outermost layer of cells covering the surface of a plant, when there are several layers of tissue.

2.2 Structural Purpose Definitions

2.2.1 *Bamboo Mat Board* — A board made of two or more bamboo mats bonded with an adhesive.

2.2.2 Bamboo Mat Corrugated Sheet — A sheet made up of adhesive soaked and coated mats assembled and pressed under specified temperature and pressure to obtain sinusoidal or other suitable corrugations.

2.2.3 Bamboo Mat Corrugated Sheets (BMCS) — Roofing sheets made from woven bamboo mats, treated and pressed with resin to form a durable, weather-resistant corrugated material.

2.2.4 Bamboo Mat Ridge Cap (BMRC) — It is a specially shaped bamboo composite designed to cover and seal the ridge line of BMCS roofs, providing a weatherproof finish to the roof structure.

2.2.5 Bamboo Mat Veneer Composite — Panel manufactured with a combination of bamboo mat and veneer. Bamboo mat can be either as outer skins or as core/cross-bands. However, the composite panel shall be balanced construction on either side of central ply.

2.2.6 Beam — A structural member which supports load primarily by its internal resistance to bending.

2.2.7 Breaking Strength — A term loosely applied to a given structural member with respect to the ultimate load it can sustain under a given set of conditions.

2.2.8 Bundle-Column — A column consisting of three or more number of culms bound as integrated unit with wire or strap type of fastenings.

NOTE — During bunching of beams/columns, binders (5 mm tie bar/wire and 15 mm bamboo dowel pins) may be carefully inserted after proper pre-drilling the holes throughout the assembly as per the requirements of structural design.

2.2.9 Centre Internode — A test specimen having its centre between two nodes.

2.2.10 Characteristic Load — The value of load which has a 95 percent probability of not exceeding during the life of the structure.

2.2.11 Characteristic Strength — The strength of the material below which not more than 5 percent of the test results are expected to fall.

2.2.12 Cleavability — The ease with which bamboo can be split along the longitudinal axis. The action of splitting is known as cleavage.

2.2.13 Column — A structural member which supports axial load primarily by inducing compressive stress along the fibres.

2.2.14 Common Rafter — A roof member which supports roof battens and roof coverings, such as boarding and sheeting.

2.2.15 Curvature — The deviation from the straightness of the culm.

2.2.16 Delamination — Separation of mats through failure of glue.

2.2.17 De-structured Bamboo — Raw material (green bamboo) reduced partially to a condition, as far as practicable, in which the element fibres remain interlocked and in their highly aligned state and which with the help of binder can be reconstituted into a valuable product of a desirable size.

2.2.18 End Distance — The distance measured parallel to the fibres of the bamboo from the centre of the fastener to the closest end of the member.

2.2.19 Flattened Bamboo — Bamboo consisting of culms that have been cut and unfolded till it is flat. The culm thus is finally spread open, the diaphragms (cross walls) at nodes removed and pressed flat.

2.2.20 Full Culm — The naturally available circular section/shape.

2.2.21 Fundamental or Ultimate Stress — The stress which is determined on a specified type/size of culms of bamboo, in accordance with standard practice and does not take into account the effects of naturally occurring characteristics and other factors.

2.2.22 Inner Diameter — Diameter of internal cavity of a hollow piece of bamboo.

2.2.23 Inside Location — Position in buildings in which bamboo remains continuously dry or protected from weather.

2.2.24 Joint — A connection between two or more bamboo structural elements.

2.2.25 Joist — A small beam directly supporting floor, ceiling or roof of a structure.

2.2.26 Length of Internode — Distance between adjacent nodes.

2.2.27 Loaded End or Compression End Distance — The distance measured from the centre of the fastener to the end towards which the load induced by the fastener acts.

2.2.28 Machette — A light cutting and slashing tool in the form of a large knife.

2.2.29 Mat — A woven sheet made using thin slivers.

2.2.30 Mortise and Tenon — A joint in which the reduced end (tenon) of one member fits into the corresponding slot (mortise) of the other.

2.2.31 Net Section — Section obtained by deducting from the gross cross-section (A), the projected areas of all materials removed by boring, grooving or other means.

2.2.32 Node — The place in a bamboo culm where branches sprout and a diaphragm is inside the culm and the walls on both sides of node are thicker.

2.2.33 Outer Diameter — Diameter of a cross-section of a piece of bamboo measured from two opposite points on the outer surface.

2.2.33.1 Mean outer diameter — The diameter taken twice on the outer surface, in direction perpendicular to each other at each cross-section on both the ends. Thus, resulting in mean of four readings for the middle of culm.

2.2.34 Outside Location — Position in building in which bamboos are occasionally subjected to wetting and drying as in case of open sheds; and outdoor exposed structures.

2.2.35 Permissible Stress — Stress obtained after applying factor of safety to the ultimate stress. This is also the working stress unless adjusted to the particular structural form in design.

2.2.36 Principal Rafter — A roof member which supports purlins.

2.2.37 Purlins — A roof member directly supporting roof covering or common rafter and roof battens.

2.2.38 Roof Battens — A roof member directly supporting tiles, corrugated sheets, slates or other roofing materials.

2.2.39 Roof Skeleton — The skeleton consisting of bamboo truss or rafter over which solid bamboo purlins are laid and lashed to the rafter or top chord of a truss by means of galvanized iron wire, cane, grass, bamboo leaves, etc.

2.2.40 Slenderness Ratio — The ratio of the length of member to the least radius of gyration is known as slenderness ratio of member (the length of the member is the equivalent length due to end conditions).

2.2.41 Splits — The pieces made from quarters by dividing the quarters radially and cutting longitudinally.

2.2.42 Taper — The ratio of difference between minimum and maximum outer diameter to length.

2.2.43 Unloaded End Distance — The end distance opposite to the loaded end.

2.2.44 Wall Thickness — Half the difference between outer diameter and inner diameter of the piece at any cross section.

2.2.44.1 Mean wall thickness — The wall thickness taken four times in the same points as the diameter has been measured. Thus, resulting in mean of eight values for the middle of culm.

2.2.45 Wet Location — Position in buildings in which the bamboos are almost continuously damp, wet or in contact with earth or water, such as piles and bamboo foundations.

2.3 Definitions Relating to Defects

2.3.1 Bamboo Bore/GHOON Hole — The defect caused by bamboo ghoon beetle (*Dinoderus spp. Bostrichidae*), which attacks felled culms.

2.3.2 Crookedness — A localized deviation from the straightness in a piece of bamboo.

2.3.3 Discolouration — A change from the normal colour of the bamboo which does not impair the strength of bamboo or bamboo composite products.

2.4 Definition Relating to Drying Degradation

2.4.1 Collapse — The defect occurring on account of excessive shrinkage, particularly in thick walled immature bamboo. When the bamboo wall shrinks, the outer layers containing a larger concentration of strong fibro-vascular bundles set the weaker interior portion embedded in parenchyma in tension, causing the latter to develop cracks. The interior crack develops into a wide split resulting in a depression on the outer surface. This defect also reduces the structural strength of round bamboo.

2.4.2 End Splitting — A split at the end of a bamboo. This is not so common a defect as drying occurs both from outer and interior wall surfaces of bamboo as well as the end at the open ends.

2.4.3 Surface Cracking — Fine surface cracks not detrimental to strength. However, the cracking which occurs at the nodes reduces the structural strength.

2.4.4 Wrinkled and Deformed Surface — Deformation in cross section, during drying, which occurs in immature round bamboos of most species; in thick walled pieces, besides this deformation the outer surface becomes uneven and wrinkled. Very often the interior wall develops a crack below these wrinkles, running parallel to the axis.

3 SYMBOLS

For the purpose of this Subsection, the following letter symbols shall have the meaning indicated against each, unless otherwise stated:

A	=	cross-sectional area of bamboo (perpendicular to the direction of the principal fibres and vessels), mm ²
	=	$\frac{\pi}{4} [D^2 - (D - 2w)^2]$;
A_B	=	area of reinforcement in bamboo reinforced cement concrete, mm ² ;
D	=	mean outer diameter of bamboo culm, mm;
d	=	inner diameter, mm
	=	$(D - 2w)$;
E	=	modulus of elasticity (MOE) in bending, N/mm ² ;
f_c	=	calculated stress in axial compression, N/mm ² ;
f_{cp}	=	permissible stress in compression along the fibres, N/mm ² ;
I	=	moment of inertia (the second moment of area) of culm section, mm ⁴
	=	$\frac{\pi}{64} [D^4 - (D - 2w)^4]$;

- ℓ = unsupported length of column, mm;
 m = moisture content, percent;
 r = radius of gyration, mm
 $= \sqrt{\frac{I}{A}}$;
 R = modulus of rupture (MOR), N/mm²;
 w = mean of wall thickness of bamboo culm, mm;
 Z = section modulus, mm³
 $= \frac{\pi}{32} \left[\frac{D^4 - (D-2w)^4}{D} \right]$; and
 δ = deflection or deformation, mm.

4 MATERIALS

The requirements shall be met by the choice of suitable materials, by appropriate design and detailing and by specifying control procedures for production, construction and use.

4.1 Species of Bamboo

More than 100 species of bamboo are native to India and a few of them are solid but most of them are hollow in structure. In all 20 species have been systematically tested so far. Sixteen species of bamboo are recommended for structural usages in round form. Some physical and mechanical properties of the same are given in [Table 1](#) both for air-dry conditions and green conditions.

4.1.1 Matured bamboo of at least 4 years of age shall be used. The bamboo shall be used after at least six weeks of felling period.

4.1.2 Bamboo shall be properly treated in accordance with the good practice [C-3B(1)].

4.1.3 Solid bamboos or bamboos whose wall thickness is comparatively more and which are generally having nodes very closer are often considered good for structural purposes.

4.1.4 Broken, damaged or collapsed bamboo shall be rejected. Dead and immature bamboos, bore/*Ghoon* holes, decay, collapse, checks more than 3 mm in depth, shall be avoided.

4.2 Grouping

4.2.1 Sixteen species of bamboo recommended for structural applications are classified in three groups on the basis of their strength properties, namely, modulus of elasticity (E) in bending in green condition and modulus of rupture (R) (extreme fibre stress in bending). For structural use in piles and mines' components, an additional criterion of compressive strength is considered in classification. The limits in ultimate strength values of these groups shall be as given in [Table 2](#) (these are corresponding to the spread of properties in structural timber).

4.2.2 Bamboo species may be identified using suitable methods.

NOTE — Methods of identification of bamboo through anatomical characters have not been perfected so far. Identification through morphological characters could be done only on full standing culm by experienced sorters.

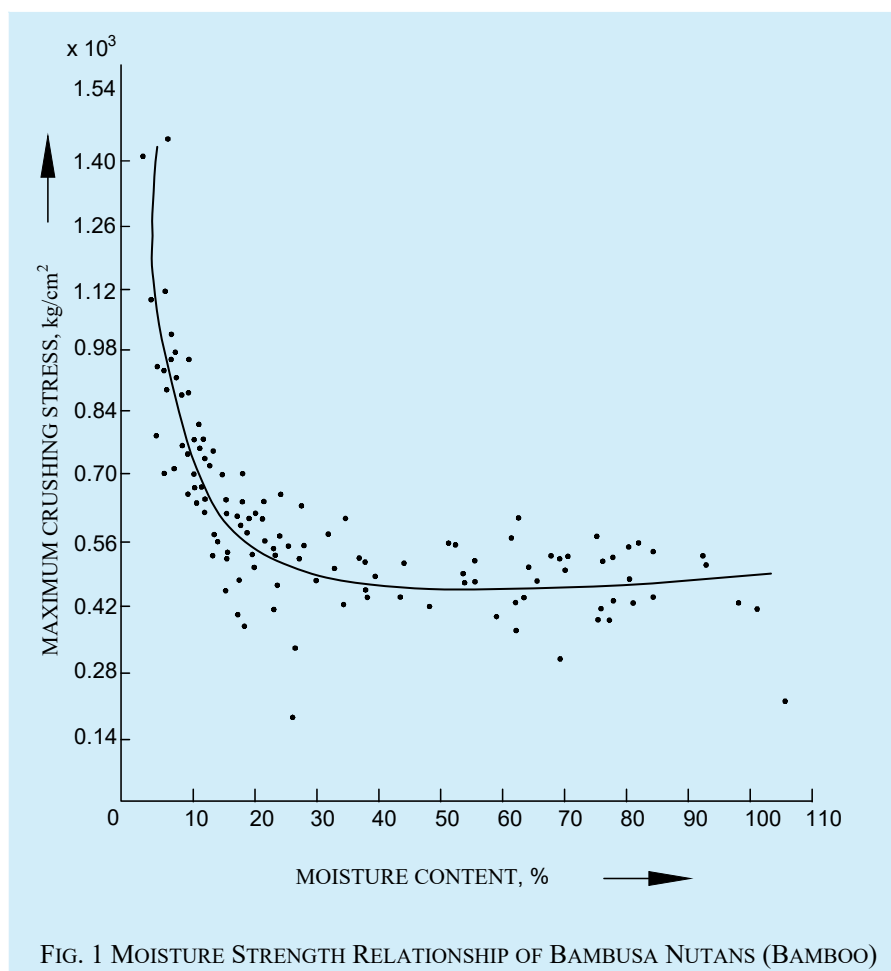
4.2.3 Species of bamboo other than those listed in the [Table 3](#) may be used, provided their strength characteristics are determined and found more than the limits specified in [Table 2](#).

4.3 Moisture Content in Bamboo

Normally only dry or seasoned bamboo shall be used otherwise special attention shall be given to dimensional changes occurring during the drying process in the joints, assemblies.

With decrease of moisture content, the strength of bamboo increases exponentially and bamboo has an intersection point (fibre saturation point) at around 25 percent moisture content depending upon the species. A typical moisture strength relationship is given at [Fig. 1](#). The moisture content of bamboo shall be determined in accordance with method given in the accepted standard [C-3B(2)]. Matured culms shall be seasoned to about 15 percent moisture content before use.

Freshly felled bamboo with shoots and leaves intact, dry fairly well in the open sun with lesser surface cracking, etc, and reduced starch contents. Rapid drying in the open can control degrades.



4.4 Grading of Structural Bamboo

Bamboo shall be graded to ensure that the properties of bamboo are satisfactory for use, particularly the strength and stiffness properties. Grading bamboo is a process that ensures its structural integrity, standardization, efficient use, and marketability. It plays a crucial role in enhancing trade and promoting sustainable harvesting by ensuring that bamboo meets the required specifications for various applications. This process is essential for ultimately producing safe and economical structures, encouraging its use in mainstream construction and other industries.

4.4.1 Grading is sorting out bamboo on the basis of characteristics important for structural utilization as under:

- a) Diameter and length of culm;
- b) Taper of culm;
- c) Straightness of culm;
- d) Inter nodal length and distribution of nodes;

- e) Wall thickness;
- f) Density and strength; and
- g) Durability and seasoning.

4.4.2 Diameter and Length of Culm

4.4.2.1 Gradation according to the mean outer diameter

For structural Group A and Group B species, culms shall be segregated in steps of 10 mm of mean outer diameter as follows:

- a) Special grade A : 100 mm < Diameter $\leq$ 130 mm
- b) Special grade B : 70 mm < Diameter $\leq$ 100 mm
- c) Grade I : 50 mm < Diameter $\leq$ 70 mm
- d) Grade II : 30 mm < Diameter $\leq$ 50 mm
- e) Grade III : Diameter $\leq$ 30 mm

For structural Group C species culms shall be segregated in steps of 20 mm of mean outer diameter as follows:

- a) Grade I : 80 mm < Diameter $\leq$ 100 mm
- b) Grade II : 60 mm < Diameter $\leq$ 80 mm
- c) Grade III : Diameter $\leq$ 60 mm

4.4.2.2 The minimum length of culms shall be preferably 4.5 m for facilitating close fittings at joints, etc.

4.4.3 Taper

The taper shall not be more than 5.8 mm per metre length (or 0.58 percent) (1 in 170) of bamboo in any grade of bamboo.

4.4.4 Curvature

The maximum curvature should not be more than 75 mm in a length of 6 m of any grade of bamboo.

NOTE — Due to the constraint of preferential selection of straighter material out of a lot, more bent material may be accommodated to avoid high cost. Following remedial options may be helpful:

- a) Adopt component assembly method to take care of some degree of curvature (bunching done after holding components straight).
- b) Cut longer bamboo components with less curvature out of taller bamboo and use cambered ones for shorter length components.

4.4.5 Wall Thickness

Preferably minimum wall thickness of 8 mm shall be used for load bearing members unless calculations and the availability dictate otherwise.

4.5 Durability and Treatability

4.5.1 Durability

The natural durability of bamboo is low and varies between 12 months and 36 months depending on the species and climatic conditions. In tropical countries the bio-deterioration is very severe. Bamboos are generally destroyed in about one to two years' time when used in the open and in contact with ground; while a service life of two to five years can be expected from bamboo when used under cover and out of contact with ground. The mechanical strength of bamboo deteriorates rapidly with the onset of fungal decay in the sclerenchymatous fibres. Split bamboo is more rapidly destroyed than round bamboo

4.5.2 Fire safety of bamboo structures shall be ensured, for which the treatment prescribed in [7.3.4](#) may be followed.

5 PERMISSIBLE STRESSES

5.1 Ultimate stress values of different species and groups of bamboo shall be determined according to the accepted standard [C-3B(2)]. These values shall then be divided by appropriate factors of safety to obtain permissible stresses to cover the effects of variability, long term loading, grade, location of use and expected dynamic loading.

NOTE — As an information, it may be noted that the method of test for split bamboos is covered in the accepted standard [C-3B(4)].

5.1.1 The strength factors for deriving safe working stresses of bamboo shall be as under:

- | | | |
|--|---|-----|
| a) Extreme fibre stress in beams | : | 4 |
| b) Modulus of elasticity | : | 4.5 |
| c) Maximum compressive stress parallel to grain/fibres | : | 3.5 |

5.2 The available data for the safe working stresses for 16 species of bamboos are given in [Table 3](#).

5.3 For change in duration of load other than continuous (long-term), the permissible stresses given in [Table 3](#) shall be multiplied by the modification factors given below:

- | | | |
|--|---|------|
| a) For permanent loads | : | 1.0 |
| b) For imposed or medium term loading (permanent + temporary load) | : | 1.25 |
| c) For short term loading (permanent + temporary load + wind load) | : | 1.50 |

Table 1 Physical and Mechanical Properties of Indian Bamboos (in Round Form)

(Clause 4.1)

SI No.	Species	Properties							
		In Green Condition				In Air Dry Condition			
		Density	Modulus of Rupture	Modulus of Elasticity	Maximum Compressive Strength	Density	Modulus of Rupture	Modulus of Elasticity	Maximum Compressive Strength
		kg/m ³	N/mm ²	× 10 ³ N/mm ²	N/mm ²	kg/m ³	N/mm ²	× 10 ³ N/mm ²	N/mm ²
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	<i>Bambusa auriculata</i>	594	65.1	15.01	36.7	670	89.1	21.41	54.3
ii)	<i>B. balcooa</i>	783	65.4	7.31	46.7	—	—	—	60.6
iii)	<i>B. bambos</i> (Syn. <i>B. arundinacea</i>)	559	58.3	5.95	35.3	663	80.1	8.96	53.4
iv)	<i>B. burmanica</i>	570	59.7	11.01	39.9	672	105.0	17.81	65.2
v)	<i>B. glaucescens</i> (Syn. <i>B. nana</i>)	691	82.8	14.77	53.9	—	—	—	—
vi)	<i>B. nutans</i>	603	52.9	6.62	45.6	673	52.4	10.72	47.9
vii)	<i>B. pallida</i>	731	55.2	12.90	54.0	—	—	—	—
viii)	<i>B. tulda</i>	658	51.1	7.98	40.7	722	66.7	10.07	68.0
ix)	<i>B. ventricosa</i>	626	34.1	3.38	36.1	—	—	—	—
x)	<i>B. vulgaris</i>	626	41.5	2.87	38.6	—	—	—	—
xi)	<i>Cephalostachyum pergracile</i>	601	52.6	11.16	36.7	640	71.3	19.22	49.4
xii)	<i>Dendrocalamus longispathus</i>	711	33.1	5.51	42.1	684	47.8	6.06	61.1

Table 1 (Concluded)

SI No.	Species	Properties							
		In Green Condition				In Air Dry Conditions			
		Density	Modulus of Rupture	Modulus of Elasticity	Maximum Compressive Strength	Density	Modulus of Rupture	Modulus of Elasticity	Maximum Compressive Strength
		kg/m ³	N/mm ²	× 10 ³ N/mm ²	N/mm ²	kg/m ³	N/mm ²	× 10 ³ N/mm ²	N/mm ²
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
xiii)	<i>D. strictus</i>	631	73.4	11.98	35.9	728	119.1	15.00	69.1
xiv)	<i>Melocanna baccifera</i>	817	53.2	11.39	53.8	751	57.6	12.93	69.9
xv)	<i>Oxytenanthera abyssinica</i>	688	83.6	14.96	46.6	—	—	—	—
xvi)	<i>Thyrsostachys oliveri</i>	733	61.9	9.72	46.9	758	90.0	12.15	58.0
NOTE — The values of stress in N/mm ² have been obtained by converting the values in kgf/cm ² by dividing the same by 10.									

Table 2 Properties of Structural Bamboo

(Clauses 4.2.1 and 4.2.3)

SI No.	Group	Modulus of Rupture (R)	Modulus of Elasticity in Bending (E)	Maximum Compressive Strength ($f_{c, \max}$)
		N/mm ²	$\times 10^3$ N/mm ²	N/mm ²
(1)	(2)	(3)	(4)	(5)
i)	A	$R > 70$	$E > 9$	$MCS > 35$
ii)	B	$70 \geq R > 50$	$9 \geq E > 6$	$35 > MCS > 30$
iii)	C	$50 \geq R > 30$	$6 \geq E > 3$	$30 > MCS > 25$

Table 3 Safe Permissible Stresses of Bamboos for Structural Designing¹⁾

(Clauses 4.2.3, 5.2 and 5.3)

SI No.	Species	Extreme Fibre Stress in Bending	Modulus of Elasticity	Allowable Compressive Stress
		N/mm ²	$\times 10^3$ N/mm ²	N/mm ²
(1)	(2)	(3)	(4)	(5)
i)	Group A:			
	a) <i>Bambusa glaucescens</i> (Syn. <i>B.nana</i>)	20.7	3.28	15.4
	a) <i>Dendrocalamus strictus</i>	18.4	2.66	10.3
	b) <i>Oxytenanthera abyssinica</i>	20.9	3.31	13.3
ii)	Group B:			
	a) <i>Bambusa balcooa</i>	16.4	1.62	13.3
	b) <i>B.pallida</i>	13.8	2.87	15.4
	c) <i>B.nutans</i>	13.2	1.47	13.0
	d) <i>B.tulda</i>	12.8	1.77	11.6
	e) <i>B.auriculata</i>	16.3	3.34	10.5
	f) <i>B.burmanica</i>	14.9	2.45	11.4
	g) <i>Cephalostachyum pergracile</i>	13.2	2.48	10.5
	h) <i>Melocanna baccifera</i> (Syn. <i>M.bambusoides</i>)	13.3	2.53	15.4
	j) <i>Thyrsotachys oliveri</i>	15.5	2.16	13.4
iii)	Group C:			
	a) <i>Bambusa arundinacea</i> (Syn. <i>B.bambos</i>)	14.6	1.32	10.1
	b) <i>B.ventricosa</i>	8.5	0.75	10.3

¹⁾ The values given pertain to testing of bamboo in green condition.

Table 3 (Concluded)

SI No.	Species	Extreme Fibre Stress in Bending	Modulus of Elasticity	Allowable Compressive Stress
(1)	(2)	N/mm ² (3)	× 10 ³ N/mm ² (4)	N/mm ² (5)
	c) <i>B.vulgaris</i>	10.4	0.64	11.0
	d) <i>Dendrocalamus longispathus</i>	8.3	1.22	12.0

NOTE — The values of stress in N/mm² have been obtained by converting the values in kgf/cm² by dividing the same by 10.

NOTE — The above properties are based on the accepted standard [C-3B(2)].

6 DESIGN CONSIDERATIONS

6.1 Basic Requirements

A structure shall be designed and constructed by personnel having the appropriate skill and experience in such a way that:

- With acceptable probability, it will remain fit for the use for which it is required, having due regard to its intended life and costs;
- With appropriate degree of reliability, it will sustain all forces and effects likely to occur during execution and use and have adequate durability in relation to maintenance costs;
- The structure will be adequately maintained;
- The structure will be used in accordance with design limit briefs;
- All structural members, assemblies or framework in a building shall be capable of sustaining, without exceeding the limits of relevant stress specified, the worst combination of all loadings;
- A fundamental aspect of design will be to determine the forces to which the structure/structural element might be subjected to, starting from the roof and working down to the soil by transferring the forces through various components and connections;
- Accepted principles of mechanics for analysis and specified design procedures shall be applied (or) the design requirements may be satisfied by prototype testing;
- Loads shall be in accordance with Part C 'Structural Design, Section 1 Loads and Load Effects' of this NBCS, keeping into consideration that permeability of walls, floors and roof made from bamboo cause internal pressure and can therefore affect the wind load factor; and
- The worst combination and location of loads shall be considered for design. Also, wind and seismic forces shall not be considered to act simultaneously.

6.2 Unlike timber, bamboo properties do not relate well to species, being dependent among other factors, on position of the culm, geographic location and age. The practice in timber engineering is to base designs on safe working stresses (*see* Note) and the same may be adopted to bamboo with the limitations that traditional experience rather than precise calculations generally govern the detailing.

NOTE — Limit states are states beyond which the structure no longer satisfies the design performance requirements. Limit states are classified into ultimate limit state relating to collapse mode and serviceability limit states associated with deformations, vibrations, etc. Design is usually based on ultimate stages with partial factors of safety.

6.3 Age old traditional experience of construction with bamboo is to be well preserved as non-standard practices for similar situations considered as an informal based on general social pattern and wisdom. Confirmation of such structures be based on reports after they have sustained the severity of earthquakes, cyclones, etc, as criteria for recommendations by the evaluation by competent engineer/builder with adequate experience in the field.

6.4 Net Section

It is determined by passing a plane or a series of connected planes transversely through the members. Least net sectional area is used for calculating load carrying capacity of a member. In the design of an intermediate or a long column, gross section shall be used in calculating load carrying capacity of column.

6.5 Structural Components

6.5.1 Main structural applications in bamboo may include roofing and flooring, shear walls, wall panelling, beams, piles, columns, arches, etc. Both from the point of view of capacity and deformation, trusses and framed skeletons are much better options in bamboo.

6.5.2 Peculiarity of Bamboo as a Design Material

This shall be based on the principles of applied mechanics involving the following assumptions:

- a) Elastic behaviour of bamboo, until failure (plastic behaviour being considered to be not significant);
- b) Bamboo culms are analyzed as hollow-tube structures (not perfectly straight) member on mean diameter and mean wall thickness basis. Designing as a variable thickness hollow-tube structure is a bit complex;
- c) Nodes do not occur at constant intervals;
- d) Structural elements of bamboo shall be appropriately supported near the nodes of culm as and where the structural system demands. The joints in the design shall be located near nodes;
- e) Bamboo structures are designed like any other conventional structural analysis taking care of details with regards to supports and joints; and the diameter, wall thickness and initial curvature; and
- f) Joints shall be considered to generally act as a hinge or as pinned, unless substantiating data justify a fixed joint.

6.6 Flexural Members

6.6.1 All flexural members may be designed using the principles of beam theory.

NOTE — The tendency of bamboo beams to acquire larger deflections under long continual loading (including creep) has to be taken care of either by giving a factor of safety to E or by considering double the design dead loading in calculation of deflections.

6.6.2 The moment of inertia, I shall be determined as follows:

- a) Outside diameter and the wall thickness shall be measured at both ends, correct up to 1 mm for diameter of culm and 0.1 mm for the wall thickness; and
- b) With these values the mean diameter and the mean thickness for the middle of the beam shall be calculated and moment of inertia determined.

6.6.3 The maximum bending stress shall be calculated and should be within the allowable stress.

6.6.4 The deflection shall be within the prescribed limits. The initial curvature shall also be considered in the calculation of the deflection.

6.6.5 The shear stress in the neutral layer at the small end shall be checked, if the length of the beam is less than 25 times the diameter at that end.

6.6.6 Forces acting on a beam, being loads or reaction forces at supports, shall act in nodes or as near to nodes as by any means possible.

6.7 Columns (Predominantly Loaded in Axial Direction)

6.7.1 Columns and struts are essential components sustaining compressive forces in a structure. They transfer load to the supporting media.

6.7.2 Design of columns shall be based on one of the following two criteria:

- a) Full scale buckling tests on the same species, size and other relevant variables; and

b) Calculations, based on the following:

- 1) Moment of inertia shall be determined as per [6.6.2](#);
- 2) For bamboo columns the best available straight bamboo culms shall be selected; and
- 3) Bending stresses due to initial curvature, eccentricities and induced deflection shall be taken into account, in addition to those due to any lateral load.

NOTE — Unlike timber, MOE has already been provided a factor of safety in case of bamboo. This is on the basis of studies on variability of MOR and MOE. In the case of timber, no factor of safety for MOE was used because there has been an understanding that there is no variability in MOE. Subsequently it was confirmed that MOE varies as the MOR and there is very strong, highly significant correlation between the two. In fact, this correlation has formed the basis of development of machine grading of timber world-over. However, somehow the practice of not giving factor of safety to MOE continued in Indian Standard. Later on when the standard on structural use of bamboo was formulated, factor of safety was given both in MOR and MOE on the basis of latest variability studies on strength of bamboo.

6.7.3 Buckling calculation shall be according to Euler, with a reduction to 90 percent of moment of inertia, to take into account the effect of the taper which shall be less than 1 : 170.

6.7.4 For strength and stability, larger diameter thick walled sections of bamboo with closely spaced nodes shall be used. Alternatively, smaller sections may be tied together as a bundle-column.

6.8 A single bamboo culm is not an optimum component to structurally design different load bearing members. Bunching of bamboo is inevitable to fabricate beams, columns, etc. There is no end to the number of culms in bunching. However, different components pose manufacturing problems during assembly, which should be appropriately taken care of (*see also* [Fig. 9D](#) and [Fig. 9E](#)).

6.8.1 Following precautions shall be taken to avoid possible split/crack in walling of bamboo:

- a) Maintaining a minimum 100 mm distance of the drills from the end of the main bamboo culms drilled;
- b) Choosing drilling points preferably within 30 mm from node;
- c) Maintaining the interference fit between bamboo body and binders as under:
 - 1) For wall thickness ≥ 8 mm : 0.25 to 0.30 mm; and
 - 2) For wall thickness < 8 mm : 0.20 to 0.25 mm.
- d) The inserting ends of the dowel pins may be dressed as tapered (conical).

6.9 Assemblies and Roof Trusses

6.9.1 Elements in structure are generally built-up in the form of assembled members for which a triangle is a simple figure of stability. Besides sloped chords, parallel chord construction is also appropriate as external profile.

6.9.2 A truss is essentially a plane structure which is very stiff in the plane of the members, that is the plane in which it is expected to carry load, but very flexible in every other direction. Roof truss generally consists of a number of triangulated frames, the members of which are fastened at ends and the nature of stresses at joints is either tensile or compressive and designed as pin-ended joints (*see* [Fig. 2A](#)). Bamboo trusses may also be formed using bamboo mat board or bamboo mat-veneer composite or plywood gusset (*see* [Fig. 2B](#)).

6.9.3 Trusses shall be analysed for the determination of axial forces in members and the deformation of joints. For the influence of eccentricities, due allowance shall be made in design.

6.9.4 The truss height shall exceed 0.15 times the span in case of a triangular truss (pitched roofing) and 0.10 times the span in case of a rectangular (parallel) truss.

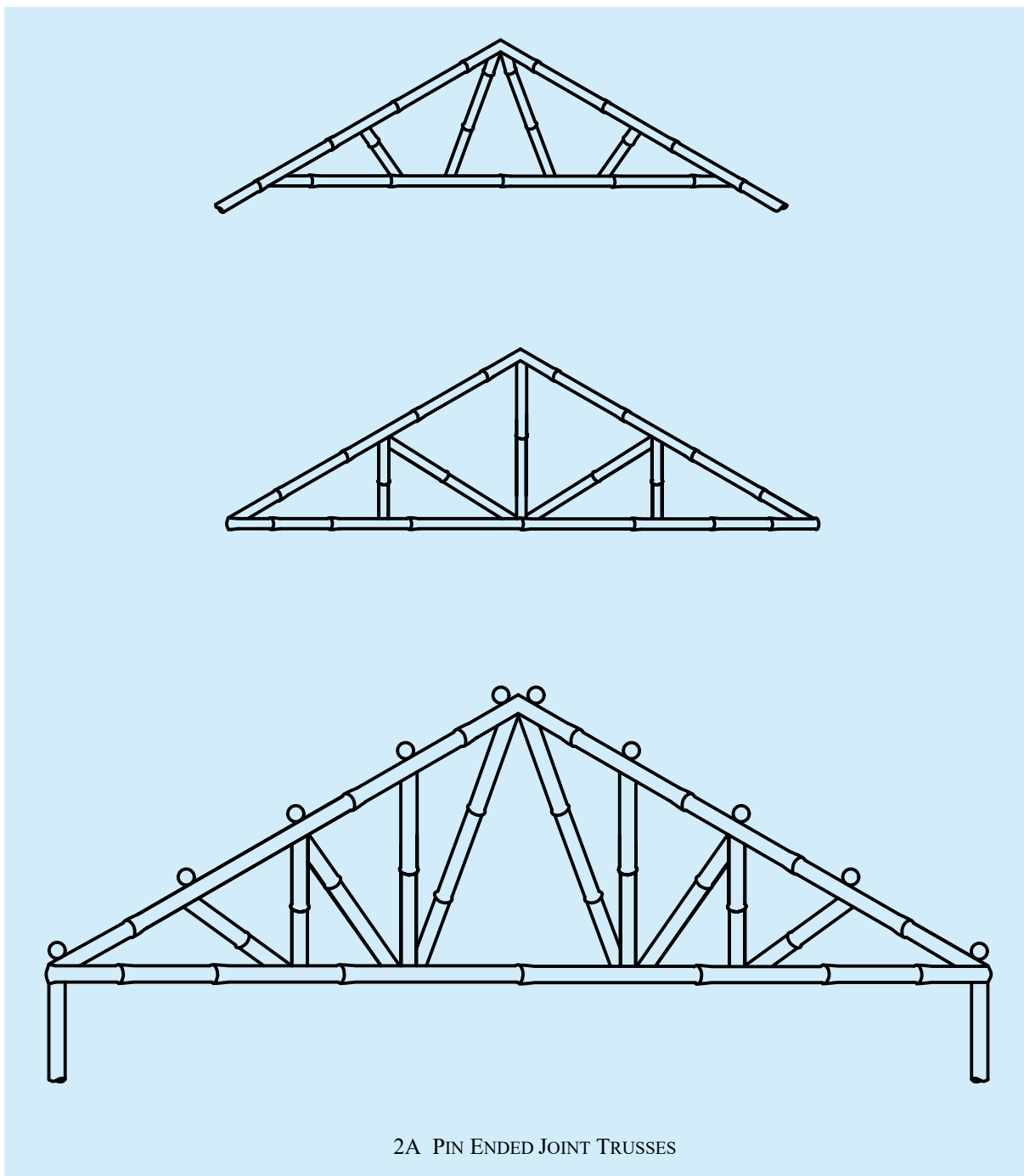
6.9.5 For members in compression, the effective length for in-plane strength verification shall be taken as the distance between two adjacent points of contraflexure. For fully triangulated trusses, effective length for simple span members without especially rigid end-connection shall be taken as the span length.

6.9.6 The spacing of trusses shall be consistent with use of bamboo purlins.

6.9.7 The ends in open beams, joists, rafters, purlins shall be suitably plugged. Bamboo roof coverings shall be considered as non-structural in function.

6.9.8 Bamboo trusses with wire bound joints using bamboo pins and/or wooden splice plates fastened with bamboo pins when tested at load for factor of safety of at least 5, result in deflection within permissible range at design load; good recovery but abnormal deflection beyond ultimate load without collapse was observed.

NOTE — This is based on the work done on prototypes of a few bamboo trusses (span 4 m) tested at Forest Research Institute, Dehradun.



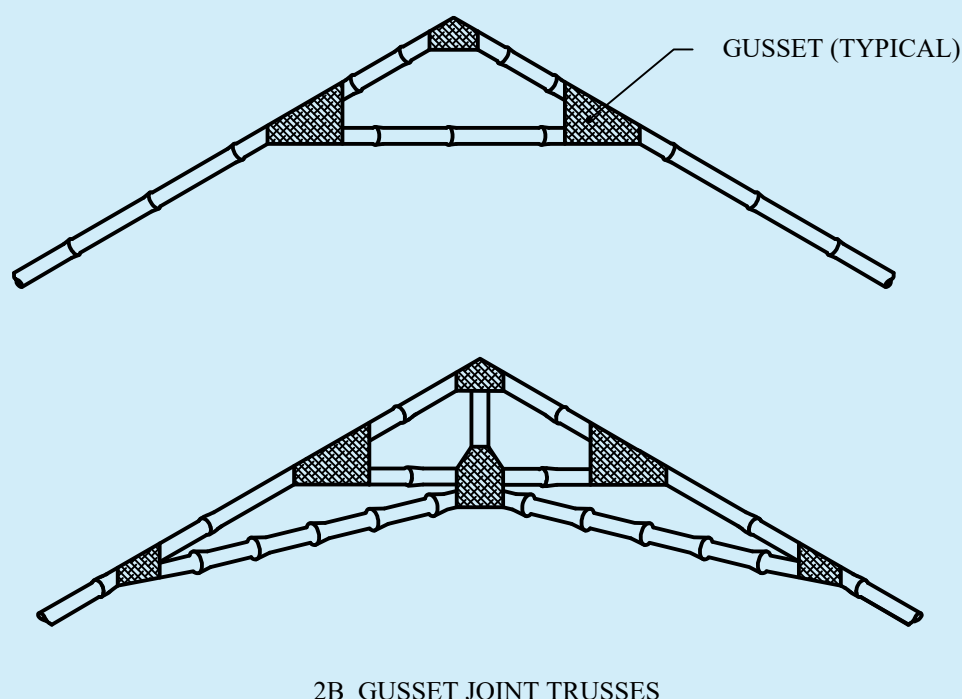


FIG. 2 SOME TYPICAL CONFIGURATIONS FOR SMALL AND LARGE TRUSSES IN BAMBOO

6.9.9 Bamboo Mat Corrugated Sheets for Building Roofs

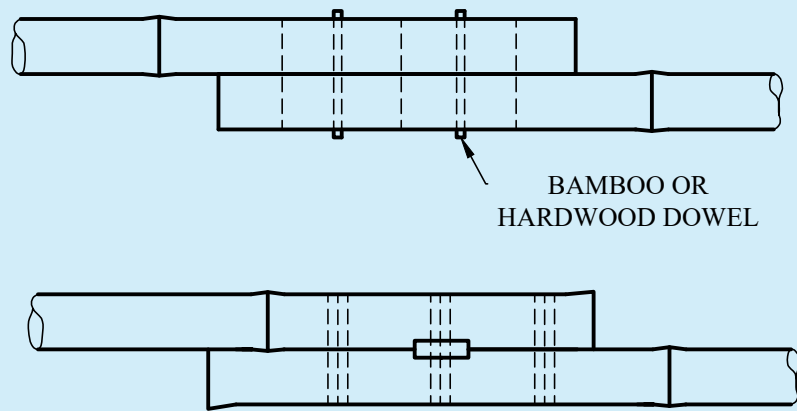
Bamboo mat corrugated sheets (BMCS) shall be used as a sustainable roofing material suitable for residential, commercial, and institutional buildings. BMCS are made from woven bamboo mats treated with phenolic resin and pressed under heat and pressure, these sheets offer a durable and eco-friendly alternative to traditional roofing materials. BMCS sheets shall provide high strength, resilience, and resistance to water, decay, and UV exposure, making them suitable for varied climates. These sheets should contribute to energy efficiency by offering thermal resistance and natural temperature regulation. BMCS shall comply with the accepted standard [C-3B(5)], meeting the necessary performance requirements for load-bearing capacity, impermeability, water absorption, and fire resistance.

6.9.10 Bamboo Mat Ridge Cap (BMRC)

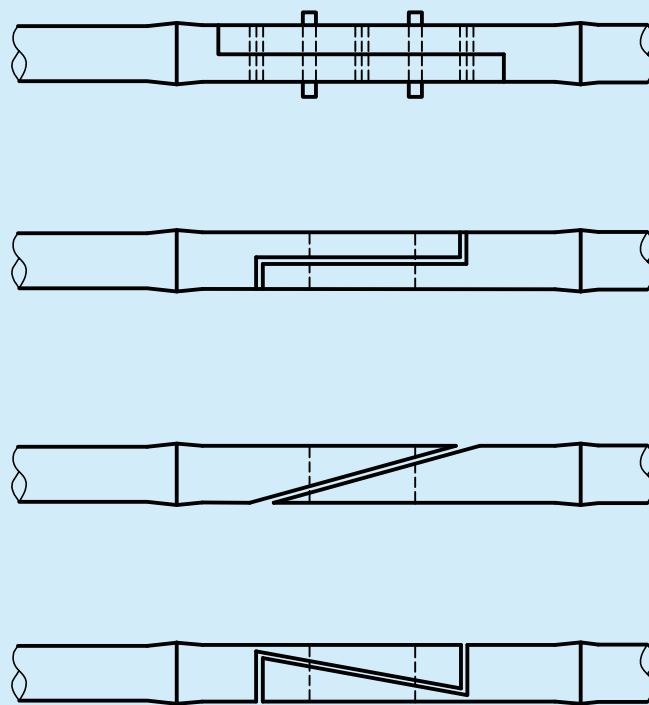
Bamboo mat ridge cap (BMRC) is a sustainable, lightweight, and eco-friendly roofing solution developed to complement bamboo mat corrugated sheets (BMCS) in building construction. Produced from bamboo mats, BMRCs are designed to cover the ridge (top edges) of BMCS roofs, providing strong physical and mechanical properties suitable for roofing applications. BMRCs are energy-efficient and cost-effective, offering a high strength-to-weight ratio, low thermal conductivity, and an aesthetically pleasing appearance.

7 DESIGN, TECHNIQUES OF JOINTS AND CONSTRUCTION PRACTICES

7.1 Connecting the load-bearing elements together for effective transfer of stress and to achieve continuity between elements with controlled displacements is one of the serious problem. As joints are a source of weakness in any bamboo structure, they have to be made as strong and rigid as possible. Joints are quite critical in assemblies, and these should be stable in relation to time. For general information, the diagrams of typical joints given in Figs. 3 to 7 may be used based on tests for adequacy of strength.



3A FULL LAPPED JOINT



3B HALF LAPPED JOINT
(REBATED/SPLAYED)

FIG. 3 SPLICED JOINT

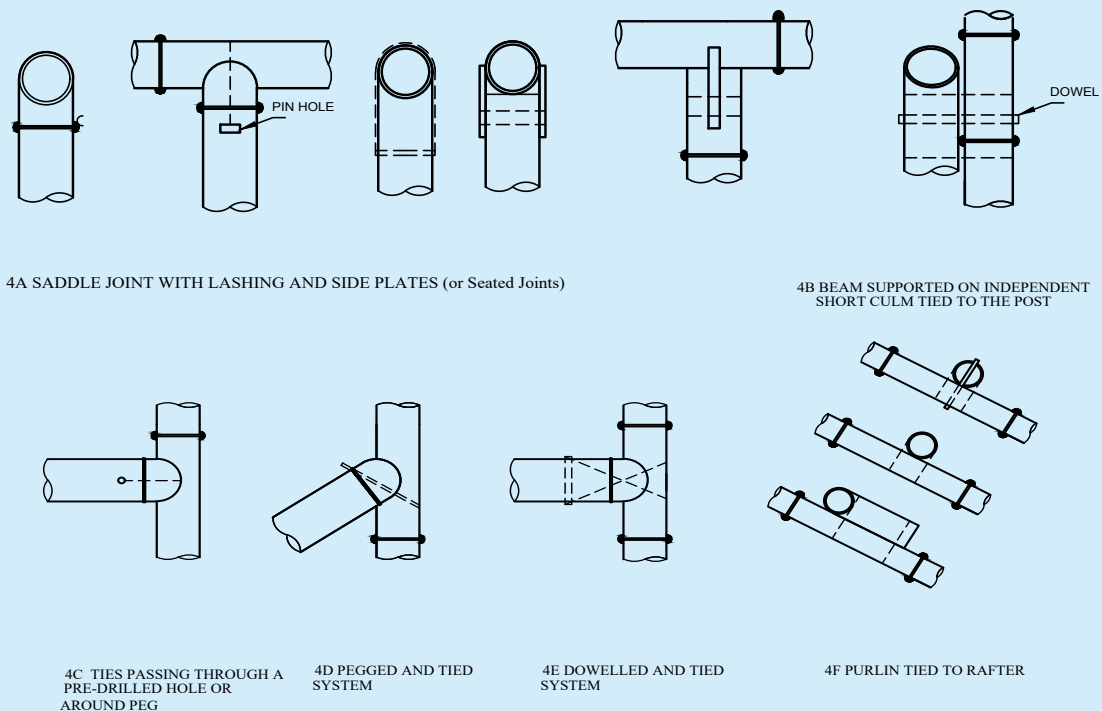


FIG. 4 BEARING JOINTS

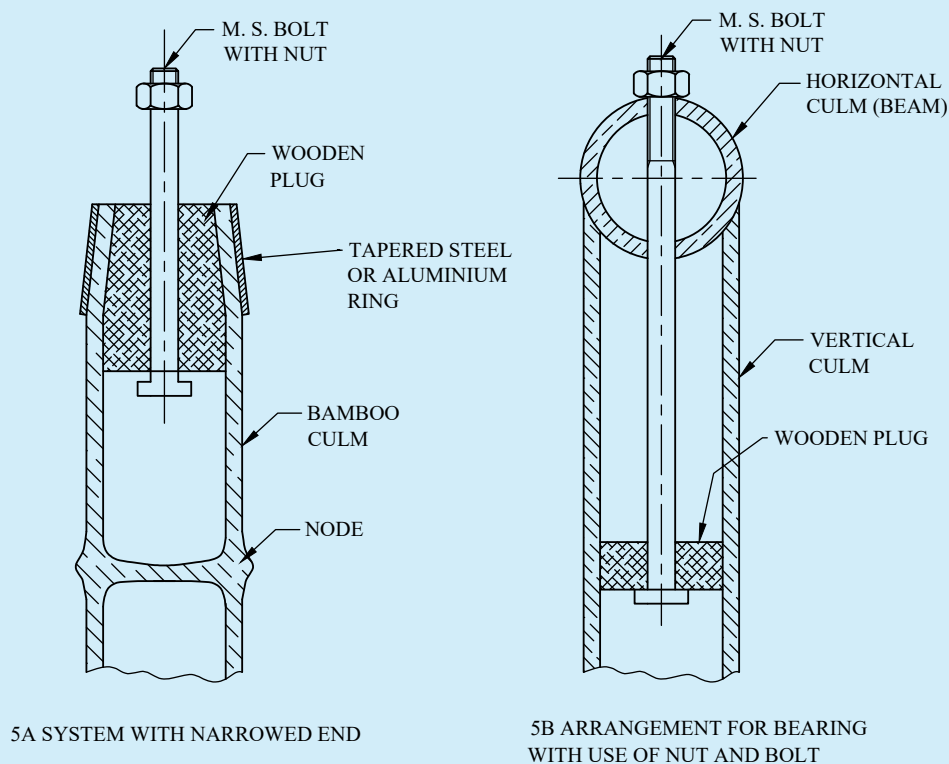
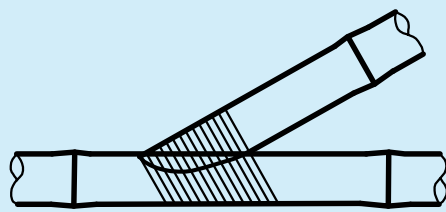
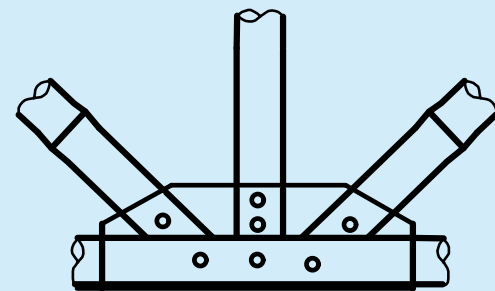
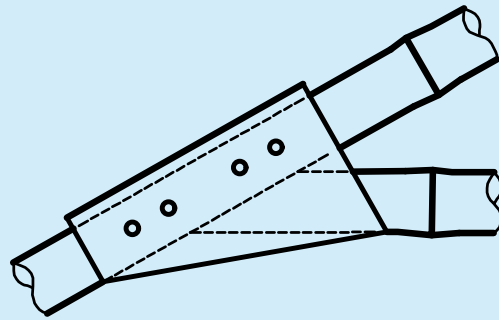


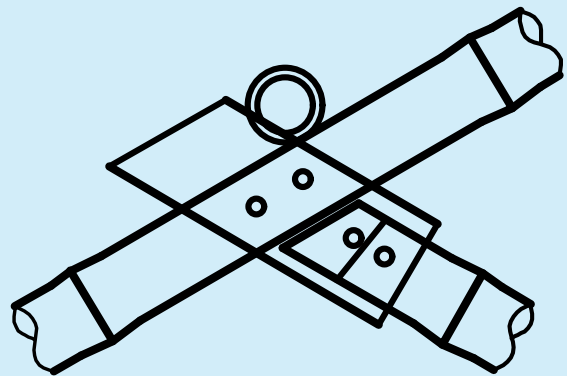
FIG. 5 SOME NUT BOLT JOINERY DETAILS



6A LASHED JOINT



6B TRUST CHORD AND WEB CONNECTIONS



6C SOLID TIMBER OR SHUTTERING GRADE
PLYWOOD PLATES FIXED WITH EITHER MS
BOLTS OR BAMBOO PEGS

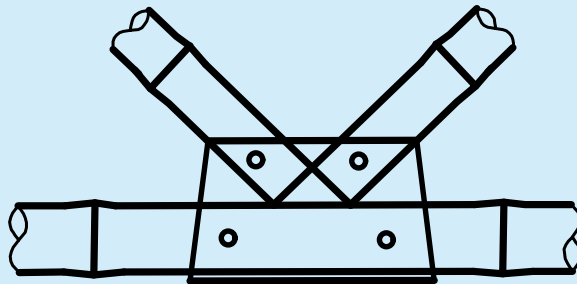
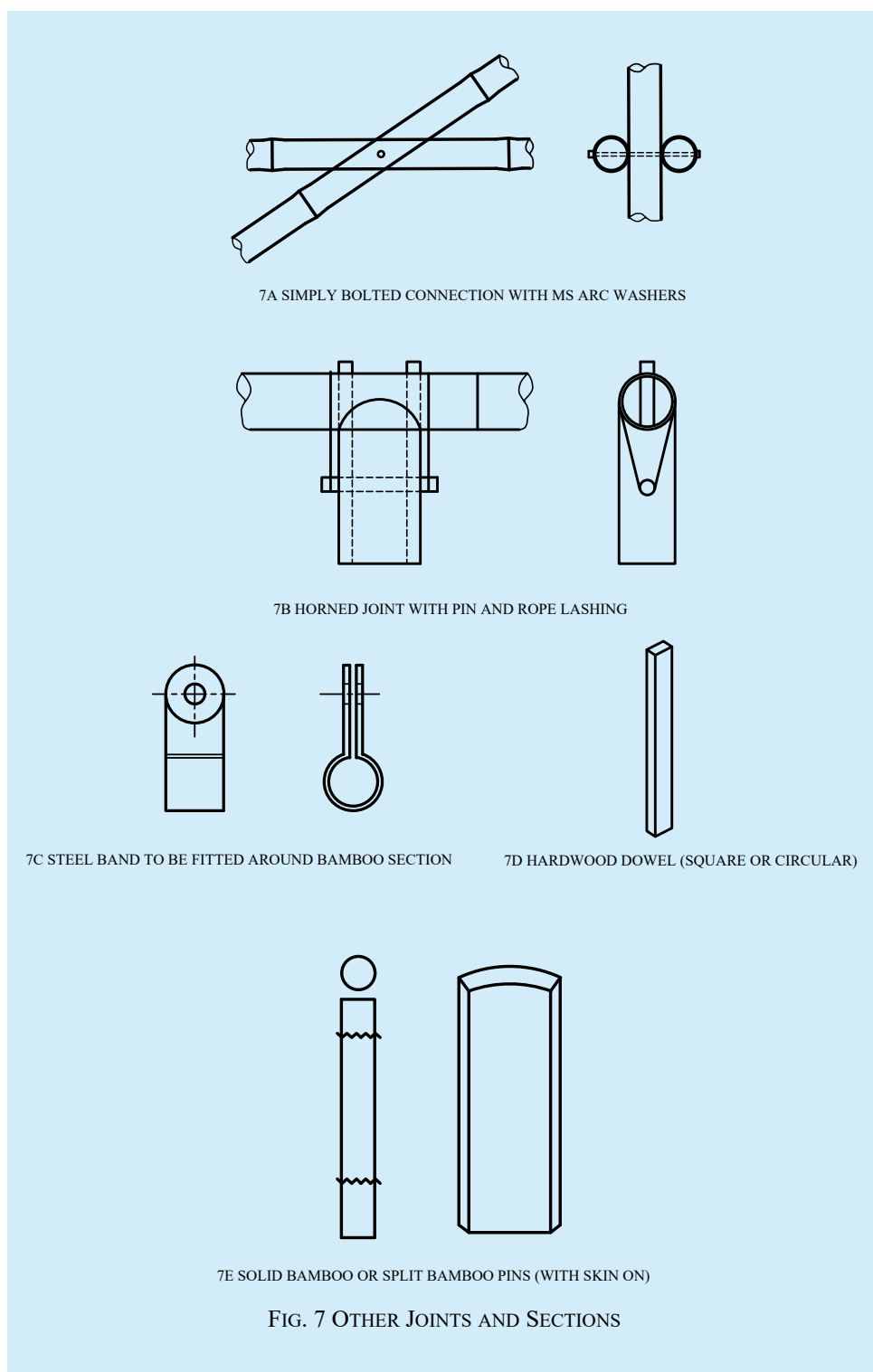


FIG. 6 GUSSET PLATED JOINTS



7.2 Bamboo Joints

Susceptibility to crushing at the open ends, splitting tendency, variation in diameter, wall thickness and straightness are some of the associated issues which have to be taken care of while designing and detailing the connections with bamboo.

Joints shall be designed to include force transmissions in a required manner, predictable deformations to be within prescribed limits and specific geometry/load direction.

Tests on full scale joints or on components shall be carried out in accordance with the good practice [C-3B(6)] and the accepted standards [C-3B(7)] as for nailed jointed timber structures.

7.2.1 Traditional Practices

Such joining methods revolve around lashing or tying by rope or string with or without pegs or dowels. Such joints lack stiffness and have low efficiency but have served the mankind for long in traditional bamboo construction.

7.2.1.1 Lengthening joints (end jointing) (see [Fig. 3](#))

7.2.1.1.1 Lap joint

End of one piece of bamboo is made to lap over that of the other in line and the whole is suitably fastened. It may be full lapping or half lapping. Full section culms are overlapped by at least one internode and tied together in two or three places. Efficiency could be improved by using bamboo or hardwood dowels.

7.2.1.1.2 Butt joints

Culms of similar diameter are butted end to end, inter-connected by means of side plates made of quarter-round culm of slightly large diameter bamboo, for two or more internode lengths. Assembly shall be fixed and tied preferably with dowel pins.

7.2.1.1.3 Sleeves and inserts

Short length of bamboo of appropriate diameter may be used either externally or internally to join two culms together.

7.2.1.1.4 Scarf joints

A scarf joint is formed by cutting a sloping plane 1 in 4 to 1 in 6 on opposite sides from the ends of two similar diameter bamboo culms to be joined. They shall be lapped to form a continuous piece and the assembly suitably fastened by means of lashings or glued.

7.2.1.2 Bearing joints (see [Fig. 4](#))

For members which either bear against the other or cross each other and transfer the loads at an angle other than parallel to the axis, bearing joints are formed.

7.2.1.2.1 Saddle joints

The simplest form consists of a horizontal member supported directly on top of a vertical member. The top of the post may be cut to form a saddle to ensure proper seating of beam for good load transfer. The saddle should be close to a node to reduce risk of splitting.

7.2.1.2.2 Angled joints

When two or more members meet or cross other than at right angles, angled joints are formed.

7.2.2 Improved Practices in Bamboo Jointing

7.2.2.1 Plywood or solid timber gusset plates may be used at joint assemblies of web and chord connection in a truss and fixed with bamboo pins or mild steel (MS) bolts. Hollow cavities of bamboo need to be stuffed with wooden plugs or otherwise by suitable means.

7.2.2.2 Use of wooden inserts to reinforce the ends of the bamboo before forming the joints. Alternatively, steel bands clamps with integral bolt/eye may be fitted around bamboo sections for jointing.

7.2.3 Fixing Methods and Fastening Devices

In case of butt joints the tie-wire may be passed through a pre-drilled hole or around hardwood or bamboo pegs or dowels inserted into preformed holes to act as horns. Pegs are driven from one side, usually at an angle to

increase strength and dowels pass right through the member, usually at right angles. Normally 1.60 mm (16 SWG) diameter galvanized iron wire may be used for tight lashing.

NOTE — Drill size needs to be specified in order to avoid possible split/crack in walling of bamboo.

7.2.3.1 Wire bound joints with or without pins

Usually 2.00 mm (14 SWG) diameter galvanized iron wire is tightened around the joints by binding the respective pieces together. At least two holes are drilled in each piece and wire is passed through them for good results. Generally, 10 mm to 12 mm diameter bamboo pins are driven and fastened to culms.

7.2.3.2 Fish plates/Gusset plated joints (see [Fig. 6](#))

At least 25 mm thick hardwood splice plate or 12 mm thick structural grade plywood are used. Solid bamboo pins help in fastening the assembly.

7.2.4 For any complete joint alternative for a given load and geometry, description of all fastening-elements, their sizes and location shall be indicated. Data shall be based on full scale tests.

7.2.5 Tests on full scale joints or on components shall be carried out in a recognized laboratory.

7.2.6 In case of high wind and seismic areas, good construction practices shall be followed taking care of joints, their damping and possible ductility. Bracings in walls shall be taken care of in bamboo structures.

7.2.7 Metal fasteners and other structural joints shall be made inherently corrosion resistant or be protected against corrosion, thereby making the joints more durable.

7.3 Construction Practices

7.3.1 Bamboo being a versatile resource characterized by high strength, low mass and ease of working with simple tools, it is desirable to increasingly make appropriate use of this material. Design of structures using bamboo should be done in accordance with [6](#). It should be ensured that the elements of structure satisfy the appropriate fire resistance requirements as specified in Part F ‘Fire and Life Safety’ of this NBCS, and quality of building materials/components used shall be in accordance with Part B ‘Building Materials’ of this NBCS. For construction using bamboo, some of the important constructional provisions given in [7.3.2](#) to [7.3.7](#) shall be followed.

7.3.2 The following sound construction practices shall be taken into account:

- a) Use of air dry bamboo; and ensure that bamboo in building remains air dry; and in case the bamboo has become wet, shall provide it the opportunity to dry again before the it deteriorates due to moisture;
- b) Permeability of walls, floors and roofs made from bamboo, cause internal pressure and can therefore change the net wind-load acting on the roof, wall and floor; and
- c) Check whether the workmanship of the labour force is according to the assumptions.

7.3.3 Working Finishing

7.3.3.1 Bamboo can be cut and split easily with very simple hand tools. Immature bamboos are soft, pliable and can be moulded to desired shape. It takes polish and paint well.

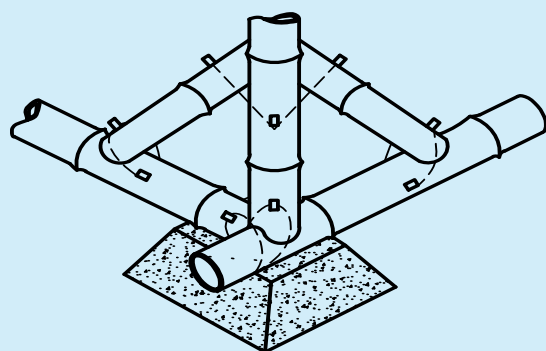
7.3.3.2 While it is possible to work with bamboo simply using a machete, a few basic tools such as, machete, hack saw, axe, hatchet, sharpening tools, adze, chisel (20 mm), drill, wood rasps, steel rod, and pliers, will greatly increase the effectiveness of the construction process.

7.3.4 For providing safety to the structure against fire, bamboo may be given fire retardant treatment using following chemicals; a few drops of concentrated Hcl shall be added to the solution to dissolve the precipitated salts:

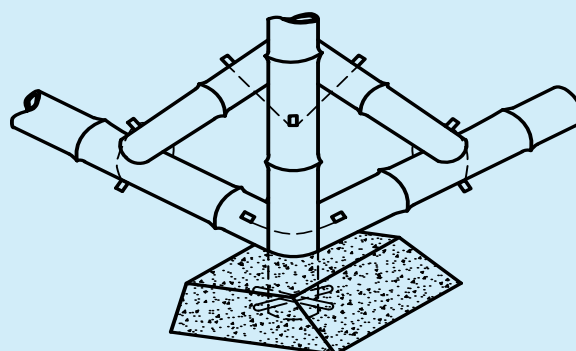
Ammonium phosphate	:	3 parts
Boric acid	:	3 parts
Copper sulphate	:	1 part
Zinc chloride	:	5 parts
Sodium dichromate	:	3 parts
Water	:	100 parts

7.3.5 Foundations

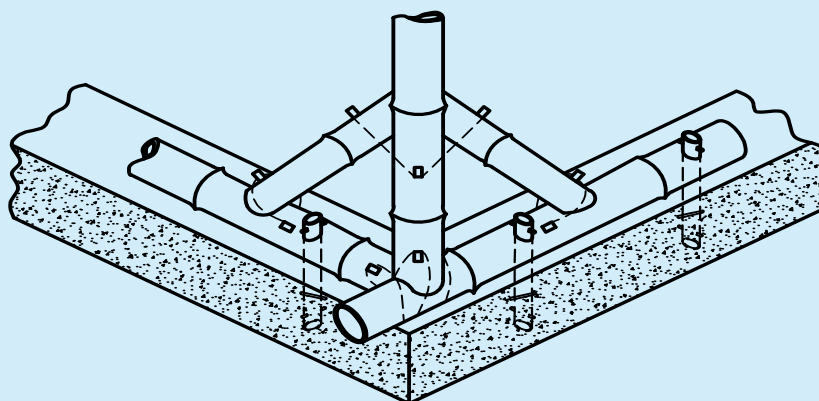
Bamboo in direct contact with ground, bamboo on rock or preformed concrete footing, bamboo incorporated into concrete, may form the foundation structure (see [Fig. 8](#)). Also, arrangements with use of metallic strap, C-clamp and bunched bamboo columns embedded in concrete may be used for foundation of bamboo structures (see [Fig. 9](#)).



8A BAMBOO ON PREFORMED
CONCRETE FOOTINGS



8B BAMBOO INCORPORATED INTO
CONCRETE FOOTINGS
(SINGLE POST FOOTING)



8C BAMBOO OUT OF GROUND CONTACT ON STRIP FOOTINGS OF CONCRETE (LARGE
DIAMETER THICK WALLED BAMBOO WITH CLOSELY SPACED NODES TO BE USED)

FIG. 8 BAMBOO FOUNDATIONS

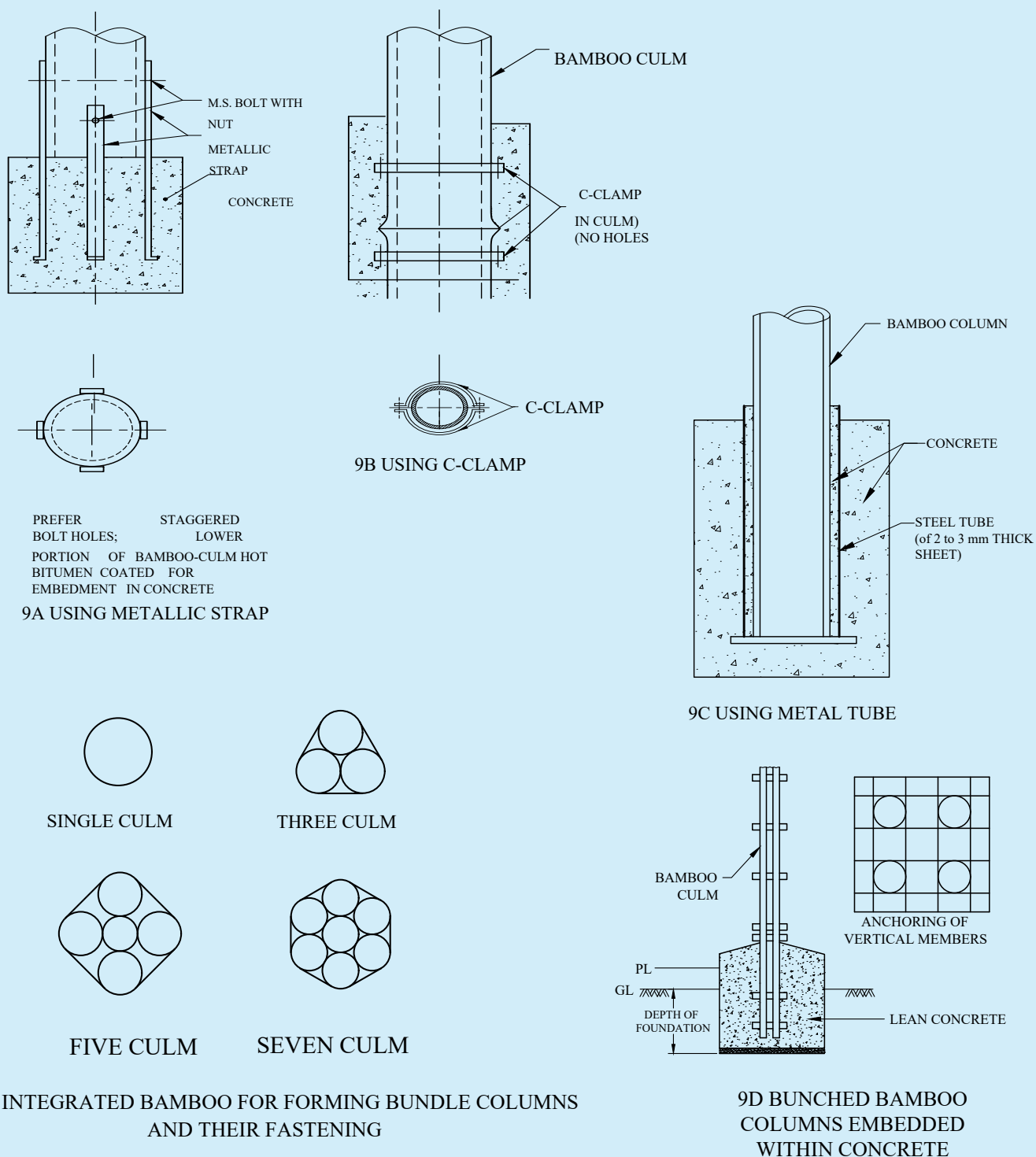


FIG. 9 BAMBOO COLUMN IN CONCRETE

7.3.6 Floors

The floor of bamboo may be at ground level with covering of bamboo matting, etc. In elevated floors, bamboo members become an integral part of structural framework of building. The floor will comprise structural bamboo elements and bamboo decking.

7.3.7 Jointing Techniques

The jointing techniques in construction using bamboo shall be in accordance with [7.2](#).

8 BAMBOO AS REINFORCEMENT IN CONCRETE

8.1 Due to its fibrous texture, bamboo possesses high tensile strength. This can act as an alternate material for reinforcement in concrete within the limitations in design and construction. The ultimate strength of some of the species of bamboo in direct tension is nearly the same as that of steel at its yield point. On an average it varies from 1 400 kg/cm² to 2 000 kg/cm². The principles adopted for design of concrete structure with steel is also applicable for the concrete member with bamboo. Strength characteristics of steel, concrete and bamboo are indicated in [Table 4](#) and the design data for balanced section in beams/slabs reinforcement is given in [Table 5](#).

Table 4 Strength Characteristics of Mild Steel, Bamboo and Concrete (For Reinforcement Purposes)

(Clause [8.1](#))

SI No.	Species	Strength Characteristics kgf/cm ²		
		Mild Steel	Bamboo (Green)	Mass Concrete
(1)	(2)	(3)	(4)	(5)
i)	Modulus of elasticity	2.1 × 10 ⁶	175 750 to 196 640 (for all locations)	–
ii)	Tensile stress	1 265.5 to 1 400	158, Bending-tension (all locations)	–
iii)	Compressive stress	–	105 (<i>D. strictus</i>)	500
iv)	Bond stress	–	3.5 (green) to 5.6 (dry) reported (likely to increase up to 10)	40 to 100
v)	Shear stress	–	115 to 180	50

Table 5 Design Data for Bamboo and Mild Steel Reinforced Cement Concrete for Balanced Section

(Clause [8.1](#))

SI No.	Design Parameter	Design Data for Balanced Section M 15 (1 : 2 : 4) Bamboo and Mild Steel Reinforced Cement	
		Mild Steel	Bamboo
(1)	(2)	(3)	(4)
i)	Modular ratio	18	1 to 1.4
ii)	Position of neutral axis from the top	0.4d	0.25d to 0.254d
iii)	Lever arm of the resisting moment (MOR)	0.87d	0.917d to 0.915d
iv)	Constant for MOR (<i>Q</i>)	137	86
	where		
	<i>Q</i> = Bending moment/ <i>bd</i> ² ,		
	<i>d</i> = depth or thickness, and		
	<i>b</i> = width.		

8.2 The whole bamboo culms are halved longitudinally and again splitted into four and strips of the size of 20 mm to 25 mm width and 9 mm thickness are prepared for use as reinforcement. Bamboo strips with triangular lugs (formed at nodes) due to conversion of bamboo into strips improve the bond strength because of the protruding

lugs beings gripped firmly in concrete. Strength properties of split bamboo out of a few species are given in [Table 6](#).

8.3 Bamboo has a tendency to absorb water from concrete which results in initial swelling and subsequent shrinkage leading to cracks in concrete. Following processes are thus recommended as mandatory:

- Chemical treatment with preference for pressure impregnation (Full cell treatment with warm coal tar creosote is likely to preserve and at the same time act as water repellent); and
- Treated strips be further given a water repellent treatment with a coating of very hot bitumen 80/100 grade, followed by a liberal coating of coarse sand. Water inhibitor will also help maintain dimensional stability of strips.

8.4 Techniques in Placing the Reinforcement (see also [Fig. 10](#))

Reinforcement shall be placed in the following manner:

- Laid edge wise in sets of twin strips for main reinforcement;
- Laid flatwise with smooth surface and rough surface alternating as distribution or subsidiary reinforcement (wedge effects increased bond);
- For negative moments, bamboo strips of length equivalent to $\frac{span}{6}$ laid flatwise in alternate rows of slab reinforcement;
- For beams and lintels, twin strips are arranged in tiers, placed one above the other depending upon requirement and held in position with the help of vertical strips;
- Split bamboo strip or preferably steel rod be used for diagonal tension to resist shear in beams, etc; and
- Bind the reinforcement with the help of galvanized iron binding wire (or nylon thread/bamboo-cane string).

Table 6 Strength Properties of Split Bamboo in Green Condition¹⁾

(Clause [8.2](#))

Sl No.	Species	Static Bending			Compression Parallel (Without Node)	Shear Parallel
		Fibre Stress at Elastic Limit	Modulus of Rupture	Modulus of Elasticity	Maximum Crushing Stress	Maximum Shearing Stress
		kg/cm ²	kg/cm ²	×10 ³ kg/cm ²	kg/cm ²	kg/cm ²
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	<i>Bambusa pallida</i> ²⁾	378	605	32.5	595	97.7
ii)	<i>B. vulgaris</i>	464	885	52.3	403	94.4
iii)	<i>Dendrocalamushami ltonii</i>	301	524	37.7	414	77.7
iv)	<i>D. giganteus</i>	472	848	50.7	360	88.5
v)	<i>Oxytenanthera abyssinica</i> ²⁾	450	831	33.6	377	112.0

¹⁾ average of bottom, middle and top portions of culms.

²⁾ average of bottom and middle portions only.

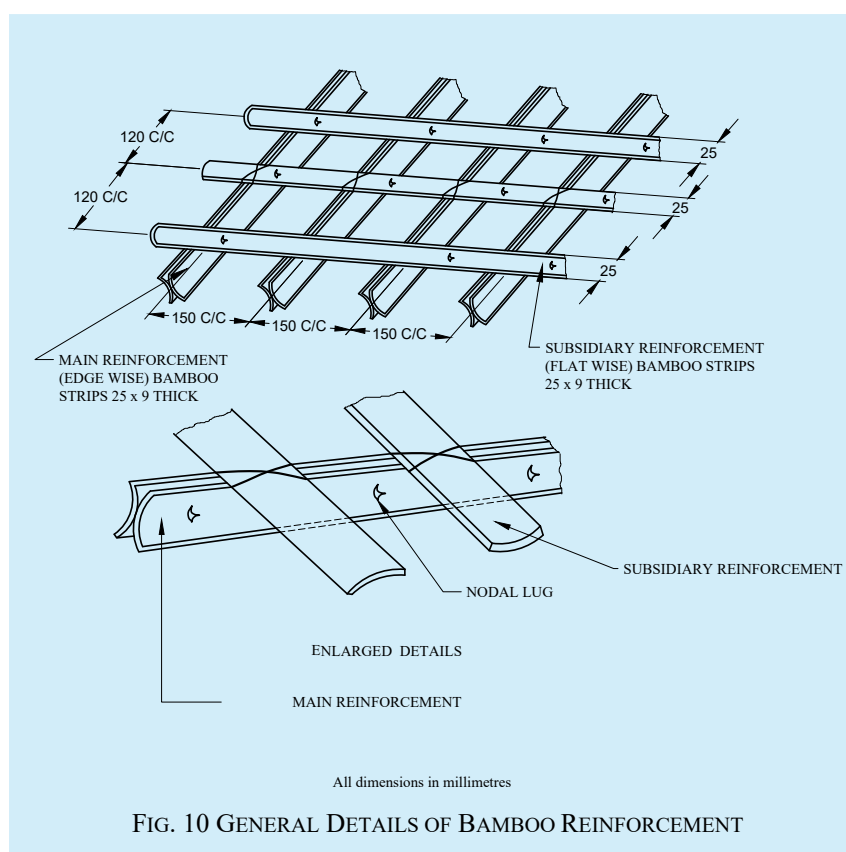


FIG. 10 GENERAL DETAILS OF BAMBOO REINFORCEMENT

9 WALLING, FLOORING, ROOFING, AND IN-FILL PANELS FOR WALL, AND COMMON ROOF COVERING/CLADDING MATERIALS

9.1 Materials

9.1.1 Bamboo Based Panel Products

The common roof covering and in-fill materials shall include bamboo mat board, bamboo mat veneer composite, bamboo mat corrugated sheet, bamboo tiles, plastered bamboo reeds, thatch, corrugated galvanized iron sheeting, asphaltic sheets, plain clay tiles, etc. The respective materials shall conform to the following:

- Bamboo mat board* — These shall conform to the accepted standard [C-3B(8)];
- Bamboo mat veneer composite* — These shall conform to the accepted standard [C-3B(9)];
- Bamboo mat corrugated sheet and bamboo mat ridge cap* — These shall conform to the accepted standard [C-3B(5)]; and
- Bamboo-jute composite corrugated and semi-corrugated sheets* — These shall conform to the accepted standard [C-3B(10)].

9.1.2 Reconstituted Wood from Destructured Bamboo

After lopping, bamboos are cross-cut into small lengths in the round and then split along the axis. The small cross-section strips are then passed between the counter revolving rollers till the material is fully destructured. This is then air dried and resin treated and suitably pressed without much disturbing the natural alignment of fibres thus retaining the directional properties of bamboo.

Some typical strength properties of reconstituted wood from destructured bamboo are given in [Table 7](#). The reconstituted product from bamboo can be sawn, nailed, moulded, bored, screwed, polished and painted. Thus, it offers a good substitute of solid wood for structural uses.

NOTE — Reconstituted boards/wood from destructured bamboo when tested for physical and mechanical properties in accordance with the accepted standards [C-3B(11)] and [C-3B(12)] yielded encouraging results. Gist of mechanical properties of these boards is excellent when compared with *Dendrocalamus strictus* (bamboo), teak and sal.

Table 7 Gist of Mechanical Properties of Reconstituted Wood from Bamboo (*Dendrocalamus strictus*)

(Clause [9.1.2](#))

Sl No.	Board Material			Density	Modulus of Rupture	Tensile Strength Parallel to Grain	Compressive Strength Parallel to Grain	Izod Impact	Shear Strength	Modulus of Elasticity	Screw Holding Power
	Set	Resin Content	Pressure								
		percent	kgf/cm ²	gm/cm ³	kgf/cm ²	kgf/cm ²	kgf/cm ²	kgf/cm	kgf/cm ²	× 10 ³ kgf/cm ²	kgf
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
i)	I	6 to 29	56	0.99 to 1.25	1 769 to 2 255	1 049 to 2 024	1 057 to 1 596	343 to 281	202 to 364	278 to 308	398 to 508
In the increasing order of resin percentage.											
ii)	II	9	21 to 28	0.62 to 0.93	1 062 to 1 615	6 99 to 1 210	675 to 1 030	186 to 202	128 to 247	155 to 232	361 to 486
Wood/Bamboo											
iii)	<i>Shorea robusta</i> (at 12 percent, dry moisture content)			0.749	1 249	—	685	124	110	152.8	369
iv)	<i>Tectona grandis</i> (at 12 percent, dry moisture content)			0.596	959	—	532	68	102	119.6	326
v)	<i>Dendrocalamus strictus</i>			0.757	1 310	—	624	—	—	180.3	—

9.1.3 Other Panel Products/Infill Materials

Other panel products/infill materials shall be in accordance with relevant Indian standards. These shall be such that they maintain their integrity and strength in the assigned service class throughout the expected life of the structure.

9.1.4 Adhesives

Adhesive shall maintain their integrity in accordance with the accepted standard [C-3B(13)].

9.2 Structural Usage of Bamboo Based Products

9.2.1 General

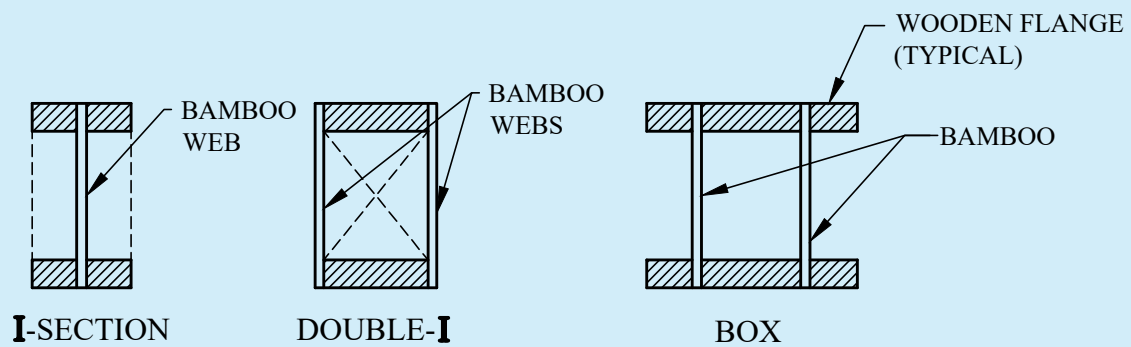
Bamboo mat board (BMB) and bamboo mat veneer composite (BMVC) have high in-plane rigidity and hence high racking strength and are more flexible than equivalent plywood. This property of BMB and BMVC can be advantageously used in many engineering applications such as wall panelling, composite beams, composite floor deck and stress skinned panels. Bamboo mat corrugated sheet (BMCS) is water proof and resistant to decay, termites/insects and fire. These sheets may be used as roofing material. Precast bamboo reinforced cement concrete and reconstituted wood from destructured bamboo can also be used in structural applications.

9.2.2 Bamboo Composites

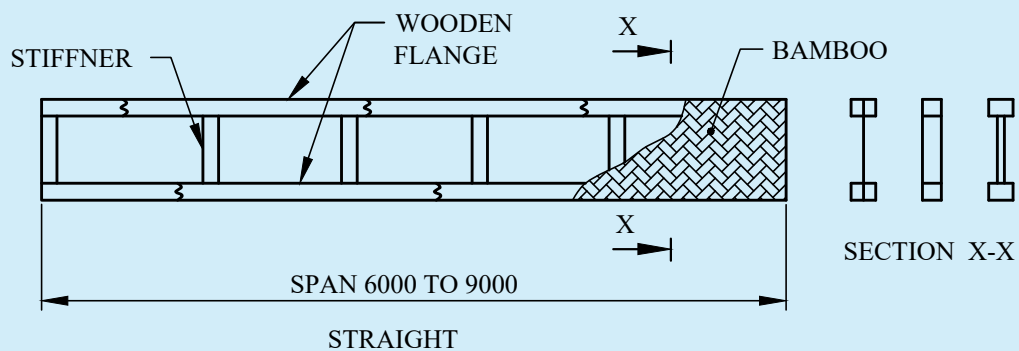
Along with bamboo, bamboo composites such as bamboo mat board, bamboo mat veneer composite, may also be successfully used in combination with timber for making efficient structural components like web because where high shear strength of bamboo mat board and bamboo mat veneer composite can be utilized as web and timber flanges for transmitting tensile and compressive stress. PF, PRF adhesives are commonly used for gluing web and flanges as shown in [Fig. 11](#). Beams can be straight, single tapered, double tapered, haunched, as shown.

Bamboo mat board and bamboo mat veneer composite can also be used for roof and wall sheathing (shear walls) and diaphragms, gussets, arch panels, barrel vaults, etc. It can also be used as skin material in sandwich construction.

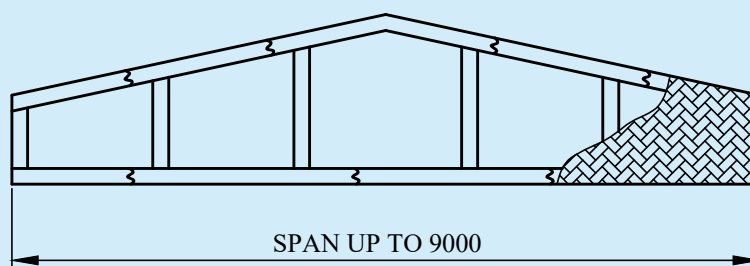
Permissible safe working stresses are given in [Table 8](#).



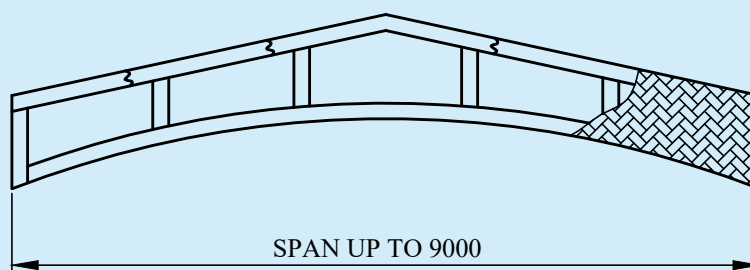
TYPICAL CROSS SECTION OF WEB BEAMS



STRAIGHT



DOUBLE TAPERED



HAUNCHED

WEB BEAM CONFIGURATIONS

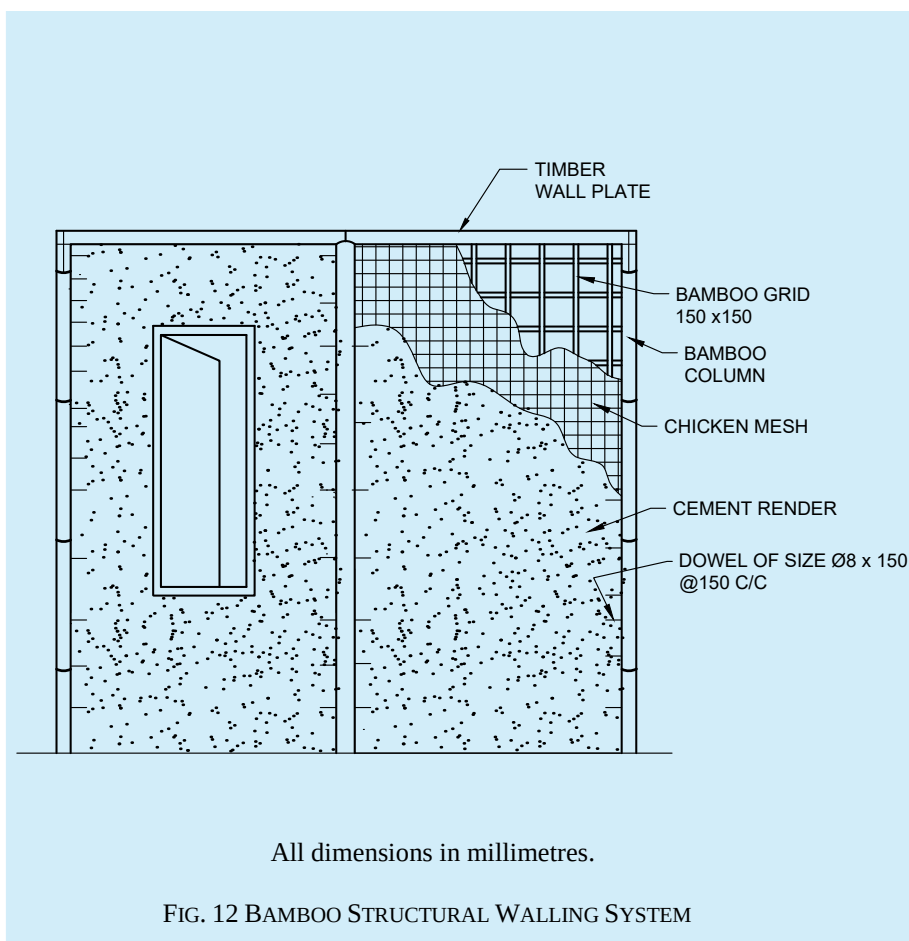
All dimensions in millimetres

FIG. 11 BAMBOO MAT BOARD WEB BEAMS

Table 8 Permissible Safe Working Stresses for Bamboo Mat Board and Bamboo Mat Veneer Composite

(Clause 9.2.2)

Sl No.	Property	Bamboo Mat Board N/mm ²	Bamboo Mat Veneer Composite in Direction	
			Along N/mm ²	Across N/mm ²
(1)	(2)	(3)	(4)	(5)
i)	Tensile strength	8.5	10	10
ii)	Compressive strength	10.80	13	6.5
iii)	Modulus of rupture	19.00	19	10.5
iv)	Modulus of elasticity	3 200	4 600	—
v)	Modulus of rigidity	4 970	1 930	—



9.2.4.1 Foundation detail

The columns pierced by steel dowels at 150 mm centre-to-centre provide connection to the bamboo grid infill. Fix 8 mm diameter MS rod of 400 mm length at 100 mm from one end of the column which has undergone treatment. For anchoring, fix two more MS rods of same diameter at 100 mm centre-to-centre starting from the first hole (see [Fig. 13](#)). Pour the concrete by giving temporary support. Foundation is limited to individual footings of size 400 mm × 400 mm × 600 mm deep for each bamboo column placed at 1 200 mm apart and two coarse concrete blocks or stone or other locally available material for infill wall foundation.

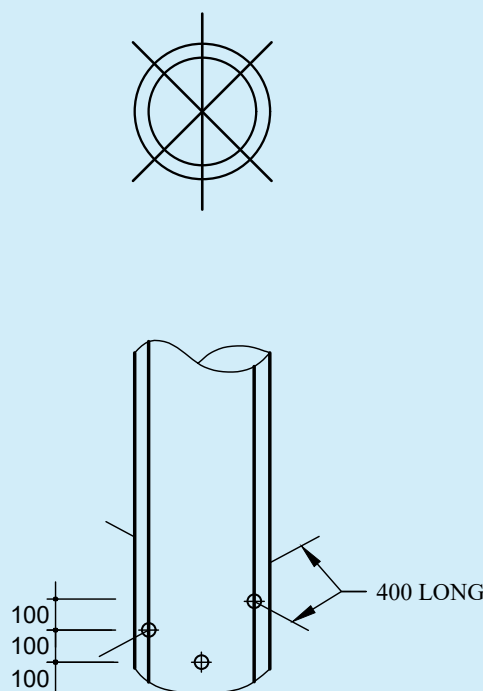
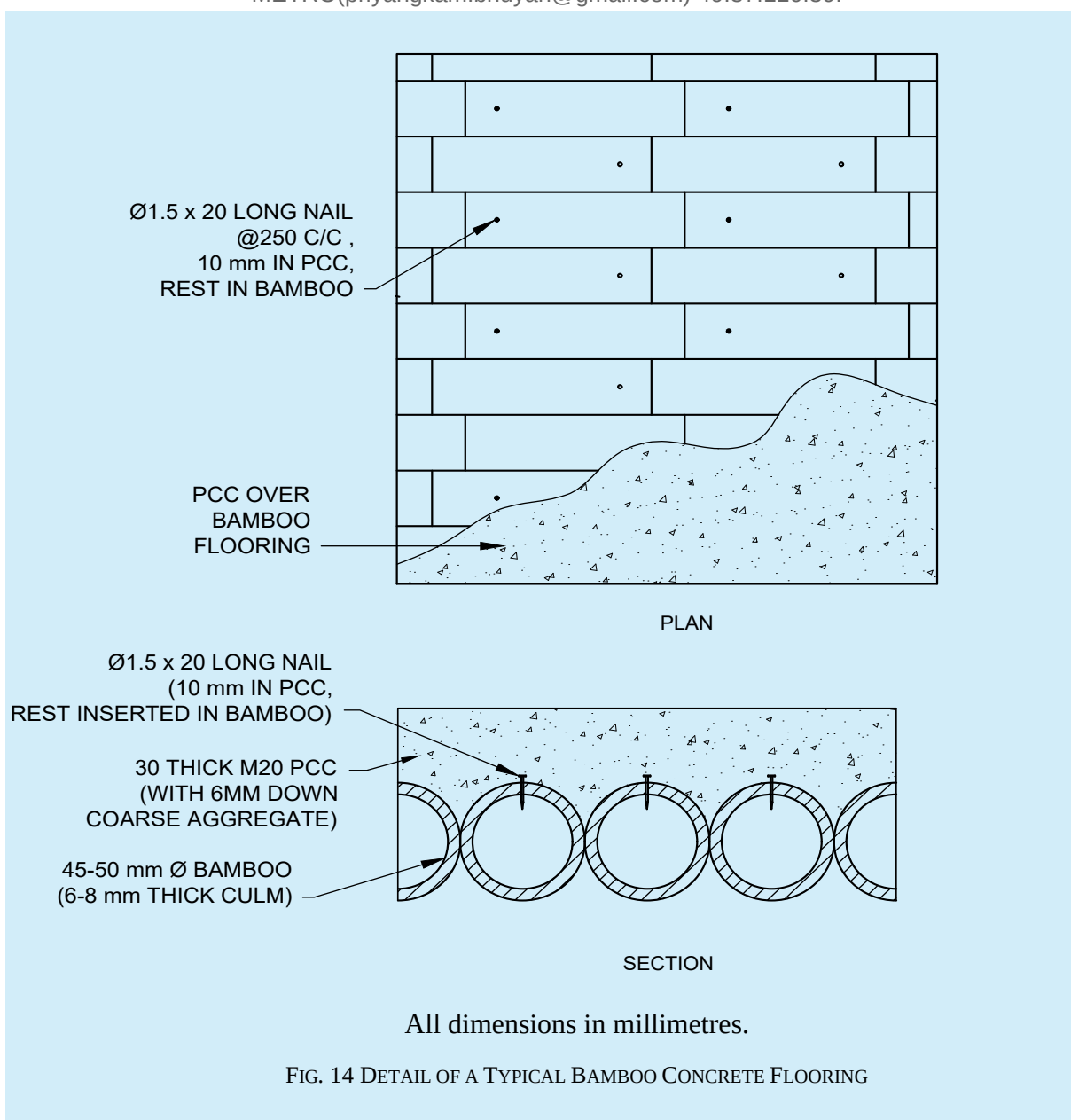


FIG. 13 DETAILS OF BOTTOM OF BAMBOO COLUMN FOR ANCHORING IN CONCRETE FOR FOUNDATION

9.2.5 Bamboo Concrete Flooring Using Round Bamboo

Technology of such a flooring (for situations needing raised flooring) is based on the method of rigidly fixing round bamboo, where 30 mm thick M20 grade PCC (with 6 mm down coarse aggregate) is placed over 6 mm to 8 mm thick culm, that are arranged side by side to form bamboo flooring.

The rigid fixing is achieved by using shear connectors in the form of 1.5 mm dia. 20 mm long MS nails partially inserted into the bamboo flooring and balance length with the heads of the nails embedded in the PCC layer, holding the two components together. The sectional properties of the combination are calculated based on the concept of flitched beams for design (see [Fig. 14](#)).



9.2.6 Bamboo Concrete Walling Using Flat Bamboo

Loads on walls and roof do not generally justify use of round bamboo as such. So, flat or split bamboo is preferred for use in these cases. The flat bamboo works like a wire-net used conventionally to hold cement plaster in critical applications. Partial plastering is done during the first day ensuring entry of plastering material in the gaps of flat bamboo. This provides the base for cement plastering or PCC laying next day. About 5 mm gap is essential for entry of plastering material of sufficient sectional area to sustain shear force. The gaps are generated by inserting a small wedge of bamboo strip and pulling it out after a part of the gap is filled by plaster. Such gaps have to be discontinuous, to retain the character of a wire-net. About 25 percent width of the flat bamboo is assumed to be gaps, part of it is the organized gap of 5 mm width as described, at an interval of about 50 mm width, to sustain shear force and the balance gap remains to be natural to flat bamboo. Minimum thickness of the flat bamboo is assumed to be 6 mm. Bamboo with thicker wall would be stronger.

Expected strength (using *Bambusa balcooa*) may be considered as follows:

- Walling of 6 mm bamboo with 18 mm thick cement sand plaster in 1 : 3 would be able to withstand wind pressure of 112 kgf/m² if used as wall, either supported continuously 1.5 m apart, or supported simply 1.25 m apart; and

- b) As a roof, the wind suction load would be partly compensated by the dead load of the roof. The imposed load of a mason sitting at the centre of span needs to be considered. This load can be accommodated by use of 18 mm M 20 PCC with 6 mm flat bamboo, supported 1.0 m apart; neatly finished.

The engineered houses for application of the above bamboo concrete walling are designed with stiffeners and diagonals in all the three planes, keeping the deformation of walls along plane of walls almost nil and deflection of columns across the plane of wall at a minimum.

10 LIFE CYCLE ASSESSMENT OF BAMBOO FOR USE IN BUILDINGS

10.1 Timber used in building structures and elements shall undergo a comprehensive life cycle assessment (LCA) to evaluate its environmental impacts across its entire lifecycle, from raw material extraction to disposal or recycling. LCA provides valuable insights into the sustainability of timber and its environmental footprint in building applications. The assessment shall cover the following stages:

- a) *Production and Construction* — This stage evaluates the energy and resources required for the extraction, transportation, and processing of raw timber into building materials. It also includes the energy consumption and waste generated during the construction process, as well as the transportation of materials to the construction site.
- b) *Use Stage* — The use stage assesses the environmental impacts of timber during the building's operational lifespan. This includes maintenance, repairs, and potential replacement of timber components, considering the durability, performance, and long-term sustainability of the material.
- c) *End-of-Life* — At the end of the building's useful life, the timber structure may undergo demolition, with the materials being processed for waste management, recycling, or reuse. This stage assesses the environmental effects of these processes and the potential for timber materials to be repurposed.
- d) *Externalized Impacts Beyond the System Boundary* — The incidental effects of recycling, reusing, and recovering energy, water, or materials beyond the system boundaries of the LCA study are included in this stage. These externalized impacts consider how recycled timber or materials might contribute to environmental benefits outside the building's lifecycle.

10.2 Environmental Impact Categories

The LCA may be used to assess the environmental impact of timber buildings in several categories, which encompass emissions and pollutants generated throughout the lifecycle. These categories include:

- a) *Global Warming Potential* — The contribution of greenhouse gas emissions to climate change;
- b) *Acidification Potential* — The potential for acidic compounds to form in the environment, affecting ecosystems and infrastructure;
- c) *Eutrophication Potential* — The impact of nutrient loading on water bodies, potentially causing excessive algal growth;
- d) *Ozone Depletion Potential* — The impact of substances that contribute to the depletion of the ozone layer.
- e) *Smog Formation Potential* — The potential for air pollution to contribute to the formation of ground-level ozone and smog; and

All relevant material and process quantities shall be included in the LCA inventory, and their impacts shall be calculated using recognized tools and methodologies.

10.3 Requirements for Life Cycle Assessment

In the context of building with timber, LCA may be required for the following purposes:

- a) *Decision-Making Support* — Providing building owners and stakeholders with data on the environmental and sustainability aspects of timber, aiding informed decision-making;
- b) *Design Evaluation* — Offering insights into material selection and environmental performance across different design possibilities;

- c) *Product Comparison* — Demonstrating the environmental advantages of timber in comparison to alternative building materials;
- d) *Environmental Benefit Evaluation* — Helping assess the ecological benefits of innovative timber products or policies; and
- e) *Performance Benchmarking* — Comparing the environmental performance of timber buildings against established benchmarks and sustainability goals.

10.4 Methodology for Conducting LCA

The LCA of timber buildings may be conducted iteratively, with increasing detail as the design or construction phases progress. The process should follow these steps:

- a) Establish the purpose and scope of the LCA, focusing on the objectives of evaluating timber's environmental impact;
- b) Gather data on the timber materials, including their types, quantities, lifespans, and lifecycle stages. Special attention should be given to materials used in the production and use stages;
- c) Using LCA tools (for example, SimaPro, MATLAB), assess the environmental impacts of timber buildings. The evaluation shall include biogenic carbon considerations, as timber stores carbon from the atmosphere, contributing to carbon sequestration during its lifecycle;
- d) Interpret the LCA results by categorizing environmental impacts by building component, material type, and lifecycle stage. Identify key contributors to the building's overall environmental footprint and visualize the outcomes; and
- e) Document all findings, including assumptions, methodology, and conclusions, and present the results in reports for stakeholders. The results shall guide sustainable construction practices and decision-making in timber building projects.

10.5 Reporting

The results of the LCA should be documented, including all assumptions, methodologies, and conclusions. Reports should be prepared in accordance with relevant standards, and may be used to guide sustainable building practices and decision-making in construction.

11 TYPICAL DETAILS OF CONVENTIONAL BAMBOO BUILDING CONSTRUCTION

Typical details of conventional bamboo building construction is given in [Annex C](#), for guidance.

ANNEX A

(Foreword)

SOURCE AND LOCAL NAMES OF SOME OF THE SPECIES OF BAMBOO

<i>Sl No.</i>	<i>Species</i>	<i>Source/Local Names</i>
(1)	(2)	(3)
i)	<i>Bambusa auriculata</i>	Assam, Bangladesh, Myanmar; introduced in Calcutta Botanical Garden.
ii)	<i>B. balcooa</i>	Asm – Baluka; Ben – Balku bans; Duars – Bora bans; Garo – Wamnah, beru; Tripura – Barak.
iii)	<i>B. bambos</i> (Syn. <i>B. arundinacea</i>)	Asm – Kotoba; Ben – Baroowa, behor, ketuas, ketwa; Manip – Saneibi; Mah – Katang bamboo, oowga; Oriya – Daba, katuig; Tel – Mulkasveduru, Mulluvedurn; English – Spiny bamboo.
iv)	<i>B. burmanica</i>	Asm – Thaikawa.
v)	<i>B. multiplex</i> <i>Syn. B. glaucescens</i> (Syn. <i>B. nana</i>)	Sans – keu-fa; Burmese – Pa-lau-pinan-wa; Malay – Bamboo tjeenah; China – Bamboo hower tjeenah.
vi)	<i>B. nutans</i>	Asm – Deobans, jotia-makal; Asm – Bidhuli, mukial; Ben – Makia; Bhutia – Jiu; Hin – Malabans; Kangra – Nal; Khasi - Seringjai; Kuki – Wa malang; Lepcha – Malubans, mahlu, mallo; Oriya – Badia bansa; Sylhet (Bangladesh) – Peechli; Tripura – Kali.
vii)	<i>B. pallida</i>	Asm – Bijli, jowa, makal, walkthai; Cachar – Bakhal, burwal; Khasi – Seskien, skhen, ineng, usker;

<i>Sl No.</i>	<i>Species</i>	<i>Source/Local Names</i>
(1)	(2)	(3)
viii)	<i>B. polymorpha</i>	Lepcha – Pashipo, pshi, pushee; Mikir – Loto; Naga – Tesero, watoi; Tripura – Makal. Asm – Jama betwa, betwa; Ben – Batua, jaibarouwa, jama; Burma – Kyathaung-wa; MP-Korku – Narangi bhas; Tripura – Basi.
ix)	<i>B. tulda</i>	Asm – Wamunna, wagi, nal-bans; Ben – Tulda, jowa; Duars – Karanti, matela; Garo – Watt; Hin – Peka; Kamrup – Bijuli, jati, joo, ghora; Tripura – Mirtinga.
x)	<i>B. vulgaris</i>	Ben and Manipuri – Bakal; Oriya – Sunarkania bans.
xi)	<i>B. wamin</i> Syn. <i>B. ventricosa</i> (Syn. <i>B. vulgaris</i> var. <i>wamin</i>)	Common name – Pitcher bamboo.
xii)	<i>Cephalostachyum pergracile</i>	MP – Bhalan bans; Manip – Wootang; Naga – Latang; Oriya – Darrgi.
xiii)	<i>Dendrocalamus giganteous</i>	English – Giant Bamboo; Asm – Worra; Manip – Maroobeb.
xiv)	<i>D. hamiltonii</i>	Nep – Tamo; Asm – Kokwa; Tripura – Pecha.
xv)	<i>D. Longispathus</i>	Tripura – Rupai.
xvi)	<i>D. membranaceus</i>	Native of Myanmar; introduced in Kerala.
xvii)	<i>D. strictus</i>	English – Male bamboo; Ben – Karail; Guj – Nakur bans;

<i>Sl No.</i>	<i>Species</i>	<i>Source/Local Names</i>
(1)	(2)	(3)
xviii)	<i>Melocanna baccifera</i>	Kan – Kiri bidiru; Mah – Male bamboo, nanvel; Oriya – Salia; Tam – Kalmungil; Tel – Sadanapa vedur; Tripura – Lathi bans; Hin – Bans kaban, Nav bans; Asm – Tarai; Ben – Muli; Cachar – Wati; Garo – Watrai; Manip – Moubi; Mikir – Artem; Naga – Turiah.
xix)	<i>Oxytenanthera abyssinicia</i>	Native of tropical Africa; cultivated at FRI, Dehradun.
xx)	<i>Thyrsostachys oliveri</i>	Native of Myanmar; Planted in Haldwani (Uttaranchal), Arunachal Pradesh, Kerala and Tamil Nadu.
NOTES		
1 The following abbreviations have been used in the above table:		
Asm	Assam	
Ben	Bengali	
Guj	Gujarati	
Hin	Hindi	
Kan	Kannada	
Mah	Maharashtra	
Manip	Manipur	
M.P.	Madhya Pradesh	
Nep	Nepali	
Sans	Sanskrit	
Tam	Tamil	
Tel	Telugu	
2 The above table does not provide an exhaustive list. It only attempts to enlist some of the information readily available in regard to species of bamboo from India and some of the neighbouring countries, and some connected information.		

ANNEX B

(Foreword)

RECOMMENDATIONS FOR BAMBOO STRUCTURES IN HIGH WIND/CYCLONE PRONE AREAS

B-1 Given below are some recommendations for bamboo structures in high wind/cyclone prone areas, which are based on rapid visual screening (RVS) of the damages of past events:

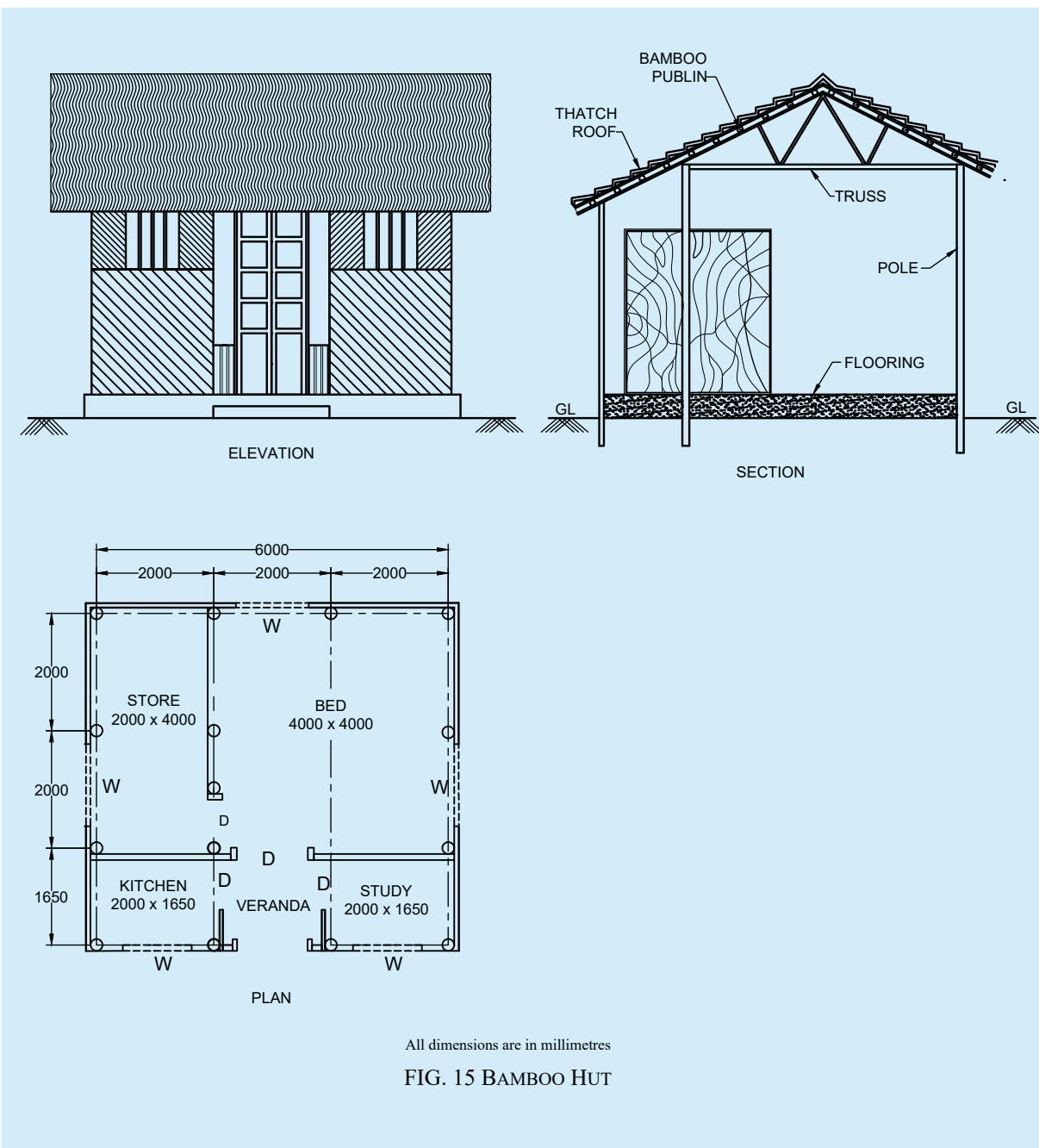
- a) Prefer a square plan of the building symmetrical to the arrangement of partitions and openings for balanced distribution of forces within the structure. Window positioning be planned in such a way that wind entering through window could pass through similar opening on opposite wall in order to reduce built-up pressure, and incorporate ridge vents;
- b) A hip-shaped, pyramidal or a conical roofing be preferred to gable end or mono-pitch ones for least wind resistance;
- c) Wide mono-pitched *verandas* abutting the entrance to residence should be structurally separated from the main building;
- d) Roof overhangs be limited to 450 mm as they contribute to higher negative suction. Large overhangs should be properly tied back to wall frame-work;
- e) Roof structure shall be properly secured with vertical load-bearing members (wooden or bamboo columns, brick piers or walling) for eliminating the chances of being blown away from the building. A continuous beam connecting all the columns would increase the framing action;
- f) Roof cladding (G.I. sheets, clay tiles or thatch) be properly connected with holding down U-bolts, or sturdy anchorages to ensure safety. Thatch roof be tied by nylon ropes in diagonal pattern;
- g) Sufficient depth for anchorage be provided; columns be embedded in pits within the foundation and secured by poured concrete. Wetting of column base be checked by raising it from the floor level;
- h) Highest importance be given to strengthening the junctions between structural elements at different levels, that is, foundation, eaves and roof. Walls be linked together suitably by a continued collar beam at floor, sill, lintel and roof level; and
- j) Fastening any two mutually perpendicular load bearing members should be done in accordance with good practice of spacing the connecting bolts with reference to loads direction and the centre of stiffness. This will increase the moment-resisting capacity of the joint and prevent hinge formation. A minimum of two bolts be provided to prevent rotation of members at a joint.

ANNEX C

(Foreword, Clause 11)

REPRESENTATIVE DESIGNS AND CONSTRUCTION DETAILS FOR BAMBOO STRUCTURES

C-1 For guidance in conventional bamboo construction, typical details are presented in the [Fig. 15](#). These details illustrate common applications of bamboo in trusses, roofing, walls, and flooring. They are intended to serve as indicative examples and not as substitutes for structural design.



C-1.1 [Figure 16](#) shows a typical bamboo truss arrangement used in conventional bamboo structures. It highlights jointing methods generally adopted in practice. The illustration is indicative and not a design substitute.

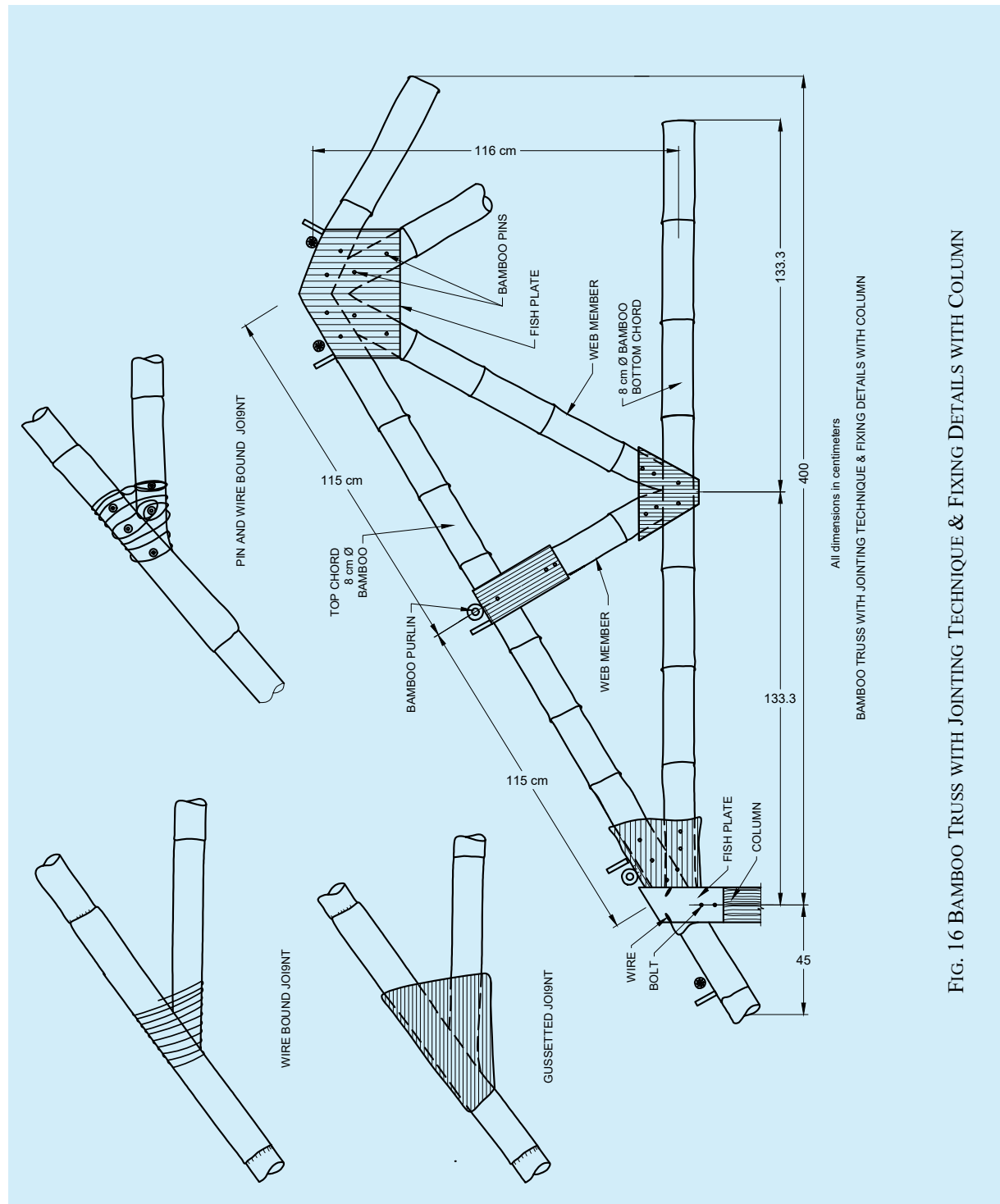
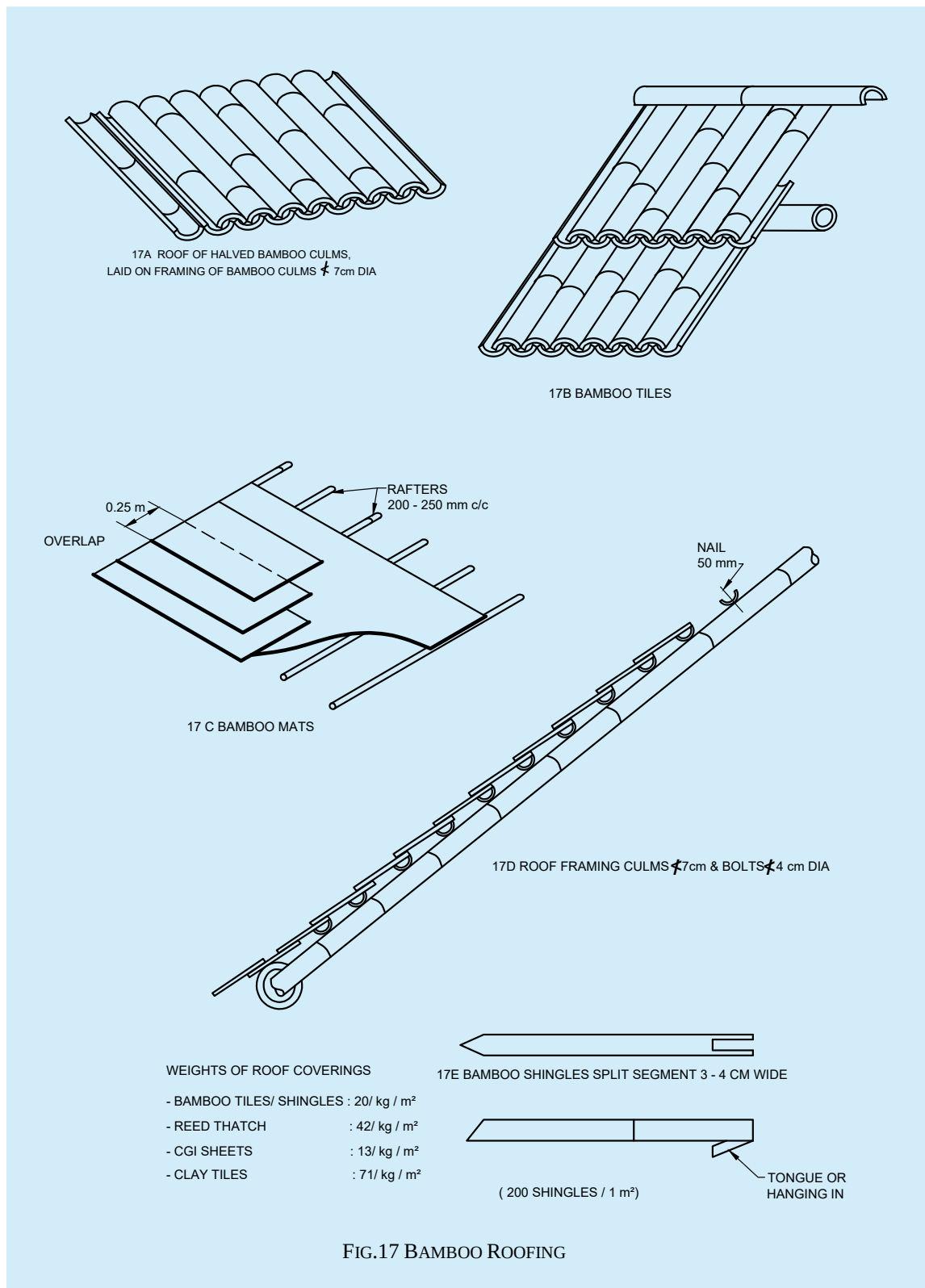
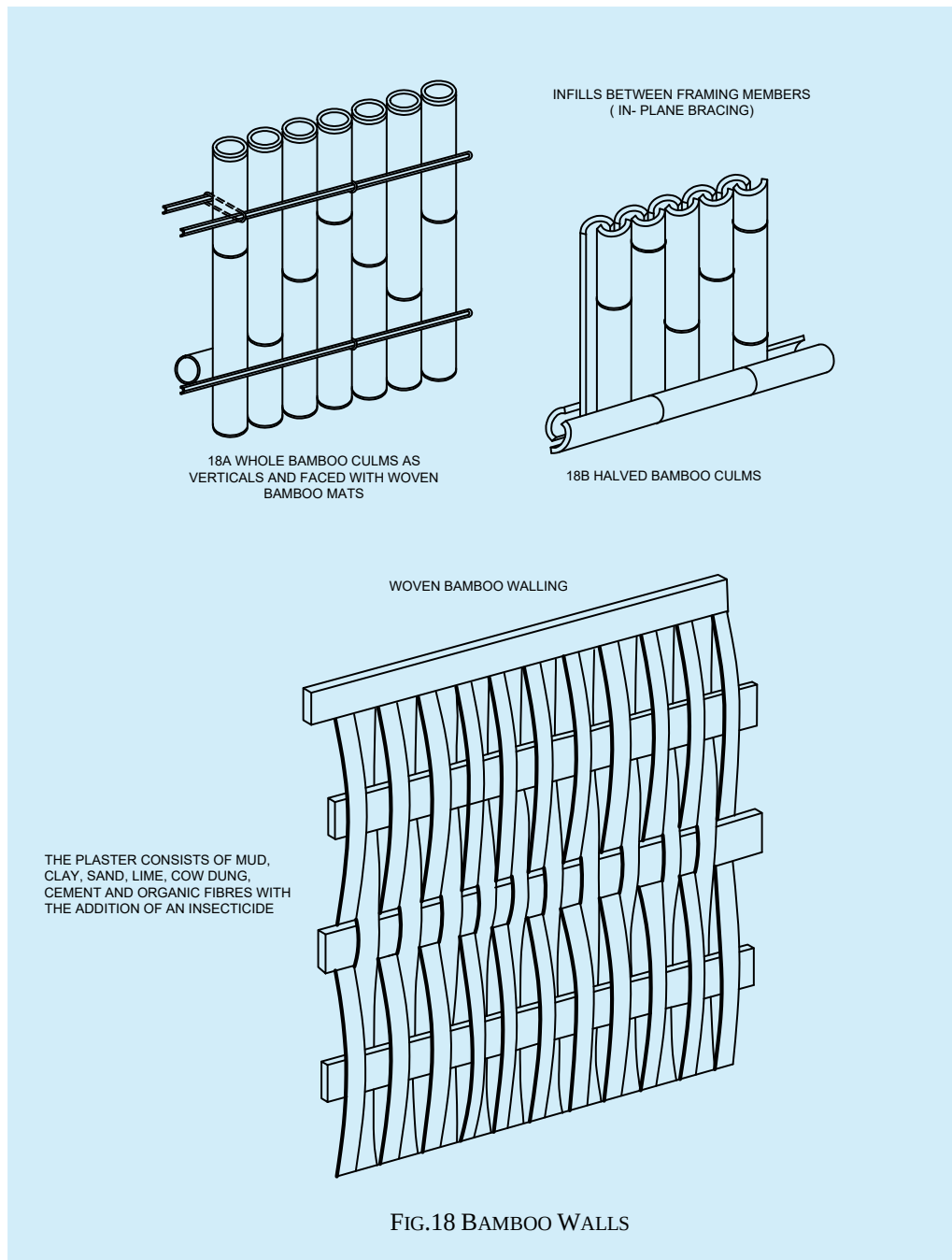


FIG. 16 BAMBOO TRUSS WITH JOINTING TECHNIQUE & FIXING DETAILS WITH COLUMN

C-1.2 [Figure 17](#) presents a representative bamboo roofing system for conventional bamboo structures. It demonstrates the use of bamboo members in roof framing. The figure is for reference only, not a structural design.



C-1.3 [Figure 18](#) depicts bamboo used as walling elements in conventional bamboo structures. It illustrates construction arrangements commonly followed. The drawing is illustrative and not a substitute for design.



C-1.4 [Figure 19](#) shows the application of bamboo in floor decking for conventional bamboo structures. It highlights a representative arrangement for structural flooring. The figure is illustrative only and not for design purposes.

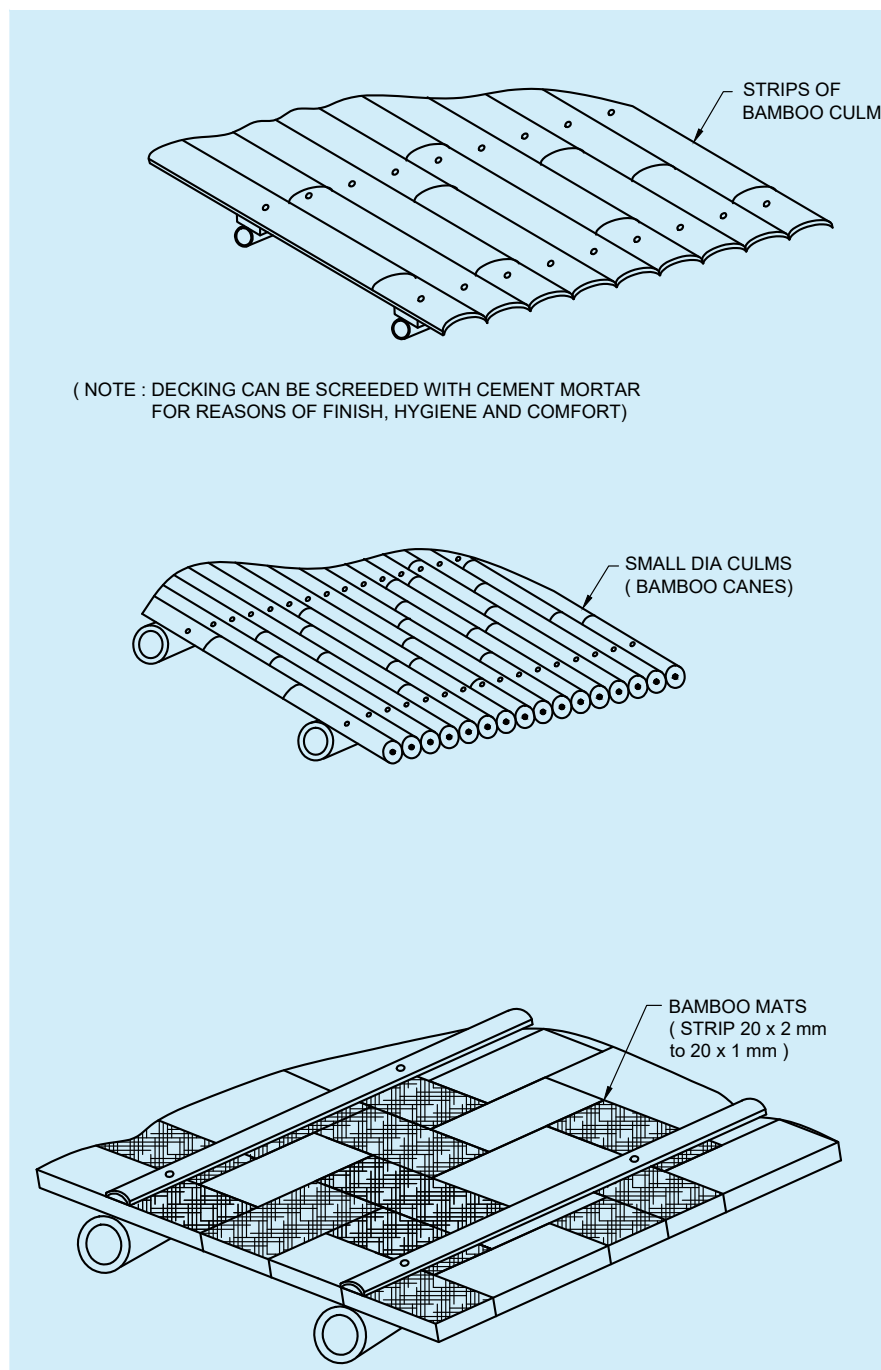


FIG. 19 BAMBOO FLOOR DECKING

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this NBCS. The latest version of the standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used for conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parenthesis indicates the number of the reference in this Subsection.

<i>IS No.</i>	<i>Title</i>
(1) IS 9096 : 2006	Preservation of bamboo for structural purposes — Code of practice (<i>first revision</i>)
(2) IS 6874 : 2008	Method of tests for bamboo (<i>first revision</i>)
(3) IS 1902 : 2006	Preservation of bamboo and cane for non-structural purposes — Code of practice (<i>second revision</i>)
(4) IS 8242 : 1976	Bamboo mat corrugated sheets — Specification
(5) IS 15476 : 2004	Specification for bamboo mat corrugated sheets
(6) IS 2366 : 1983	Code of practice for nail-jointed timber construction (<i>first revision</i>)
(7) IS 4924	Method of test for nail-jointed timber trusses
(Part 1) : 1968	Destructive test
(Part 2) : 1968	Proof test
(8) IS 13958 : 1994	Bamboo mat board for general purposes — Specification
(9) IS 14588 : 1999	Bamboo mat-veneer composite for general purposes — Specification
(10) IS 15972 : 2012	Bamboo-jute composite corrugated and semi-corrugated sheets — Specification
(11) IS 1734 (Parts 1 to 20) : 1983	Methods of test for plywood (<i>second revision</i>)
(12) IS 1708 (Parts 1 to 18) : 1986	Methods of testing of small clear specimens of timber (<i>second revision</i>)
(13) IS 848 : 2025	Synthetic resin adhesives for plywood (phenolic, aminoplastic and bio-materials based) — Specification (<i>third revision</i>)

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NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 4 MASONRY

BUREAU OF INDIAN STANDARDS

CONTENTS

1	SCOPE	... 7
2	TERMINOLOGY	.. 7
3	MATERIALS	... 9
4	DESIGN CONSIDERATIONS	...12
5	STRUCTURAL DESIGN	... 26
6	GENERAL REQUIREMENTS	...32
7	REINFORCED BRICK AND REINFORCED BRICK CONCRETE FLOORS AND ROOFS	...34
8	SPECIAL CONSIDERATION FROM EARTHQUAKE POINT OF VIEW	...34
9	CONFINED MASONRY WALLS (CMW)	... 72
10	REINFORCED MASONRY	... 91
11	GUIDELINES FOR IMPROVING EARTHQUAKE RESISTANCE OF LOW STRENGTH MASONRY BUILDINGS	... 102
12	MASONRY WALLS USING RAT-TRAP BOND TECHNOLOGY	... 111
13	SUSTAINABILITY IN MASONRY CONSTRUCTION	... 111
14	NOTATIONS AND SYMBOLS	... 111
ANNEX A	SOME GUIDELINES FOR ASSESSMENT OF ECCENTRICITY OF LOADING ON WALLS...	112
ANNEX B	CALCULATION OF BASIC COMPRESSIVE STRESS OF MASONRY BY PRISM TEST	... 112
ANNEX C	GUIDELINES FOR DESIGN OF MASONRY SUBJECTED TO CONCENTRATED LOADS	... 113
ANNEX D	GUIDELINES FOR APPROXIMATE DESIGN OF NON-LOAD BEARING WALL	... 114
ANNEX E	TYPICAL ITERATIVE METHOD FOR WALLS SUBJECTED TO FLEXURE AND AXIAL LOAD	... 116
ANNEX F	DISTRIBUTION OF LATERAL FORCES IN THE PIERS OF A MASONRY WALL	... 122
ANNEX G	MASONRY WALLS USING RAT-TRAP BOND	... 125
ANNEX H	NOTATIONS, SYMBOLS AND ABBREVIATIONS	... 132
	LIST OF STANDARDS	... 134

National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part C/Section 4) covers the structural design of unreinforced and reinforced masonry elements in buildings. Provisions on reinforced brick and reinforced brick concrete floors and roofs have also been dealt with.

This Section was first published in 1970 and subsequently revised in 1983, 2005 and 2016 as a chapter of National Building Code of India. The first revision of this Section was modified in 1987 through Amendment No. 2 to bring this Section in line with the latest revised masonry code. In the amendment, certain provisions were updated following the revision of IS 1905 'Code of practice for structural use of unreinforced masonry' on which the earlier version was based. In the amendment, requirements of masonry element for stability were modified; in the design of free standing wall, provisions were made for taking advantage of the tensile resistance in masonry under certain conditions; provision regarding effective height of masonry wall between openings was modified; method of working out effective height of wall with a membrane type DPC was modified; the criteria for working out effective length of wall having openings was modified; some general guidelines for dealing with concentrated loads for design of walls were included; and provision of cutting and chases in walls were amplified. The Section was thereafter revised in 2005.

In the second revision of this Section in 2005, the provisions of special considerations in earthquake zones were aligned in line with the revised Indian Standard, IS 4326 : 1993 'Code of practice for earthquake resistant design and construction of buildings'; a new clause covering guidelines for improving earthquake resistance of low strength masonry buildings was added; and reference to design of reinforced brick and reinforced brick concrete floors and roofs was included.

In the third revision of this Section in 2016, the detailed provisions relating to reinforced masonry and durability criteria for selection of masonry mortars had been included. Also, the number of storeys of masonry construction had been limited to 4, in line with the seismic design standards. A new masonry wall construction using rat-trap bond had been included.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

As a result of experience gained in the implementation of this Section and feedback received, as well as in view of formulation of new varieties of cement and new types of masonry blocks, a need to revise this Section was again felt. This section has, therefore, been revised to incorporate the following significant changes:

- 1) New masonry blocks have been included as masonry unit.
- 2) The admissibility of different structural systems in the earthquake zones is clarified. The following structural systems are admissible to construct earthquake resistant masonry buildings:
 - a) Masonry walls with prescriptive bands (MWB);
 - b) Masonry walls with prescriptive bands and vertical reinforcement (MWBR);
 - c) Confined masonry walls (CMW); and
 - d) Reinforced masonry walls (RMW).
- 3) The clause on confined masonry has been updated.
- 4) The clause on reinforced masonry has been updated.
- 5) Reference to all the concerned Indian Standards has been updated.

Structural design requirements of this section are largely based on following Indian Standards:

- a) IS 1905 : 1987 'Code of practice for structural use of unreinforced masonry (*third revision*)';
- b) IS 4326 : 2013 'Earthquake resistant design and construction of buildings — Code of practice (*third revision*)';

- c) IS 10440 :1983 ‘Code of practice for construction of RB and RBC floors and roofs’; and
- d) IS 17848 : 2022 ‘Confined masonry for earthquake resistance — Code of practice’.

Buildings with MWB and MWBR systems shall not be of height more than 15 m subject to a maximum of four storeys, measured from the base of the buildings to the roof slab or ridge level.

A reference to SP 20 : 1991 ‘Handbook on masonry design and construction (*first revision*)’, may be useful. In the formulation of this Section, inputs were also derived from the IITK-GSDMA’s publication, ‘Guidelines for structural use of reinforced masonry’.

All standards, whether given herein above or cross-referred to in the main text of this section, are subject to revision. The parties to agreement based on this section are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Section is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 ‘Rules for rounding off numerical values (*second revision*)’. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section.

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Important Explanatory Note for Users of the NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to ‘**accepted standard**’ in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

- a) Accepted standards [C-4(1)] refers to the Indian Standard(s) give at serial number (1) of the list of standards given at the end of this Part C/Section 4, that is, IS 1077 : 1992 ‘Specification for common burnt clay building bricks (*fifth revision*)’, IS 1725 : 2023 ‘Specification for stabilized soil blocks used in general building construction (*third revision*)’, IS 2180 : 1988 ‘Specification for heavy duty burnt clay building bricks (*third revision*)’, IS 2185 (Part 1) : 2005 ‘Concrete masonry units — Specification: Part 1 Hollow and solid concrete blocks (*third revision*)’, IS 2185 (Part 2) : 1983 ‘Concrete masonry units — Specification: Part 2 Hollow and solid light weight concrete blocks (*first revision*) (superseding IS 3590)’, IS 2185 (Part 3) : 1984 ‘Concrete masonry units— Specification: Part 3 Autoclaved cellular (aerated) concrete blocks (*first revision*)(superseding IS 5482)’, IS 2185 (Part 4) : 2008 ‘Concrete masonry units — Specification: Part 4 Cellular preformed foam cellular concrete blocks’, IS 2222 : 1991 ‘Specification for burnt clay perforated building bricks (*fourth revision*)’, IS 2849 : 1983 ‘Specification for non-load bearing gypsum partition blocks (solid and hollow types)(*first revision*)’, IS 3115 : 1992 ‘Specification for lime based blocks (*second revision*)’, IS 3316 : 1974 ‘Specification for structural granite (*first revision*)’, IS 3620 : 1979 ‘Specification for laterite stone block for masonry (*first revision*)’, IS 3952 : 2013 ‘Burnt clay hollow bricks and blocks for walls and partitions — Specification (*third revision*)’, IS 4139 : 1989 ‘Specification for calcium silicate bricks (*second revision*)’, IS 12440 : 1988 ‘Specification for precast concrete stone masonry blocks’, IS 12894 : 2002 ‘Specification for pulverized fuel ash lime bricks (*first revision*)’, IS 13757 : 1993 ‘Specification for burnt clay fly ash building bricks’ and IS 16720 : 2018 ‘Pulverized fuel ash-cement bricks — Specification’.
- b) Good practice [C-4(4)] refers to the Indian Standard(s) give at serial number (4) of the list of standards given at the end of this Part C/Section 4, that is, IS 2250 : 1981 ‘Code of practice for preparation and use of masonry mortars (*first revision*)’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 4 MASONRY

1 SCOPE

1.1 This NBCS (Part C/Section 4) covers the structural design aspects of unreinforced load bearing and non-load bearing walls, constructed with masonry units permitted in accordance with this Section.

1.2 This Section also deals with the selection of materials, special features of design and construction for masonry construction using rectangular masonry units. It also covers guidelines regarding earthquake resistant buildings constructed using masonry of low strength.

1.3 This Section also covers provisions for design of reinforced brick and reinforced brick concrete floors and roofs.

1.4 This Section also gives the recommendations for structural design aspects of reinforced load bearing and non-load bearing walls, constructed with different types of bricks and blocks.

1.5 The recommendations of the Section do not apply to walls constructed in mud mortars.

2 TERMINOLOGY

For the purpose of this Section, the following definitions shall apply.

2.1 Bed Block — A block bedded on a wall, column or pier to disperse a concentrated load on a masonry element.

2.2 Bond — Arrangement of masonry units in successive courses to tie the masonry together both longitudinally and transversely; the arrangement is usually worked out to ensure that no vertical joint of one course is exactly over the one in the next course above or below it, and there is maximum possible amount of lap.

2.3 Column, Pier and Buttress

- a) *Column* — An isolated vertical load bearing member, width of which does not exceed four times the thickness;
- b) *Pier* — A thickened section forming integral part of a wall placed at intervals along the wall, to increase the stiffness of the wall or to carry a vertical concentrated load. Thickness of a pier is the overall thickness including the thickness of the wall or, when bonded into a leaf of a cavity wall, the thickness obtained by treating that leaf as an independent wall (see [Fig. 1](#))
- c) *Buttress* — A pier of masonry built as an integral part of wall and projecting from either or both surfaces, decreasing in cross-sectional area from base to top.

2.4 Cross-Sectional Area of Masonry Unit — Net cross-sectional area of a masonry unit shall be taken as the gross cross-sectional area minus the area of cellular space. Gross cross-sectional area of cored units shall be determined to the outside of the coring but cross-sectional area of grooves shall not be deducted from the gross cross-sectional area to obtain the net cross-sectional area.

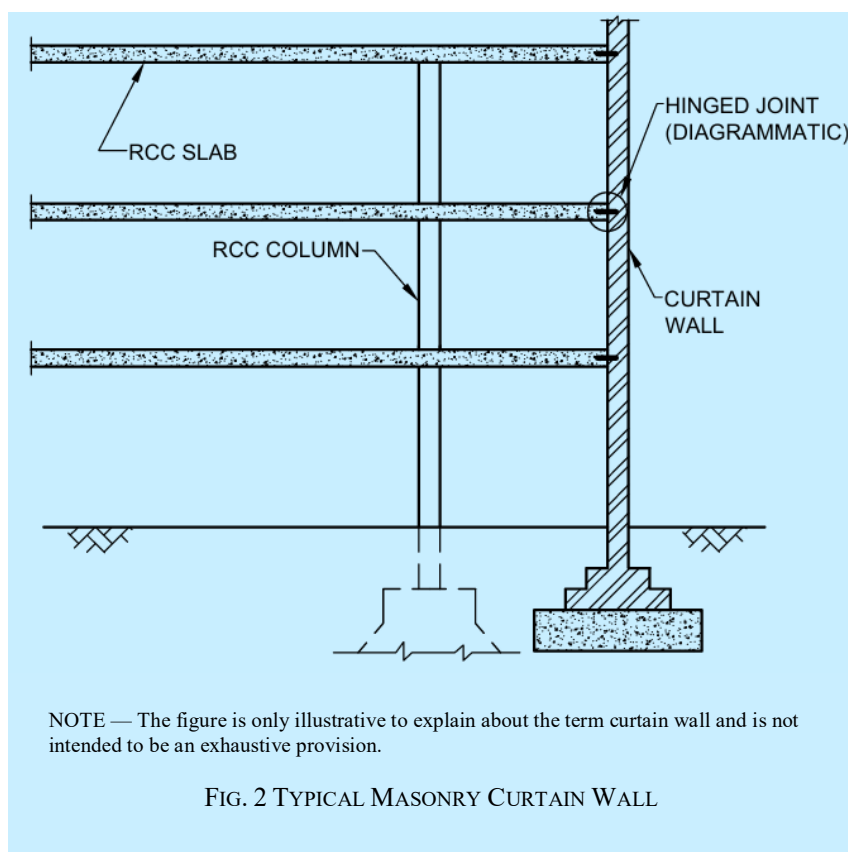
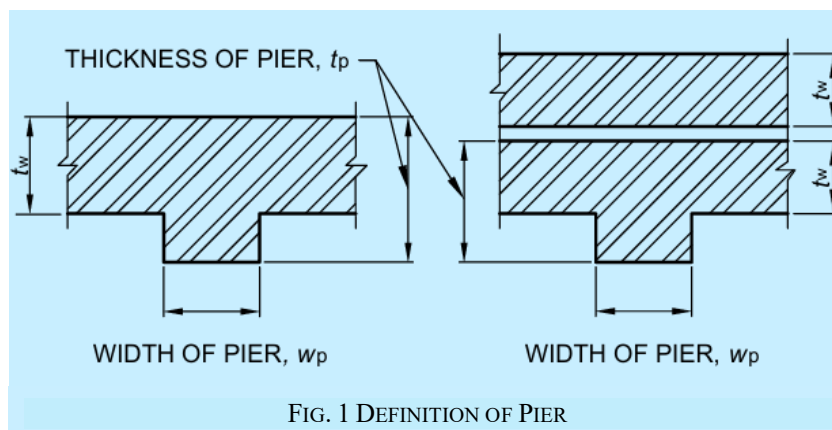
2.5 Curtain Wall — A non-load bearing wall subject to lateral loads. It may be laterally supported by vertical or horizontal structural members, where necessary (see [Fig. 2](#)).

2.6 Effective Height — The height of a wall or column, to be considered for calculating slenderness ratio.

2.7 Effective Length — The length of a wall to be considered for calculating slenderness ratio.

2.8 Effective Thickness — The thickness of a wall or column to be considered for calculating slenderness ratio.

2.9 Hollow Unit — A masonry unit of which net cross-sectional area in any plane parallel to the bearing surface is less than 75 percent of its gross cross-sectional area measured in the same plane.



2.10 Grout — Mortar of pourable consistency.

2.11 Joint — A junction of masonry units, which can be classified as:

- Bed Joint* — A horizontal mortar joint upon which masonry units are laid;
- Cross Joint* — A vertical joint, normal to the face of the wall; and
- Wall Joint* — A vertical joint parallel to the face of the wall.

2.12 Leaf — Inner or outer section of a cavity wall.

2.13 Lateral Support — A support which enables a masonry element to resist lateral load and/or restrains lateral deflection of a masonry element at the point of support.

2.14 Load Bearing Wall — A wall designed to carry an imposed vertical load in addition to its own weight, together with any lateral load.

2.15 Masonry — An assemblage of masonry units properly bonded together with mortar.

2.16 Masonry Unit — Individual units which are bonded together with the help of mortar to form a masonry element such as wall, column, pier, buttress, etc.

2.17 Partition Wall — An interior non-load bearing wall, one storey or part storey in height.

2.18 Panel Wall — An exterior non-load bearing wall in framed construction, wholly supported at each storey but subjected to lateral loads.

2.19 Shear Wall — A wall designed to carry horizontal forces acting in its plane with or without vertical imposed loads.

2.20 Slenderness Ratio — Ratio of effective height or effective length to effective thickness of a masonry element.

2.21 Types of Walls

- a) *Cavity Wall* — A wall comprising two leaves, each leaf being built of masonry units and separated by a cavity and tied together with metal ties or bonding units to ensure that the two leaves act as one structural unit, the space between the leaves being either left as continuous cavity or filled with a non-load bearing insulating and water-proofing material;
- b) *Faced Wall* — A wall in which facing and backing of two different materials are bonded together to ensure common action under load (see [Fig. 3](#)).

NOTE — To ensure monolithic action in faced walls, shear strength between the facing and the backing shall be provided by toothing, bonding or other means; and

- c) *Veneered Wall* — A wall in which the facing is attached to the backing but not so bonded as to result in a common action under load.

2.22 Reinforced Brick Slab (RB Slab) — Reinforced brick slab is particularly same as reinforced concrete slab in all its essential features except that brickwork is substituted partly for cement concrete. See [Fig. 4\(A\)](#) for details.

2.23 Reinforced Brick Concrete Slab (RBC Slab) — These are reinforced brick slab using concrete in the joints and on the top of bricks. See [Fig. 4\(B\)](#) for details.

3 MATERIALS

3.1 General

The materials used in masonry construction shall be in accordance with Part B ‘Building Materials’ of this NBCS. Storage of materials shall be in accordance with good practice [C-4(2)].

3.2 Masonry Units

Masonry units used in construction shall conform to accepted standards [C-4(1)].

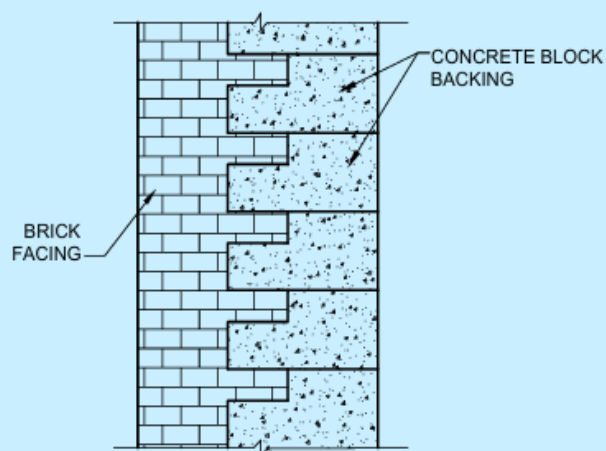
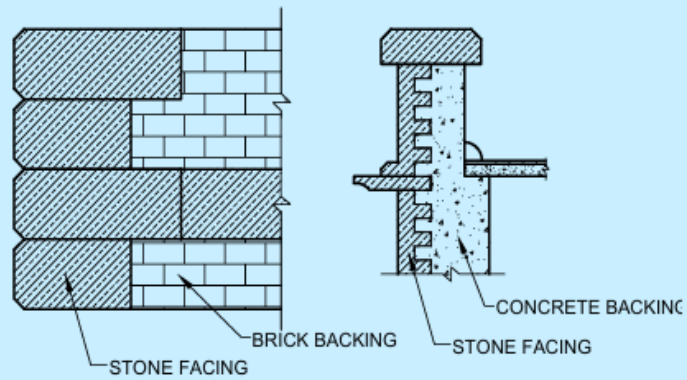


FIG. 3 TYPICAL FACED WALL

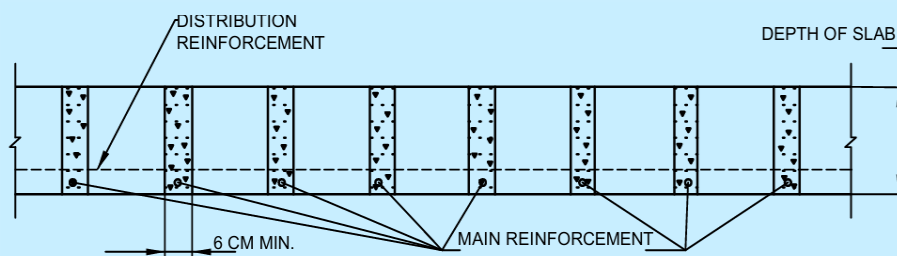


FIG. 4(A) CROSS-SECTION OF RB SLAB

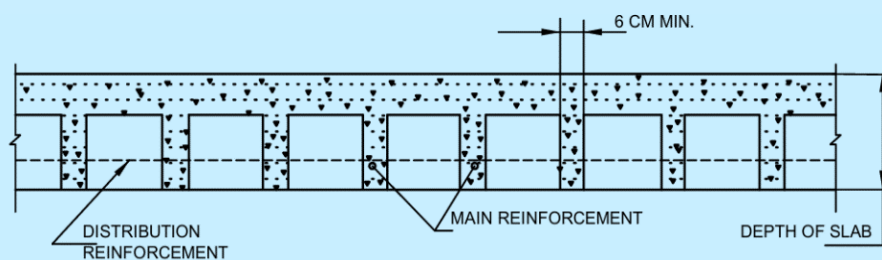


FIG. 4(B) CROSS-SECTION OF RBC SLAB

FIG. 4 CROSS-SECTIONS OF RB AND RBC SLABS

3.2.1 Masonry units may be of the following types:

- a) Common burnt clay building bricks;
- b) Burnt clay fly ash building bricks;
- c) Pulverized fuel ash lime bricks;
- d) Pulverized fuel ash cement bricks;
- e) Stones (in regular sized units);
- f) Sand-lime bricks;
- g) Concrete blocks (solid and hollow);
- h) Lime based blocks;
- j) Burnt clay hollow blocks;
- k) Gypsum partition blocks,
- m) Autoclaved cellular concrete blocks;
- n) Stabilized soil blocks; and
- p) Concrete stone masonry blocks.

NOTES

- 1 Gypsum partition blocks are used only for construction of non-load bearing partition walls.
- 2 Use of other masonry units such as precast stone blocks, fly-ash-lime-gypsum bricks, stabilized soil blocks and other bricks/blocks not covered by the above specifications may also be permitted based on test results.

3.2.2 Masonry units that have been previously used shall not be reused in brickwork or blockwork construction, unless they have been thoroughly cleaned and conform to this Section for similar new masonry units.

3.3 Mortar

Mortar for masonry shall conform to accepted standard [C-4(3)].

3.3.1 Minimum compressive strength for different grades of mortar on the basis of mix proportions as per [3.3.2](#) are given in [Table 1](#).

3.3.2 Mix proportions and compressive strengths of some of the commonly used mortars are given in [Table 2](#).

Table 1 Compressive Strength for Different Grades of Mortar <i>(Clauses 3.3.1, 3.4.1, 5.4.3, G-4.1 and Table 9)</i>		
SI No.	Grade of Mortar	Minimum Compressive Strength at 28 Days N/mm ²
(1)	(2)	(3)
i)	H1	10
ii)	H2	6.0
iii)	M1	5.0
iv)	M2	3.0
v)	M3	1.5
vi)	L1	0.7
vii)	L2	0.5

3.4 Durability Criteria for Selection of Masonry Mortars

3.4.1 The selection of masonry mortars from durability considerations shall cover both the loading and exposure conditions of the masonry. The requirements for masonry mortar shall generally be as given in [3.4.2](#) to [3.4.6](#) (see also [Table 1](#)).

3.4.2 In the case of masonry exposed frequently to rains and where there is further protection by way of plastering or rendering or other finishes, the grade of mortar shall not be less than M3 but shall preferably be of grade M2. Where no protection is provided, the grade of mortar for external walls shall not be less than M2.

3.4.3 In the case of load bearing internal walls, the grade of mortar shall preferably be M3 or more for high durability but in no case less than L1.

3.4.4 In the case of masonry in foundations laid below damp-proof course, the grades of mortar for use in masonry shall be as specified below:

- a) Where soil has little moisture, masonry mortar of grade not less M3 shall be used;
- b) Where soil is very damp, masonry mortar of grade preferably M2 or more shall be used. But in no case shall the grade of mortar be less than L1 and M3; and
- c) Where soil is saturated with water, masonry mortar of grade M1 shall be used but in no case shall the grade of mortar be less than M2.

3.4.5 For masonry in buildings subject to vibration of machinery, the grade of mortar shall not be less than M1.

3.4.6 For parapets, where the height is greater than three times the thickness, the grade of masonry mortar used shall not be less than M1. In the case of low parapets, the grade of mortar shall be the same as that used in the wall masonry below.

3.4.7 The grade of mortar for bedding joints in masonry with large concrete blocks shall not be less than M1.

4 DESIGN CONSIDERATIONS

4.1 General

Masonry structures gain stability from the support offered by cross walls, floors, roof and other elements such as piers and buttresses. Load bearing walls are structurally more efficient when the load is uniformly distributed and the structure is so planned that eccentricity of loading on the members is as small as possible. Avoidance of eccentric loading by providing adequate bearing of floor/roof on the walls providing adequate stiffness in slabs and avoiding fixity at the supports, etc, is especially important in load bearing walls in multistorey structures. These matters should receive careful consideration during the planning stage of masonry structures.

4.2 Lateral Supports and Stability

4.2.1 Lateral Supports

Lateral supports for a masonry element such as load bearing wall or column are intended,

- a) to limit slenderness of a masonry element so as to prevent or reduce possibility of buckling of the member due to vertical loads; and
- b) to resist horizontal components of forces so as to ensure stability of a structure against overturning.

4.2.1.1 Lateral support may be in the vertical or horizontal direction, the former consisting of floor/roof bearing on the wall or properly anchored to the same and latter consisting of cross walls, piers or buttresses.

Table 2 Mix Proportions and Strength of Mortars for Masonry

(Clause 3.3.2 and Table 23)

SI No.	Mix Proportions (By Mass)					Minimum Compressive Strength at 28 Days in N/mm ² (with Cement Compressive Strength at 28 Days, in N/mm ² , between 33 and 43)	Minimum Compressive Strength of Cement : Sand Mortar at 28 Days in N/mm ² (with Cement Compressive Strength at 28 Days, in N/mm ² ≥ 43)
	Cement	Lime	Lime - Pozzolana Mixture	Pozzolana	Sand		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1	1	¼ C or B	0	0	3	10	—
2(a)	1	¼ C or B	0	0	4	7.5	—
2(b)	1	½ C or B	0	0	4 ½	6	—
2(c)	1	0	0	0	4	6	7
3(a)	1	0	0	0	5	5	6
3(b)	1	1 C or B	0	0	6	3	—
3(c)	0	0	1(LP-40)	0	1 ½	3	—
4(a)	1	0	0	0	6	3	4
4(b)	1	2B	0	0	9	2	—
4(c)	0	1A	0	0	2	2	—
4(d)	0	1 B	0	1	1	2	—
4(e)	0	1 C or B	0	2	0	2	—
4(f)	0	0	1(LP-40)	0	1 ¾	2	—
5(a)	1	0	0	0	7	2	3
5(b)	1	3 B	0	0	12	1.5	—
5(c)	0	1 A	0	0	3	1.5	—
5(d)	0	1 B	0	2	1	1.5	—

Sl No.	Mix Proportions (By Mass)					Minimum Compressive Strength at 28 Days in N/mm ² (with Cement Compressive Strength at 28 Days, in N/mm ² , between 33 and 43)	Minimum Compressive Strength of Cement : Sand Mortar at 28 Days in N/mm ² (with Cement Compressive Strength at 28 Days, in N/mm ² ≥ 43)
	Cement	Lime	Lime - Pozzolana Mixture	Pozzolana	Sand		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)

5(e)	0	1 C or B	0	3	0	1.5	—
5(f)	0	0	1(LP-40)	0	2	1.5	—
6(a)	1	0	0	0	8	1	2
6(b)	0	1 B	0	1	2	0.7	—
6(c)	0	1 C or B	0	2	1	0.7	—
6(d)	0	0	1(LP-40)	0	2 ¼	0.7	—
6(e)	0	0	1(LP-20)	0	1 ½	0.7	—
7(a)	0	1 B	0	0	3	0.5	—
7(b)	0	1 C or B	0	1	2	0.5	—
7(c)	0	0	1(LP-7)	0	1 ½	0.5	—

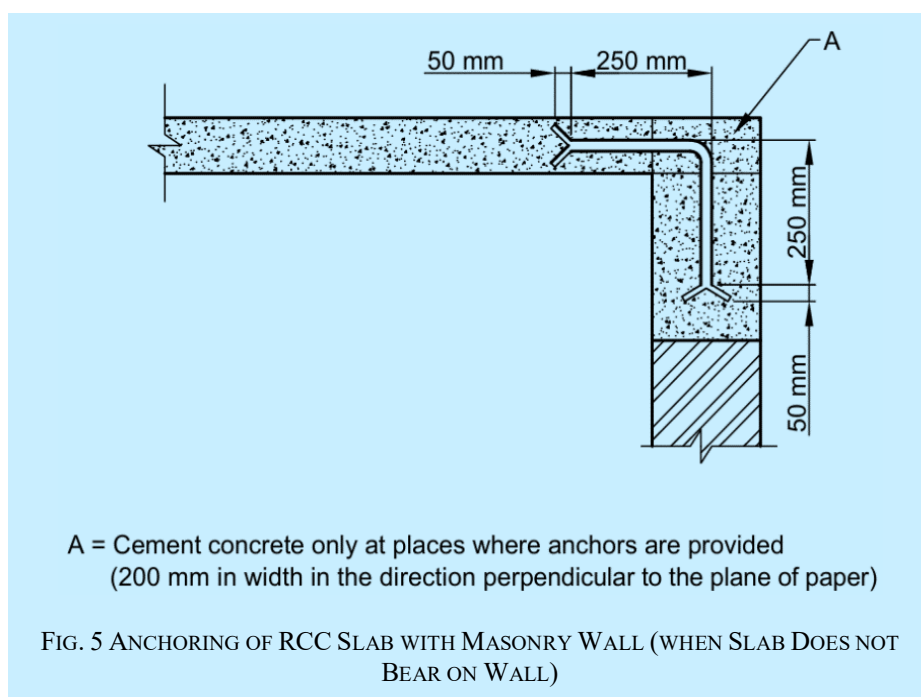
1	Sand for making mortar should be well graded. In case sand is not well graded, its proportions shall be reduced in order to achieve the minimum specified strength.
2	For mixes in SI No. 1 and SI No. 2, use of lime is not essential from consideration of strength as it does not result in increase in strength. However, its use is highly recommended since it improves workability.
3	For mixes in SI No. 3(a), 4(a), 5(a) and 6(a) either lime C or B to the extent of ¼ part of cement (by mass) or some plasticizer should be added for improving workability.
4	For mixes in SI No. 4(b) and 5(b), lime and sand should first be ground in mortar mill and then cement added to coarse stuff.
5	It is essential that mixes in SI No. 4(c), 4(d), 5(d), 5(e), 6(b), 6(c), 7(a) and 7(b) are prepared by grinding in a mortar mill.
6	A, B and C denote eminently hydraulic lime, semi-hydraulic lime and fat lime respectively, as specified in appropriate standards listed in Part B 'Building Materials' of this NBCS.

4.2.1.2 Requirements of **4.2.1(a)** from consideration of slenderness may be deemed to have been met with, if:

- a) in case of a wall, where slenderness ratio is based on effective height, any of the following constructions are provided;
 - 1) RCC floor/roof slab (or beams and slab) irrespective of the direction of span, bears on the supported wall as well as cross walls, to the extent of at least 90 mm;
 - 2) RCC floor/roof slab not bearing on the supported wall or cross wall is anchored to it with non-corrodible metal ties of 600 mm length and of section not less than $6 \text{ mm} \times 30 \text{ mm}$, and at intervals not exceeding 2 m, as shown in [Fig. 5](#); and
 - 3) Timber floor/roof, anchored by non-corrodible metal ties of length 600 mm and of minimum section $6 \text{ mm} \times 30 \text{ mm}$, securely fastened to joists and built into walls as shown in [Fig. 6](#) and [Fig. 7](#). The anchors shall be provided in the direction of span of timber joists as well as in its perpendicular direction, at intervals of not more than 2 m in buildings up to two storeys and 1.25 m for buildings more than two storeys in height.

NOTES

- 1 In case precast RCC units are used for floors and roofs, it is necessary to interconnect them and suitably anchor them to the cross walls so that they can transfer lateral forces to the cross walls.
 - 2 In case of small houses of conventional designs, not exceeding two storeys in height, stiffening effect of partitions and cross walls is such that metal anchors are normally not necessary in case of timber floor/roof and precast RCC floor/roof units.
- b) in case of a wall, when slenderness ratio is based on its effective length; a cross wall/pier/buttress of thickness equal to or more than half the thickness of the supported wall or 90 mm, whichever is more, and length equal to or more than one fifth of the height of wall, is built at right angle to the wall (see [Fig. 8](#)) and bonded to it according to provision of **4.2.2.2 (d)**;
 - c) in case of a column, an RCC or timber beam/RS joist/roof truss, is supported on the column. In this case, the column will not be deemed to be laterally supported in the direction at right angle to it; and
 - d) in case of a column, an RCC beam forming a part of beam and slab construction is supported on the column, and slab adequately bears on stiffening walls. This construction will provide lateral support to the column, in the direction of both horizontal axes.



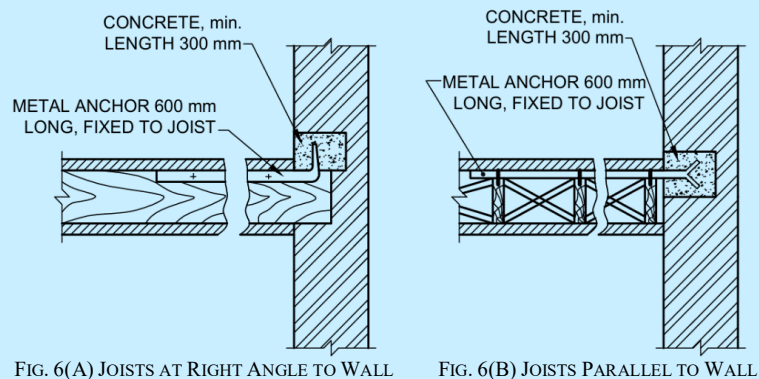


FIG. 6 TYPICAL DETAILS FOR ANCHORAGE OF SOLID WALLS

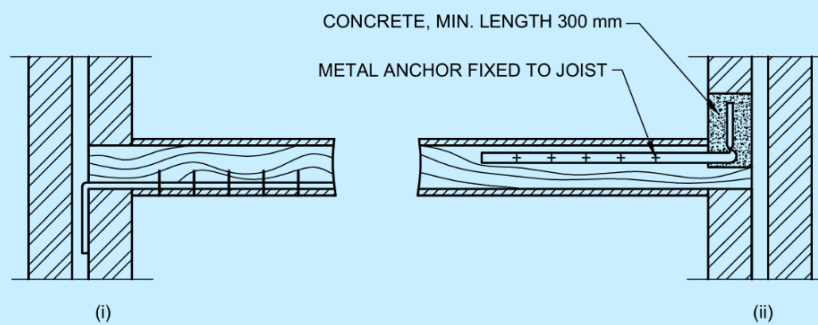


FIG. 7(A) TIMBER JOISTS AT RIGHT ANGLES TO WALL

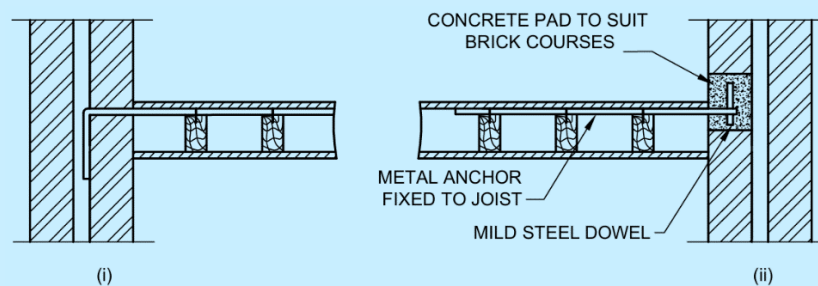


FIG. 7(B) TIMBER JOISTS PARALLEL TO WALL

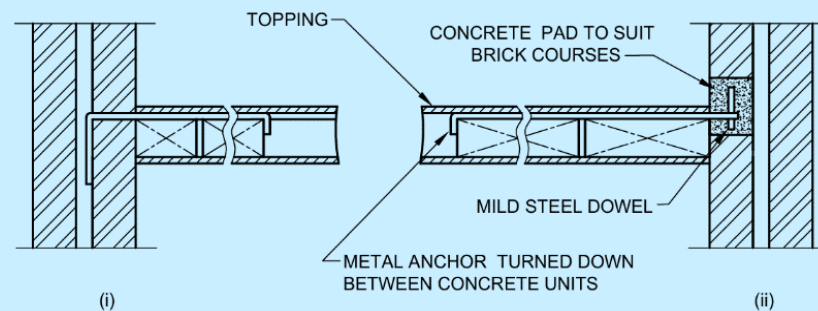
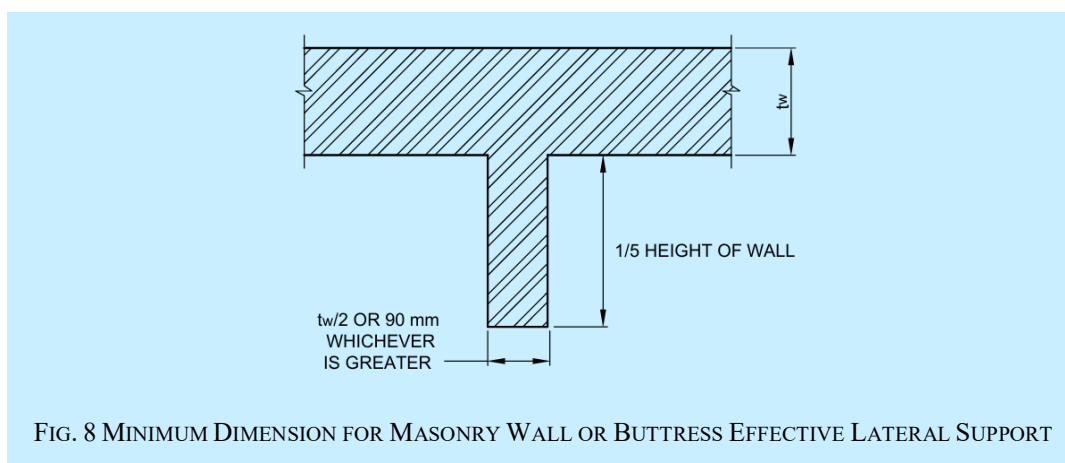


FIG. 7(C) PRECAST CONCRETE FLOOR UNITS PARALLEL TO WALL

FIG. 7 TYPICAL DETAILS FOR ANCHORAGE OF CAVITY WALLS



4.2.2 Stability

A wall or column subjected to vertical and lateral loads may be considered to be provided with adequate lateral support from consideration of stability, if the construction providing the support is capable of resisting some of the following forces:

- Simple static reactions at the point of lateral support to all the lateral loads; plus
- 2.5 percent of the total vertical load that the wall or column is designed to carry at the point of lateral support.

4.2.2.1 For the purpose specified in [4.2.2](#), if the lateral supports are in the vertical direction, these should meet the requirements given in [4.2.1.2 \(a\)](#) and should also be capable of acting as horizontal girders duly anchored to the cross wall so as to transmit the lateral loads to the foundations without exceeding the permissible stresses in the cross walls.

4.2.2.2 In case of load bearing buildings up to four storeys, stability requirements of [4.2.2](#) may be deemed to have been met with, if:

- height to width ratio of building does not exceed 2;
- cross walls acting as stiffening walls continuous from outer wall to outer wall or outer wall to a load bearing inner wall, and of thickness and spacings as given in [Table 3](#) are provided. If stiffening wall or walls that are in a line, are interrupted by openings, length of solid wall or walls in the zone of the wall that is to be stiffened shall be at least one-fifth of height of the opening as shown in [Fig. 9](#);
- floors and roof either bear on cross walls or are anchored to those walls as in [4.2.1.2](#) such that all lateral loads are safely transmitted to those walls and through them to the foundation; and
- cross walls are built jointly with the bearing walls and are jointly mortared, or the two interconnected by toothing. Alternatively, cross walls may be anchored to walls to be supported by ties of non-corrodible metal of minimum section $6 \text{ mm} \times 35 \text{ mm}$ and length 600 mm with ends bend at least 50 mm; maximum vertical spacing of ties being 1.2 m (see [Fig. 10](#)).

4.2.2.3 In case of halls exceeding 8.0 m in length, safety and adequacy of lateral supports shall always be checked by structural analysis.

4.2.2.4 A trussed roofing may not provide lateral support unless special measures are adopted to brace and anchor the roofing. However, in case of residential and similar buildings of conventional design with trussed roofing having cross walls, it may be assumed that stability requirements are met with by the cross walls and structural analysis for stability may be dispensed with.

4.2.2.5 Capacity of a cross wall, also called shear wall, sometimes to take horizontal loads and consequently bending moments increases, when parts of bearing walls act as flanges to the cross wall. Maximum overhanging length of bearing wall which could effectively function as a flange should be taken as $12t$ or $H/6$, whichever is less in case of T/I shaped walls, and $6t$ or $H/16$, whichever is less in case of L/U shaped walls, where t is the thickness of bearing wall and H is the total height of wall above the level being considered, as shown in [Fig. 11](#).

4.2.2.6 External walls of basement and plinth

In case of external walls of basement and plinth, stability requirements of [4.2.2](#) may be deemed to have been met with, if:

- bricks used in basement and plinth have a minimum crushing strength of 5 N/mm^2 and mortar used in masonry is of Grade M1 or better;
- clear height of ceiling in basement does not exceed 2.6 m;
- walls are stiffened according to provisions of [4.2.2.1](#);
- in the zone of action of soil pressure on basement walls, traffic load excluding any surcharge due to adjoining buildings does not exceed 5 kN/m^2 and terrain does not rise; and
- minimum thickness of basement walls is in accordance with [Table 4](#). However, if percentage opening in the basement wall for purposes like ventilation are more than 25 percent, then stability analysis/check need to be performed particularly for resistance to seismic loads.

NOTE — In case there is surcharge on basement walls from adjoining buildings, thickness of basement walls shall be based on structural analysis.

4.2.2.7 Walls mainly subjected to lateral loads

- Free standing wall* — A free standing wall such as compound wall or parapet wall is acted upon by wind force which tends to overturn it. This tendency to over-turning is resisted by gravity force due to self-weight of wall, and also by flexural moment of resistance on account of tensile strength of masonry. Free standing walls shall thus be designed as in [5.5.2.1](#). If mortar used for masonry cannot be relied upon for taking flexural tension (see [5.4.2](#)), stability of free standing wall shall be ensured such that stability moment of wall due to self-weight equals or exceeds 1.5 times the overturning moment; and
- Retaining wall* — Stability for retaining walls shall normally be achieved through gravity action but flexural moment of resistance could also be taken advantage of under special circumstances at the discretion of the designer (see [5.4.2](#)).

Table 3 Thickness and Spacing of Stiffening Walls

[Clause [4.2.2.2\(b\)](#) and [G-4.6.2](#)]

Sl No.	Thickness of Load Bearing Wall to be Stiffened	Height ¹⁾ of Storey Not to Exceed	Stiffening Wall ¹⁾		
			Minimum Thickness		Maximum Spacing
			For 1 to 3 Storeys	For 4 Storeys	
	mm	m	mm	mm	m
(1)	(2)	(3)	(4)	(5)	(6)
i)	100	3.2	100	—	4.5
ii)	200	3.2	100	200	6.0
iii)	300	3.4	100	200	8.0
iv)	Above 300	5.0	100	200	8.0

¹⁾ Storey height and maximum spacings as given are centre-to-centre dimensions.

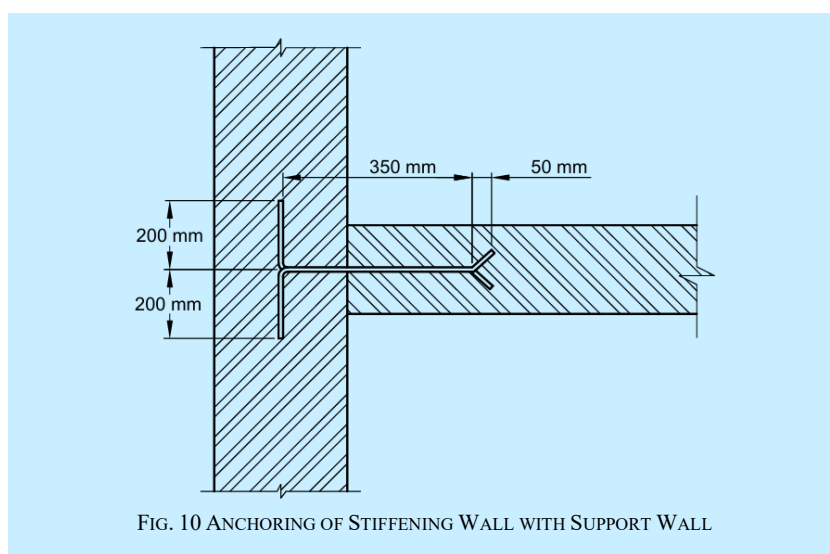
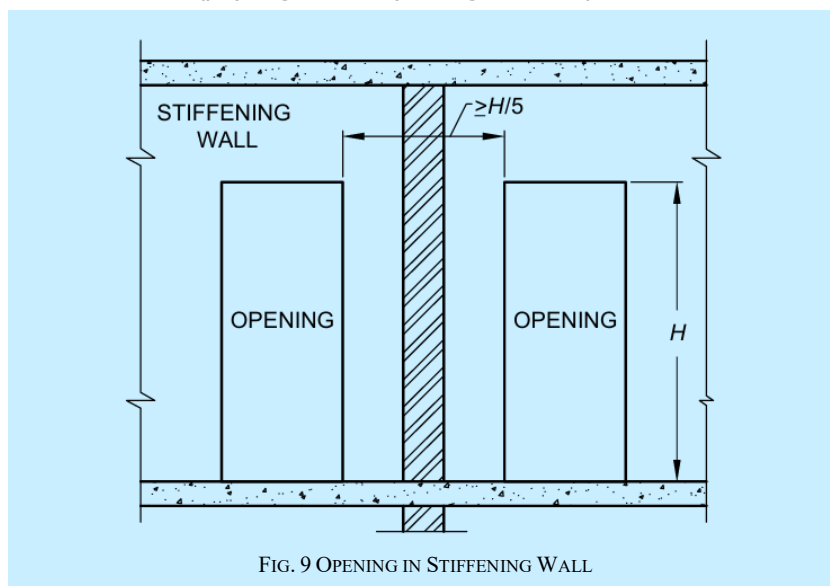


Table 4 Minimum Thickness of Basement Walls			
[Clause 4.2.2.6(c)]			
Sl No.	Height of the Ground above Basement Floor Level		Minimum Nominal Thickness of Basement Wall
	Wall loading (permanent load) less than 50 kN/m	Wall loading (permanent load) more than 50 kN/m	
(1)	(2)	(3)	(4)
i)	Up to 1.4 m	Up to 1.75 m	300 mm
ii)	Up to 2 m	Up to 2.5 m	400 mm

4.3 Effective Height

4.3.1 Wall

Effective height of a wall shall be taken as shown in [Table 5](#) (see [Fig. 12](#)).

NOTE — A roof truss or beam supported on a column meeting the requirements of [4.2.2.1](#) is deemed to provide lateral support to the column only in the direction of the beam/truss.

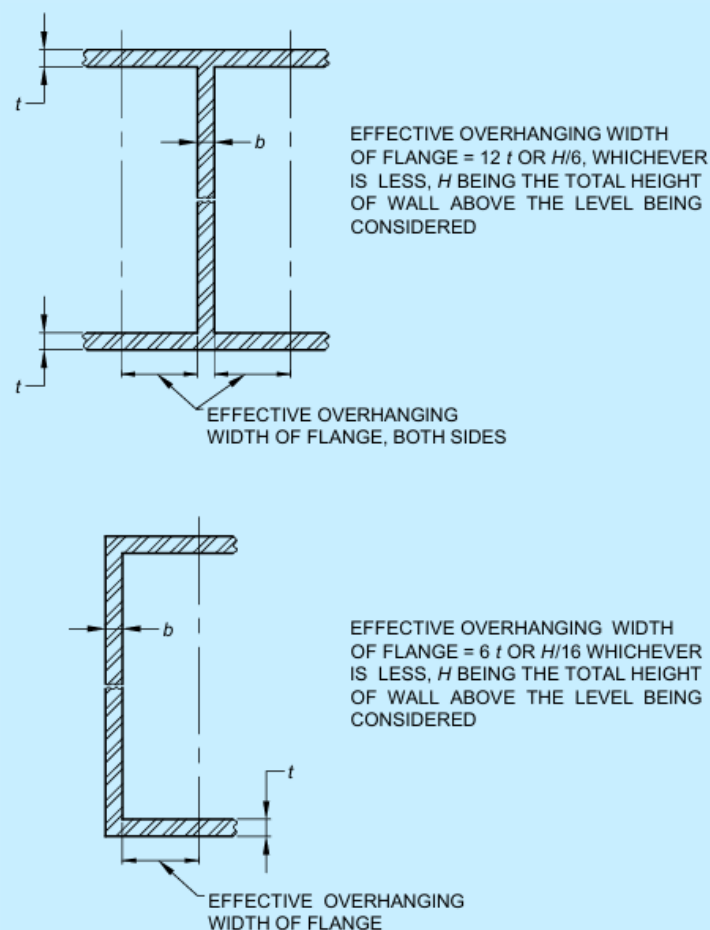


FIG. 11 TYPICAL DETAILS FOR ANCHORAGE OF SOLID WALLS

Table 5 Effective Height of Walls

(Clause 4.3.1, 8.7.9.3, Table 22 and Table 23)

SI No.	Condition of Support	Effective Height
(1)	(2)	(3)
i)	Lateral as well as rotational restraint (that is, full restraint) at top and bottom. For example, when the floor/roof spans on the walls so that reaction to load of floor/roof is provided by the walls, or when an RCC floor/roof has bearing on the wall (minimum 90 mm), irrespective of the direction of the span foundation footings of a wall give lateral as well as rotational restraint	$0.75 H$
ii)	Lateral as well as rotational restraint (that is, full restraint) at one end and only lateral restraint (that is, partial restraint) at the other. For example, RCC floor/roof at one end spanning or adequately bearing on the wall and timber floor/roof not spanning on wall, but adequately anchored to it, on the other end	$0.85 H$
iii)	Lateral restraint, without rotational restraint (that is, partial restraint) on both ends. For example, timber floor/roof, not spanning on the wall but adequately anchored to it on both ends of the wall, that is, top and bottom	$1.00 H$
iv)	Lateral restraint as well as rotational restraint (that is, full restraint) at bottom but have no restraint at the top. For example, parapet walls with RCC roof having adequate bearing on the lower wall, or a compound wall with proper foundation on the soil.	$1.50 H$

Table 5 (Concluded)

NOTES

- 1 H is the height of wall between centres of support in case of RCC slabs and timber floors. In case of footings or foundation block, height (H) is measured from top of footing or foundation block. In case of roof truss, height (H) is measured up to bottom of the tie beam. In case of beam and slab construction, height should be measured from centre of bottom slab to centre of top beam. All these cases are illustrated by means of examples shown in Fig. 12.
- 2 For working out effective height, it is assumed that concrete DPC, when properly bonded with masonry, does not cause discontinuity in the wall.
- 3 Where membrane type damp-proof course or termite shield causes a discontinuity in bond, the effective height of wall may be taken to be greater of the two values calculated as follows:
 - a) Consider H from top of footing ignoring DPC and take effective height as $0.75 H$; and
 - b) Consider H from top of DPC and take effective height as $0.85 H$.
- 4 When assessing effective height of walls, floors not adequately anchored to walls shall not be considered as providing lateral support to such walls.
- 5 When thickness of a wall bonded to a pier is at least two-thirds of the thickness of the pier measured in the same direction, the wall and pier may be deemed to act as one structural element.

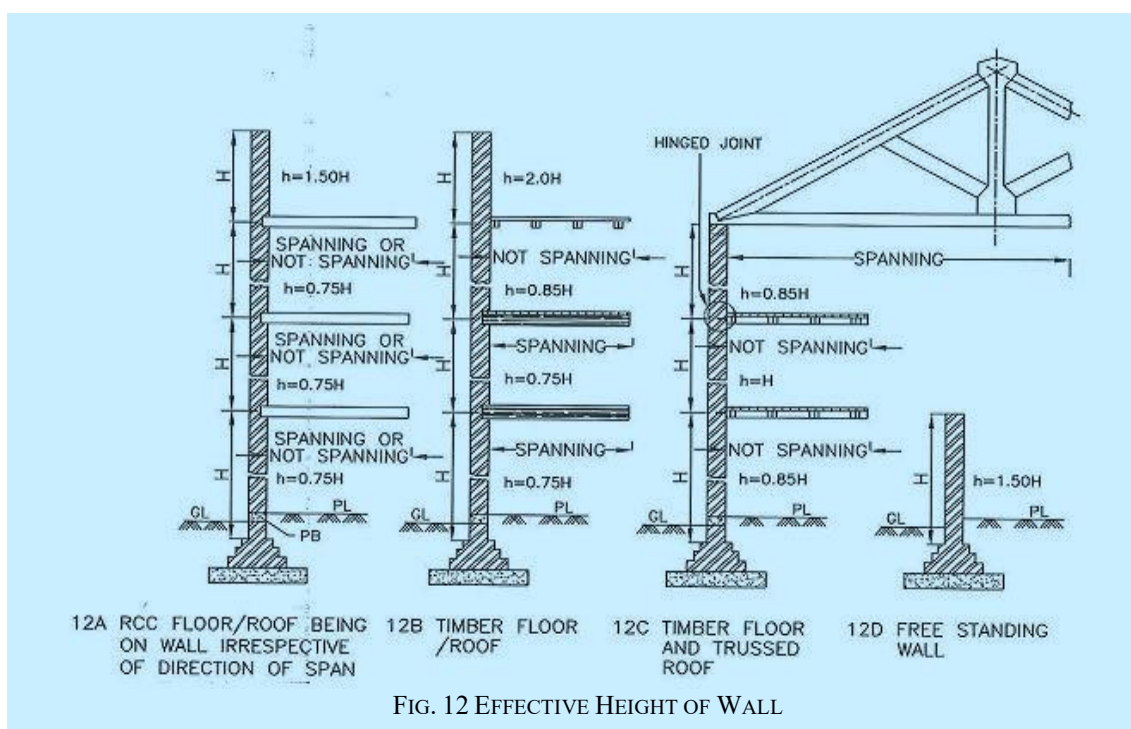
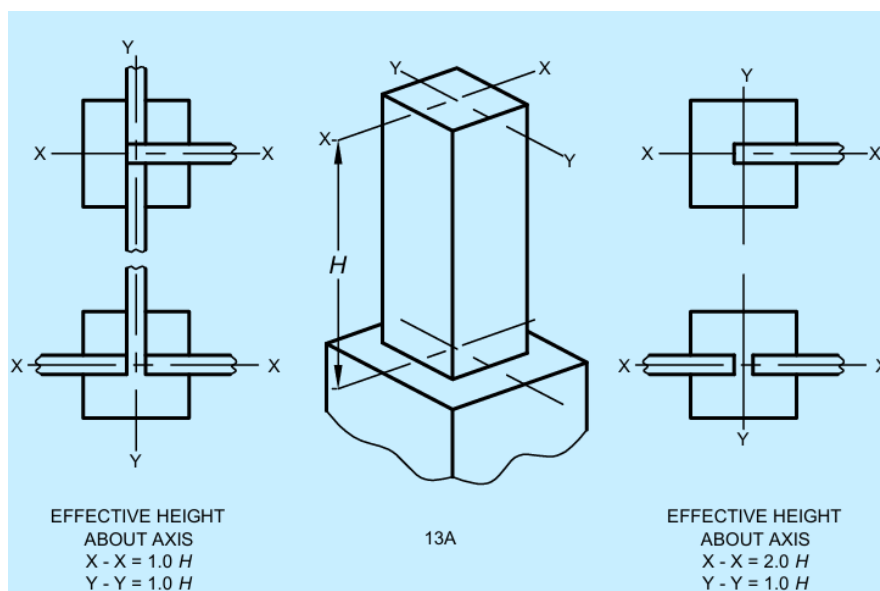
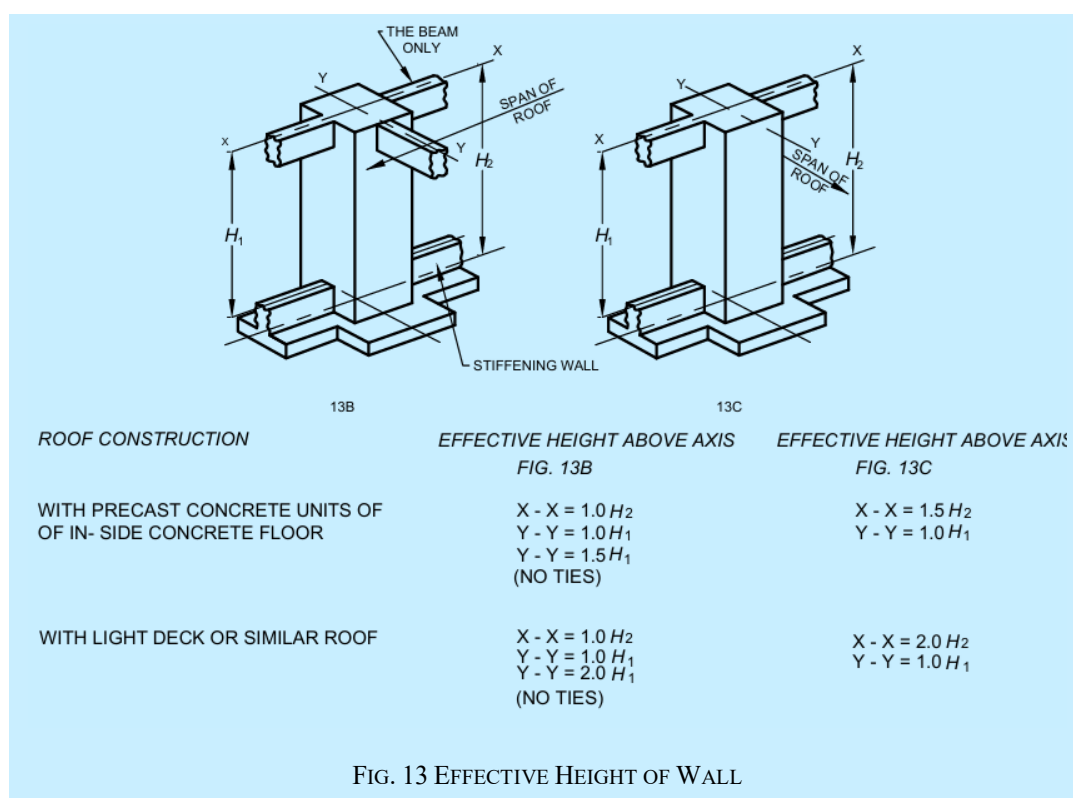


FIG. 12 EFFECTIVE HEIGHT OF WALL

NOTE — If Plinth band (PB) is equal to or more than 150 mm, height can be taken from above the plinth band.





4.3.2 Column

In case of a column, effective height shall be taken as actual height for the direction it is laterally supported and as twice the actual height for the direction it is not laterally supported (see Fig. 13).

NOTES

- 1 A roof truss or beam supported on a column meeting the requirements of 4.2.2.1 is deemed to provide lateral support to the column only in the direction of the beam/truss.
- 2 When floor or roof consisting of RCC beams and slabs is supported on columns, the columns would be deemed to be laterally supported in both directions.

4.3.3 Openings in Walls

When openings occur in a wall such that masonry between the openings is by definition a column, effective height of masonry between the openings shall be reckoned as follows:

- a) *When wall has full restraint at the top:*
 - 1) Effective height for the direction perpendicular to plane of wall equals $0.75 H$ plus $0.25 H_1$, where H is the distance between supports and H_1 is the height of the taller opening; and
 - 2) Effective height for the direction parallel to the wall equals H , that is, the distance between the supports.
- b) *When wall has partial restraint at the top and bottom:*
 - 1) Effective height for the direction perpendicular to plane of wall equals H when height of neither opening exceeds $0.5 H$ and it is equal to $2 H$ when height of any opening exceeds $0.5 H$; and
 - 2) Effective height for the direction parallel to the plane of the wall equals $2 H$.

4.4 Effective Length

Effective length of a wall shall be as given in Table 6.

4.5 Effective Thickness

Effective thickness to be used for calculating slenderness ratio of a wall or column shall be obtained as in [4.5.1](#) to [4.5.5](#).

4.5.1 For solid walls, faced walls or columns, effective thickness shall be the actual thickness.

4.5.2 For solid walls adequately bonded into piers, buttresses, effective thickness for determining slenderness ratio based on effective height shall be the actual thickness of wall multiplied by stiffening coefficient as given in [Table 7](#). No modification in effective thickness, however, shall be made when slenderness ratio is to be based on effective length of walls.

Table 6 Effective Length of Walls

(Clauses [4.4](#))

SI No. (1)	Conditions of Support (See Fig. 14) (2)	Effective Length (3)
i)	Where a wall is continuous and is supported by cross wall and there is no opening within a distance of $H/8$ from the face of cross wall (see Fig. 14) or Where a wall is continuous and is supported by piers/buttresses conforming to 4.2.1.2 (b)	$0.8 L$
ii)	Where a wall is supported by a cross wall at one end and continuous with cross wall at other end or Where a wall is supported by a pier/buttress at one end and continuous with pier/buttress at other end conforming to 4.2.1.2 (b)	$0.9 L$
iii)	Where a wall is supported at each end by cross wall or Where a wall is supported at each end by a pier/buttress conforming to 4.2.1.2 (b)	$1.0 L$
iv)	Where a wall is free at one end and continuous with a pier/buttress at the other end or Where a wall is free at one end and continuous with a pier/buttress at the other end conforming to 4.2.1.2 (b)	$1.5 L$
v)	Where a wall is free at one end and supported at the other end by a cross wall or Where a wall is free at one end and supported at the other end by a pier/buttress conforming to 4.2.1.2 (b) where L = length of wall from or between centres of cross wall, piers or buttress; and H = actual height of wall between centres of adequate lateral support.	$2.0 L$
NOTE — In case there is an opening taller than $0.5 H$ in a wall, ends of the wall at the opening shall be considered as free. Cross walls shall conform to 4.2.2.2 (d) .		

4.5.3 For solid walls or faced walls stiffened by cross walls, appropriate stiffening coefficient may be determined from [Table 7](#) on the assumption that the cross walls are equivalent to piers of width equal to the thickness of the cross wall and of thickness equal to three times the thickness of stiffened wall.

4.5.4 For cavity walls with both leaves of uniform thickness throughout, effective thickness shall be taken as two-thirds of the sum of the actual thickness of the two leaves.

4.5.5 For cavity walls with one or both leaves adequately bonded into piers, buttresses or cross walls at intervals, the effective thickness of the cavity wall shall be two-thirds of the sum of the effective thickness of each of the two leaves; the effective thickness of each leaf being calculated using [4.5.1](#) or [4.5.2](#) as appropriate.

Table 7 Stiffening Coefficient for Walls Stiffened by Piers, Buttresses or Cross Walls

(Clauses [4.5.2](#) and [4.5.3](#))

Sl No.	Ratio $\frac{S_p}{w_p}$	Stiffening Coefficient		
		$\frac{t_p}{t_w} = 1$	$\frac{t_p}{t_w} = 2$	$\frac{t_p}{t_w} = 3 \text{ or more}$
(1)	(2)	(3)	(4)	(5)
i)	6	1.0	1.4	2.0
ii)	8	1.0	1.3	1.7
iii)	10	1.0	1.2	1.4
iv)	15	1.0	1.1	1.2
v)	20 or more	1.0	1.0	1.0

where

S_p = centre-to-centre spacing of the pier or cross wall;

t_p = thickness of pier as defined in [2.3 \(b\)](#) (see [Fig. 1](#));

t_w = actual thickness of the wall proper (see [Fig. 1](#)); and

w_p = width of the pier in the direction of the wall or the actual thickness of the cross wall.

NOTE — Linear interpolation between the values given in this table is permissible but not extrapolation outside the limits given.

4.6 Slenderness Ratio

4.6.1 Walls

For a wall, slenderness ratio shall be effective height divided by effective thickness or, effective length divided by the effective thickness, whichever is less. In case of a load bearing wall, slenderness ratio shall not exceed that given in [Table 8](#). To reduce the slenderness ratio one and a half brick thick wall shall be up to ground level.

4.6.2 Columns

For a column, slenderness ratio shall be taken to be the greater of the ratios of effective heights to the respective effective thickness, in the two principal directions. Slenderness ratio for a load bearing column shall not exceed 12.

4.7 Eccentricity

Eccentricity of vertical loading at a particular junction in a masonry wall shall depend on factors, such as extent of bearing, magnitude of loads, stiffness of slab or beam, fixity at the support and constructional details at junctions. No exact calculations are possible to make accurate assessment of eccentricity. Extent of eccentricity

under any particular circumstances has, therefore, to be decided according to the best judgment of the designer. Some guidelines for assessment of eccentricity are given in [Annex A](#).

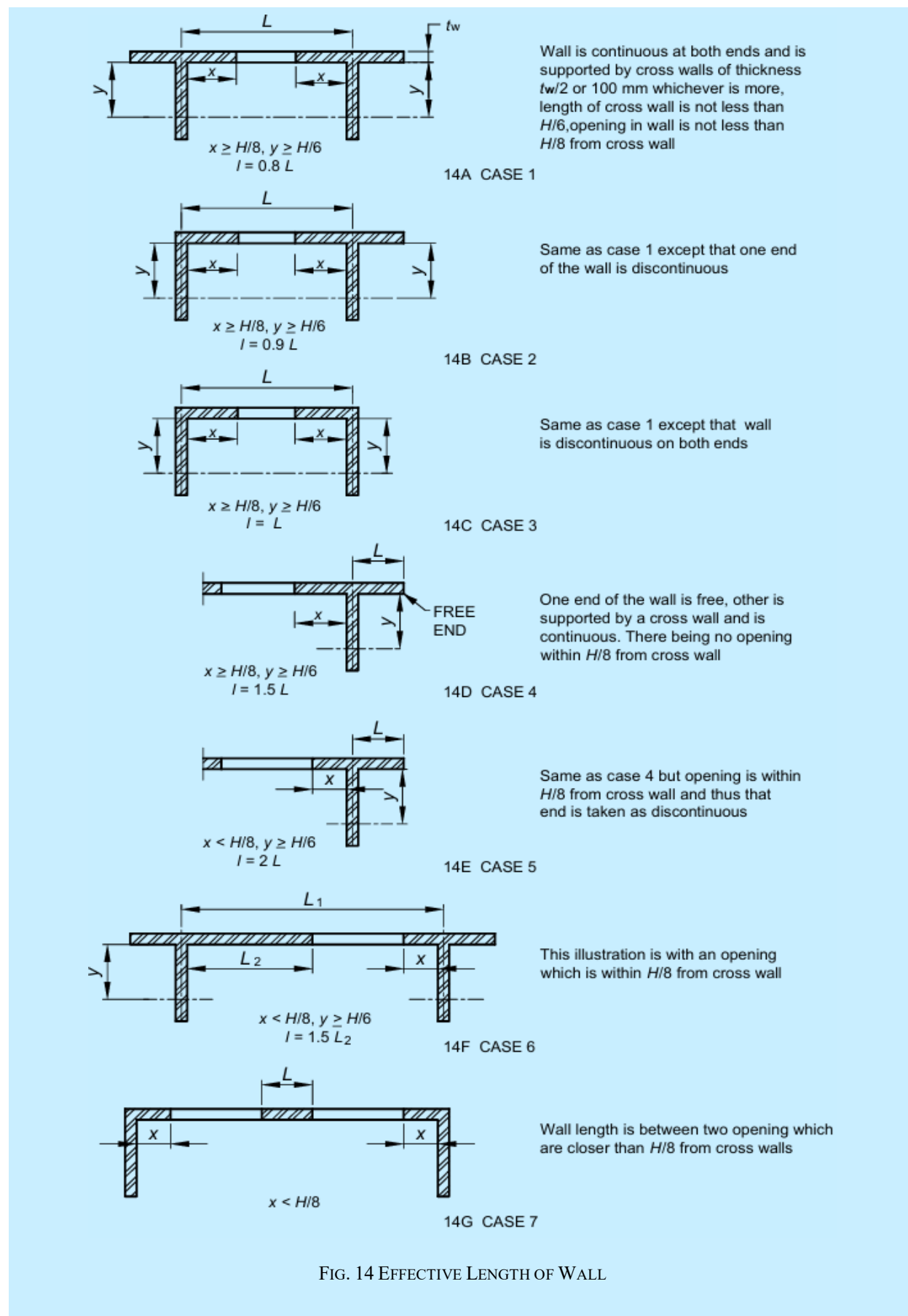


FIG. 14 EFFECTIVE LENGTH OF WALL

Table 8 Maximum Slenderness Ratio for a Load Bearing Wall

(Clause 4.6.1)

Sl No.	Number of Storeys	Maximum Slenderness Ratio	
		Using Portland Cement or Portland Pozzolana Cement in Mortar	Using Lime Mortar
(1)	(2)	(3)	(4)
i)	Not exceeding 2	27	20
ii)	Exceeding 2	27	13

5 STRUCTURAL DESIGN

5.1 General

The building as a whole shall be analysed by accepted principles of mechanics to ensure safe and proper functioning in service of its component parts in relation to the whole building. All component parts of the structure shall be capable of sustaining the most adverse combinations of loads, which the building may be reasonably expected to be subjected to during and after construction.

5.2 Design Loads

Loads to be taken into consideration for designing masonry components of a structure are:

- dead loads of walls, columns, floors and roofs;
- imposed loads of floors and roof;
- wind loads on walls and sloping roof; and
- seismic forces.

NOTE — When a building is subjected to other loads, such as vibration from railways; machinery, etc, these should be taken into consideration accordingly to the best judgment of the designer (*see also* Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS).

5.2.1 The design loads and other forces to be taken for the design of masonry structures shall conform to Part C/Section 1 of this NBCS.

NOTE — During construction, suitable measures shall be taken to ensure that masonry is not liable to damage or failure due to action of wind forces, back filling behind walls or temporary construction loads.

5.3 Load Dispersion

5.3.1 General

The angle of dispersion of vertical load on walls shall be taken as not more than 30° from the vertical.

5.3.2 Arching Action

Account may also be taken of the arching action of well-bonded masonry walls supported on lintels and beams, in accordance with established practice. Increased axial stresses in the masonry associated with arching action in this way, shall not exceed the permissible stresses given in 5.4.

5.3.3 Lintels

Lintels that support masonry construction shall be designed to carry loads from masonry (allowing for arching and dispersion where applicable), and loads received from any other part of the structure. Length of bearing of lintel at each end shall not be less than 90 mm or one-tenth of the span, whichever is more, and area of the bearing

shall be sufficient to ensure that stresses in the masonry (combination of wall stresses, stresses due to arching action and bearing stresses from the lintel) do not exceed the stresses permitted in [5.4](#) (see [Annex C](#)).

5.4 Permissible Stresses

5.4.1 Permissible Compressive Stress

Permissible compressive stress in masonry shall be based on value of basic compressive stress (f_b) as given in [Table 9](#) and multiplying this value by factors known as stress reduction factor (k_s), area reduction factor (k_a) and shape modification factor (k_p) as detailed in [5.4.1.1](#) to [5.4.1.3](#). Values of basic compressive stress given in [Table 9](#) take into consideration crushing strength of masonry unit and grades of mortar, and hold good for values of slenderness ratio not exceeding 6, zero eccentricity and masonry unit having height to width ratio (as laid) equal to 0.75 or less.

Alternatively, basic compressive stress may be based on results of prism test given in [Annex B](#) on masonry made from masonry units and mortar to be actually used in a particular job.

Table 9 Basic Compressive Stresses for Masonry (After 28 days)

(Clauses [5.4.1](#), [5.4.5](#), [6.3.1](#) and [Table 14](#))

Sl No.	Mortar Type (Ref to, Table 1)	Basic Compressive Stresses, in N/mm ² , Corresponding to Masonry Units of which Height to Width Ratio does not Exceed 0.75 and Crushing Strength, in N/mm ² , is not Less than											
		3.5	5.0	7.5	10	12.5	15	17.5	20	25	30	35	40
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
i)	H1	0.35	0.50	0.75	1.00	1.16	1.31	1.45	1.59	1.91	2.21	2.5	3.05
ii)	H2	0.35	0.50	0.74	0.96	1.09	1.19	1.30	1.41	1.62	1.85	2.1	2.5
iii)	M1	0.35	0.50	0.74	0.96	1.06	1.13	1.20	1.27	1.47	1.69	1.9	2.2
iv)	M2	0.35	0.44	0.59	0.81	0.94	1.03	1.10	1.17	1.34	1.51	1.65	1.9
v)	M3	0.25	0.41	0.56	0.75	0.87	0.95	1.02	1.10	1.25	1.41	1.55	1.78
vi)	L1	0.25	0.36	0.53	0.67	0.76	0.83	0.90	0.97	1.11	1.26	1.4	1.06
vii)	L2	0.25	0.31	0.42	0.53	0.58	0.61	0.65	0.69	0.73	0.78	0.85	0.95

NOTES

- 1 The table is valid for slenderness ratio up to 6 and loading with zero eccentricity.
- 2 The values given for basic compressive stress are applicable only when the masonry is properly cured.
- 3 Linear interpolation is permissible for units having crushing strengths between those given in the table.
- 4 The permissible stress for random rubble masonry may be taken as 75 percent of the corresponding stress for coarsed walling of similar materials.
- 5 The strength of ashlar masonry (natural stone masonry of massive type with thin joints) is closely related to intrinsic strength of the stone and allowable working stress in excess of those given in the table may be allowed for such masonry at the discretion of the designer.
- 6 For calculation of basic compressive stress of stabilized soil block having thickness 100 mm or more, reference to specialist literature may be made.
- 7 If the work is inadequately supervised, strength should be reduced to three-fourth or less at the discretion of the designer.

5.4.1.1 Stress reduction factor

This factor, as given in [Table 10](#), takes into consideration the slenderness ratio of the element and also the eccentricity of loading.

5.4.1.2 Area reduction factor

This factor takes into consideration smallness of the sectional area of the element and is applicable when sectional area of the element is less than 0.2 m². The factor $k_a = 0.7 + 1.5A$, A being the area of section, in m².

Table 10 Stress Reduction Factor for Slenderness Ratio and Eccentricity

(Clause 5.4.1.1, 9.2.2.3, 9.2.2.9 and 10.4.1)

Sl No.	Slenderness Ratio	Eccentricity of Loading Divided by the Thickness of the Member					
		0	1/24	1/12	1/6	1/4	1/3
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	6	1.00	1.00	1.00	1.00	1.00	1.00
ii)	8	0.95	0.95	0.94	0.93	0.92	0.91
iii)	10	0.89	0.88	0.87	0.85	0.83	0.81
iv)	12	0.84	0.83	0.81	0.78	0.75	0.72
v)	14	0.78	0.76	0.74	0.70	0.66	0.66
vi)	16	0.73	0.71	0.68	0.63	0.58	0.53
vii)	18	0.67	0.64	0.61	0.55	0.49	0.43
viii)	20	0.62	0.59	0.55	0.48	0.41	0.34
ix)	22	0.56	0.52	0.48	0.40	0.32	0.24
x)	24	0.51	0.47	0.42	0.33	0.24	—
xi)	26	0.45	0.40	0.35	0.25	—	—
xii)	27	0.43	0.38	0.33	0.22	—	—

NOTES
1 Linear interpolation between values is permitted.
2 Where in special cases the eccentricity of loading lies between 1/3 and 1/2 of the thickness of the member, the stress reduction factor should vary linearly between unity and 0.20 for slenderness ratio of 6 and 20, respectively.
3 Slenderness ratio of a member for sections within 1/8 of the height of the member above or below a lateral support may be taken to be 6.

5.4.1.3 Shape modification factor

This factor takes into consideration the shape of the unit, that is, height to width ratio (as laid) and is given in [Table 11](#). This factor is applicable for units of crushing strength up to 15 N/mm².

5.4.1.4 Increase in permissible compressive stresses allowed for eccentric vertical loads, lateral loads under certain conditions

In members subjected to eccentric and/or lateral loads, increase in permissible compressive stress is allowed as follows:

- When resultant eccentricity ratio exceeds 1/24 but does not exceed 1/6, 25 percent increase in permissible compressive stress is allowed in design; and
- When resultant eccentricity ratio exceeds 1/6, 25 percent increase in permissible stress is allowed but the area of the section under tension shall be disregarded for computing the load carrying capacity of the member.

NOTE — When resultant eccentricity ratio of loading is 1/24 or less, compressive stress due to bending shall be ignored and only axial stress need to be computed for the purpose of design.

5.4.1.5 Increase in permissible compressive stress for walls subjected to concentrated loads

When a wall is subjected to a concentrated load (a load being taken to be concentrated when area of supporting wall equals or exceeds three times the bearing area), certain increase in permissible compressive stress may be allowed because of dispersal of the load. Since, according to the present state of art, there is diversity of views in regard to manner and extent of dispersal, design of walls subjected to concentrated loads may, therefore, be worked out as per the best judgment of the designer. Some guidelines in this regard are given in [Annex C](#).

Table 11 Shape Modification Factor for Masonry Units

(Clause 5.4.1.3)

Sl No.	Height to Width Ratio of Units (as Laid)	Shape Modification Factor (k_p) For Units Having Crushing Strength in N/mm ²			
		Up to 5.0	7.5	10.0	15.0
(1)	(2)	(3)	(4)	(5)	(6)
i)	Up to 0.75	1.0	1.0	1.0	1.0
ii)	1.0	1.2	1.1	1.1	1.0
iii)	1.5	1.5	1.3	1.2	1.1
iv)	2.0 to 4.0	1.8	1.5	1.3	1.2

NOTE — Linear interpolation between values is permissible.

5.4.2 Permissible Tensile Stress

As a general rule, design of masonry shall be based on the assumption that masonry is not capable of taking any tension. However, in case of lateral loads normal to the plane of wall, which causes flexural tensile stress, as for example, panel, curtain partition and free standing walls, flexural tensile stresses as follows may be permitted in the design for masonry:

a) *Grade M1 or better mortar:*

- 1) 0.07 N/mm² for bending in the vertical direction where tension developed is normal to bed joints; and
- 2) 0.14 N/mm² for bending in the longitudinal direction where tension developed is parallel to bed joints, provided crushing strength of masonry units is not less than 10 N/mm².

b) *Grade M2 mortar:*

- 1) 0.05 N/mm² for bending in the vertical direction where tension developed is normal to bed joints; and
- 2) 0.10 N/mm² for bending in the longitudinal direction where tension developed is parallel to bed joints, provided crushing strength of masonry units is not less than 7.5 N/mm².

NOTES

1 No tensile stress is permitted in masonry in case of water-retaining structures in view of water in contact with masonry. Also no tensile stress is permitted in earth-retaining structures, in view of the possibility of presence of water at the back of such walls.

2 Allowable tensile stress in bending in the vertical direction may be increased to 0.1 N/mm² for M1 mortar and 0.07 N/mm² for M2 mortar in case of boundary walls/compound at the discretion of the designer, since there is not much risk to life and property in the event of failure of such walls.

5.4.3 Permissible Shear Stress

In case of walls built in mortar not leaner than Grade M1 (*see* Table 1) and resisting horizontal forces in the plane of the wall, permissible shear stress calculated on the area of bed joints, shall not exceed the value obtained by the formula given below, subject to a maximum of 0.5 N/mm²:

$$f_s = 0.1 + f_d/6$$

where

f_s = permissible shear stress, in N/mm²; and

f_d = compressive stress due to dead loads, in N/mm².

5.4.4 If there is tension in any part of a section of masonry, the area under tension shall be ignored while working out shear stress on the section.

5.4.5 Specified Compressive Strength

The specified compressive strength f_m (in MPa) shall be determined by:

- a) testing prism specimens as per [Annex B](#); and
- b) the expression:

$$f_m = 0.433 f_b^{0.64} f_{m0}^{0.36}$$

where

f_b = compressive strength (in MPa) of clay brick as per [Table 9](#); and

f_{m0} = compressive strength of mortar as per [5.4.1](#).

- c) as per [5.4.1](#).

5.4.6 Elastic Modulus

Elastic modulus, E_m of masonry shall be determined by testing masonry prisms as per [5.4.5\(a\)](#); the chord modulus of elasticity shall be taken as the slope of the chord between 0.05 Max. and 0.33 of the Max. compressive strength of each prism:

The expressions:

$$E_m = \begin{cases} 550 f_m & \text{clay brick masonry} \\ 850 f_m & \text{concrete block masonry} \end{cases}$$

5.4.7 Shear Modulus

The shear modulus, G_m shall be determined by:

$$G_m = 0.2 E_m$$

5.5 Design Thickness/Cross-Section

5.5.1 Walls and Columns Subjected to Vertical Loads

Walls and columns bearing vertical loads shall be designed on the basis of permissible compressive stress. Design consists in determining thickness in case of walls and section in case of columns in relation to strength of masonry units and grade of mortar to be used, taking into consideration various factors, such as slenderness ratio, eccentricity, area of section, workmanship, quality of supervision, etc, subject further to provisions of [5.5.1.1](#) to [5.5.1.4](#).

5.5.1.1 Solid walls

Thickness used for design calculation shall be the actual thickness of masonry computed as the sum of the average dimensions of the masonry units specified in the relevant standard, together with the specified joint thickness. In masonry with raked joints, thickness shall be reduced by the depth of raking of joints for plastering/pointing.

5.5.1.2 Cavity walls

- a) Thickness of each leaf of a cavity wall shall not be less than 75 mm;
- b) Where the outer leaf is half masonry unit in thickness, the uninterrupted height and length of this leaf shall be limited so as to avoid undue loosening of ties due to differential movements between the two leaves. The outer leaf shall, therefore, be supported at least at every third storey or at every 10 m of height, whichever is less and at every 10 m or less along the length;

- c) Where the load is carried by both leaves of a wall of a cavity construction, the permissible stress shall be based on the slenderness ratio derived from the effective thickness of the wall as given in [4.5.4](#) or [4.5.5](#). The eccentricity of the load shall be considered with respect to the centre of gravity of the cross-section of the wall; and
- d) Where the load is carried by one leaf only, the permissible stress shall be the greater of values calculated by the following two alternative methods:
 - 1) The slenderness ratio is based on the effective thickness of the cavity wall as a whole as given in [4.5.4](#) or [4.5.5](#) and on the eccentricity of the load with respect to the centre of gravity of the cross-section of the whole wall (both leaves) (This is the same method as where the load is carried by both the leaves but the eccentricity will be more when the load is carried by one leaf only); and
 - 2) The slenderness ratio is based on the effective thickness of the loaded leaf only using [4.5.1](#) and [4.5.2](#), and the eccentricity of the load will also be with respect to the centre of gravity of the loaded leaf only.

In either alternative, only the actual thickness of the load bearing leaf shall be used in arriving at the cross-sectional area resisting the load (see [5.5.1.1](#)).

5.5.1.3 Faced wall

The permissible load per length of wall shall be taken as the product of the total thickness of the wall and the permissible stress in the weaker of the two materials. The permissible stress shall be found by using the total thickness of the wall when calculating the slenderness ratio.

5.5.1.4 Veneered wall

The facing (veneer) shall be entirely ignored in calculations of strength and stability. For the purpose of determining the permissible stress in the backing, the slenderness ratio shall be based on the thickness of the backing alone.

5.5.2 Walls and Columns Mainly Subjected to Lateral Loads

5.5.2.1 Free standing walls

- a) Free standing walls, subjected to wind pressure or seismic forces shall be designed on the basis of permissible tensile stress in masonry or stability as in [4.2.2.4](#). However, in seismic Zones II, free-standing walls may be apportioned without making any design calculations with the help of [Table 12](#) provided the mortar used is of grade not leaner than M1; and
- b) If there is a horizontal damp-proof course near the base of the wall, that is, not capable of developing tension vertically, the minimum wall thickness should be the greater of that calculated from either:
 - 1) the appropriate height to thickness ratio given in [Table 12](#) reduced by 25 percent, reckoning the height from the level of the damp-proof course; or
 - 2) the appropriate height to thickness ratio given in [Table 12](#) reckoning the height from the lower level at which the wall is restrained laterally.

Table 12 Height to Thickness Ratio of Free Standing Walls Related to Wind Speed

[Clause [5.5.2.1\(a\)](#) and [\(b\)](#)]

Sl No.	Design Wind Pressure N/mm ²	Height to Thickness Ratio
(1)	(2)	(3)
i)	Up to 285	10
ii)	575	7
iii)	860	5
iv)	1 150	4
NOTES		
1 For intermediate values, linear interpolation is permissible.		
2 Height is to be reckoned from 150 mm below ground level or top of footing/foundation block, whichever is higher, and up to the top edge of the wall.		
3 The thickness should be measured including the thickness of the plaster.		

5.5.2.2 Retaining walls

Normally masonry of retaining walls shall be designed on the basis of zero-tension, and permissible compressive stress. However, in case of retaining walls for supporting horizontal thrust from dry materials, retaining walls may be designed on the basis of permissible tensile stress at the discretion of the designers.

5.5.3 Walls and Columns Subjected to Vertical as well as Lateral Loads

For walls and columns, stress worked out separately for vertical loads as in [5.5.1](#) and lateral loads as in [5.5.2](#) shall be combined and elements designed on the basis of permissible stress.

5.5.4 Walls Subjected to In-Plane Bending and Vertical Loads (Shear Walls)

Walls subjected to in-plane bending and vertical loads, that is, shear walls shall be designed on the basis of no tension with permissible shear stress and permissible compressive stress.

5.5.5 Non-Load Bearing Walls

Non-load bearing walls, such as panel walls, curtain walls and partition walls which are mainly subjected to lateral loads, according to present state of art, are not capable of precise design and only approximate methods based on some tests are available. Guidelines for approximate design of these walls are given in [Annex D](#).

6 GENERAL REQUIREMENTS

6.1 Methods of Construction

6.1.1 General

Construction of the following types of load bearing and non-load bearing masonry walls shall be carried out in accordance with good practice [C-4(4)]:

- a) brickwork;
- b) stone masonry;
- c) hollow concrete block masonry;
- d) gypsum partition blocks;
- e) autoclaved cellular concrete block masonry; and
- f) lightweight concrete block masonry.

6.1.2 Construction of Buildings in Seismic Zones

Special features of construction for earthquake resistant masonry buildings in zones III and above shall be applicable according to [8](#).

6.2 Minimum Thickness of Walls from Consideration Other than Structural

Thickness of walls determined from consideration of strength and stability may not always be adequate in respect of other requirements, such as resistance to fire, thermal insulation, sound insulation and resistance to damp penetration for which reference may be made to the appropriate Parts/Sections of the NBCS, and thickness suitably increased, where found necessary.

6.3 Workmanship

6.3.1 General

Workmanship has considerable effect on strength of masonry and bad workmanship may reduce the strength of brick masonry to as low as half the intended strength. The basic compressive stress values for masonry as given in [Table 9](#) holds good for commercially obtainable standards of workmanship with reasonable degree of supervision.

6.3.2 Bedding of Masonry Units

Masonry units shall be laid on a full bed of mortar with frog, if any, upward such that cross-joints and wall joints are completely filled with mortar. Masonry units which are moved after initial placement shall be re-laid in fresh mortar, discarding the disturbed mortar.

6.3.3 Bond

Cross-joints in any course of one brick thick masonry wall shall be not less than one-fourth of a masonry unit in horizontal direction from the cross-joints in the course below. In masonry walls more than one brick in thickness, bonding through the thickness of wall shall be provided by either header units or by other equivalent means in accordance with good practice [C-4(5)].

6.3.4 Verticality and Alignment

All masonry shall be built true and plumb within the tolerances prescribed below; care shall be taken to keep the perpend properly aligned:

- a) deviation from vertical within a storey shall not exceed 6 mm per 3 m height;
- b) deviation in verticality in total height of any wall of a building more than one storey in height shall not exceed 12.5 mm.;
- c) deviation from position shown on plan of any brickwork shall not exceed 12.5 mm;
- d) relative displacement between load bearing walls in adjacent storeys intended to be in vertical alignment shall not exceed 6 mm;
- e) deviation of bed-joint from horizontal in a length of 12 m shall not exceed 6 mm subject to a maximum deviation of 12 mm; and
- f) deviation from the specified thickness of bed joints, cross-joints and perpend shall not exceed one-fifth of the specified thickness.

NOTE — These tolerances have been specified from the point of view of their effect on the strength of masonry. The permissible stress recommended in [5.3](#) may be considered applicable only, if these tolerances are adhered to.

6.4 Joints to Control Deformation and Cracking

Special provision shall be made to control or isolate thermal and other movements so that damage to the fabric of the building is avoided and its structural sufficiency preserved. Design and installation of joints shall be done according to the appropriate recommendations in accordance with good practice [C-4(6)].

6.5 Chases, Recesses and Holes

6.5.1 Chases, recesses and holes are permissible in masonry only if these do not impair strength and stability of the structure.

6.5.2 In masonry, designed by structural analysis, all chases, recesses and holes shall be considered in structural design and detailed in building plans.

6.5.3 When chases, recesses and holes have not been considered in structural design and are not shown in drawings, these may be provided, subject to the constraints and precautions specified in [6.5.3.1](#) to [6.5.3.10](#).

6.5.3.1 As far as possible, services should be planned with the help of vertical chases and use of horizontal chases should be avoided.

6.5.3.2 For load bearing walls, depth of vertical and horizontal chases shall not exceed one-third and one-sixth of the wall thickness, respectively.

6.5.3.3 Vertical chases shall not be closer than 2 m in any stretch of wall and shall not be located within 345 mm of an opening or within 230 mm of a cross wall that serves as a stiffening wall for stability. Width of a vertical chase shall not exceed thickness of wall in which it occurs.

6.5.3.4 When unavoidable horizontal chases of width not exceeding 60 mm in a wall having slenderness ratio not exceeding 15 may be provided. These shall be located in the upper or lower middle third height of wall at a distance not less than 600 mm from a lateral support. No horizontal chase shall exceed 1 m in length and there shall not be more than 2 chases in any one wall. Horizontal chases shall have minimum mutual separation distance of 500 mm. Sum of lengths of all chases and recesses in any horizontal plane shall not exceed one-fourth the length of the wall.

6.5.3.5 Holes for supporting put-logs of scaffolding shall be kept away from bearings of beams, lintels, and other concentrated loads. If unavoidable, stresses in the affected area shall be checked to ensure that these are within safe limits.

6.5.3.6 No chase, recess or hole shall be provided in any stretch of a masonry wall, the length of which is less than four times the thickness of wall, except when found safe by structural analysis.

6.5.3.7 Masonry directly above a recess or a hole, if wider than 300 mm, shall be supported on a lintel. No lintel, however, is necessary in case of a circular recess or hole exceeding 300 mm in diameter provided upper half of the recess or hole is built as a semi-circular arch of adequate thickness and there is an adequate length of masonry on the sides of openings to resist the horizontal thrust.

6.5.3.8 As far as possible, chases, recesses and holes in masonry should be left (inserting sleeves, where necessary) at the time of construction of masonry so as to obviate subsequent cutting. If cutting is unavoidable, it should be done without damage to the surrounding or residual masonry. It is desirable to use such tools for cutting which depend upon rotary and not on heavy impact for cutting action.

6.5.3.9 No chase, recess or hole shall be provided in half-brick load bearing wall, excepting the minimum number of holes needed for scaffolding.

6.5.3.10 Chases, recesses or holes shall not be cut into walls made of hollow or perforated units, after the units have been incorporated in masonry.

6.6 Corbelling

6.6.1 Where corbelling is required for the support of some structural element, maximum projection of masonry unit should not exceed one-half of the height of the unit or one-half of the built-in part of the unit and the maximum horizontal projection of the corbel should not exceed one-third of the wall thickness.

6.6.2 The load per unit length on a corbel shall not be greater than half of the load per unit length on the wall above the corbel. The load on the wall above the corbel, together with four times the load on the corbel, shall not cause the average stress in the supporting wall or leaf to exceed the permissible stresses given in [5.4](#).

6.6.3 It is preferable to adopt header courses in the corbelled portion of masonry from considerations of economy and stability.

7 REINFORCED BRICK AND REINFORCED BRICK CONCRETE FLOORS AND ROOFS

The construction and design of reinforced brick and reinforced brick concrete floors and roof shall be in accordance with good practices [C-4(7)].

8 SPECIAL CONSIDERATION FROM EARTHQUAKE POINT OF VIEW

The design and construction of masonry buildings shall be governed by provisions of [4](#), except as modified by the provisions of this section for those elements participating in lateral force resistance. The design and construction for earthquake resistant masonry buildings are given in [8](#), [9](#) and [10](#). The earthquake resistant design and detailing of masonry buildings having the following structural systems:

- a) Masonry Walls with Prescriptive Bands (MWB);
- b) Masonry Walls with Prescriptive Bands and Vertical Reinforcements (MWBR);
- c) Confined Masonry Walls (CMW) (*see* [9](#)); and
- d) Reinforced Masonry Walls (RMW) (*see* [10](#)).

Buildings with MWB and MWBR systems shall not be of height more than 15 m subject to a maximum of four storeys, measured from the base of the buildings to the roof slab or ridge level.

8.1 For the purpose of **8** and **9**, the following definitions shall apply in addition to those given in **2**.

8.1.1 *Separation Section*

The gap of specified width between adjacent buildings or parts of the same building either left uncovered or covered suitably to permit movement to avoid pounding due to earthquake.

8.1.1.1 *Crumple section*

The separation gap filled with appropriate material that crumples or fractures in the event of an earthquake.

8.1.2 *Centre of Rigidity*

The point at the floor level of a structure where application of lateral force in a particular principal direction produces equal deflections of the floor components along that direction.

8.1.3 *Load Bearing Wall*

A wall designed to resist axial force, shear force in its own plane and bending moment about its major axis.

8.1.4 *Box System*

A building made of masonry load bearing walls and horizontal floors.

8.1.5 *Band*

A wooden (in low strength masonry buildings), reinforced concrete or reinforced brick runner provided in the walls to tie them together and to impart horizontal bending strength in them.

8.1.6 *Earthquake Zone*

The earthquake zones as classified in **5** of Part C/Section 1 of this NBCS.

8.1.7 *Design Horizontal Acceleration Coefficient*

The horizontal acceleration coefficient A_h computed considering the importance factor and soil-foundation system as specified in **5** of Part C/Section 1 of this NBCS.

8.1.8 *Concrete Strength*

The 28-day compressive strength (in MPa) of concrete cubes of 150 mm size.

8.1.9 *Cross-Sectional Area of Masonry Unit*

The net cross-sectional area of a masonry unit shall be taken as the gross cross-sectional area minus the area of cellular space (if any). Gross cross-sectional area of cored units shall be determined to the outside of the coring, but cross-sectional area of grooves (if any) shall not be deducted from the gross cross-sectional area to obtain the net cross-sectional area.

8.1.10 *Grout*

A mixture of cement, sand, and water of pourable consistency for filling small voids.

8.1.11 *Grouted Masonry*

Masonry in which the masonry units with holes are filled with a mixture of cement and sand.

8.1.11.1 Grouted hollow-unit masonry

A form of grouted masonry construction in which certain designated cells of hollow units are continuously filled with grout.

8.1.11.2 Grouted multi-wythe masonry

A form of grouted masonry construction in which the space between the wythes is solidly or periodically filled with grout.

8.1.12 Joint Reinforcement

A prefabricated reinforcement in the form of lattice truss which has been hot dip galvanized after fabrication and is to be laid in the mortar bed joint.

8.1.13 Pier

An isolated vertical member whose horizontal dimension measured at right angles to its thickness is not less than 4 times its thickness and whose height is less than 5 times its length.

8.1.14 Prism

An assemblage of masonry units bonded by mortar with or without grout used as a test specimen for determining properties of masonry.

8.1.15 Reinforced Masonry

Masonry which is reinforced (as specified in [2](#)) and grouted (if necessary) so that the two materials act together in resisting forces.

8.1.16 Grouted Cavity Reinforced Masonry

Masonry with two parallel single-leaf walls spaced at least 50 mm apart, and effectively tied together with wall ties. The intervening cavity contains steel reinforcement and is filled with infill concrete so as to result in common action with masonry under load.

8.1.17 Pocket type Reinforced Masonry

Masonry reinforced primarily to resist lateral loading where the main reinforcement is concentrated in vertical pockets formed in the tension face of the masonry and is surrounded by in situ concrete.

8.1.18 Quetta Bond Reinforced Masonry

Masonry that is at least one and half units thick, in which vertical pockets containing reinforcement and mortar or concrete infill occur at intervals along its length.

8.1.19 Specified Compressive Strength of Masonry

Minimum compressive strength (in MPa) is force per unit of net cross-section area, which is required of the masonry used in construction and upon which the design is based.

8.1.20 Wall Tie

A metal fastener, which connects wythes of masonry to each other or to other materials.

8.1.21 Wythe

A continuous vertical layer of masonry wall of one unit in thickness.

NOTE — For low strength masonry buildings, wooden runners may also be used as runner.

8.2 General Principles

The general principles given in [8.2.1](#) to [8.2.9](#) shall be observed in construction of earthquake resistant buildings.

8.2.1 Lightness

Since the earthquake force is a function of mass, the weight of the building shall be as minimum as possible, consistent with structural safety and functional requirements. Roofs and upper storeys of buildings, in particular, should be designed as light as possible.

8.2.2 Continuity of Construction

8.2.2.1 As far as possible, the parts of the building should be tied together in such a manner that the building acts as one unit.

8.2.2.2 For parts of buildings between separation or crumple sections to expansion joints, floor slabs shall be continuous throughout as far as possible. Concrete slabs shall be rigidly connected or integrally cast with the support beams.

8.2.2.3 Additions to the structures shall be accompanied by the provision of separation or crumple sections between the new and the existing structures as far as possible, unless positive measures are taken to establish continuity between the existing and the new construction.

8.2.2.4 Alteration to the building structure shall be done by maintaining its structural stability by ensuring proper load path.

8.2.3 Projecting and Suspended Parts

8.2.3.1 Projecting parts shall be avoided as far as possible. If the projecting parts cannot be avoided, they shall be properly reinforced and firmly tied to the main structure, and their design shall be in accordance with **5** of Part C/Section 1 of this NBCS.

8.2.3.2 Ceiling plaster shall preferably be avoided. When it is unavoidable, the plaster shall be as thin as possible.

8.2.3.3 Suspended ceiling shall be avoided as far as possible. Where provided they shall be light, adequately framed and secured to which electrical fixtures shall be fully secured.

8.2.4 Building Configuration

8.2.4.1 Masonry buildings shall have walls placed uniformly along the two principal plan directions. Buildings with the irregularities listed in **5** of Part C/Section 1 of this NBCS perform poorly during earthquake shaking, and should be avoided. When these irregularities appear in a building, the guidance given in **5** of Part C/Section 1 of this NBCS for the respective irregularity shall be followed. In order to minimize torsion and stress concentration, provisions given in [8.2.4.2](#) to [8.2.4.4](#) should be complied with as relevant.

8.2.4.2 The building should have a simple rectangular plan and be symmetrical both with respect to mass and rigidity so that the centre of mass and rigidity of the building coincide with each other in which case no separation sections other than expansion joints are necessary. For provision of expansion joints reference may be made to [C-4(6)].

8.2.4.3 If symmetry of the structure is not possible in plan, elevation or mass, provision shall be made for torsional and other effects due to earthquake forces in the structural design or the parts of different rigidities may be separated through crumple sections. The length of such building between separation sections shall not preferably exceed three times the width.

NOTE — As an alternative to separation section to reduce torsional moments, the centre of rigidity of the building may be brought close or coincident to the centre of mass by adjusting the locations and/or sizes of columns and walls.

8.2.4.4 Buildings having plans with shapes like L, T, E and Y shall preferably be separated into rectangular parts by providing separation sections at appropriate places. Typical examples are shown in [Fig. 15](#).

NOTES

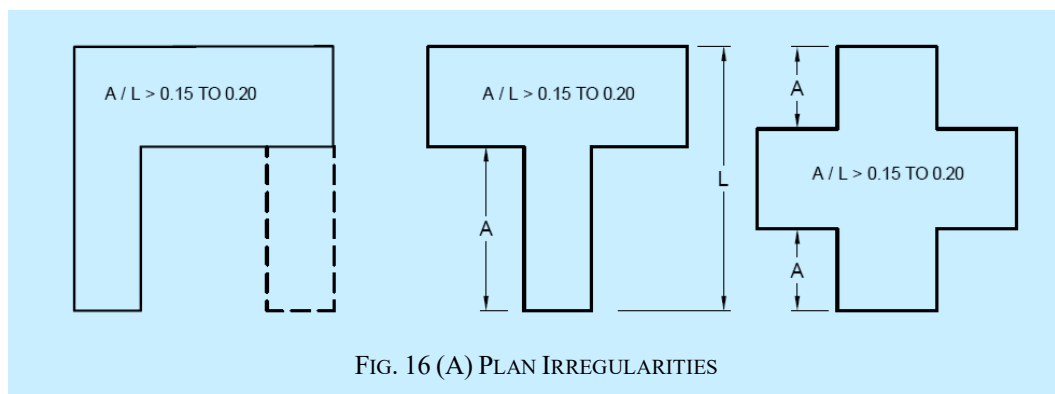
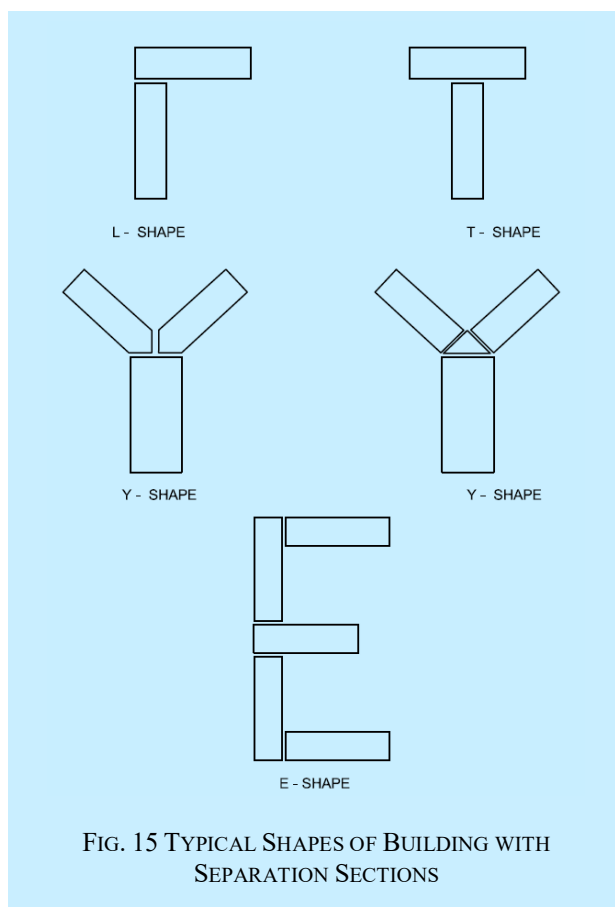
- 1 The buildings with small lengths of projections forming L, T, E or Y shapes need not be provided with separation section. In each cases the length of the projection may not exceed 15 percent to 20 percent of the total dimension of the building in the direction of the projection (see Fig. 16).
- 2 For buildings with minor asymmetry in plan and elevation, separation sections may be omitted.

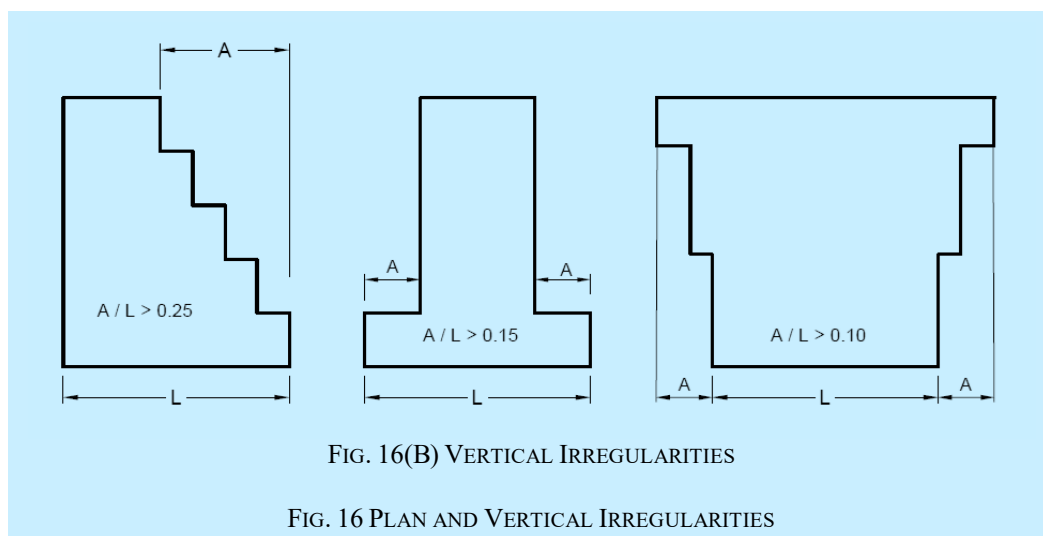
8.2.5 Strength in Various Directions

The structure shall be designed to have adequate strength against earthquake effects along both the horizontal axes. The design shall also be safe considering the reversible nature of earthquake forces.

8.2.6 Foundations

The structure shall not be founded on such soils which shall subside or liquefy during an earthquake, resulting in large differential settlements (see also 8.3.3.3).





8.2.7 Ductility

The main structural elements and their connection shall be designed to have a ductile failure. This shall enable the structure to absorb energy during earthquakes to avoid sudden collapse of the structure. Providing reinforcing steel in masonry at critical sections, as specified in this Section shall not only increase strength and stability but also ductility. The details for achieving ductility in reinforced concrete structures is given in Part C 'Structural Design, Section 5 Structural Concrete' of this NBCS.

8.2.8 Damage to Non-Structural Parts

Suitable details shall be worked out to connect the non-structural parts with the structural framing so that the deformation of the structural frame leads to minimum damage of the non-structural elements.

8.2.9 Fire Safety

Fire frequently follows an earthquake and therefore, buildings shall be constructed to make them fire resistant in accordance with the provisions of Part F 'Fire and Life Safety' of this NBCS.

8.3 Special Construction Features

8.3.1 Separation of Adjoining Structures

8.3.1.1 Separation of adjoining structures or parts of the same structure is required for structures having different total heights or storey heights and different dynamic characteristics. This is to avoid collision during an earthquake.

8.3.1.2 Minimum width of separation gaps as mentioned in [8.3.1.1](#), shall be as specified in [Table 13](#). The design seismic coefficient to be used shall be in accordance with **5** of Part C/Section 1 of this NBCS.

8.3.1.3 Where separation is necessary, a complete separation of the parts shall be made except below the plinth level. The plinth beams, foundation beams and footings may be continuous. Where separation sections are provided in a long building, they shall take care of movement owing to temperature changes also.

8.3.2 Separation or Crumple Section

8.3.2.1 In case of framed construction, members shall be duplicated on either side of the separation or crumple section. As an alternative, in certain cases, such duplication may not be provided, if the portions on either side can act as cantilevers to take the weight of the building and other relevant loads.

8.3.2.2 Typical details of separation and crumple sections are shown in [Fig. 17](#). For other types of joint details, reference may be made to [C-4(5)].

Table 13 Gap Width for Adjoining Structures

(Clause [8.3.1.2](#))

Sl No.	Buildings Categories for Earthquake Resisting Feature	Gap Width per Storey for Adjoining Masonry Building
(1)	(2)	mm (3)
i)	Frames with shear walls	15
ii)	Category B ¹⁾	10
iii)	Category C ¹⁾	10
iv)	Category D ¹⁾	15
v)	Category E ¹⁾	20

NOTE — Minimum total gap shall be 25 mm. For any other value of A_h the gap width shall be determined proportionately.
¹⁾ See [Table 14](#).

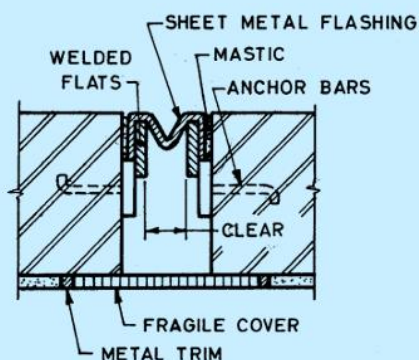


FIG. 17(A) BRICK WALL

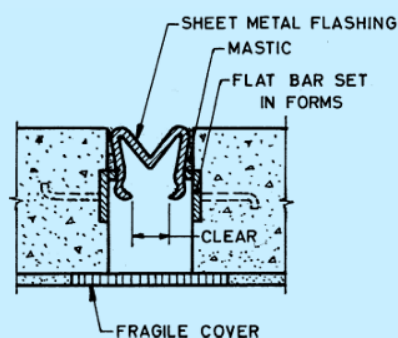


FIG. 17(B) CONCRETE WALLS

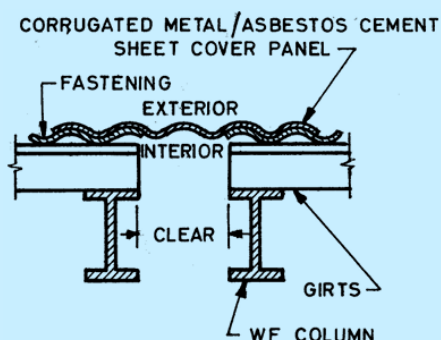


FIG. 17(C) METAL SLIDING INDUSTRIAL WORK

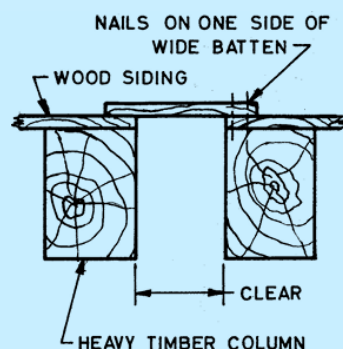
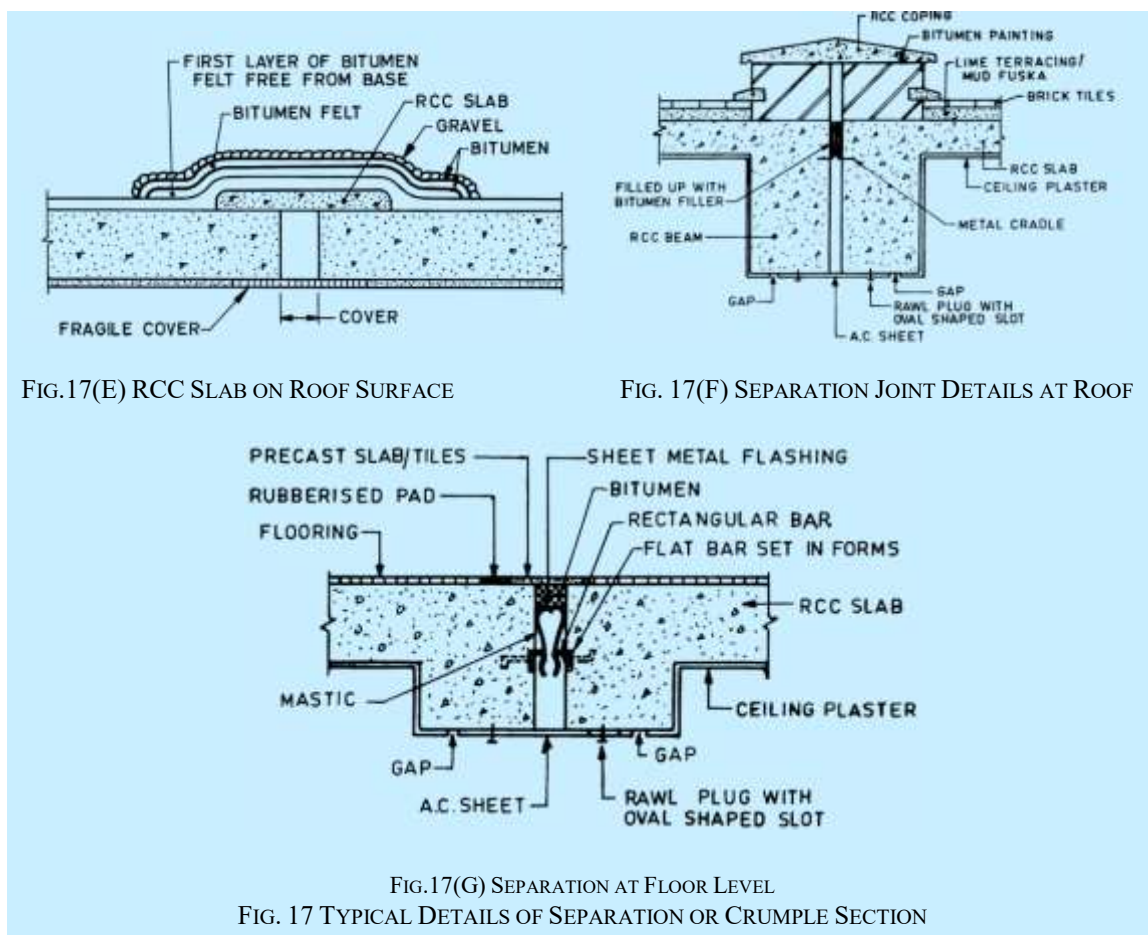


FIG. 17(D) WOOD SHEATHING INDUSTRIAL WORK



8.3.3 Foundations

8.3.3.1 For the design of foundations, the provisions of Part C/Section 1 and Section 2 of this NBCS shall generally be followed.

8.3.3.2 The subgrade below the entire area of the building shall preferably be of the same type of the soil. Wherever this is not possible, a suitably located separation or crumple section shall be provided.

8.3.3.3 Loose fine sand, soft silt and expansive clays should be avoided. If unavoidable, the building shall rest either on a rigid raft foundation or on piles taken to a firm stratum. However, for light constructions the following measures may be taken to improve the soil on which the foundation of the building may rest:

- a) sand piling; and
- b) soil stabilization.

8.3.3.4 Isolated footings for columns

All the individual footings or pile caps where used in Type III Soft soils, shall be connected by reinforced concrete ties at least in two directions approximately at right angles to each other. For buildings with no basement, the ties may be placed at or below the plinth level and for buildings with basement they may be placed at the level of basement floor. They shall need to be designed to carry the load of the panel walls also.

NOTE — The ties may not be necessary where structural floor connects the columns at or below the plinth level.

8.3.3.4.1 Where ties are used, their sections shall be designed to carry in tension as well as in compression, an axial load not less than the earthquake force in the direction the tie acting on the heavier of the columns connected, but the sections shall not be less than 200 mm × 200 mm with M20 concrete reinforced with 4 bars of 12 mm diameter plain mild steel bars or 10 mm diameter high strength deformed bars, one at each corner, bound by 8 mm diameter stirrups not more than 150 mm apart.

NOTE — In working out the buckling strength of ties, the lateral support provided by the soil may be taken into account. Calculations show that for such buried ties, lateral buckling is not a problem and the full section of the tie may be taken effective as a short column.

8.3.3.4.2 In the case of reinforced concrete slab, the thickness shall not be less than $1/50^{\text{th}}$ of the clear distance between the footings, but not less than 100 mm in any case. It shall be reinforced with not less than 0.15 percent mild steel bars or 0.12 percent high strength deformed bars in each direction placed symmetrically at top and bottom.

8.3.4 Roofs and Floors

8.3.4.1 Flat roof or floor shall not preferably be made of tiles of ordinary bricks supported on steel, timber or reinforced concrete joists, nor they shall be of a type which in the event of an earthquake is likely to be loosened and parts or all of which may fall. If this type of construction cannot be avoided, the joists should be blocked at ends and bridged at intervals such that their spacing is not altered during an earthquake.

8.3.4.1.1 For pitched roofs, corrugated iron or asbestos sheets shall be used in preference to country, Allahabad or Mangalore tiles or other loose roofing units. All roofing materials shall be properly tied to the supporting members. Heavy roofing materials shall generally be avoided.

8.3.4.2 Pent roofs

8.3.4.2.1 All roof trusses shall be supported on reinforced concrete or reinforced brick band (*see* [8.5](#)). The holding down bolts shall have adequate size and length as required for earthquake forces in accordance with **5** of Part C/Section 1 of this NBCS.

Where a trussed roof adjoins a masonry gable, the ends of the purlins shall be carried on and secured to a plate or bearer which shall be adequately bolted to reinforced concrete or reinforced brick band at the top of gable end masonry.

NOTE — Hipped roof in general have shown better structural behaviour during earthquakes than gable ended roofs.

8.3.4.2.2 At tie level all the trusses and the gable end shall be provided with diagonal braces in plan so as to transmit the lateral shear due to earthquake force to the gable walls acting as shear walls at the ends as specified in [8.5](#).

8.3.4.3 Jack arches

Jack arched roofs or floors, where used shall be provided with mild steel ties in all spans along with diagonal braces in plan to ensure diaphragm actions.

8.3.5 Staircases

8.3.5.1 The inter-connection of the stairs with the adjacent floors should be appropriately treated by providing sliding joints at the stairs to eliminate their bracing effect on the floors. Large stair halls shall preferably be separated from the rest of the building by means of separation or crumple sections.

8.3.5.2 Three types of stair construction may be adopted as described below:

- a) *Separated staircases* — One end of the staircase rests on a wall and the other end is carried by columns and beams which have no connection with the floors. The gap at the vertical joints between the floor and the staircase may be covered either with a tread plate attached to one side of the joint and sliding on the other side, or covered with some appropriate material which could crumple or fracture during an earthquake without causing structural damage. The supporting members, columns or walls, are isolated from the surrounding floors by means of separation or crumple sections. A typical example is shown in [Fig. 18](#);
- b) *Built-in staircase* — When stairs are built monolithically with floors, they can be protected against damage by providing rigid walls at the stair opening. An arrangement, in which the staircase is enclosed by two walls, is given in [Fig. 19](#). In such cases, the joints, as mentioned in respect of separated staircases, shall not be necessary;

The two walls mentioned above, enclosing the staircase, shall extend through the entire height of the stairs and to the building foundations; and

- c) *Staircases with sliding joints* — In case it is not possible to provide rigid walls around stair openings for built-in staircase or to adopt the separated staircases, the staircases shall have sliding joints so that they shall not act as diagonal bracing.

8.3.6 Framing of Thin Load Bearing Walls (see [Fig. 20](#))

Load bearing walls can be made thinner than 200 mm say 150 mm inclusive of plastering on both sides. Reinforced concrete framing columns and collar beams shall be necessary to be constructed to have full bond with the walls. Columns are to be located at all corners and junctions of walls and spaced not more than 1.5 m apart but so located as to frame up the doors and windows. The horizontal bands or ring beams are located at all floors, roof as well as lintel levels of the openings. The sequence of construction between walls and columns shall be first to build the wall up to 4 to 6 courses height leaving toothed gaps (tooth projection being about 40 mm only) for the columns and second to pour M20 (1 : 1½ : 3) concrete to fill the columns against the walls using wood forms only on two sides. The columns steel should be accurately held in position all along. The band concrete should be cast on the wall masonry directly so as to develop full bond with it.

Such construction may be limited to only two storeys maximum in view of its vertical load carrying capacity. The horizontal length of walls between cross walls shall be restricted to 7 m and the storey height to 3 m.

8.3.7 Reinforcing Details for Hollow Block Masonry

The following details may be followed in placing the horizontal and vertical steel in hollow block masonry using cement-sand or cement-concrete blocks.

8.3.7.1 Horizontal band

U-shaped blocks may be used for construction of horizontal bands at various levels of the storeys as shown in [Fig. 21](#), where the amount of horizontal reinforcement shall be taken 25 percent more than that given in [Table 18](#) and provided by using four bars and 6 mm diameter stirrups. Other continuity details shall be followed, as shown in [Fig. 28](#).

NOTES

- 1 Detailing of steel around the window openings will be same as for door openings with additional 75 mm band at sill level.
- 2 Figures are for just explanation purposes and the same shall not be used as design guide.

8.3.7.2 Vertical reinforcement

Bars, as specified in [Table 19](#) shall be located inside the cavities of the hollow blocks, one bar in each cavity (see [Fig. 29](#)). Where more than one bar is planned these can be located in two or three consecutive cavities. The cavities containing bars are to be filled by using micro-concrete 1 : 2 : 3 or cement coarse sand mortar 1 : 3, and properly rodded for compaction. The vertical bars should be spliced by welding or overlapping for developing full tensile strength. For proper bonding, the overlapped bars should be tied together by winding the binding wire over the lapped length. To reduce the number of overlaps, the blocks may be made U-shaped as shown in [Fig. 29](#) which shall avoid lifting and threading of bars into the hollows.

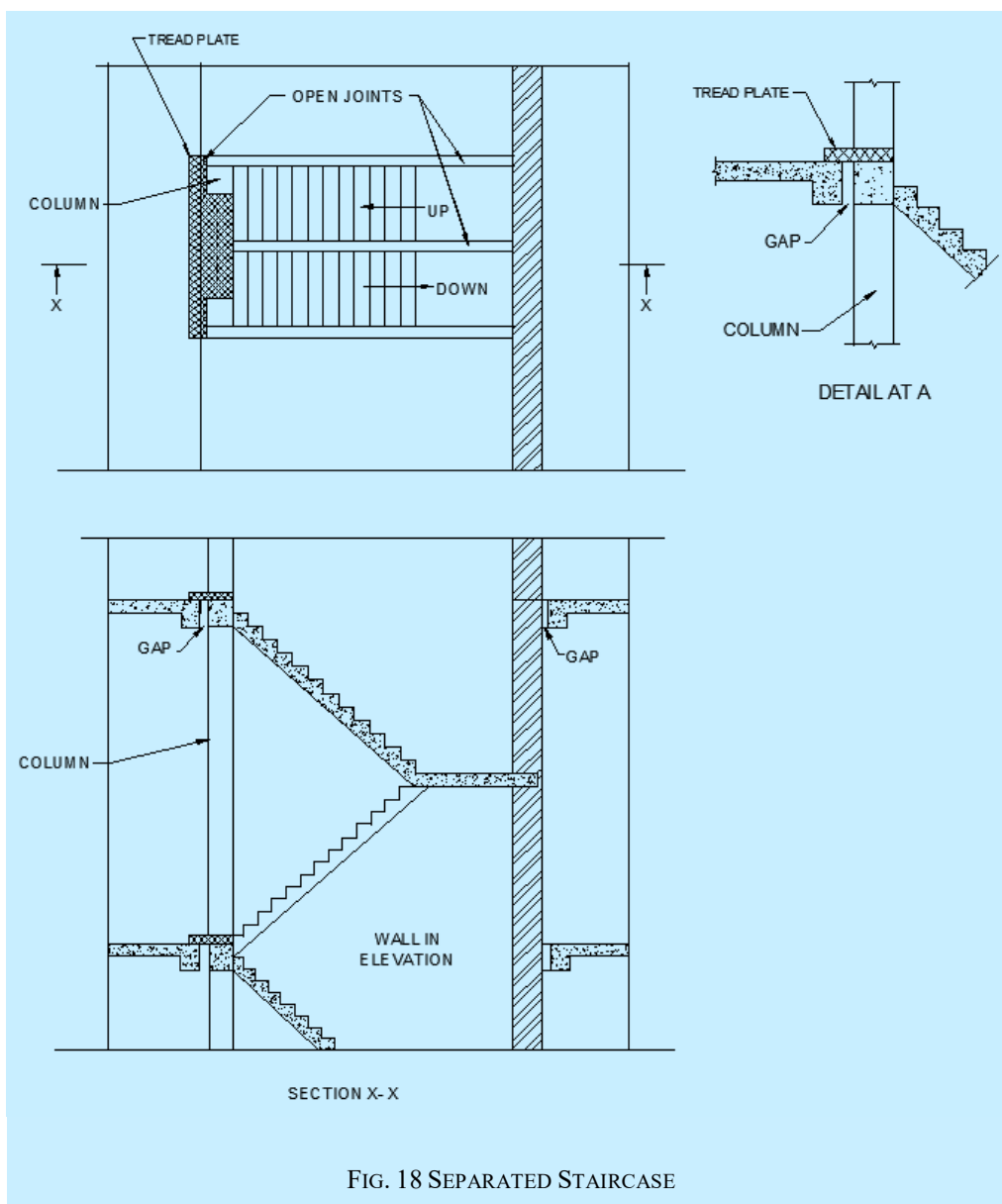


FIG. 18 SEPARATED STAIRCASE

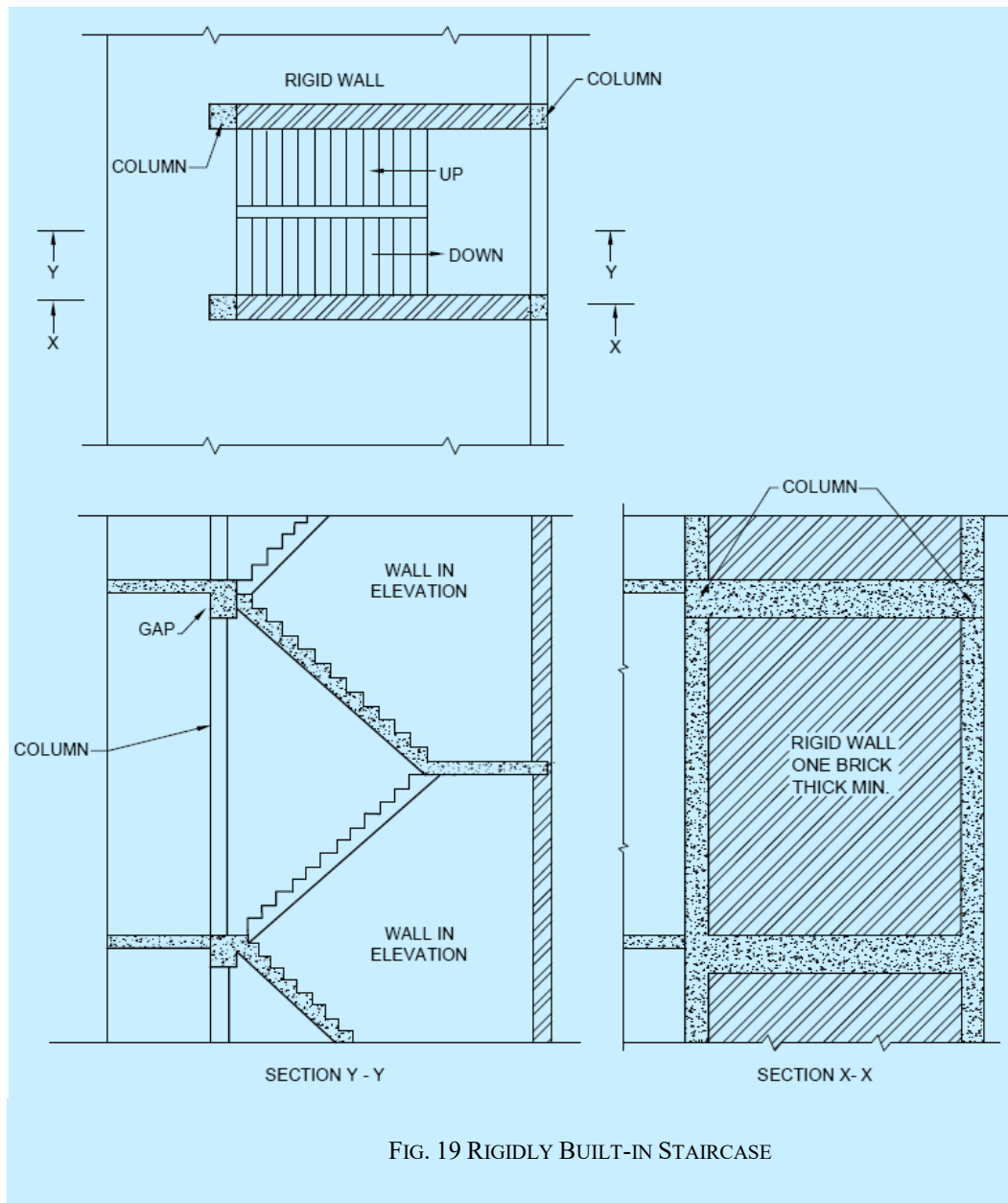
8.4 DESIGN CONSIDERATIONS

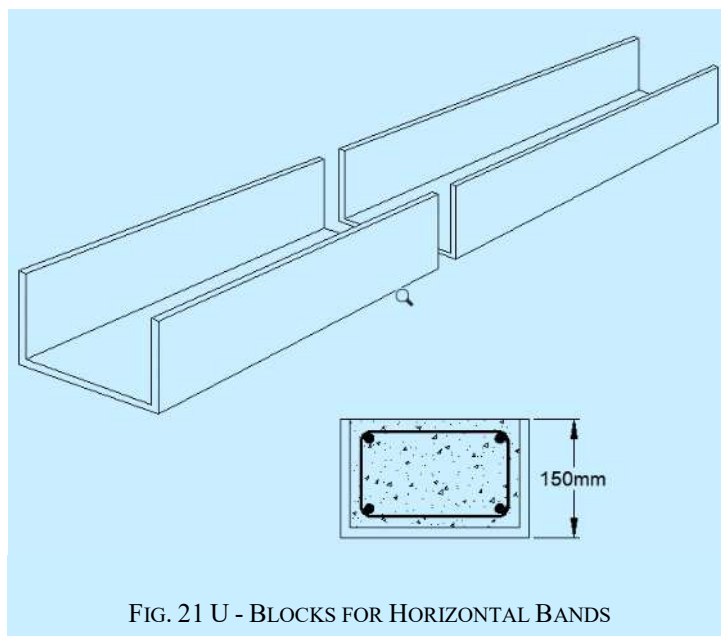
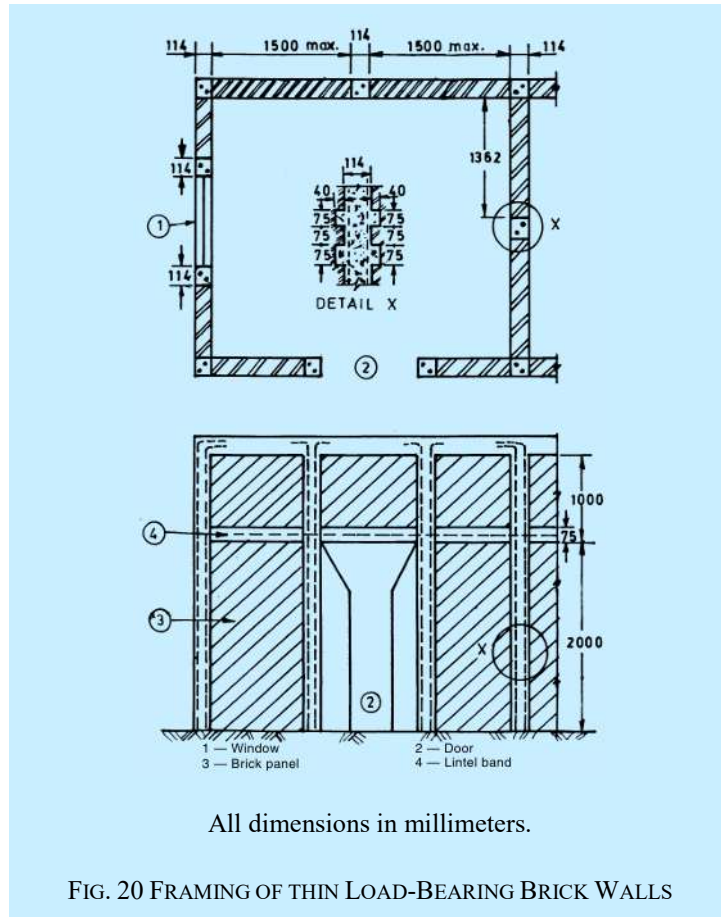
8.4.1 Design Philosophy

The design approach adopted for masonry buildings is the working stress method at serviceability loads, with structural materials assumed to behave in linear elastic manner. Safety is ensured by restricting stresses in material induced by design loads imposed on the structure as per the load combinations specified in 5 of Part C/Section 1 of this NBCS.

8.4.1.1 Working stress method

In the working stress method of design of members, permissible stresses given in [Table 14](#) shall govern the earthquake resistant design of members of the masonry building systems specified in this section.





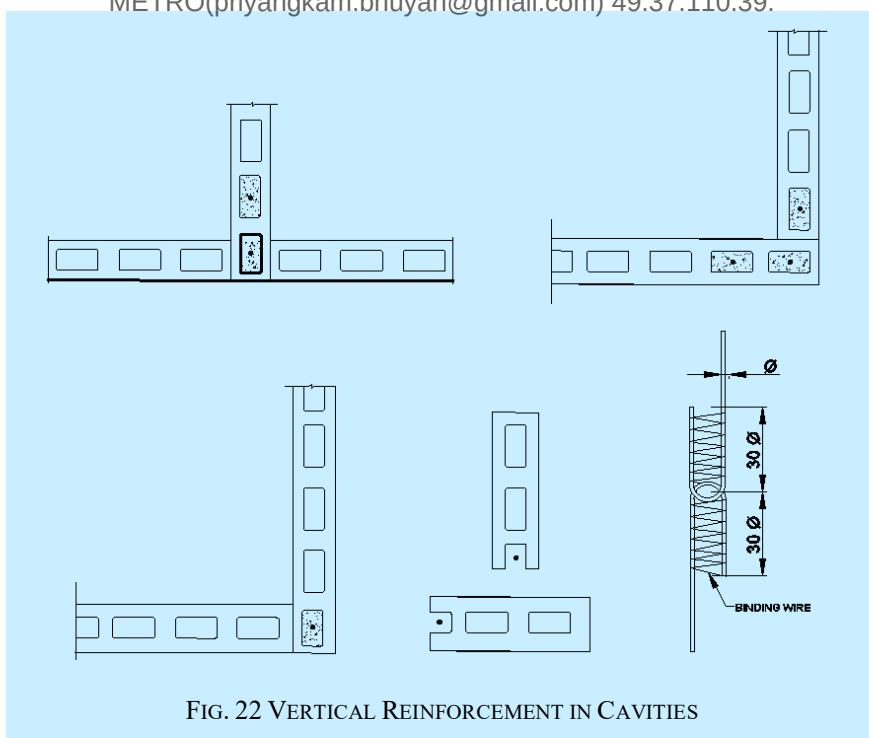


FIG. 22 VERTICAL REINFORCEMENT IN CAVITIES

Table 14 Permissible Stresses in Materials to be Used in the Design of Members of Masonry Buildings
(Clauses 8.4.1.1, 11.5 and 11.6.1)

Sl No.	Compression		Tension	Shear
	Axial	Axial and Flexural	Axial and Flexural	
(1)	(2)	(3)	(4)	(5)
Masonry Wall with Prescriptive Bands (MWB) Masonry Wall with Prescriptive Bands and Vertical Reinforcements (MWBR)				
i)	5.4.1 and Table 9 apply	Increase permissible compressive stress permitted as per 5.4.1.4 applies	5.4.2 applies. Masonry walls subjected to in-plane bending and vertical loads shall be designed based on no-tension, permissible shear, and compressive stresses	5.4.3 applies.
Confined Masonry Wall (CMW)				
ii)	Compressive force in confined masonry due to axial load shall not exceed: $P_0 = k_s(0.25f_m A_m + 0.65f_s A_t + f_{c0} A_c)$	Compressive stress in masonry due to combined action of axial load and bending shall not exceed $1.25 f_a$	5.4.2 applies.	Allowable shear stress shall be according to 5.4.3
Reinforced Masonry Wall (RMW)				
iii)	Compressive force due to axial load shall not exceed: $P_0 = k_s(0.25f_m A_m + 0.65f_s A_t)$	Compressive stress in masonry due to combined action of axial load and bending shall not exceed $1.25 f_a$	5.4.2 applies.	For flexural members: a) without web reinforcement: $f_v = 0.083\sqrt{f_m}$ $\leq 0.75 \text{ MPa}$

Table 14 (Concluded)

Table 14 Permissible Stresses in Materials to be Used in the Design of Members of Masonry Buildings (Clauses 8.4.1.1 , 11.5 and 11.6.1)				
SI No.	Compression		Tension	Shear
	Axial	Axial and Flexural	Axial and Flexural	
(1)	(2)	(3)	(4)	(5)
				b) with web reinforcement: $f_v = 0.25\sqrt{f_m} \leq 0.75\text{MPa}$ For walls, shall be as per 5.4.3

8.4.2 Bands and Vertical Elements

Masonry buildings, other than those with reinforced masonry walls and confined masonry walls, shall be provided with horizontal bands and vertical reinforcing elements:

- the horizontal bands shall run along the full perimeter of the external and internal walls, within the full thickness of the walls; and
- the vertical reinforcing elements shall pass through vertically at all corners and adjoining all openings (door, window, and ventilator).

8.4.2.1 Types

Five types of bands are admissible, namely:

- Gable band (when masonry buildings have gable walls);
- Roof band;
- Lintel band;
- Sill band; and
- Plinth band.

8.4.2.2 Geometry

The width of RC band shall be same as the thickness of the wall. Wall thickness shall be at least 200 mm. A clear cover of 20 mm shall be maintained for the reinforcement.

The vertical thickness of RC Band shall be at least:

- 75 mm, where two longitudinal bars are specified, one on each face; and
- 150 mm, where four bars are specified, one at each corner.

8.4.2.3 Ductile detailing

The bands shall be made of reinforced concrete of grade not leaner than M25 or reinforced brickwork in cement mortar not leaner than 1 : 3. In coastal areas, the filling mortar with water proofing admixture shall be used.

8.4.3 Floor Slabs

The roof and floor slabs in MWBs, MWBR, CMBs and RMBs shall be designed as beam supported slabs as per the provisions of Part C/Section 5 of this NBCS.

8.4.4 Building Structure Type

Masonry buildings shall be categorized into four types (Table 15), namely Types B, C, D and E, based on the earthquake and the category of the building and the importance of building I, where I is importance factor applicable to the building. Type E buildings shall not be more than 3 storeys tall.

Table 15 Building Categories of Building for Earthquake Resisting Features					
(Clause 8.4.4)					
SI No.	Importance Factor	Earthquake Zone			
		II	III	IV	V
(1)	(2)	(3)	(4)	(5)	(6)
i)	1.0	B	C	D	E
ii)	1.5	C	D	E	E

8.4.4.1 Slenderness of walls

The masonry walls shall meet the slenderness limit specified in Table 16.

Table 16 Allowable Effective Height to Effective Thickness Ratios of Unreinforced Masonry Walls					
(Clause 8.4.4.1)					
SI No.	Wall	Building Structure Type			
		B	C	D	E
(1)	(2)	(3)	(4)	(5)	(6)
i)	MWB and MWBR	15	15	12	12
ii)	CMW	15	15	15	15
iii)	RMW	15	15	15	15
iv)	URM parapet walls	12	12	6	6

8.4.5 Footings

Strip footings under load bearing walls shall be designed as per the provisions of [C-4(8)] in conjunction with Part C/Section 1 of this NBCS.

8.5 Masonry Walls with Prescriptive Bands (MWB), and Masonry Walls with Prescriptive Bands and Vertical Reinforcement (MWBR)

8.5.1 Masonry Walls

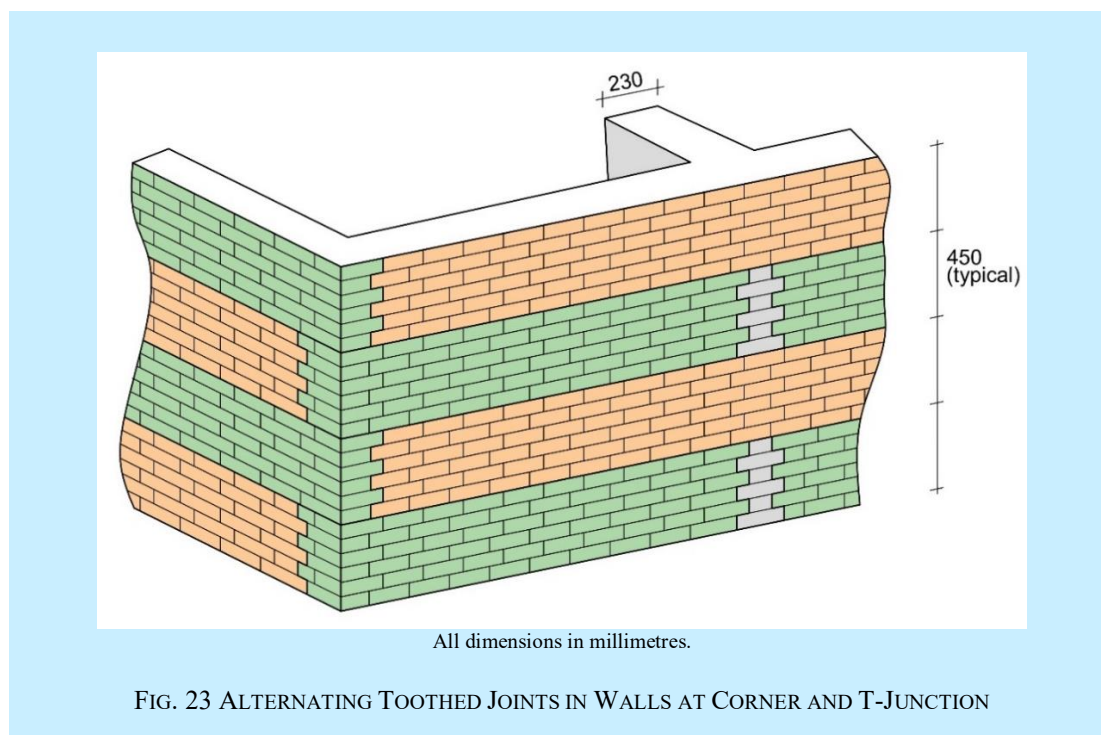
8.5.1.1 The bearing walls in both directions shall be straight and symmetrical in plan.

8.5.1.2 The wall panels formed between cross walls and floors, or roof shall be checked for their strength in bending as a plate or as a vertical strip subjected to the earthquake force acting on its own mass. This check need not be complied with in walls of 200 mm or larger thickness and of storey height less than 3.5 m

8.5.1.3 Masonry bond

For achieving full strength of masonry, the usual bonds specified for masonry should be followed so that the vertical joints are broken properly from course to course. To obtain full bond between perpendicular walls, it is necessary to make a slopping (stepped) joint by making the corners first to a height of 600 mm and then building

the wall in between them. Otherwise, the toothed joint should be made in both the walls alternatively in lifts of about 450 mm ([Fig. 23](#)).



8.5.2 Openings in Bearing Walls

8.5.2.1 Door and window openings reduce the lateral load resistance of walls, and hence, should preferably be small and more centrally located in the wall panels. The guidelines on the size and position of opening are given in [Table 17](#) and [Fig. 24](#).

8.5.2.2 Openings in any storey shall have their top at the same level to accommodate a continuous band over them, including the lintels throughout the building. In addition, openings in different storeys in a masonry wall shall be vertically aligned, and not staggered.

8.5.2.3 Where openings do not comply with the guidelines of [Table 17](#), they shall be strengthened by providing reinforced concrete or reinforcing the brickwork ([Fig. 25](#)) with high strength deformed bars of 8 mm diameter but the quantity of steel shall be increased at the jambs to comply with [8.5.3.9](#), if so required.

8.5.2.4 If a window or ventilator is to be projected out, the projection shall be in reinforced masonry or concrete and well anchored.

8.5.2.5 If an opening is tall (from bottom to almost top of a storey, thus dividing the wall into two portions), the adjoining portions shall be reinforced with horizontal reinforcement of 6 mm diameter bars at not more than 450 mm intervals, one on inner and one on outer face, properly tied to vertical steel at jambs, corners or junction of walls, where used.

8.5.2.6 The use of arches to span over the openings is a source of weakness and shall be avoided. When compelled to use, steel ties shall be provided.

8.5.3 Earthquake Strengthening Arrangements

8.5.3.1 All masonry buildings shall be strengthened by the methods specified for different types of buildings, as listed in [Table 18](#), and detailed in subsequent clauses. [Fig. 26](#) and [Fig. 27](#) show, schematically, the overall strengthening arrangements to be adopted for types D and E buildings, which consist of horizontal bands of reinforcement at critical levels, vertical reinforcing bars at corners, junctions of walls and jambs of openings.

In four-storey buildings of type B, the requirements of vertical steel may be checked through an earthquake analysis using a design horizontal acceleration coefficient equal to 4 times that given in **5** of Part C/Section 1 of this NBCS. If this analysis shows that vertical steel is not required, the designer may take the decision accordingly.

Table 17 Size and Position of Openings in Bearing Walls

(Clauses [8.5.2.1](#), [8.5.2.3](#) and [Fig. 24](#))

Sl No.	Position of Opening	Details of Opening		
		B	C	D, E
(1)	(2)	(3)	(4)	(5)
i)	Minimum Distance b_5 (mm) from inside corner of outside wall	0	230	450
ii)	Total length of openings shall be such that the ratio $(b_1 + b_2 + b_3)/l_1$ or $(b_6 + b_7)/l_2$ does not exceed:			
	a) 1-storeyed building	0.60	0.55	0.50
	b) 2-storeyed building	0.50	0.46	0.42
	c) 3- or 4-storeyed building	0.42	0.37	0.33
iii)	Minimum width b_4 (mm) of pier between consecutive openings	340	450	560
iv)	Minimum vertical distance h_3 (mm) between two openings (one above the other)	600	600	600
v)	Maximum Width b_8 (mm) of opening of ventilator	900	900	900

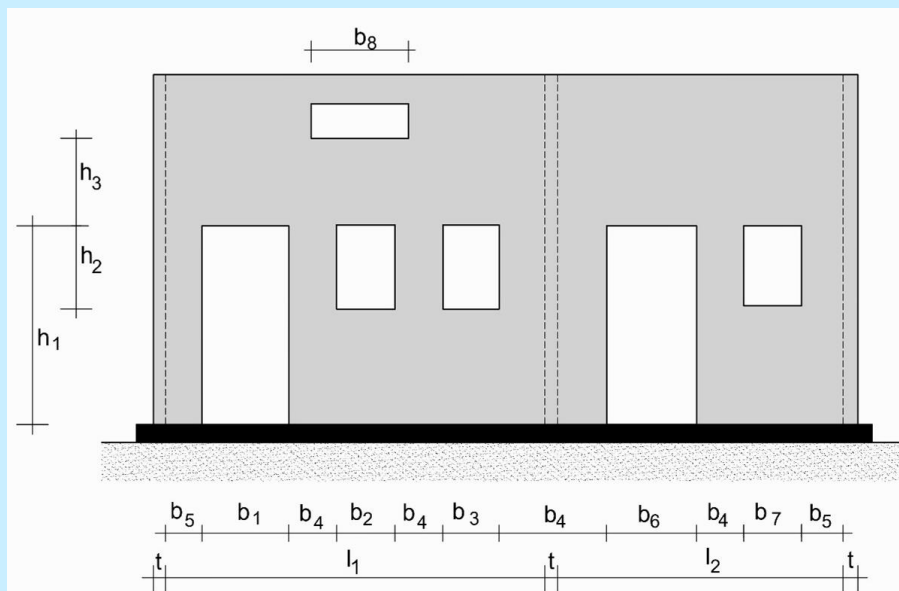


FIG. 24 DIMENSIONS OF OPENINGS AND PIERS FOR RECOMMENDATIONS IN TABLE 17

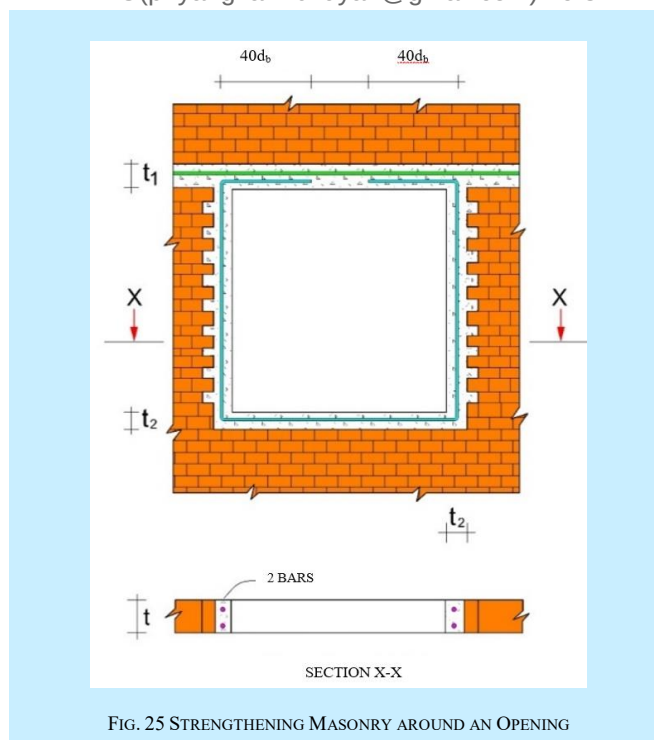


Table 18 Measures Recommended in Masonry Buildings Made with Rectangular Masonry Units

(Clauses [8.3.7.1](#), [8.5.3.1](#) and [8.6.2.3](#))

Sl No.	Structure Type	Number of Storeys	Measures to be Provided in all Storeys
(1)	(2)	(3)	(4)
i)	B	1 to 3	a, b, c, f, g
		4	a, b, c, d, f, g
ii)	C	1 and 2	a, b, c, f, g
		3 and 4	a, b, c, d, e, f, g
iii)	D	1 and 2	a, b, c, d, e, f, g
		3 and 4	a, b, c, d, e, f, g, h
iv)	E	1 to 3	a, b, c, d, e, f, g, h

where

- a = masonry mortar (as per [5.4.6](#));
- b = lintel band (as per [8.5.3.2](#));
- c = roof band and gable band where necessary (as per [8.5.3.3](#) and [8.5.3.4](#), respectively);
- d = vertical steel at corners and junctions of walls (as per [8.5.3.8](#));
- e = vertical steel at jambs of openings (as per [8.5.3.9](#));
- f = bracing in plan at tie level of roofs;
- g = plinth band where necessary (as per [8.5.3.6](#)); and
- h = dowel bars (as per [8.5.3.7](#)).

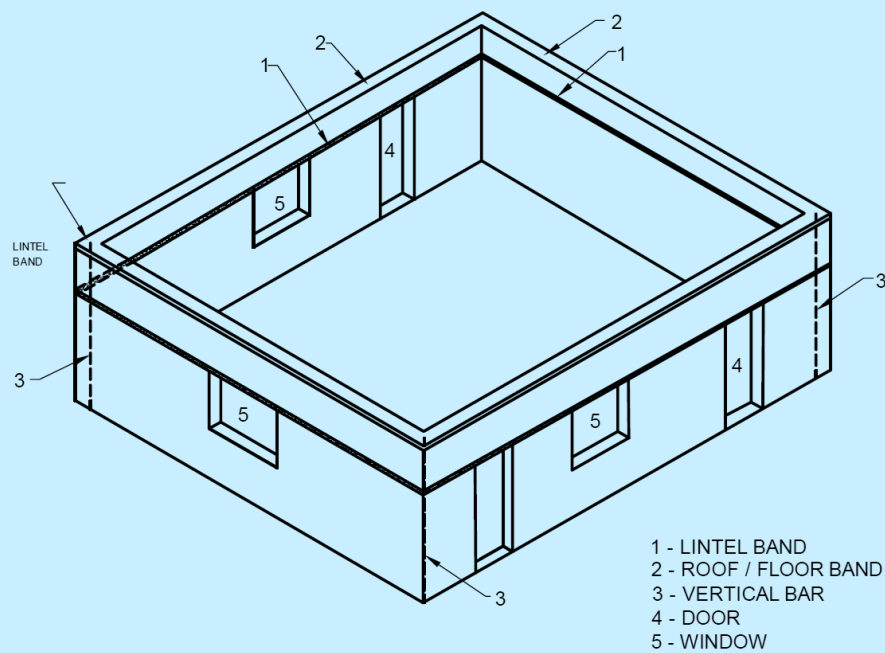
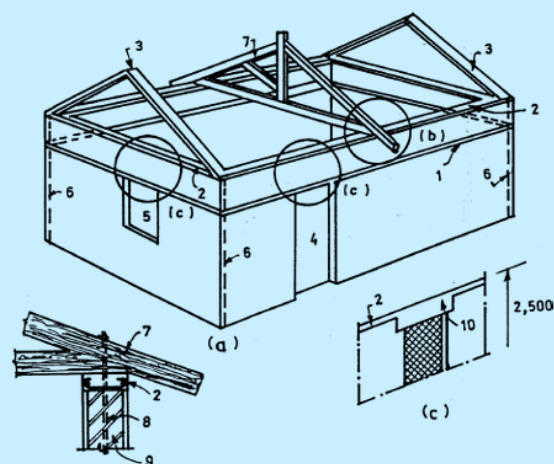


FIG. 26 OVERALL ARRANGEMENT OF REINFORCING MASONRY BUILDINGS



- | | |
|-----------------------------|---|
| 1 — Lintel band | 9 — brick/stone wall |
| 2 — Eaves level (roof) band | 10 — Door lintel integrated with roof band: |
| 3 — Gable band | a) Perspective view |
| 4 — Door | b) Details of truss connection with wall |
| 5 — Window | c) Detail of integrating door lintel with roof band |
| 6 — Vertical steel bar | |
| 7 — Rafter | |
| 8 — Holding down bolt | |

FIG. 27 OVERALL ARRANGEMENT OF REINFORCING MASONRY BUILDING HAVING PITCHED ROOF

8.5.3.2 Lintel band shall be provided at the lintel level on all load bearing internal, external longitudinal and cross walls, as per [8.5.3.5](#). This band provided in panel or partition walls improves their stability during strong earthquake shaking.

8.5.3.3 Roof band shall be provided immediately below the roof or floors. The specifications of the band are given in [8.5.3.5](#). Such a band need not be provided underneath reinforced concrete or brick-work slabs resting on bearing walls, provided that the slabs are continuous over the intermediate wall up to the crumple sections, if any, and cover the width of end walls, fully or at least three-fourths of the wall thickness.

8.5.3.4 Gable band is provided at the top of gable masonry below the purlins. The specifications of the band are given in [8.5.3.5](#). This band shall be made continuous with the roof band at the eaves level.

8.5.3.5 Section and reinforcement of band

8.5.3.5.1 The band shall be as per [8.4.2](#).

8.5.3.5.2 The bands shall be of the full width of the wall not less than 75 mm in depth and reinforced with longitudinal steel ([Table 19](#)). The longitudinal steel bars shall be held in position by steel links or stirrups 6 mm diameter spaced 150 mm apart. With respect to [Table 19](#):

- a) span of wall shall be the distance between centrelines of its cross walls or buttresses. For spans greater than 8 m intermediate pilasters or buttresses shall be provided or the adequacy of such long-span walls shall be ascertained through detailed analysis and full-scale test.
- b) the numbers and diameter of bars given above pertain to high strength deformed bars.
- c) the geometry of the bands shall be as per [8.4.2.2](#).
- d) concrete used in the bands shall be as per [8.3.4](#).
- e) the longitudinal steel bars shall be held in position by steel links or stirrups 6 mm diameter spaced at maximum 150 mm apart.

8.5.3.5.3 In reinforced brickwork, the thickness of joints containing steel bars shall be increased to have at least a minimum mortar cover of 10 mm around the bar. In bands of reinforced brickwork, the area of steel provided should be at least equal to that specified in [Table 19](#).

8.5.3.5.4 For full integrity of walls at corners and junctions of walls and effective horizontal bending resistance of bands continuity of reinforcement is essential. The details as shown in [Fig. 28](#) are recommended.

Table 19 Recommended Longitudinal Steel in Reinforced Concrete Bands

(Clauses [8.3.7.2](#), [8.5.3.5.2](#) and [8.5.3.5.3](#))

Sl No.	Span	Building Structure Type							
		B		C		D		E	
		Number of bars	Diameter	Number of bars	Diameter	Number of bars	Diameter	Number of bars	Diameter
	m		mm		mm		mm		mm
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	5 or less	2	8	2	8	2	8	2	10
ii)	6	2	8	2	8	2	10	2	12
iii)	7	2	8	2	10	2	12	4	10
iv)	8	2	10	2	12	4	10	4	12

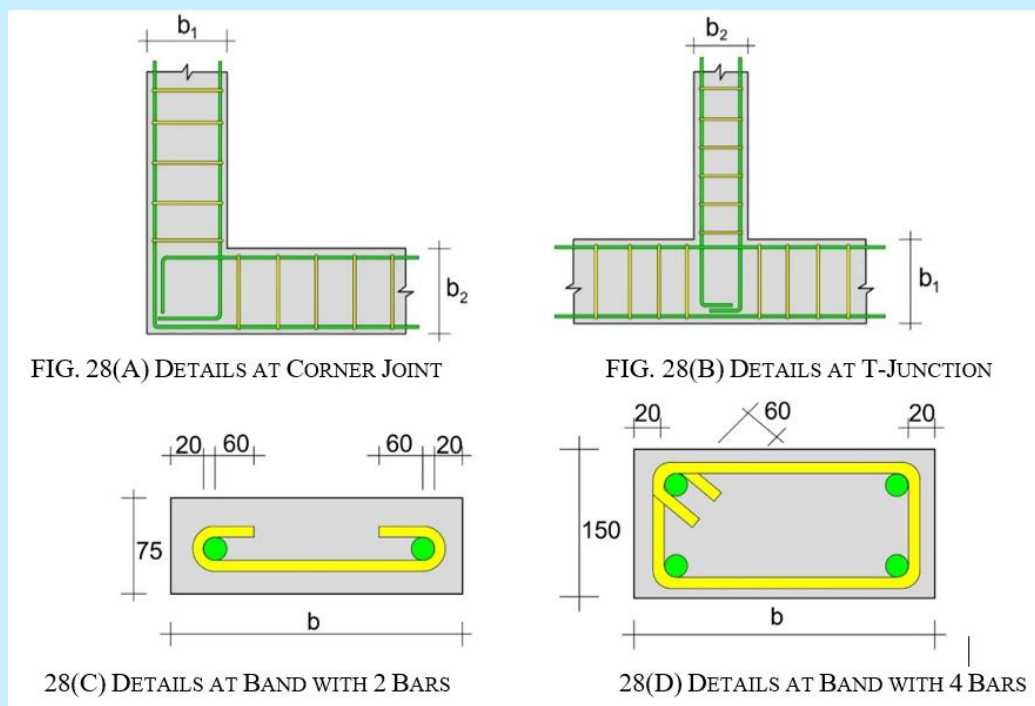


FIG. 28 REINFORCEMENT AND BENDING DETAIL IN RC BAND

8.5.3.6 Plinths band is provided at plinth level of walls on top of the foundation wall. This is to be provided where strip footings of masonry (other than reinforced concrete or reinforced masonry) are used and the soil is either soft or uneven in its properties. Where used, its section may be kept the same as in [8.5.3.5](#). This band shall serve as damp proof course as well.

8.5.3.7 In Type D and E buildings, to further enhance the box action of walls, steel dowel bars may be used at corners and T-junctions of walls at the sill level of windows to length of 900 mm from the inside corner in each wall. Such dowel may be in the form of U stirrups 8 mm diameter. Where used, such bars must be laid in 1 : 3 cement sand mortar with a minimum clear cover of 10 mm on all sides to minimize corrosion.

8.5.3.8 Vertical reinforcement

Vertical steel at corners and junctions of walls, which are up to 340 mm (1½ brick) thick, shall be provided as specified in [Table 20](#). For walls thicker than 340 mm, the area of the bars shall be proportionately increased. With respect to [Table 20](#):

- the diameters given above are for reinforcing steels of grade Fe415 or higher; and
- typical details of providing vertical steel in brickwork masonry with rectangular solid units at corners and T-junctions are shown in [Fig. 29](#). The vertical bars shall be covered with concrete of grade M20 or 1 : 3 mortar in suitably created pockets around the bars ([Fig. 29](#)). This shall ensure their safety from corrosion and good bond with masonry.

Table 20 Diameter (mm) of Single Vertical Steel Reinforcement in Masonry Walls with Rectangular Masonry Units

(Clause [8.5.3.8](#), [8.5.3.9](#) and [8.6.2.3](#))

Sl No.	Number of Storeys	Storey	Buildings Structure Type			
			Type B	Type C	Type D	Type E
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	1	—	Nil	Nil	10	12
ii)	2	Top	Nil	Nil	10	12
		Bottom	Nil	Nil	12	16
iii)	3	Top	Nil	10	10	12
		Middle	Nil	10	12	16
		Bottom	Nil	12	12	16
iv)	4	Top	10	10	10	Not permitted
		Third	10	10	12	
		Second	10	12	16	
		Bottom	12	12	20	

8.5.3.8.1 The vertical reinforcement shall be properly embedded in the plinth masonry of foundations and roof slab or roof band to develop its tensile strength in bond. It shall pass through the lintel bands and floor slabs or floor level bands in all storeys.

8.5.3.8.2 Bars in different storeys shall be lapped as per the requirements of Part C/Section 5 of this NBCS.

8.5.3.9 Vertical reinforcement of jambs of window and door openings shall be provided as per [Table 20](#). It may start from the foundation of floor and terminate in lintel band ([Fig. 29](#)).

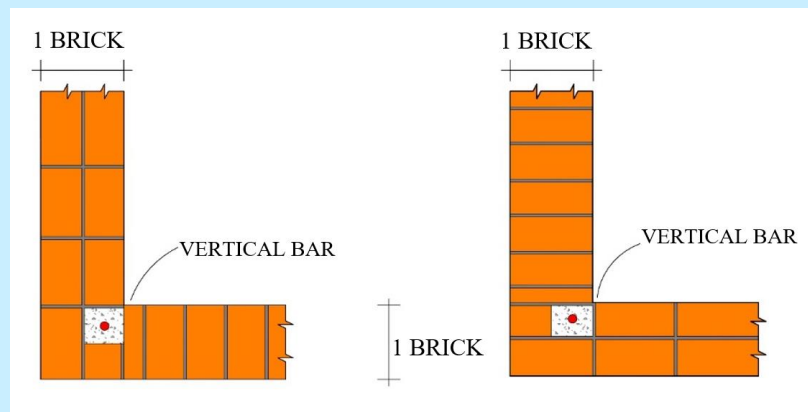


FIG. 29(A) SINGLE BRICK WALL CORNER JOINT DETAIL

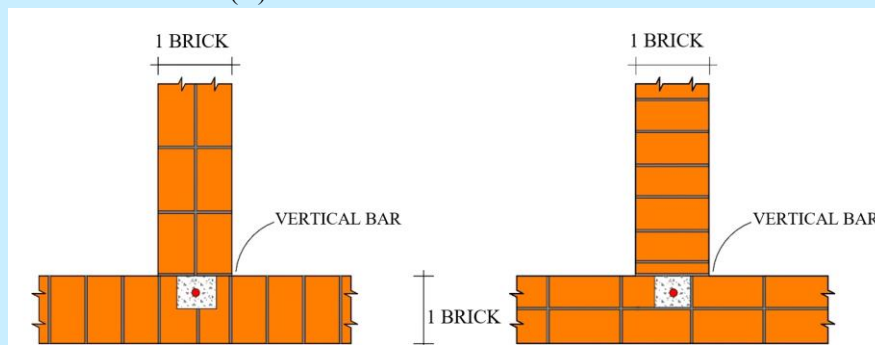


FIG. 29(B) SINGLE BRICK WALL T-JOINT DETAIL

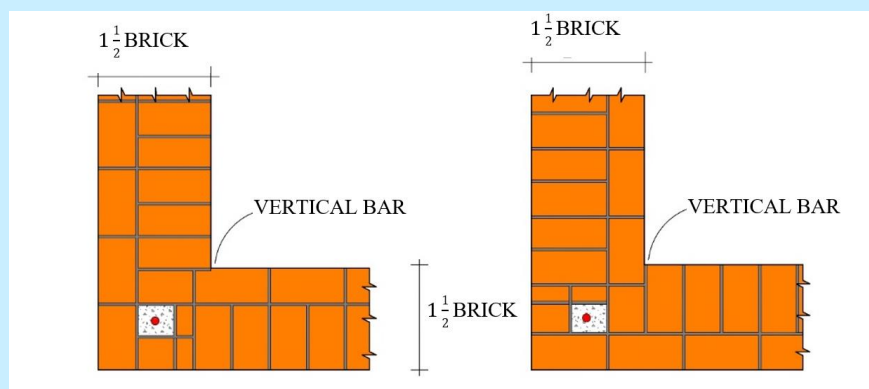


FIG. 29(C) 1 1/2 BRICK WALL CORNER JOINT DETAIL

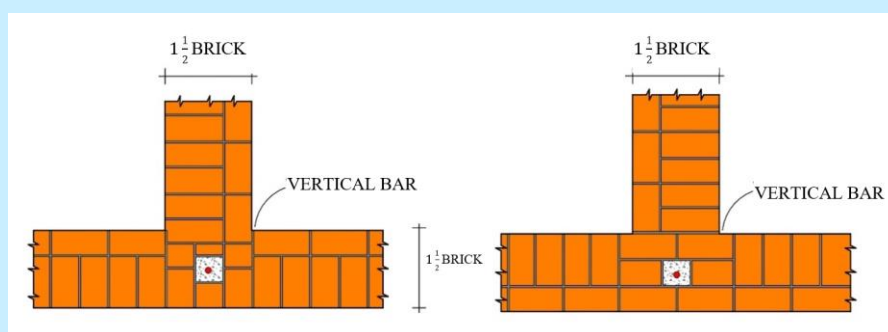


FIG. 29(D) 1 1/2 BRICK WALL T-JOINT DETAIL

FIG. 29 TYPICAL DETAILS OF PROVIDING VERTICAL STEEL BARS IN BRICK MASONRY

8.6 Floors/Roofs with Small Precast Components

8.6.1 Types of Precast Floors/Roofs

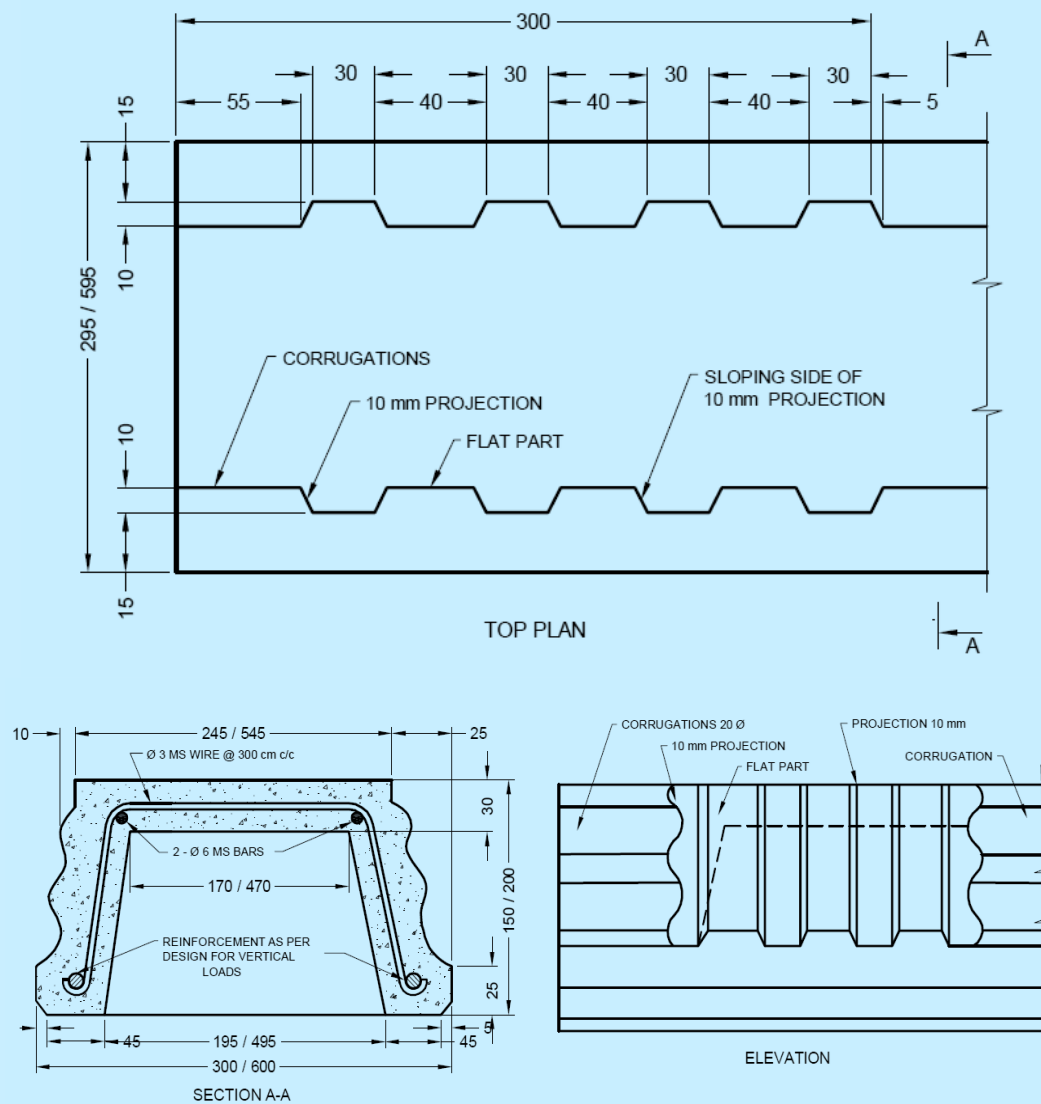
Earthquake resistance measures for floors and roofs with small precast components, as covered in this Section, have been dealt with as typical examples.

8.6.1.1 Precast reinforced concrete unit roof/floor

The unit is a precast reinforced concrete component, channel (inverted trough) shaped in section (see Fig. 30). The nominal width of the unit varies from 300 mm to 600 mm, its height from 150 mm to 200 mm and a minimum flange thickness of 30 mm. Length of unit shall vary according to room dimensions, but the maximum length is restricted to 4.2 m from stiffness considerations. Horizontal corrugations are provided on the two longitudinal faces of the units so that the structural roof/floor acts monolithic after concrete grouted in the joints between the units attains strength (see Fig. 31).

8.6.1.2 Precast reinforced concrete cored unit roof/floor

The unit is a reinforced concrete component having a nominal width of 300 to 600 mm and thickness of 130 to 150 mm having two circular hollows 90 mm diameter, throughout the length of the unit (see Fig. 32). The minimum flange/web thickness of the unit shall be 20 mm. Length of unit varies according to room dimensions, but the maximum length shall be restricted to 4.2 m from stiffness considerations. Horizontal corrugations are provided on the two longitudinal faces of the units so that the structural roof/floor acts monolithic after concrete grouted in the joints between the units attains strength (see Fig. 33).



All dimensions in millimetres

FIG. 30 CHANNEL UNITS

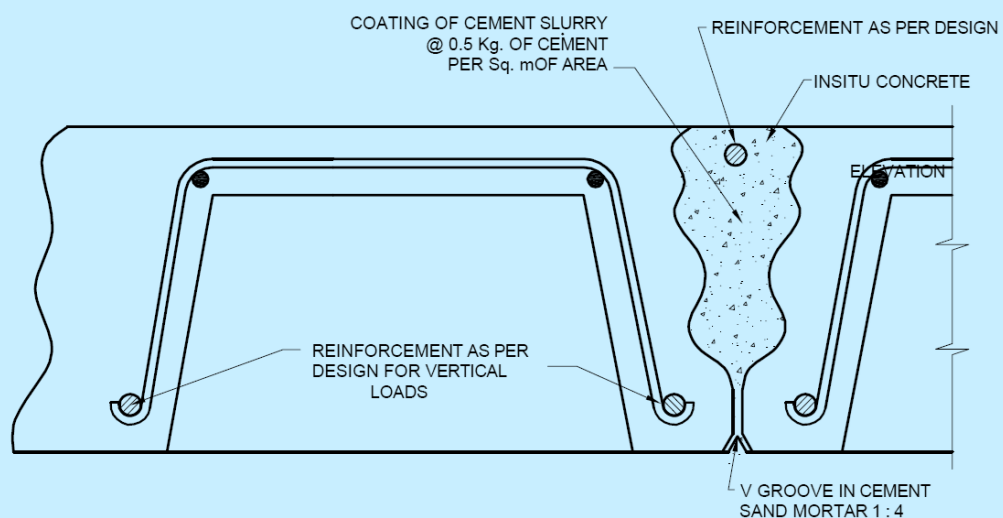


FIG. 31 CHANNEL UNITS

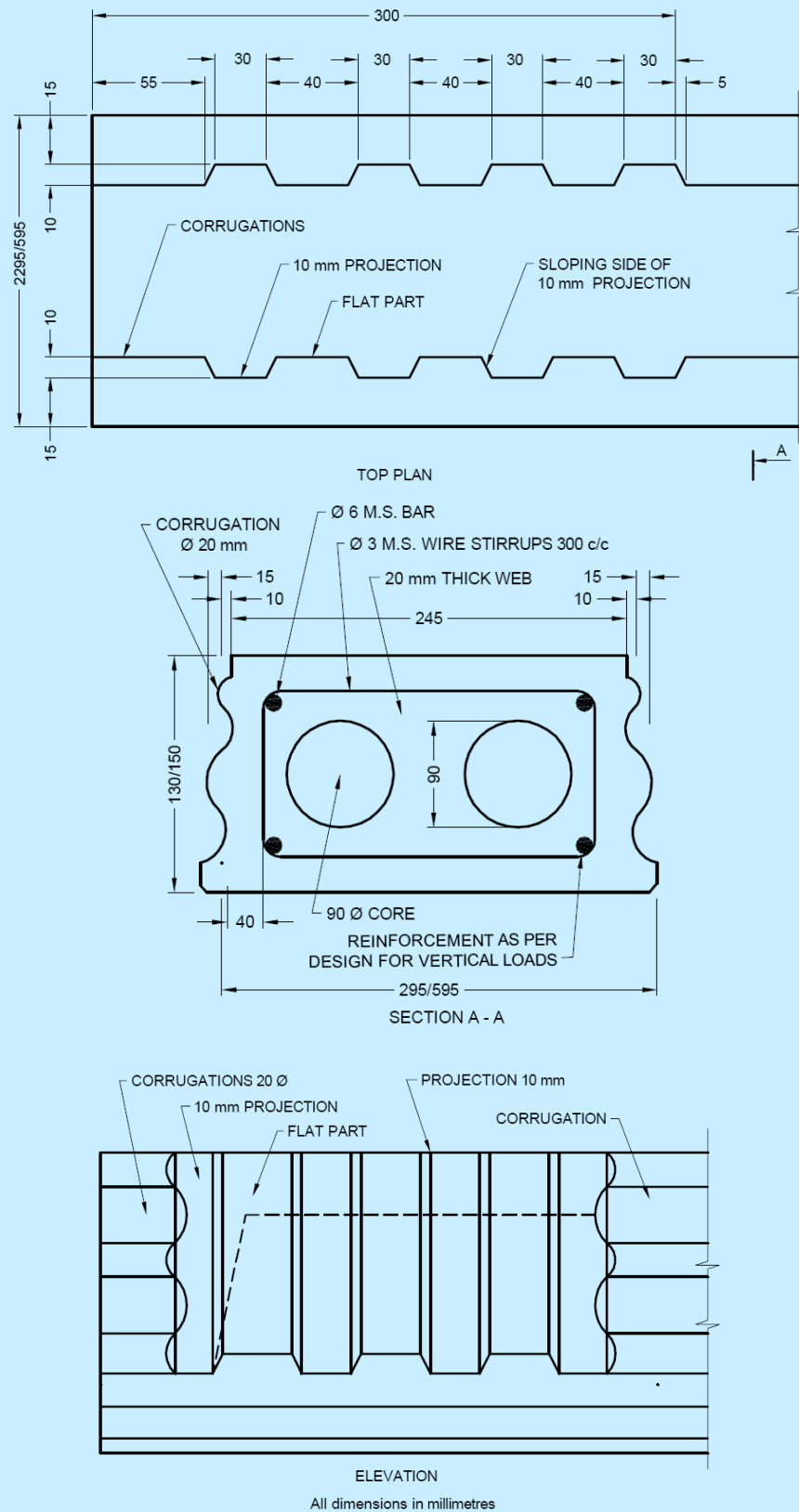


FIG. 32 CORE UNITS

8.6.1.3 Precast reinforced concrete plank and joist scheme for roof/floor

Technical drawing of a concrete beam specimen for testing. The beam is 1500 mm long, 300 mm wide, and 30 mm high. It features a central longitudinal slot 100 mm wide and 30 mm deep. The drawing includes dimensions for the overall size, the slot, and the reinforcement layout.

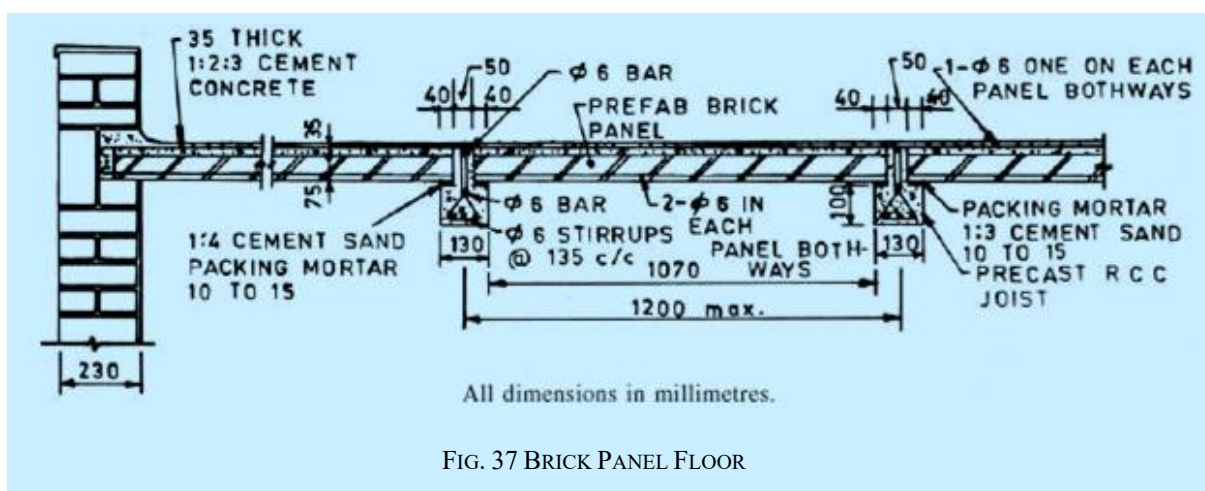
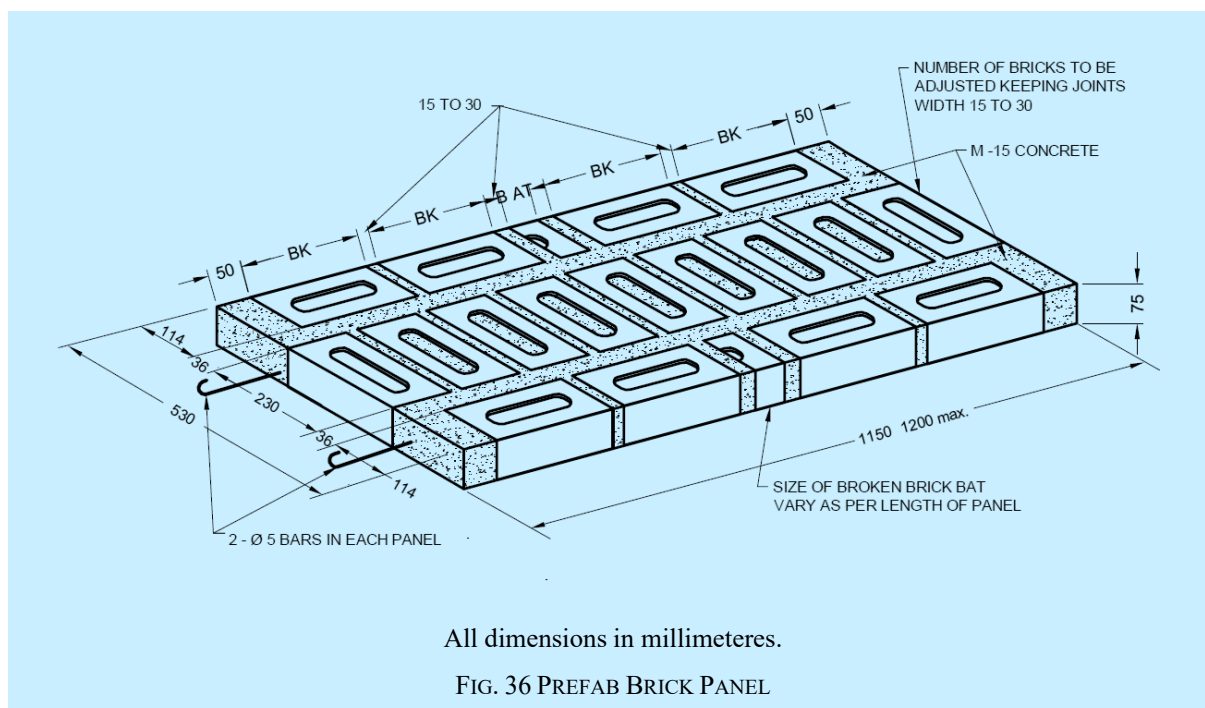
SECTION OF FLOOR

Labels and Dimensions:

- INSITU CONCRETE
- FLOOR FINISH
- 300
- 300
- 70
- 300
- 40
- 40
- 50
- 12 THICK SMOOTH CEMENT PLASTER
- WALL / BEAM
- PARTIALLY PRECAST JOIST
- 60
- 150
- 150
- REINFORCEMENT AS PER DESIGN FOR VERTICAL LOADS

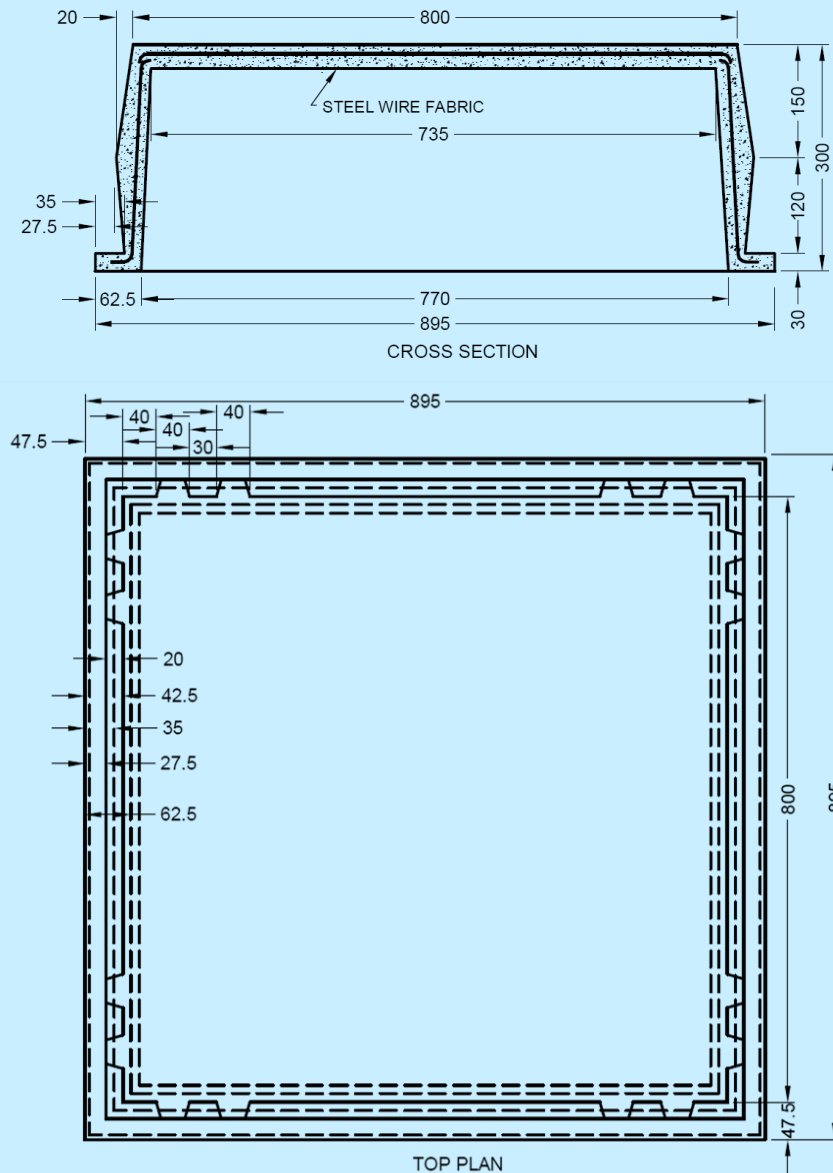
FIG. 35 PRECAST REINFORCED PLANK FLOOR

It consists of prefabricated reinforced brick panels (*see Fig. 36*) supported on Precast reinforced concrete joists with nominal reinforced 35 mm thick structural deck concrete over the brick panels and joists (*see Fig. 37*). The width of the brick panels shall be 530 mm for panels made of bricks of conventional size and 450 mm for panels made of bricks of modular size. The thickness of the panels shall be 75 mm or 90 mm, respectively depending upon whether conventional or modular bricks are used. The length of the panels shall vary depending upon the spacing of the joists, but the maximum length shall not exceed 1.2 m.



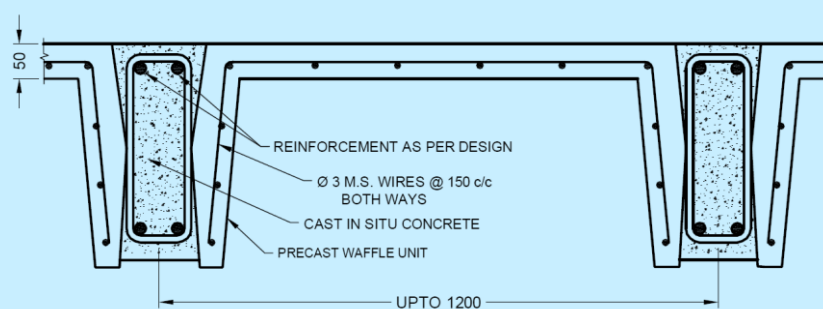
8.6.1.5 Precast reinforced concrete waffle unit roof/floor

Waffle units are of the shape of inverted troughs, square or rectangular in plan, having lateral dimensions up to 1.2 m and depth depending upon the span of the roof/floor to be covered (see [Fig. 38](#) and [Fig. 39](#)). The minimum thickness of flange/web shall be 35 mm. Horizontal projections may be provided on all the four external faces of the unit and the unit shall be so shaped that it shall act monolithic with *in-situ* concrete to ensure load transfer. Vertical castellation, called shear keys, shall be provided on all the four external faces of the precast units to enable them to transfer horizontal shear force from one unit to adjacent unit through *in-situ* concrete filled in the joints between the units. The waffle units shall be laid in a grid pattern with gaps between two adjacent units, and reinforcement, as per design, and structural concrete shall be provided in the gaps between the units in both the directions. The scheme is suitable for two way spanning roofs and floors of buildings having large spans.



All dimensions in millimeters.

FIG. 38 WAFFLE UNITS



All dimensions in millimeters.

FIG. 39 WAFFLE UNIT FLOOR

8.6.2 Seismic Resistance Measures

8.6.2.1 All floors and roofs to be constructed with small precast components shall be strengthened as specified for various categories of buildings in [Table 21](#). The strengthening measures are detailed in [8.6.2.3](#) and [8.6.2.8](#).

8.6.2.2 Vertical castellation, called shear keys, shall be provided on the longitudinal faces of the channel, cored and waffle units to enable them to transfer horizontal shear force from one unit to the adjacent unit through the *in-situ* concrete filled in the joints between the units. The minimum percentage of area of shear keys as calculated below, on each face of the unit, shall be 15.

Shear keys shall have a minimum width of 40 mm at its root with the body of the component and shall be to the full height of the component and preferably at uniform spacing. Percentage of area of shear keys shall be calculated as:

$$\frac{\text{No. of shear keys on one face of the component} \times 40}{\text{Length of the face of the component, in mm}} \times 100$$

Table 21 Strengthening Measures for Floors/Roofs with Small Precast Components

(Clauses [8.6.2.1](#), [8.6.2.4](#), [8.6.2.5](#), [8.6.2.6](#), [8.6.2.7](#) and [8.6.2.8](#))

Sl No.	Building Category	Number of Storeys	Strengthening to be Provided in Floor/Roof with			
			Channel/Cored Unit	R.C. Planks and Joists	Brick Panels and Joists	Waffle Units
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	B	1 to 34	aa, c	aa, c	aa, d	aa
ii)	C	1 and 2 3 and 4	a, ba, b, c	aa, c	aa, d	aa, e
iii)	D	1 to 4	a, b c	a, c	a, d	a, c, e
iv)	E	1 to 3	a, b, c	a, c	a, d	a, c, e
NOTES						
1 Tie beam as per 8.6.2.3 ;						
2 Reinforcing bars of precast unit and tied to tie beam reinforcement as per 8.6.2.4 ;						
3 Reinforced deck concrete as per 8.6.2.5 ;						
4 Reinforced deck concrete as per 8.6.2.6 ; and						
5 Reinforced bars in joint between precast waffle units tied to tie beam reinforcement as per 8.6.2.7 .						

8.6.2.3 Tie beam (*see* [Table 20](#)) is a beam provided all round the floor or roof to bind together all the precast components to make it a diaphragm. The beams shall be to the full width of the supporting wall or beam less the bearing of the precast components. The depth of the beam shall be equal to the depth of the precast components plus the thickness of structural deck concrete, where used over the components. The beam shall be made of cement concrete of grade not leaner than M15 and shall be reinforced as indicated in [Table 18](#). If depth of tie is more than 75 mm, equivalent reinforcement shall be provided with one bar of minimum diameter 8 mm at each corner. Tie beams shall be provided on all longitudinal and cross walls. Typical details of the beams are shown in [Fig. 40](#) to [Fig. 44](#).

NOTE — Adequate edge support say 60 mm, shall be provided to precast element on the wall so as to avert its slippage during seismic ground motion.

8.6.2.4 Top reinforcement in the channel or cored units (*see* [Table 21](#)) shall be projected out at both the ends for anchorage length and tied to tie beam reinforcement.

8.6.2.5 Structural deck concrete (*see Table 21*) of grade not leaner than M15 shall be provided over precast components or act monolithic. Wherever, deck concrete is to be provided, the top surface of the components shall be finished rough. Cement slurry with 0.5 kg of cement/m² of the surface area shall be applied over the components immediately before laying the deck concrete and the concrete shall be compacted using plate vibrators. The minimum thickness of deck concrete shall be 35 mm or 40 mm reinforced with 6 mm diameter bars and 150 mm apart both ways and anchored into the tie beam placed all round. The maximum size of coarse aggregate used in deck concrete shall not exceed 12 mm.

NOTE — Under conditions of economic constraints, the deck concrete itself could serve as floor finish. The concrete is laid in one operation (*see Fig. 44*) without joints.

8.6.2.6 The deck concrete normally used over the brick panel with joist floor shall be reinforced with 6 mm diameter bars spaced 150 mm apart both ways (*see Table 21*).

8.6.2.7 For floors/roofs with precast waffle units, two 16 mm diameter high strength deformed bars shall be provided as top reinforcement in the joints between waffle units, in addition to reinforcement required for taking bending moment for vertical loads. This reinforcement (*see Table 21*) shall be fixed to tie beam reinforcement.

8.6.2.8 In case of floors/roofs with precast components other than those indicated in *Table 21*, the buildings shall be analysed for maximum expected seismic forces and the floor/roof shall be designed to act as diaphragm and take care of the resulting forces.

8.7 Timber Construction

8.7.1 Timber has higher strength per unit weight and is, therefore, very suitable for earthquake resistant construction. Materials, design and construction in timber shall generally conform to good practice [C-4(9)].

8.7.2 Timber (with masonry in-fill) construction shall generally be restricted to two storeys with or without the attic floor.

8.7.3 In timber (with masonry in-fill) construction attention shall be paid to fire safety against electric short circuiting, kitchen fire, etc.

8.7.4 The superstructure of timber buildings shall be made rigid against deformations by adopting suitable construction details at the junctions of the framing members and in wall panels as given in *8.7.6* to *8.7.10* so that the construction as a whole behaves as one unit against earthquake forces.

8.7.5 Foundations

8.7.5.1 Timber construction shall preferably start above the plinth level, the portion below shall be in masonry or concrete.

8.7.5.2 The superstructure may be connected with the foundation in one of the two ways as given below:

- a) The superstructure may simply rest on the plinth masonry, or in the case of small buildings of one storey having plan area less than about 50 m², it may rest on firm plane ground so that the building is free to slide laterally during ground motion; and

NOTES

1 Past experience has shown that superstructure of the buildings not fixed with the foundation escaped collapse even in a severe earthquake although they were shifted sideways.

2 Where fittings for water supply or water borne sanitation from the house are to be installed, proper attention should be given to permit movement so as to avoid fracture of damage to pipes.

- b) The superstructure may be rigidly fixed into the plinth masonry or concrete foundation as given in *Fig. 45* or in case of small building having plan area less than 50 m², it may be fixed to vertical poles embedded into the ground. In each case the building is likely to move along with its foundation. Therefore, the superstructure shall be designed to carry the resulting earthquake shears.

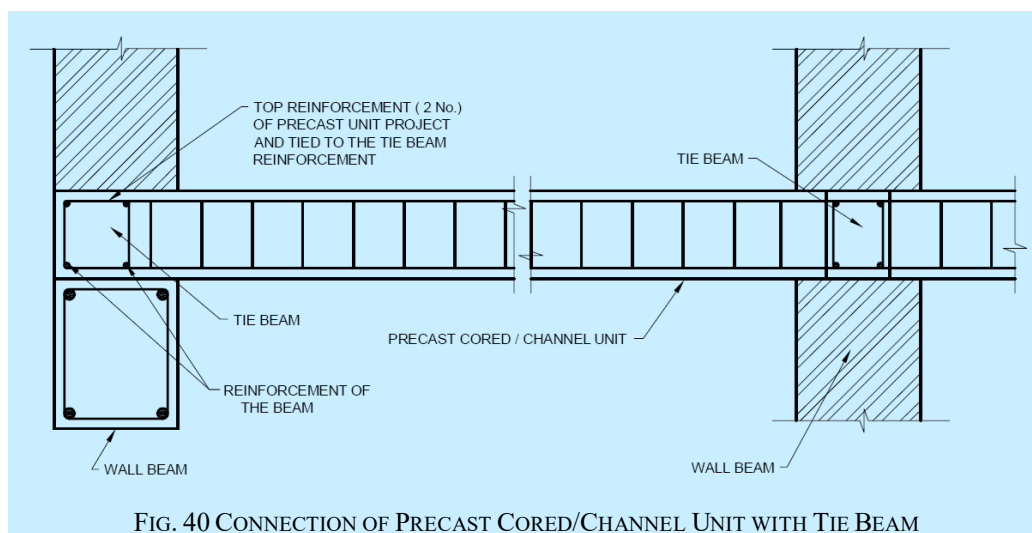


FIG. 40 CONNECTION OF PRECAST CORED/CHANNEL UNIT WITH TIE BEAM

8.7.6 Types of Framing

The types of construction usually adopted in timber building are as follows:

- a) Stud wall construction; and
- b) Brick nogged timber frame construction.

8.7.7 Stud Wall Construction

8.7.7.1 The stud wall construction consists of timber studs and corner posts framed into sills, top plates and wall plates. Horizontal struts and diagonal braces are used to stiffen the frame against lateral loads. The wall covering may consist of *EKRA*, timber or like. Typical details of stud walls are shown in [Fig. 46](#). Minimum sizes and spacing of various members used are specified in [8.7.7.2](#) to [8.7.7.10](#).

8.7.7.2 The timber studs for use in load bearing walls shall have a minimum finished size of 40 mm × 90 mm and their spacing shall not exceed those given in [Table 22](#).

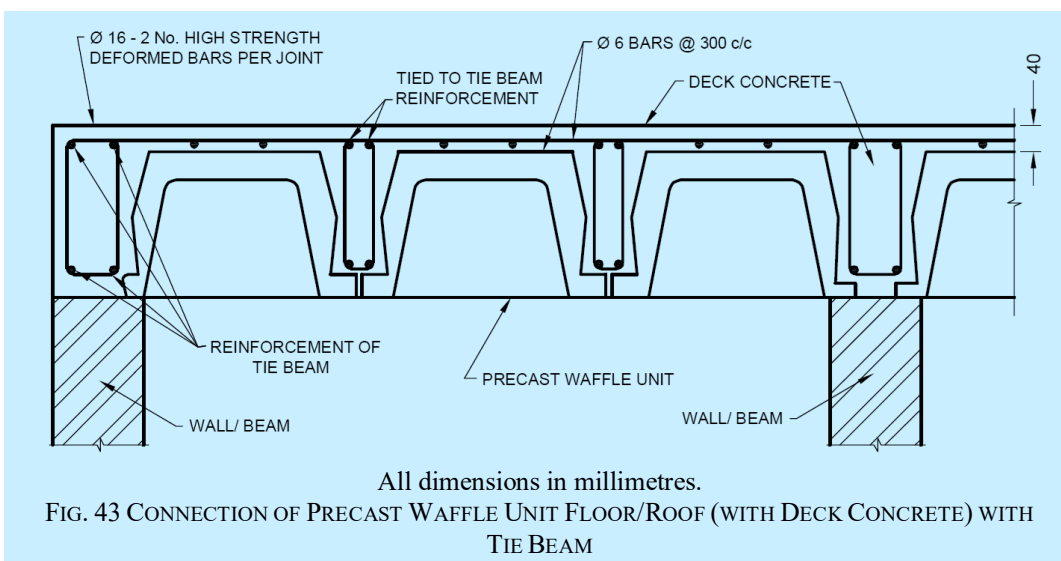
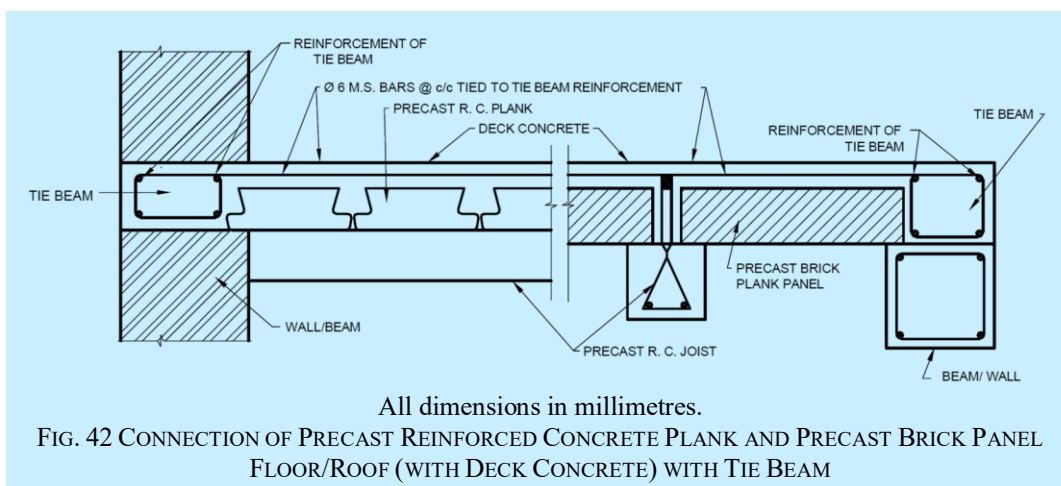
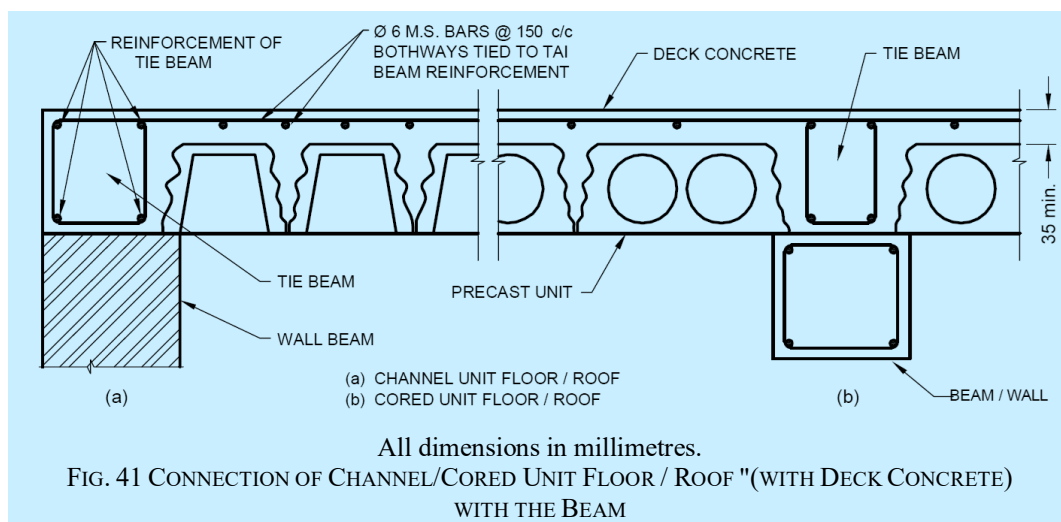
8.7.7.3 The timber studs in non-load bearing walls shall not be less than 40 mm × 70 mm in finished cross section. Their spacing shall not exceed 1 m.

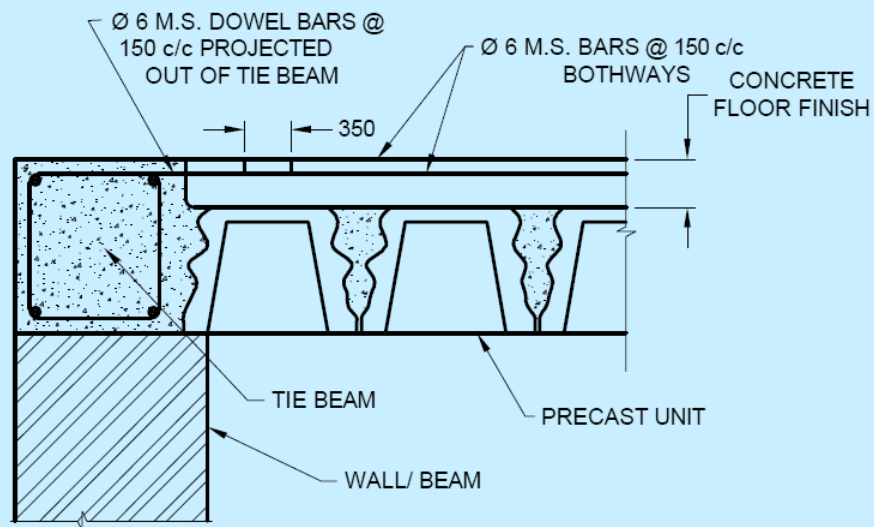
8.7.7.4 There shall be at least one diagonal brace for every 1.6 m × 1 m area of load bearing walls. Their minimum finished sizes shall be in accordance with [Table 23](#).

8.7.7.5 The horizontal struts shall be spaced not more than 1 m apart. They shall have a minimum size of 30 mm × 40 mm for all locations.

8.7.7.6 The finished sizes of the sill, the wall plate and top plate shall not be less than the size of the studs used in the wall.

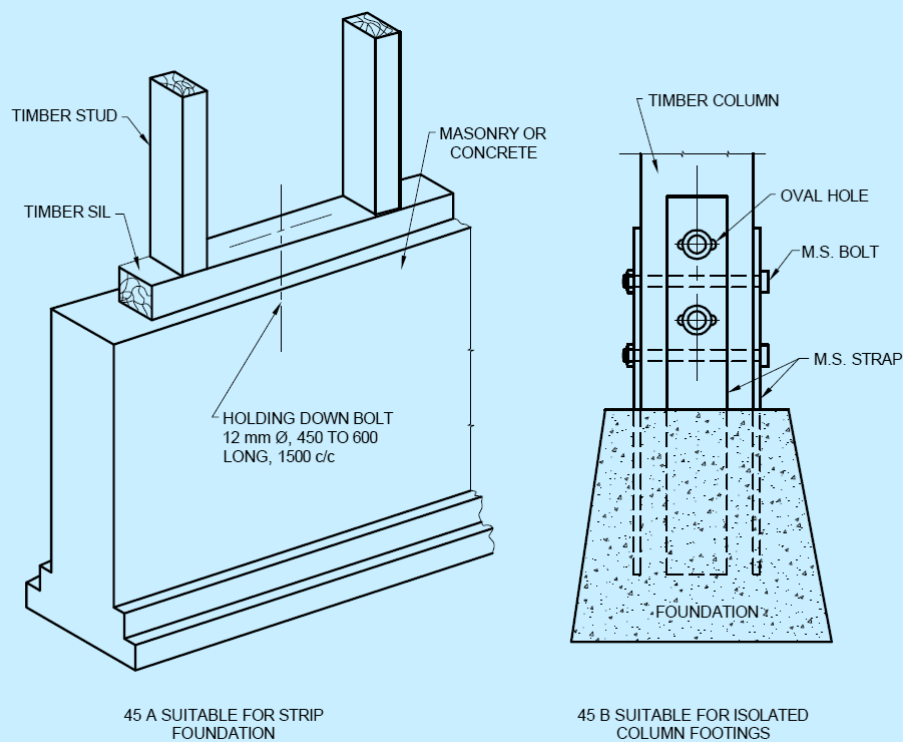
8.7.7.7 The corner posts shall consist of three timbers, two being equal in size to the studs used in the walls meeting at the corner and the third timber being of a size to fit so as to make a rectangular section (*see* [Fig. 46](#)).





All dimensions in millimetres.

FIG. 44 PROVISION OF REINFORCEMENT IN CONCRETE FLOOR FINISH TIE BEAM



All dimensions in millimetres.

FIG. 45 DETAILS OF CONNECTION OF COLUMN WITH FOUNDATION

Table 22 Maximum Spacing of 40 mm × 90 mm Finished Size Studs in Stud Wall Construction
(Clauses [8.7.7.2](#))

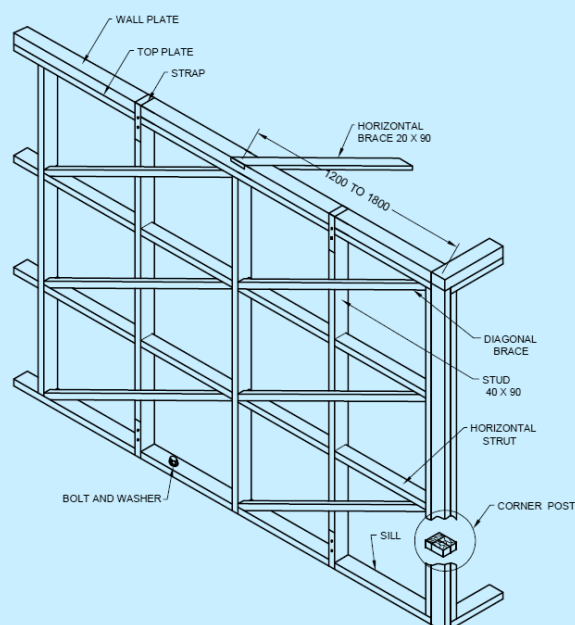
Sl No.	Group of Timber (Grade I ¹⁾)	Single Storeyed or First Floor of the Double Storeyed Buildings		Ground Floor of Double Storeyed Buildings	
		Exterior Wall mm	Interior Wall mm	Exterior Wall mm	Interior Wall mm
(1)	(2)	(3)	(4)	(5)	(6)
i)	Group A, B	1000	800	500	400
ii)	Group C	1000	1000	500	500

¹⁾ Grade I timbers as defined in Table 5 of [C-4(9)].

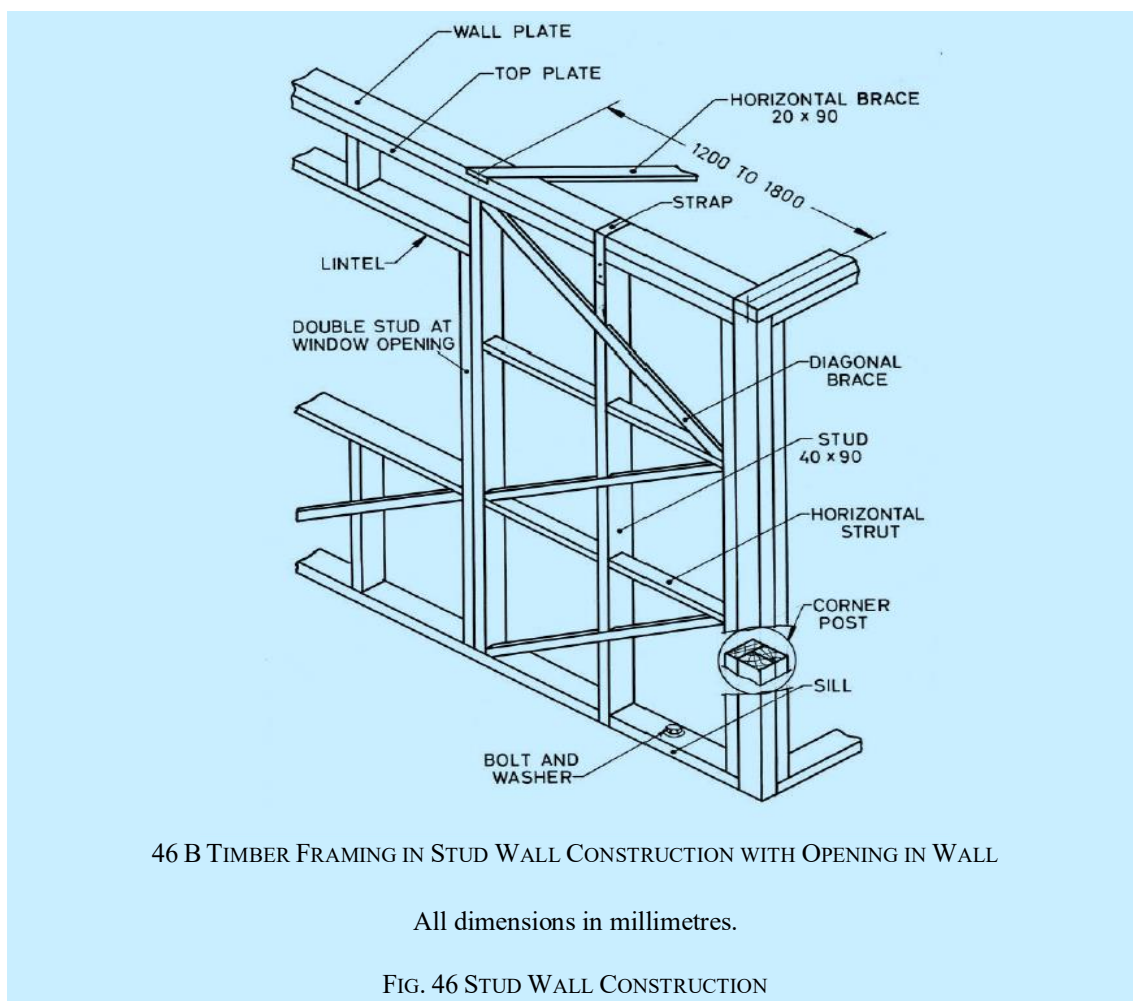
Table 23 Minimum Finished Sizes of Diagonal Braces
(Clauses [8.7.7.4](#), [8.7.8.2](#) and [8.7.8.4](#))

Sl No.	Building Category	Group of Timber (Grade I ¹⁾)	Single Storeyed or First Floor of Double Storeyed Buildings		Ground Floor of Double Storeyed Building	
			Exterior Wall mm × mm	Interior Wall mm × mm	Exterior Wall mm × mm	Interior Wall mm × mm
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	B, C	All	20 × 40	20 × 40	20 × 40	20 × 40
ii)	D and E	Group A and Group B	20 × 40	20 × 40	20 × 40	30 × 40
		Group C	20 × 40	30 × 40	30 × 40	30 × 40

¹⁾ Grade I timber as defined in Table 5 of [C-4(9)].



46 A TIMBER FRAMING IN STUD WALL CONSTRUCTION WITHOUT OPENING IN WALL



8.7.7.8 The diagonal braces shall be connected at their ends with the stud wall members by means of wire nails having 6 gauge (4.88 mm diameter) and 100 mm length. Their minimum number shall be 4 nails for 20 mm × 40 mm braces and 6 nails for 30 mm × 40 mm braces. The far end of nails may be clutched as far as possible.

8.7.7.9 Horizontal bracing shall be provided at corners of T-junctions of walls at sill, first floor and eaves levels. The bracing members shall have a minimum finished size of 20 mm × 90 mm and shall be connected by means of wire nails to the wall plates at a distance between 1.2 m and 1.8 m measured from the junction of the walls. There shall be a minimum number of six nails of 6 gauge (4.88 mm diameter) and 10 cm length with clutching as far ends.

8.7.7.10 Unsheathed studding shall not be used adjacent to the wall of another building. The studding shall be sheathed with close jointed 20 mm or thicker boards.

8.7.8 Brick Nogged Timber Frame Construction

8.7.8.1 The brick nogged timber frame consists of intermediate verticals, columns, sills, wall plates, horizontal nogging members and diagonal braces framed into each other and the space between framing members filled with tight-fitting brick masonry in stretcher bond. Typical details of brick nogged timber frame construction are shown in [Fig. 47](#). Minimum sizes and spacing of various elements used are specified in [8.7.8.2](#) to [8.7.8.9](#).

8.7.8.2 The vertical framing members in brick nogged load bearing walls shall have minimum finished sizes as specified in [Table 23](#).

8.7.8.3 The minimum finished size of the vertical members in non-load bearing walls shall be 40 mm × 100 mm spaced not more than 1.5 m apart.

8.7.8.4 The sizes of diagonal bracing members shall be the same as in [Table 23](#).

8.7.8.5 The horizontal framing members in brick-nogged construction shall be spaced not more than 1 m apart. Their minimum finished sizes shall be in accordance with [Table 24](#) and [Table 25](#).

8.7.8.6 The finished sizes of the sill, wall plate and top plate shall be not less than the size of the vertical members used in the wall.

8.7.8.7 Corner posts shall consist of three vertical timbers as described in [8.7.7.7](#).

8.7.8.8 The diagonal braces shall be connected of their ends with the other members of the wall by means of wire nails as specified in [8.7.7.8](#).

8.7.8.9 Horizontal bracing members of corners of T-junctions of wall shall be as specified in [8.7.7.9](#).

Table 24 Minimum Finished Sizes of Vertical in Brick Nogged Timber Frame Construction						
(Clause 8.7.8.5)						
SI No.	Spacing	Group of Timber (Grade I ¹⁾)	Single Storeyed or First Floor of Double Storeyed Buildings		Ground Floor of Double Storeyed Buildings	
	m		Exterior Wall	Interior Wall	Exterior Wall	Interior Wall
(1)	(2)	(3)	mm × mm	mm × mm	mm × mm	mm × mm
			(4)	(5)	(6)	(7)
i)	1	Group A, B	50 × 100	50 × 100	50 × 100	50 × 100
		Group C	50 × 100	70 × 100	70 × 100	90 × 100
ii)	1.5	Group A, B	50 × 100	70 × 100	70 × 100	80 × 100
		Group C	70 × 100	80 × 100	80 × 100	100 × 100

¹⁾ Grade I timber as defined in Table 5 of [C-4(9)].

Table 25 Minimum Finished Size of Horizontal Nogging Members		
(Clause 8.7.8.5)		
SI No.	Spacing of Verticals	Size
(1)	(2)	(3)
i)	1.5	70 × 100
ii)	1	50 × 100

8.7.9 Notching and Cutting

8.7.9.1 Timber framing frequently requires notching and cutting of the vertical members. The notching or cutting should in general be limited to 20 mm in depth unless steel strips are provided to strengthen the notched face of the members. Such steel strips, where necessary shall be at least 1.5 mm thick and 35 mm wide extending at least 150 mm beyond each side of the notch or cut and attached to the vertical member by means of bolts or screws at each end.

8.7.9.2 The top plate, the wall plate or the sill of a wall may be notched or cut, if reinforcing strip of iron is provided as specified in [8.7.9.1](#). In case the member is notched or cut not to exceed 40 mm in depth, such reinforcing strip may be placed along the notched edge only. Where the notch or cut is more than 40 mm in depth or the member is completely cut through, such reinforcing strips shall be placed on both edges of the member. The details of notching and cutting are shown in [Fig. 48](#).

8.7.9.3 Joints in timber shall preferably be bound by metallic fasteners.

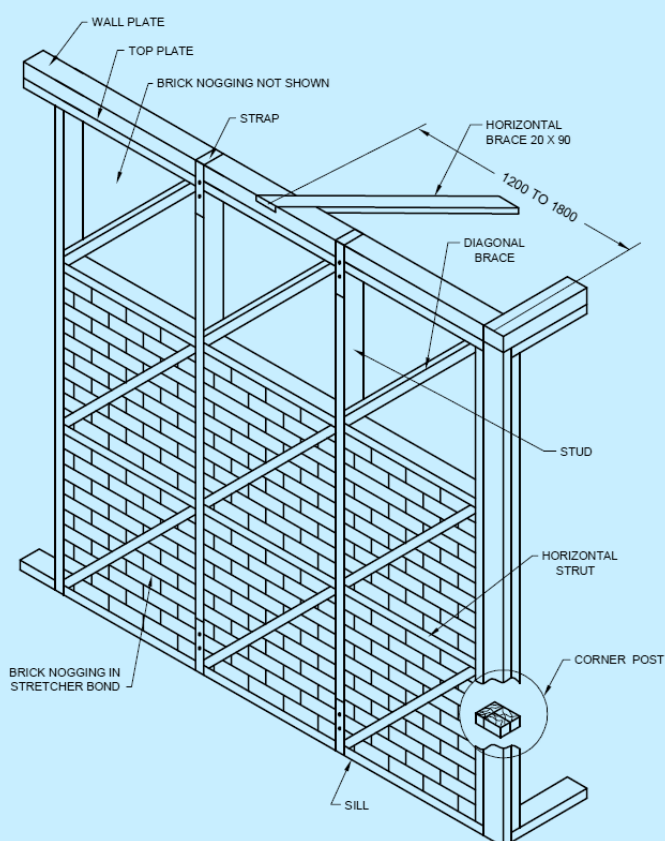
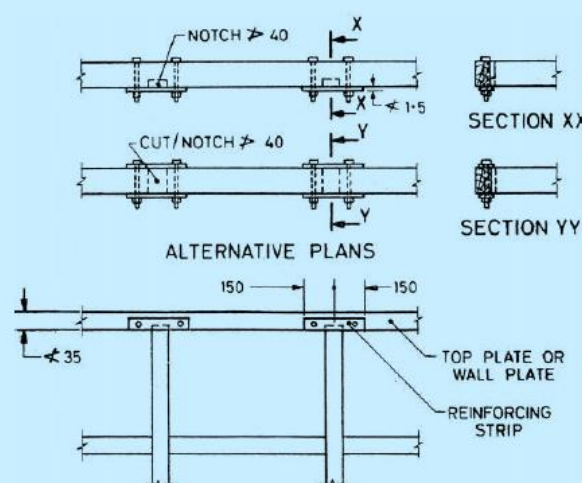


FIG. 47 BRICK NOGGED TIE FRAME CONSTRUCTION



All dimensions in millimetres.

FIG. 48 NOTCHING AND CUTTING

8.7.10 Bridging and Blocking

8.7.10.1 All wooden joists shall have at least one row of cross bridging for every 3.5 m length of span. The cross-section of the bridging member shall be a minimum of 40 mm × 70 mm and the member shall be screwed or nailed to the joists.

8.7.10.2 All spaces between joists shall be blocked at all bearing with solid blocks not less than 40 mm thick and the full depth of the joists. The block shall be screwed or nailed to the joists as well as to the bearings.

9 CONFINED MASONRY WALLS (CMW)

9.1 Design Considerations

9.1.1 General

The provisions given in [9.1.2](#) to [9.1.6](#) apply in addition to those specified in [4](#).

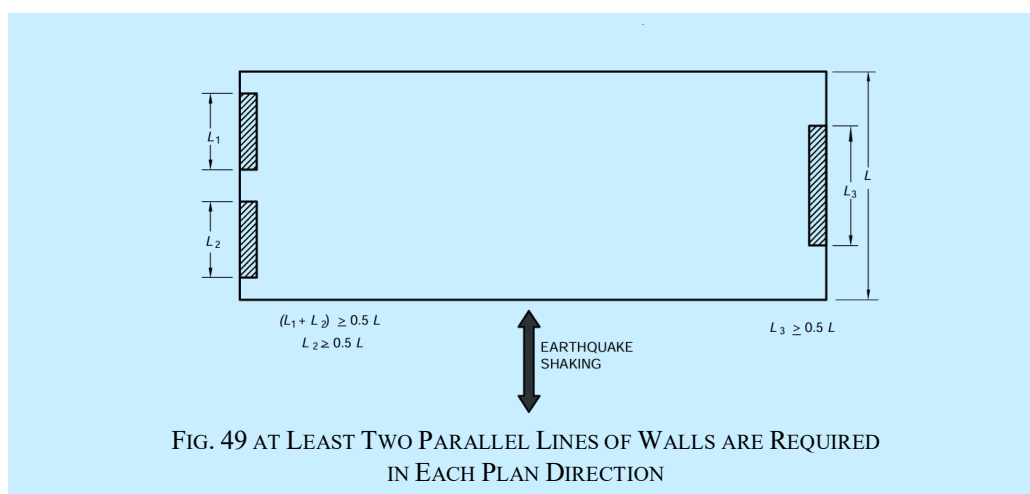
9.1.2 Structural Integrity

Intersecting walls shall be joined together to resist the effect of gravity and lateral loads. The walls shall be adequately connected to elements which provide lateral support, such as floors and roofs.

9.1.3 Building Configuration

A regular building configuration is one of the key requirements for satisfactory earthquake performance. The following recommendations related to building plan shape shall be followed:

- The building's length-to-width ratio in plan shall not exceed 4 and storey height shall be less than 4 m;
- The walls shall be built in a symmetrical manner with regard to the horizontal axes through the centre of the building plan. Some walls shall be placed on the perimeter also to increase resistance of the building to twisting (torsion) during an earthquake;
- There shall be at least two lines of walls in each orthogonal direction of the building plan, and the walls along each line shall extend over at least 50 percent of the building dimension in the direction considered at each storey level ([Fig. 49](#));
- The walls shall be continuous up to the full building height. There shall be no vertical offsets in the walls along the building height;
- The openings (doors and windows) shall be placed in the same position on each floor;
- The total cross-sectional area of all walls in two adjacent floors shall not be different by more than 30 percent; and
- The structural system in a building shall be the same in all storeys and in both principal plan directions.



9.1.4 Minimum Design Dimensions and Placements of Confining Elements

Requirements regarding spacing of tie-columns and tie-beams are presented in Fig. 50. Minimum dimension of tie-column and tie-beam shall be 150 mm along the wall direction and as much as the thickness of the wall in the direction perpendicular to it.

9.1.4.1 Placement of reinforced concrete tie-columns and tie-beams

Tie-columns should be provided at:

- intersections of walls;
- intermediate locations in longer walls, where spacing shall not exceed lesser than $1.5 h$ (where h is the clear wall height between floors) or 4 m; and
- free ends of wall panels that provide lateral load resistance to the building (Fig. 51).

Also, reinforced concrete tie-beams must be provided at the top of each wall and spacing between tie-beams should be preferably less than 4 m.

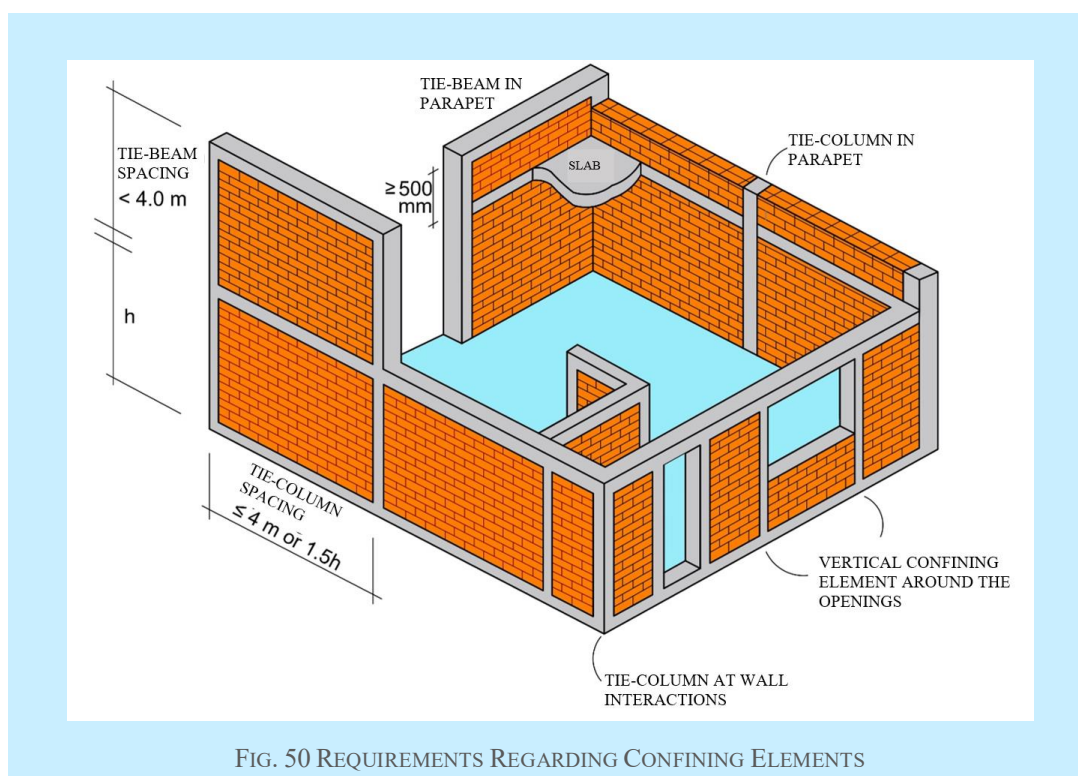
9.1.4.2 Horizontal lintel bands

When the unsupported height of the wall h is greater than 2.5 m, a continuous horizontal lintel band shall be provided over the openings. A continuous horizontal band is not essential for $h < 2.5$ m, but a lintel beam shall be provided above the openings.

9.1.5 Minimum Dimensions of Masonry Walls

The following shall be applicable:

- the wall thickness (t) shall not be less than 110 mm; and
- the maximum wall height-to-thickness (h/t) ratio shall not exceed 25, where h is the height between floors.



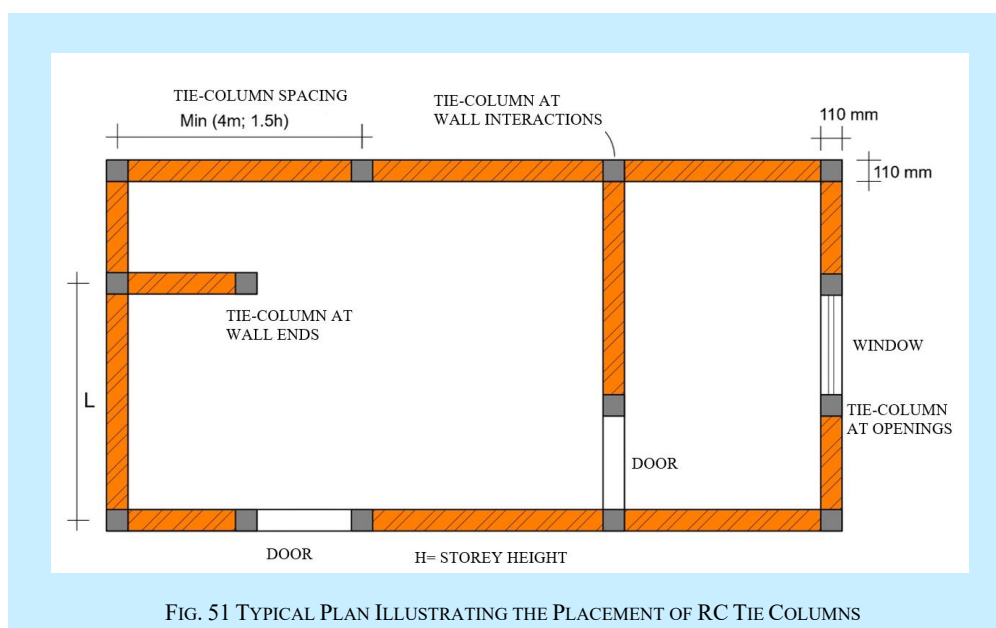


FIG. 51 TYPICAL PLAN ILLUSTRATING THE PLACEMENT OF RC TIE COLUMNS

9.1.5.1 Aspect Ratio of Confined Masonry Walls

The height-to-width ratio of a wall that is a part of the lateral force-resisting system shall be less than or equal to 2.

9.1.5.2 Parapets

RC tie-columns and tie-beams shall extend to the top of the parapet [Fig. 51](#). When a parapet is not confined by tie-beams, its height shall not be more than 3 times its thickness. For parapet walls with top tie-beams, the height shall not be more than 1.2 m.

9.1.5.3 Gable Walls

The top of gable shall be confined with reinforced concrete tie-beams, and tie-columns located at the middle of the gable wall shall be extended from the lower floor to the top of gable wall (whenever applicable), as shown in [Fig. 52\(A\)](#). Alternatively, a gable portion of the wall can be made of timber or other light-weight material that is continuous [Fig. 52\(B\)](#).

9.1.6 Walls with Openings

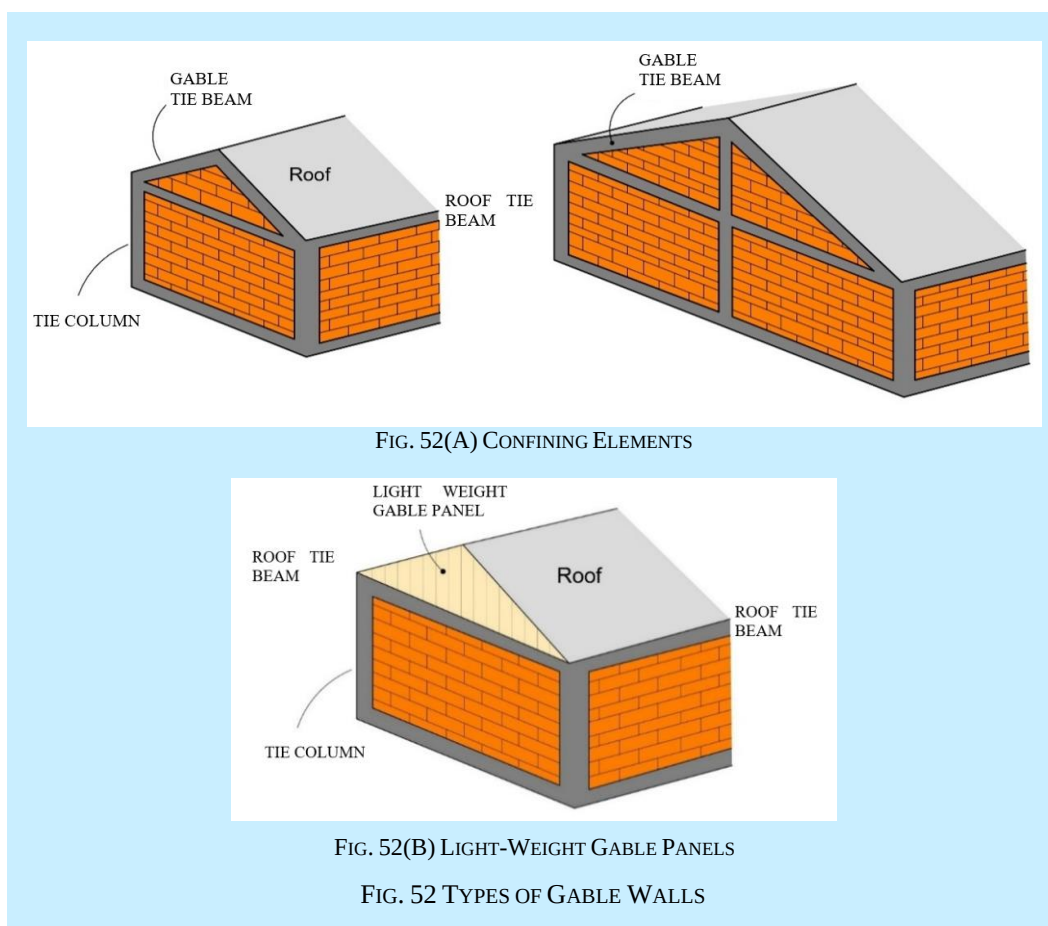
9.1.6.1 Size of opening

The presence of large openings may have a negative effect on earthquake performance of confined masonry buildings, especially if openings are not confined. A large opening has a total area greater than 10 percent of the wall panel area, while a small opening has a total area less than or equal to 10 percent of the wall panel area.

9.1.6.2 Walls with large openings

The following three approaches shall be followed in walls with large openings:

- when reinforced concrete tie-columns are not provided at the ends of an opening, the panel shall not be considered as confined and its contribution to earthquake resistance of the building shall be disregarded but shall be strengthened [see [Fig. 53\(A\)](#)];
- when reinforced concrete tie-columns are provided at the opening, and the confined masonry panels are considered to contribute to earthquake resistance of the building, the aspect ratio H/L of these panels shall be less than 2.0 [Fig. 53\(B\)](#). Better performance can be achieved by providing both sill and lintel bands below and above the openings, respectively; and
- if total area of openings is greater than 25 percent, both openings and masonry piers shall be confined with horizontal and vertical confining elements [Fig. 53\(C\)](#).



9.1.6.3 Walls with small and single openings

The following three approaches shall be followed in walls with small openings:

- when the opening is located outside the diagonals [Fig. 54\(a\)](#), it shall be ignored, and the entire wall cross-sectional area considered for earthquake resistance;
- when an opening is located at the intersection of the panel diagonals [[Fig. 54\(b\)](#)], the panel cross-sectional area (A_T) considered for earthquake resistance should exclude the opening length; and
- when an opening is located close to one end of the panel, the panel cross-sectional area (A_T) considered in earthquake resistance shall use the larger pier length [[Fig. 54\(c\)](#)].

9.2 Structural Design

9.2.1 Design Criteria

Design of structural elements shall be performed as per the working stress method.

9.2.2 Structural Design as per Working Stress Method

9.2.2.1 Load combinations

The working stress method shall be employed for the structural design of SEs of confined masonry buildings for the load combinations specified in 5 of Part C/Section 1 of this NBCS.

9.2.2.2 Transformed section properties

Confined masonry wall panel shall be treated as a composite structural element which consists of a masonry wall and reinforced concrete tie-columns, and designed as a transformed section. Modular ratio (m) shall be taken as the ratio of elastic modulus for concrete and masonry. The modulus of elasticity of concrete can be taken as per

Part C/Section 5 of this NBCS and modulus of elasticity of masonry as per 5.4.6. Transformed section properties of a confined masonry panel used in design are illustrated in Fig. 55.

9.2.2.3 Permissible compressive force

Compressive force in confined masonry due to axial load shall not exceed that given by:

$$P_o = (0.25 f_m A_m + f_{cc} A_c + 0.65 A_s f_s) k_s,$$

where

- A_c = cross-sectional area of concrete excluding reinforcing steel;
- A_m = net area of masonry;
- A_s = area of steel;
- f_{cc} = permissible stress in concrete in direct compression as per Part C/Section 5 of this NBCS;
- f_m = compressive strength of masonry as per 5.4.5;
- f_s = permissible steel tensile stress as per Part C/Section 5 of this NBCS; and
- k_s = stress reduction factor as per Table 10.

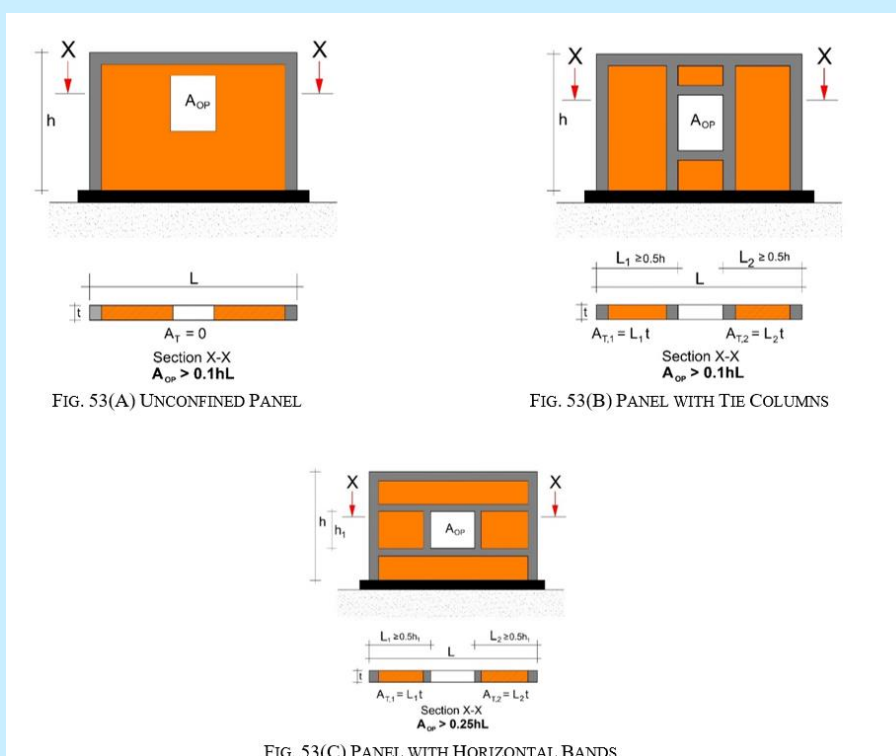


FIG. 53 MASONRY WALLS WITH LARGE OPENING

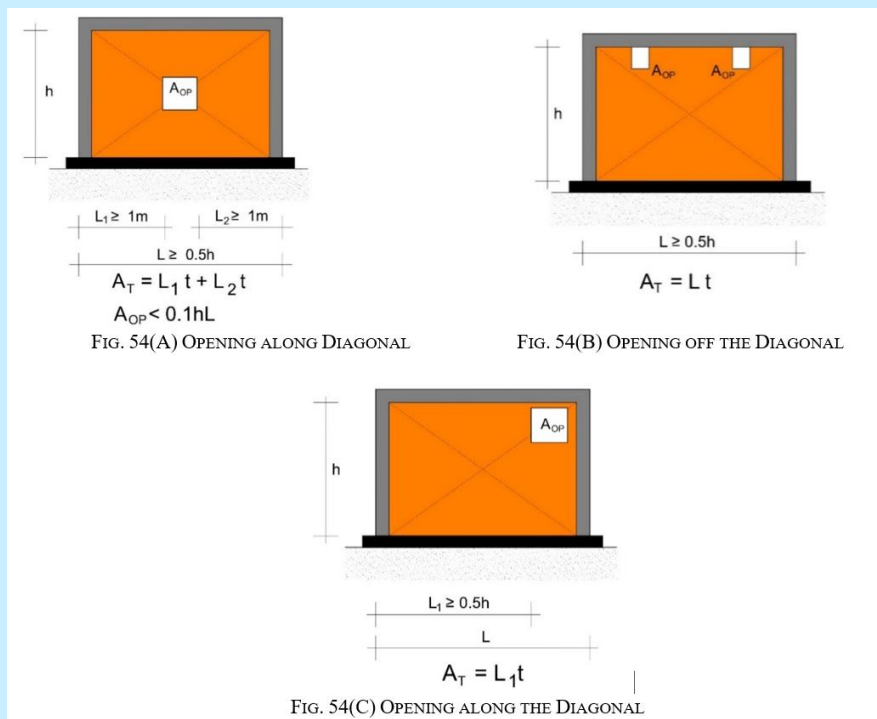
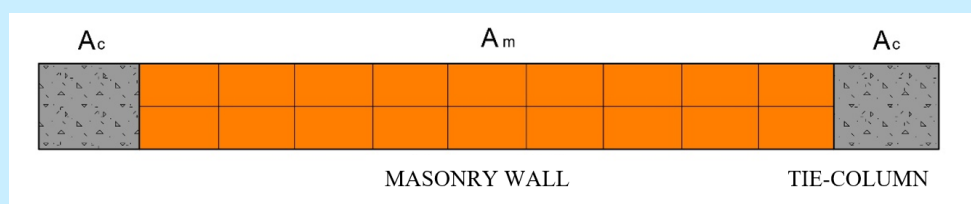
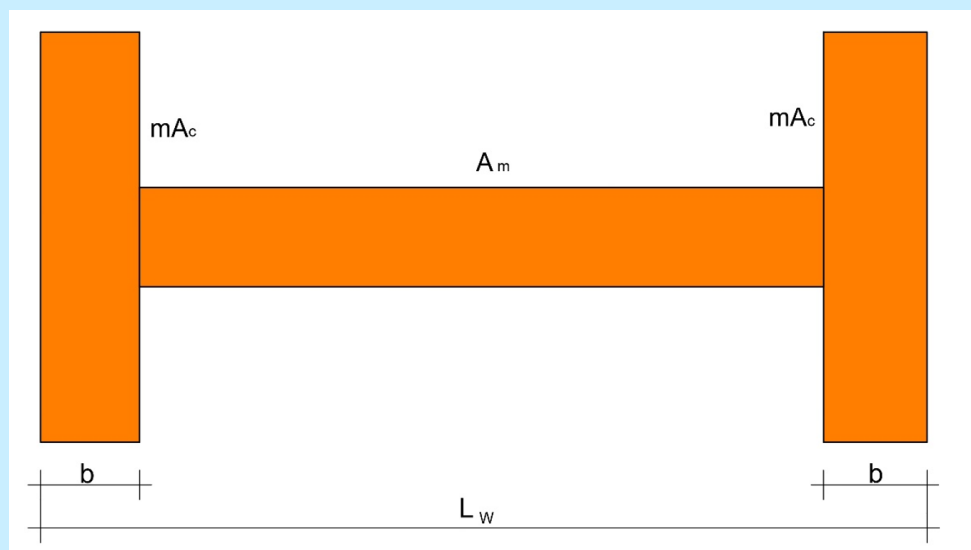


FIG. 54 CONFINED MASONRY WALL PANEL WITH SMALL OPENING



55A) MASONRY WALL



55B) WALL TO BE CONSIDERED IN DESIGN

FIG. 55 TRANSFORMED SECTION PROPERTIES OF A CONFINED MASONRY PANEL

Table 26 Maximum Allowable Shear Stress τ_{cm} of Confined Masonry

(Clause 9.2.2.6)

Sl No.	$\left(\frac{M}{Vd}\right)$	f_t (MPa)	τ_{cm} (MPa)
(1)	(2)	(3)	(4)
i)	< 1	$\frac{1}{36}\left(4 - \frac{M}{Vl_w}\right)$	$0.4 - 0.2\left(\frac{M}{Vl_w}\right)$
ii)	> 1	$0.083\sqrt{f_m}$	0.4

9.2.2.4 Combined permissible axial and flexural compressive stress

For walls subjected to combined axial load and flexure, the compressive stress in masonry due to combined action of axial load and bending shall not exceed $1.25 f_a$ and compressive stress in masonry due to axial load only shall not exceed f_a .

9.2.2.5 Permissible tensile stress

Provisions of 5.4.2 shall apply.

9.2.2.6 Permissible shear stress

The permissible shear stress τ_{cm} for confined masonry walls shall be as per Table 26. If there is tension in any part of a section of masonry, the area under tension shall be ignored while estimating the shear stress demand on the cross-section.

When designing buildings as per the working stress method, the load combinations given in Table 27 shall be used. These combinations are provided considering short-term effects. When assessing the long-term effects due to creep, the dead load, and that part of imposed load likely to be permanent alone shall be used.

Table 27 Load Combinations to be Considered in Design of Buildings as per Working Stress Method

(Clause 9.2.2.6)

Sl No.	Combination Case	Serviceability Load Combination
(1)	(2)	(3)
i)	1	DL + IL
ii)	2	DL + EL
iii)	3	DL + 0.8 IL + 0.8 EL

9.2.2.7 Design assumptions

The following assumptions shall be made in design of confined masonry wall panels subjected to axial and/or flexural loads:

- masonry behaves like a homogeneous material;
- strain distribution across the cross-section of the wall varies linearly;
- tensile stresses are resisted only by reinforcing steel (masonry and concrete tensile resistance is ignored);
- bond is perfect between steel reinforcement and adjacent concrete;
- maximum compressive strains in masonry (ϵ_m) and concrete (ϵ_c) shall be taken equal to 0.004 0 and 0.003 0, respectively; and

the stress-strain curve of masonry is linear, unless a more accurate relation is determined by experimental study of masonry prisms.

9.2.2.8 Stress-strain curves for concrete, masonry and steel

The relationship between the compressive stress distribution in concrete and the strain in concrete may be assumed to be rectangle, trapezoid, parabola, or any other shape which results in prediction of strength in substantial agreement with the results of test. An acceptable stress-strain curve is given in Part C/Section 5 of this NBCS. But the ultimate strain in concrete shall be taken as 0.003 0 in place of 0.003 5. For design purposes, the compressive strength of concrete in the structure shall be assumed to be 0.67 times the characteristic strength. The partial safety factor γ_m of 1.5 shall be applied in addition to this.

The stresses in the reinforcement are derived from representative stress-strain curve for the type of steel used. Typical curves are given in Part C/Section 5 of this NBCS. For design purposes, the partial safety factor γ_m of 1.15 shall be applied.

The relation between the compressive stress-strain in masonry may be assumed to be rectangle, trapezoid, parabola, or any other shape which results in prediction of strength in substantial agreement with the results of test. An acceptable stress-strain curve is given in [Fig. 56](#). For design purposes the partial safety factor γ_m of 2.0 shall be applied.

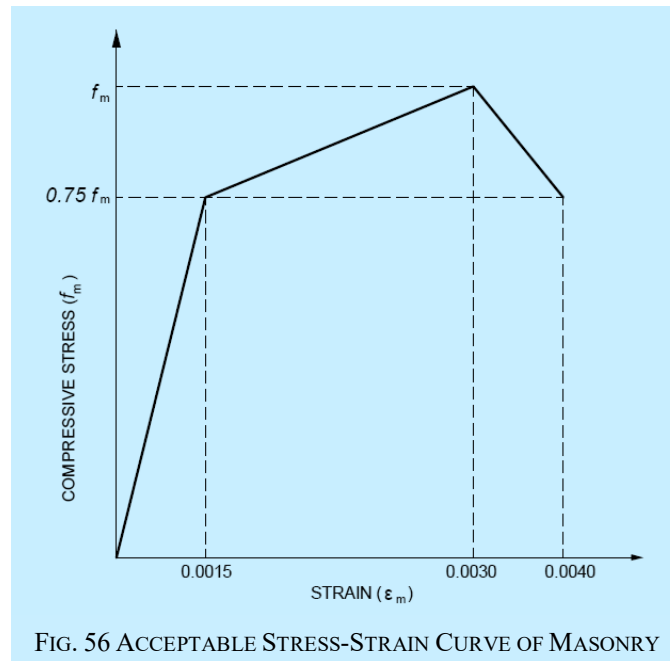


FIG. 56 ACCEPTABLE STRESS-STRAIN CURVE OF MASONRY

9.2.2.9 Axial load resistance

Axial load resistance of a confined masonry wall (P_u) shall be determined considering the contribution of masonry and longitudinal steel reinforcement in tie-columns assuming that the steel has yielded in compression, that is:

$$P_u = (0.4 f_m A_m + 0.45 f_{ck} A_c + 0.75 f_y A_s) k_s,$$

where

- f_m = compressive strength of masonry;
- A_m = net area of masonry;
- f_{ck} = characteristic cube compressive strength of concrete;
- A_c = cross-sectional area of concrete excluding reinforcing steel;

- f_y = characteristic yield strength of the reinforcing steel;
 A_s = area of steel; and
 k_s = stress reduction factor as in [Table 10](#).

9.2.2.10 Design of confined masonry walls for combined axial load and out-of-plane bending

Confined masonry walls need to be designed for combined effects of axial load and out-of-plane bending (bending perpendicular to the wall surface). Bending moments may be due to eccentric gravity loads or lateral loads acting perpendicular to wall plane (wind and earthquake). Generally, confined masonry walls have sufficient resistance against earthquake force acting in out-of-plane direction if slenderness limits in [9.4.1](#) are satisfied. For out-of-plane forces, only one-way bending may be considered, that is, bending in the vertical direction as shown in [Fig. 57](#).

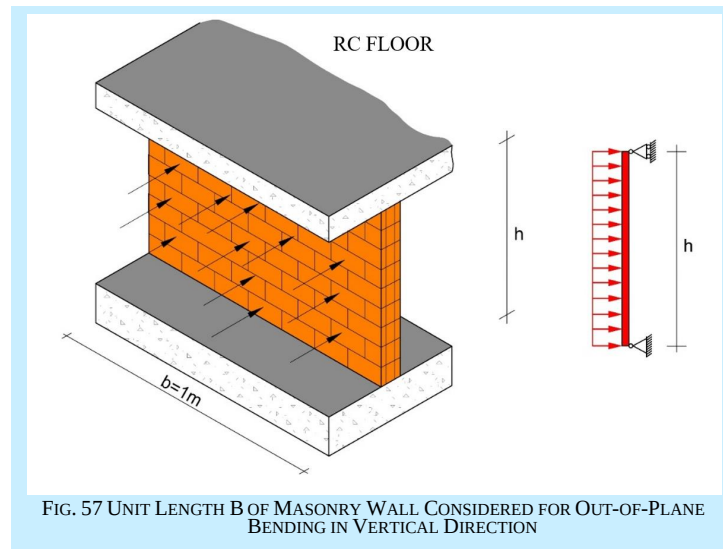


FIG. 57 UNIT LENGTH B OF MASONRY WALL CONSIDERED FOR OUT-OF-PLANE BENDING IN VERTICAL DIRECTION

The interaction of compression due to axial stresses and bending stresses should be within the following limits:

$$\frac{P}{P_u} + \frac{M}{M_u} \leq 1.0$$

Design bending moment demand M due to design wind pressure p_d [as per Part C/Section 1 of this NBCS] acting out-of-plane over unit length b can be calculated as follows:

$$M = \frac{p_d h^2}{8}$$

where h is floor-to-floor height. The axial resistance P_u shall be calculated as per [9.2.2.9](#) by ignoring the contribution of concrete and reinforcement in tie-column and considering the net area of masonry per unit length A'_m . The moment of resistance M_u per unit length shall be estimated for bending about horizontal axis normal to the length of wall as follows:

$$M_u = \left(\frac{f_t}{\gamma_m} + \frac{P}{A'_m} \right) S'_m$$

where axial load P should not exceed self-weight of the unit length of panel plus $0.15A'_m$ (in MPa),

where

- f_t = flexural tensile strength normal to bed joints;

$$= \begin{cases} 0.3 \text{ MPa} & \text{Grade M1 or better Mortar} \\ 0.2 \text{ MPa} & \text{Grade M2 Mortar} \end{cases}$$

$$A'_m = \text{net area of masonry per unit length (mm}^2\text{); and}$$

$$S'_m = \text{section modulus per unit length (mm}^3\text{).}$$

9.2.2.11 Moment resistance of confined masonry walls due to combined axial load and in-plane bending

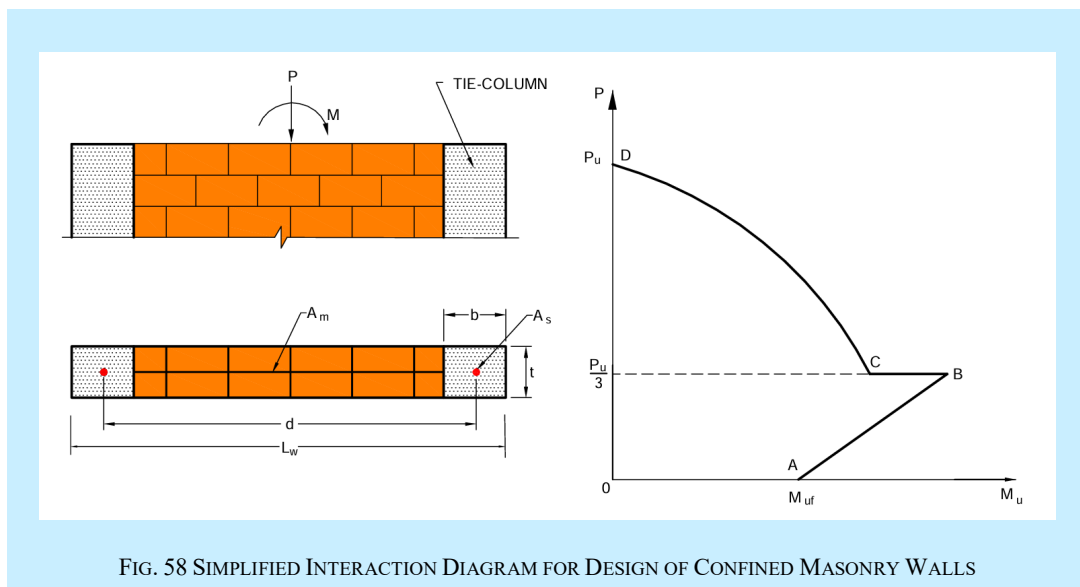
- a) *General approach* — Moment resistance M_u for a section of a confined masonry wall subjected to the design axial load P shall be determined from equilibrium of internal forces acting on the section, by following the assumptions stated in 9.2.2.7; and
- b) *Simplified axial load and bending moment interaction diagram* — For walls with longitudinal bars placed symmetrically in tie-columns, the following equations define a simplified axial load-bending moment interaction diagram (Fig. 58):

$$M_u = \begin{cases} \begin{aligned} &0.3Pd + M_{uf} \\ &0.10Pd + M_{uf} \\ &\frac{2}{3}(0.05Pd + 1.5M_{uf}) \end{aligned} & \begin{aligned} &\text{Points A-B} :: 0 \leq P < (P_u/3) \\ &\text{Point B} :: P = (P_u/3) \\ &\text{Point C} :: P = (P_u/3) \end{aligned} \\ \left(1 - \frac{P}{P_u}\right)(0.15Pd + 1.5M_{uf}) & \text{Points A-B} :: P > (P_u/3) \end{cases}$$

where

$M_f = 0.87f_y A_s (l_w - b)$ is the moment resistance corresponding to pure bending load condition, and d the effective depth of the wall section.

- c) *Alternate method* — The P-M interaction curve for the confined masonry section can also be determined by sectional analysis method employed in reinforced concrete members.



9.2.2.12 Design for shear

Shear resistance of a confined masonry wall is due to combined contribution of masonry and horizontal wall reinforcement. Although reinforced concrete tie-columns contribute to shear resistance of a confined masonry wall, their contribution shall be ignored in the design of confined masonry structure.

a) *Masonry shear resistance*

- 1) *Empirical expression method* — Shear strength of masonry wall may be estimated by any of the following methods:

Shear resistance V_u provided by masonry wall shall be determined as:

$$V_u = 0.8(0.5v_m A_T + 0.4P_d)f \leq 1.5v_m A_T$$

where

A_T = area of cross-section of confined masonry wall including tie-columns;

f = corrected factor for the aspect ratio H/L of the wall;

$$f = \begin{cases} 1.55 & \left(\frac{H}{L}\right) \leq 0.2 \\ 1.70 - 0.7\left(\frac{H}{L}\right) & 0.2 < \left(\frac{H}{L}\right) \leq 1.0 \\ 1.00 & 1.0 < \left(\frac{H}{L}\right) \end{cases}$$

P_d = design compressive axial load which shall include permanent loads only and with the partial safety factor of 1.0;

v_m = masonry shear strength; and

$$= 0.16 \sqrt{f_m} \leq 0.6 \text{ MPa.}$$

- 2) *Test method* — Shear resistance V_u of masonry wall shall be determined as:

$$V_u = v_m A_T$$

where, v_m can be obtained from the standard test method for diagonal tension (shear) in masonry assemblages as given hereunder:

- i) When shear strength v_m of masonry is to be established by tests, it shall be done in advance of the construction, using masonry specimens built of similar materials under the same conditions with the same bonding arrangement as for the structure. In making the specimens, moisture content of the units at the time of laying, the consistency of the mortar, the thickness of mortar joints and workmanship shall be the same as will be used in the structure.

The walls specimens (Fig. 59) shall have a length at least two and a half times the length of the masonry unit in one direction and adequate number of courses in perpendicular direction so that the specimen has approximately square shape. The sides of the specimen may not be less than 600 mm. The walls specimens shall be tested by loading them in compression along one diagonal. Tests shall be conducted on at least three specimens constructed with the same size and type of masonry units, mortar, and workmanship.

Specimens shall be tested after 28 days and two steel loading shoes (Fig. 59) shall be used to apply the machine load to the specimen. For the distribution of the applied load P_s , the ratio of the length of bearing, l_b and the width or height of specimen must be equal to or greater than 0.2. The load shall be applied at the rate of 50 kN/min to 100 kN/min and the load at failure shall be recorded. Load shall be applied at a uniform rate so that the maximum load is reached in not less than 1 min and not more than 2 min. Minimum thickness of steel plate used for loading shoe shall be 12 mm.

- ii) The shear stress strength v_m of the specimen as:

$$v_m = \frac{P_s}{A_{sn}\sqrt{2}}$$

where

$$\begin{aligned}
 A_{sn} &= \text{net area (mm}^2\text{) of the wall specimen;} \\
 &= \left(\frac{w_s + n_s}{2} \right) t_s; \\
 P_s &= \text{applied design axial load (N);} \\
 h_s &= \text{height (mm);} \\
 t_s &= \text{total thickness (mm) of the wall specimen; and} \\
 w_s &= \text{width (mm).}
 \end{aligned}$$

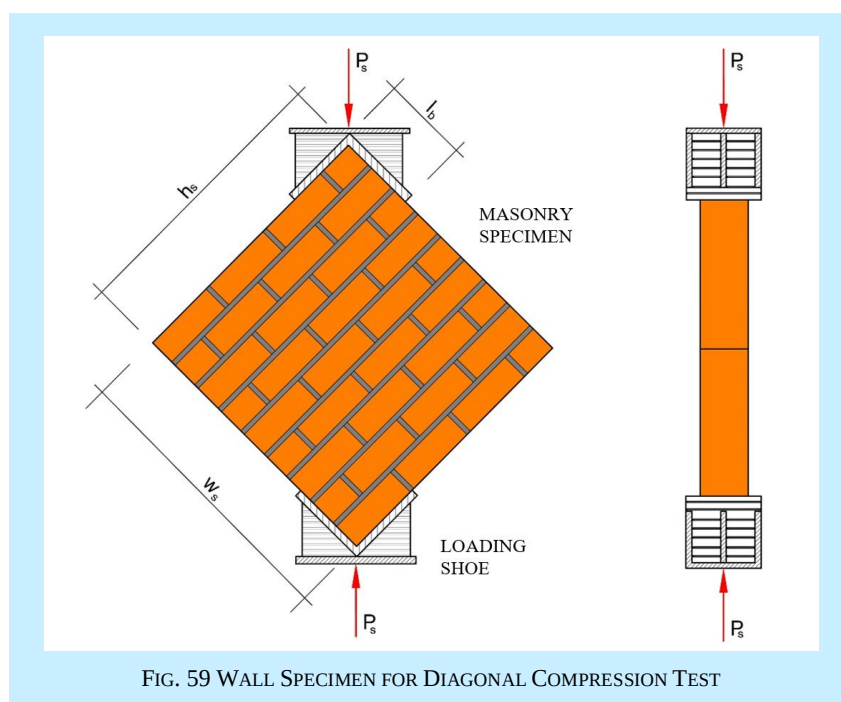


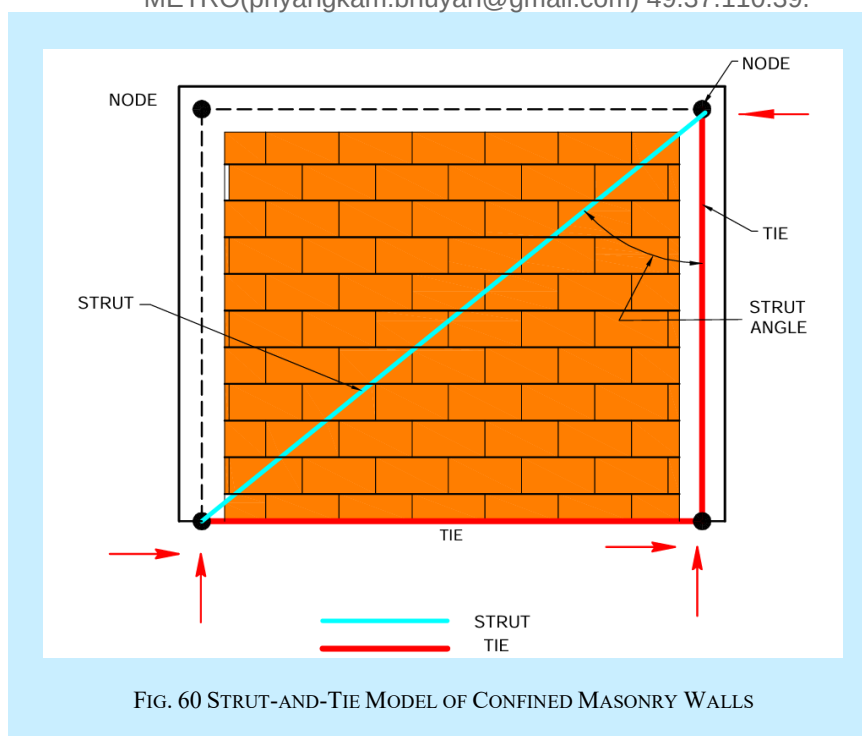
FIG. 59 WALL SPECIMEN FOR DIAGONAL COMPRESSION TEST

b) *Alternate method (Strut-and-Tie method)*

The Strut-and-Tie method can be used for determining the shear capacity of confined masonry walls. This method provides a rational and consistent design approach by idealizing complex structural members with an appropriate simplified truss model. According to this procedure, based on the knowledge of direction of principal stresses, load paths are drawn through the wall in form of a truss which is analyzed for the design loads. A possible strut-and-tie model for the confined masonry wall based on its load resisting mechanism is illustrated in Fig. 60; the broken and solid lines represent the strut and tie, respectively. If the opening distribution is irregular or complex in elevation, strut and tie method can be used to estimate the in-plane resistance of the confined masonry wall.

The basic prerequisites for the strut-and-tie model are as follows:

- The equilibrium of forces at nodes should be maintained for a given set of loads;
- Tension in concrete and masonry is neglected and adequate detailing of anchorage shall be provided for tie reinforcement;
- The member forces in the struts and ties are uniaxial which shall not exceed the corresponding member strength at every section; and
- Sufficient ductility shall be available to make the transition from elastic to plastic behaviour which will enable redistribution of internal forces in the members.



9.2.2.13 Design of tie-columns and tie-beams

a) Minimum amount of longitudinal reinforcement

Longitudinal reinforcement in tie-columns and tie-beams shall be proportioned to resist the corresponding vertical and horizontal components of the compression strut that develops in masonry when resisting combined gravity and lateral loads. The total area of reinforcement shall be not less than 0.8 percent of the gross cross-section area of the column. For a 2-storey height, the minimum area of reinforcement can be 0.6 percent;

b) Minimum amount of transverse reinforcement (Ties)

Tie-columns and tie-beams shall have transverse reinforcement in the form of closed stirrups (ties) with the minimum total cross-sectional area A_{sc} (mm²) of stirrup legs effective in shear equal to:

$$A_{sc} = 0.0012 sh_c,$$

where

s = tie spacing, in mm; and

h_c = Dimension, in mm of tie-column or tie-beam in the wall plane.

c) Spacing of transverse reinforcement

Tie spacing shall not exceed the lesser of 200 mm and $1.5t$.

9.2.3 Foundations and Plinth Construction

The foundation shall be constructed in a similar manner as for traditional masonry construction. Either a brick masonry footing or an RC strip footing can be used. An RC plinth band shall be constructed on top of the foundation. A plinth band shall be provided to fully confine wall panels along their bases and prevent excessive wall damage due to building settlement in soft soil areas. Note that the longitudinal reinforcement shall be extended from an RC tie-column into the plinth band, and whenever possible, into the foundation.

9.3 Earthquake Resistant Design

9.3.1 Basis

Building shall be designed for earthquake forces determined as per 5 of Part C/Section 1 of this NBCS. Also, the load combinations shall be as per Part C/Section 1 of this NBCS.

9.3.2 Methods of Earthquake Analysis

9.3.2.1 General considerations

9.3.2.1.1 For determining internal forces and moments acting on confined masonry walls, the building may be analyzed for estimating the effects of earthquake shaking using either the simplified method or the earthquake analysis methods specified in **5** of Part C/Section 1 of this NBCS. The simplified method can be used for buildings up to three storeys high, provided that the requirements stipulated in **9.3.2.2** have been met.

The effect of openings shall be considered in lateral stiffness and strength calculations. The structure shall be modelled to adequately simulate the behaviour of its critical structural elements. Modulus of elasticity E_m and shear modulus G_m of masonry shall be used to reflect the axial and shear stiffness expected in the structure.

9.3.2.1.2 Design forces and moments shall be obtained from the analysis using design loads and the corresponding load combinations specified in **5** of Part C/Section 1 of this NBCS. A confined masonry wall shall be designed for the effects of gravity loads, and the effects of lateral loads, including shear force, in-plane bending, and out-of-plane bending moments. When the simplified method is used, design for lateral loads may be limited to the effects of shear force.

9.3.2.2 Applicability of the simplified method

9.3.2.2.1 The simplified method is based on an idealized distribution of lateral earthquake forces in regular load bearing wall buildings with rigid diaphragms. The method compares the shear capacity of all walls at the base of the building and the earthquake base shear force determined from **5** of Part C/Section 1 of this NBCS. It is suitable for verifying adequacy of regular buildings with symmetrical wall layout and predominant shear behaviour. The method does not consider torsional effects, arising from eccentricity between centre of mass and centre of resistance. Due to its assumptions, application of this method is restricted to buildings with a height not exceeding three storeys.

9.3.2.2.2 The design can be simplified further for small buildings by applying the concept of wall index, as per procedure in **9.3.2.3**, which can be used for the design of buildings with the total ground plan area not exceeding 200 m² and the height not exceeding two storeys or 7 m.

9.3.2.3 Wall index method

9.3.2.3.1 Wall Index (WI), also known as wall density, shall be used as an indicator of earthquake load-resisting capacity of earthquake resistant design of low-rise buildings with regular plan shape and elevation. The design according to the simplified method shall be deemed satisfactory provided that the actual wall index (WI_{floor}) per floor in each direction of the building plan is greater than or equal to the required Wall Index (WI_{req}) for specific building.

Wall index WI_{floor} at the ground floor shall be taken as ratio of the sum of cross-sectional areas of all confined masonry walls at that level in the considered direction of earthquake shaking, given by:

$$WI_{\text{Floor}} = \frac{A_{wV}}{A_p}$$

where

- A_w = cross-sectional area of all confined walls in one direction at the ground floor level; the cross-sectional area of a confined masonry wall is the product of its length (including the reinforced concrete tie-columns and the masonry wall) and thickness; and
- A_p = plan area of the floor diaphragm (floor slab) at the ground floor level; the area of any cutouts (openings) in the floor slab shall be deducted in the calculation of A_p .

Parameters for wall index calculations are illustrated in [Fig. 61](#). The wall index shall be estimated along each principal plan direction (X and Y) of the building. Only confined masonry walls aligned in the direction under consideration (shown shaded in [Fig. 61](#)) shall be considered in the A_w calculations for the specific direction.

9.3.2.3.2 The required wall index WI_{req} in buildings in different earthquake zones shall be taken as per [Table 28](#) or estimated as follows:

$$WI_{req} \geq \left(\frac{1.5A_h w}{v_m / \gamma_m} \right),$$

where

- A_h = design horizontal acceleration coefficient determined as per **5** of Part C/Section 1 of this NBCS for different zones;
- n = number of floor levels in a building;
- v_m = masonry shear strength as per [9.2.2.12](#);
- γ_m = partial safety factor for masonry as per [9.2.2.8](#); and
- w = average floor weight, including the self-weight of floor slab, floor finish, walls and 25 percent of the imposed load.

9.3.2.4 Earthquake analysis methods

Earthquake analysis of confined masonry buildings of four-storey or higher and buildings up to three-storey high that do not meet requirements for the application of simplified method shall be analysed by the method specified in **5** of Part C/Section 1 of this NBCS. Lateral load effects induced by earthquake shall be determined based on the relative stiffness of walls and wall segments, by considering the effect of both shear and flexure in stiffness calculations. The analysis model shall consider stiffness of floor and roof systems, and any other restraints that may influence wall rotations.

9.3.3 Other Prescriptive Requirements

9.3.3.1 Wall spacing

The maximum spacing of transverse walls in buildings with flexible diaphragms shall not exceed 4 m. For spacing more than 4 m, the tie-beam at roof level should have a width of $L/20$.

9.3.3.2 Spacing of transverse reinforcement (ties) in tie-columns

In earthquake zones IV and above reduced tie spacing ($s/2$) shall be provided of the transverse reinforcement at the ends of tie-columns, where s is the spacing of transverse reinforcement specified in [9.2.2.13\(c\)](#) as shown in [Fig. 62](#). The length over which the reduced tie spacing is required shall not exceed the largest of $(2h_c)$, $(2t)$ and $(h_o/6)$, where h_c is the depth of the tie-column, h_o the clear height of tie-column between floors, and t the width of the tie-column.

9.4 General Requirements

9.4.1 Reinforcement Detailing

9.4.1.1 Concrete cover

- a) The minimum concrete cover to ties (transverse reinforcement) in tie-columns and tie-beams shall not be less than 20 mm; and
- b) When horizontal wall reinforcement is provided, the minimum clear distance between a horizontal bar and the exterior wall surface shall not be less than 20 mm.

9.4.1.2 Bar size and number in tie-columns and tie-beams

Longitudinal reinforcement in tie-columns and tie-beams shall consist of a minimum of 4 reinforcing bars with the minimum 10 mm diameter.

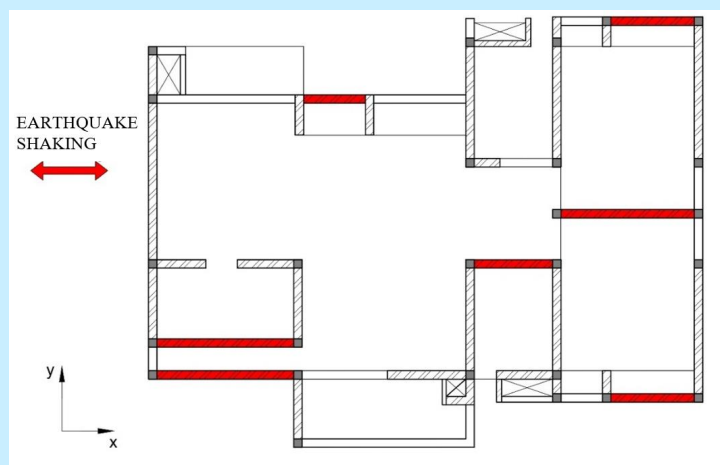


FIG. 61(A) X-DIRECTION

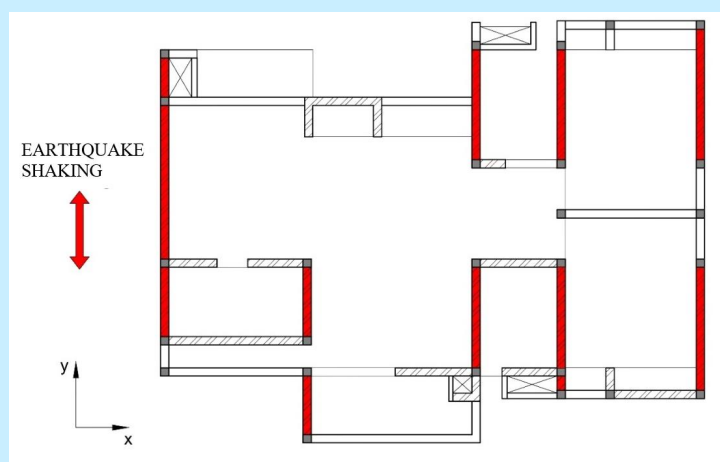


FIG. 61(B) Y-DIRECTION

FIG. 61 WALLS TO BE CONSIDERED IN ESTIMATION OF WALL INDEX ALONG PRINCIPAL PLAN DIRECTIONS OF A BUILDING

Table 28 $W_{I_{req}}$ for Different Earthquake Zones

(Clause [9.3.2.3.2](#))

Sl No.	Earthquake Zone	II	III	IV	V
(1)	(2)	(3)	(4)	(5)	(6)
i)	$W_{I_{req}}$ (percent)	0.7	1.1	1.6	2.4

9.4.1.3 Ties in tie-columns and tie-beams

Minimum 6 mm diameter bars shall be used for ties in tie-columns and tie-beams ([Fig. 63](#)). Ties shall have 135° hooked ends. Hooks shall be staggered as shown in [Fig. 63](#).

9.4.1.4 Anchorage of longitudinal bars

- Longitudinal reinforcement in tie-columns and tie-beams shall be anchored to develop the full specified steel yield stress. Longitudinal bars in tie-columns and tie-beams shall have a 90° hooked anchorage at intersections. The lap length of the hook tails shall be the largest of 20 times the bar diameter or 500 mm ([Fig. 64](#)); and
- Tie-column longitudinal bars at the roof level shall be bent by 90° and lapped with the tie-beam longitudinal reinforcement ([Fig. 65](#)).

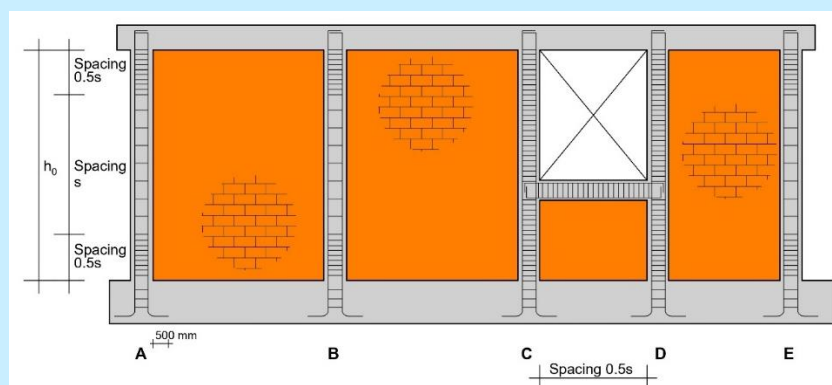


FIG. 62 TIE-COLUMN TRANSVERSE REINFORCEMENT REQUIREMENTS IN EARTHQUAKE ZONES IV AND ABOVE

9.4.1.5 Lap Splices for longitudinal reinforcement

Lap splices for longitudinal reinforcement should be at least 40 times the bar diameter. Longitudinal reinforcing bars shall be spliced within the middle third of the column height or beam span. The splices shall be staggered so that not more than 2 bars are spliced at any one location.

9.4.1.6 Tie-column-to-tie-beam joints

Continuity of longitudinal tie-beam reinforcement through the joint must be ensured. An example of a continuous longitudinal reinforcement is shown in Fig. 66. First tie at the ends of tie-columns (top and bottom) shall be placed as close to the joint as possible. When tie-beam depth exceeds 300 mm, vertical reinforcement in an RC tie-column must be confined by the ties, below and above the joint. An additional U-shaped stirrup shall be placed in the tie-beam at mid-height (Fig. 67).

Reinforced concrete tie-columns shall be provided at the openings as shown in Fig. 54(B) and Fig. 54(C). A nominal reinforcement around openings as detailed in Fig. 53 shall be provided. When reinforced concrete tie-columns are not provided at the ends of an opening then they should be strengthened as per good practice of [C-4(10)] or minimum two vertical bars of 10 mm diameter with 6 mm ties at 150 mm spacing should be provided.

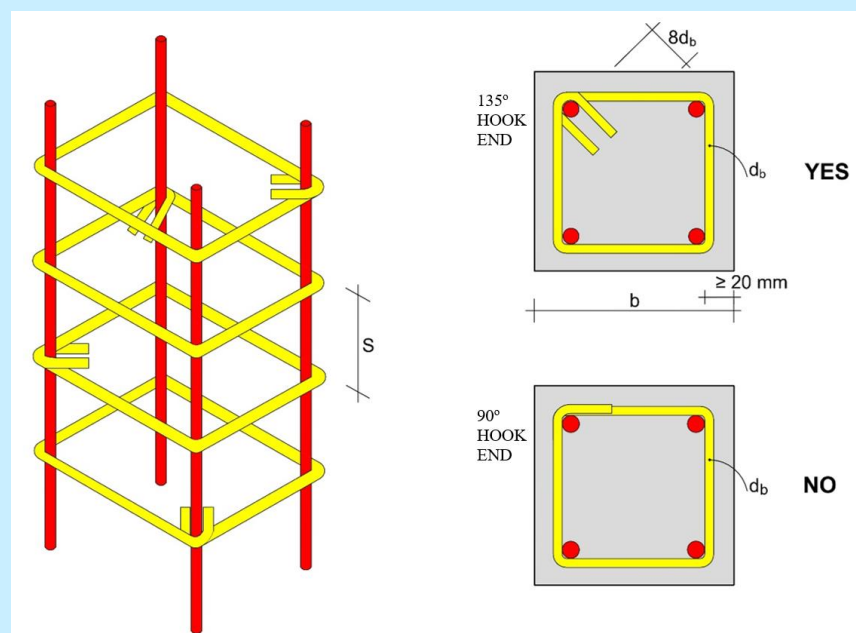


FIG. 63 LAYOUT AND DETAILING OF TIE-COLUMNS AND TIE-BEAMS

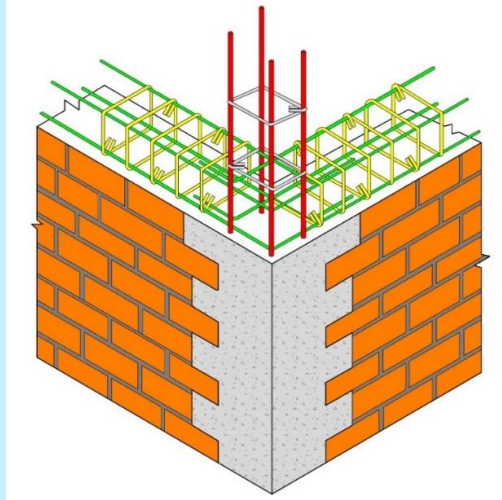


FIG. 64(A) WALL INTERSECTION

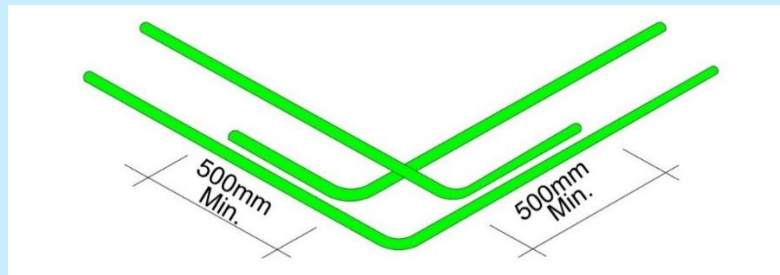


FIG. 64(B) HOOKED ANCHORAGE OF LONGITUDINAL REINFORCEMENT AT WALL INTERSECTIONS

FIG. 64 TIE-BEAM CONSTRUCTION

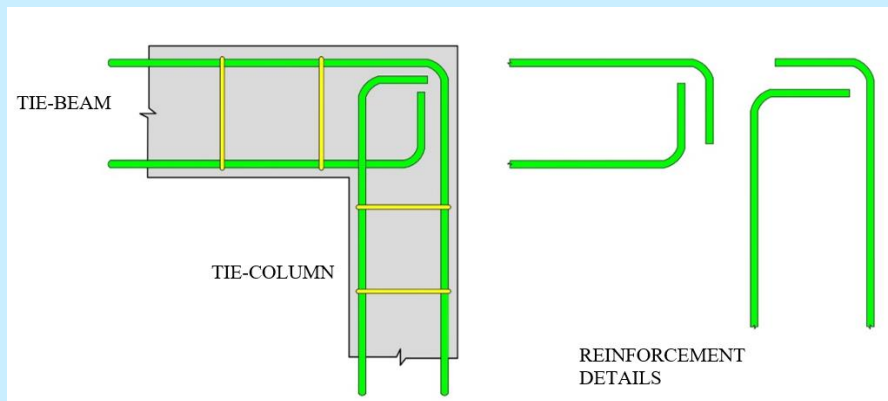


FIG. 65 ANCHORAGE OF LONGITUDINAL REINFORCEMENT OF TIE-COLUMN AND TIE-BEAM

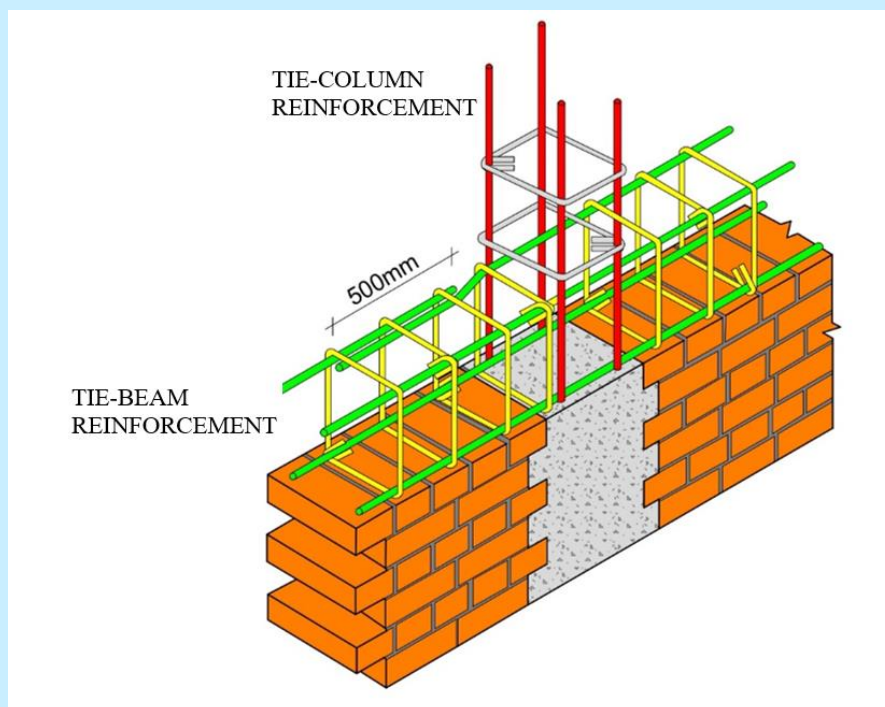


FIG. 66 CONTINUITY OF TIE-BEAM REINFORCEMENT THROUGH JOINT

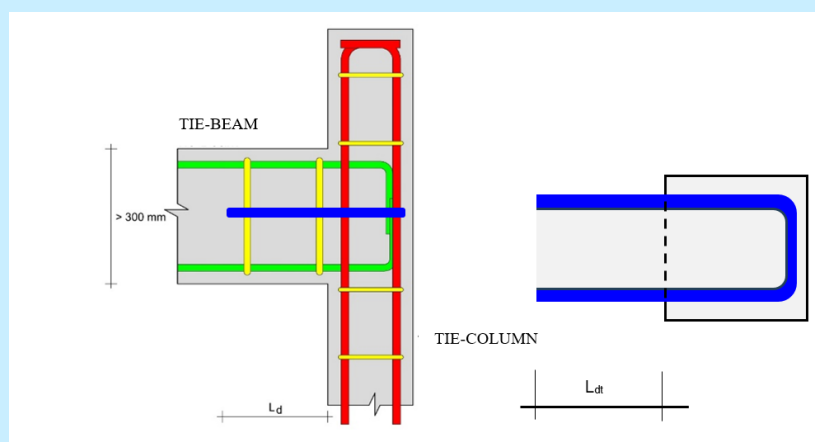


FIG. 67 ADDITIONAL CONFINEMENT OF VERTICAL REINFORCEMENT IN TIE-BEAMS AND TIE-COLUMNS IN EXTERIOR JOINT

9.4.2 Construction

Construction of confined masonry buildings shall satisfy the conditions specified hereunder.

9.4.2.1 Masonry wall construction

- a) *General* — During construction, in addition to the requirements of the sections above, the following shall be complied:
 - 1) all intersecting walls shall be connected, unless measures to assure stability and good performance are taken;
 - 2) surfaces of construction joints shall be clean and rough. The joint shall be moistened before construction when clay units are used;
 - 3) fulfill the requirements of excellent masonry work, Flemish bond or English bond in brickwork shall be adopted; and

- 4) During construction, necessary caution to assure wall stability at the job, shall be taken, possible horizontal pressure and loads, including wind and earthquake shall be considered.

b) *Masonry units*

The shape and dimension of masonry units, construction practices, including methods of positioning of reinforcement, placing and compacting of grout, as well as design and detailing shall be, such as to promote homogeneity of structural members, development of the bond between the grout to both reinforcement and masonry units and avoidance of corrosion of reinforcement.

c) *Mortar joint thickness*

Joint mortar shall totally cover the horizontal and vertical sides of the masonry unit. Joint thickness shall be the minimum to get a uniform layer of mortar and good alignment of the units. If industrialized units are used, the thickness of the horizontal joints shall not exceed 12 mm if horizontal reinforcement is placed in the joints, and 10 mm without horizontal reinforcement. Minimum thickness shall be 6 mm.

d) *Toothing*

The vertical interface of masonry with adjacent tie-columns shall be detailed to transfer shear forces. It shall be accepted to indent the masonry (referred to as toothing) or else, to place steel dowels or horizontal reinforcement as shown in [Fig. 68](#).

Toothed edges shall be left on each side of the wall at the interface with the tie-columns. Tothing length shall be equal to one-quarter of the masonry unit length, but not less than 50 mm, as shown in [Fig. 68\(A\)](#). When hand-made bricks are used, it is desirable to cut the brick edges, as shown in [Fig. 68\(B\)](#). It is important to clean the surfaces of 'toothed' masonry units before the concrete has been poured. Horizontal reinforcement anchored into RC tie-columns, also known as dowels, can be used as an alternative to toothing, as shown in [Fig. 68\(C\)](#). The dowels are not necessary when toothed edges are used. 6 mm diameter bars shall be preferably used as dowels.

9.4.2.2 Reinforcement

Reinforcement shall be tightly secured before casting.

9.4.2.3 Concrete

Casting of tie-columns shall be cast after the masonry wall or the corresponding part has been constructed. The masonry around the tie-columns shall be saturated before the concrete has been poured.

9.4.2.4 Piping and ducts

Piping and ducts shall be installed without damaging the masonry. If solid or grouted hollow units are used, grooves in the wall shall be permitted to embed piping and ducts, but the following shall be satisfied:

- a) Groove depth shall not exceed one fourth the thickness of the masonry wall ($t/4$);
- b) Groove shall be vertical; and
- c) Groove shall not be longer than one half the free height of the wall ($H/2$). If hollow units are used, pipes or ducts shall not be placed in cells with reinforcement. Cells with pipes and ducts shall be grouted.

It shall not be permitted to place piping and ducts in tie-columns with a structural function, whether external or internal.

10 REINFORCED MASONRY WALLS (RMW)

10.1 General

The requirements for structural design aspect of reinforced load bearing/walls, constructed with solid or perforated burnt clay bricks, sand-lime bricks, stones, concrete blocks, lime-based blocks or burnt clay hollow blocks shall be specified hereunder related to the materials to be used, the permissible stresses and methods of design.

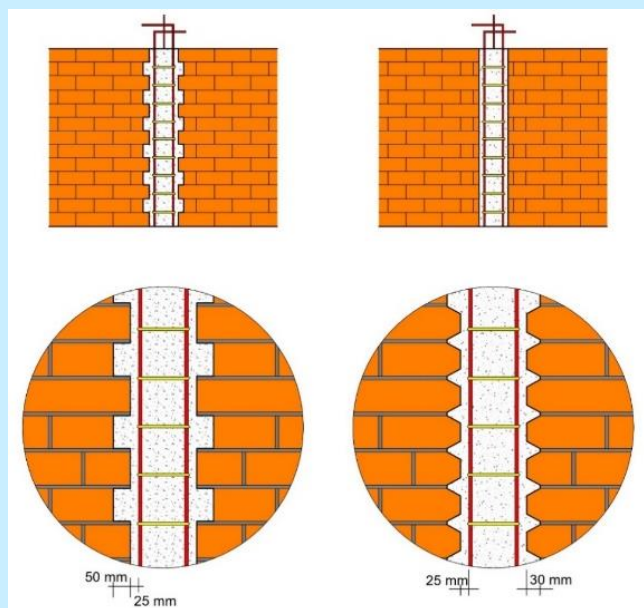


FIG. 68(A) FULL TOTHING

FIG. 68(B) PARTIAL TOTHING

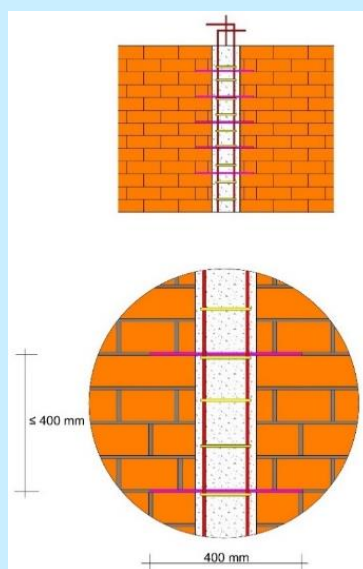


FIG. 68(C) NO TOTHING

FIG. 68 REINFORCEMENT DETAILS OF TOTHING IN CONFINED MASONRY WALLS

10.2 Design Considerations

10.2.1 General

Design considerations shall be as per geometrical requirements of 4 along with the following provisions.

10.2.2 Structural Continuity

Intersecting structural elements intended to act as a unit shall be joined together to resist the design forces. Walls shall be joined together to all floors, roofs or other elements which provide lateral support for the wall. Where floors or roofs are designed to transmit horizontal forces to walls, the anchorages to the walls shall be designed to resist the horizontal forces.

10.2.3 Effective Span

10.2.3.1 Effective span of simply supported or continuous members shall be taken as the smaller of the following:

- a) Distance between centers of supports; and
- b) Clear distance between supports plus an effective depth d .

10.2.3.2 Effective span of a cantilever shall be taken as greater of:

- a) distance between the end of cantilever and;
- b) centre of its support; and
- c) face of support plus half its effective depth.

10.2.4 Slenderness Ratio

10.2.4.1 Wall

The slenderness ratio (ratio of effective height h_{eff} to effective thickness t_{ef}) shall not exceed 27 for vertically loaded reinforced masonry walls in their plane. For reinforced masonry members, such as walls subjected to out-of-plane bending and for spandrel masonry beams that are a part of wall subjected to bending in the plane of the wall, the maximum effective span to effective depth ratio shall be as given in [Table 29](#).

10.2.4.2 Columns

For a column, slenderness ratio shall be taken to be the greater of the ratios of effective heights to the respective effective thickness in the two principal directions. Slenderness ratio for a load bearing unreinforced column shall not exceed 15 whereas for reinforced column the slenderness ratio should be limited to 20.

10.2.5 Minimum Design Dimensions

10.2.5.1 Minimum thickness of load bearing walls columns

The nominal thickness of masonry bearing walls in building shall not be less than 230 mm.

10.2.5.2 Parapet wall

Parapet walls shall be at least 200 mm thick and height shall not exceed 3 times the thickness. The parapet wall shall not be thinner than the wall below. The tie-columns shall be extended into the parapet wall and terminated at the top of the parapet in a tie-beam.

10.2.6 Eccentricity in Columns

Columns shall be designed for a minimum eccentricity of 10 percent of side dimension for each axis in addition to applied loads.

10.3 Requirements Governing Reinforcement and Detailing

10.3.1 General

This section provides requirements for the working (allowable) stress design of masonry structure neglecting the contribution of tensile strength of masonry.

Table 29 Maximum Ratio Of Effective Span to Effective Depth For Walls Subjected to Out-Of-Plane Bending

(Clause [10.2.4.1](#))

Sl No.	Boundary Conditions	Maximum Permissible Ratios	
		Effective Span to Effective Thickness (l_{ef}/t_{ef}) of Wall Subjected to Out-of-Plane Bending	Effective Span to Effective Depth (l_{ef}/d) of Beam Part of Wall Subjected to In-Plane Bending
(1)	(2)	(3)	(4)
i)	Simply supported	35	20
ii)	Continuous	45	26
iii)	Spanning in 2 directions	45	—
iv)	Cantilevered	18	7

10.3.1.1 Members are designed for composite action. Stresses shall be computed using transformed area concept of linear elastic analysis as follows:

$$A_t = A_b + m A_s$$

where

A_t = total transformed cross-sectional area of the member;

A_b = cross sectional area of brick;

m = modular ratio of steel reinforcement and brick; and

A_s = cross sectional area of reinforcement.

10.3.1.2 Stiffness calculation shall be based on un-cracked section properties.

10.3.2 Steel reinforcement-allowable stresses

10.3.2.1 Tension

Tensile stress in reinforcement shall not exceed the following:

- 140 MPa in case of Mild Steel (MS) bars of diameter ≤ 20 mm and 130 MPa in case of MS bars of diameter > 20 mm; and
- $0.55f_y$ in case high strength bars, where f_y is the characteristic strength of steel.

10.3.2.2 Compression

Compressive stress in reinforcement shall not exceed the following:

- 130 MPa in case of MS bars; and
- 190 MPa in case high strength bars.

10.3.3 Size of Reinforcement

- The maximum size of reinforcement used in masonry shall be 25 mm diameter bars and minimum size shall not be less than 8 mm; and
- The diameter of reinforcement shall not exceed one-half the least clear dimension of the cell, bond beam, or collar joint in which it is placed.

10.3.4 Spacing of Reinforcement

- a) In beams, clear distance between parallel bars shall not be less than the diameter of the bars, or less than 25 mm; and
- b) In columns and pilasters, clear distance between vertical bars shall not be less than 1.5 times the bar diameter, nor less than 35 mm.

10.3.5 Anchorage

10.3.5.1 Development length of bars

The development length L_{dtm} shall be taken as:

$$L_{dtm} = \begin{cases} 0.25d_b F_s \geq 300 \text{ mm} & \text{Fe415, Fe500 and Fe550} \\ 0.40d_b F_s \geq 300 \text{ mm} & \text{Fe250} \end{cases}$$

where

d_b = nominal diameter of bar (mm); and

F_s = permissible tensile/compressive stress in steel (MPa).

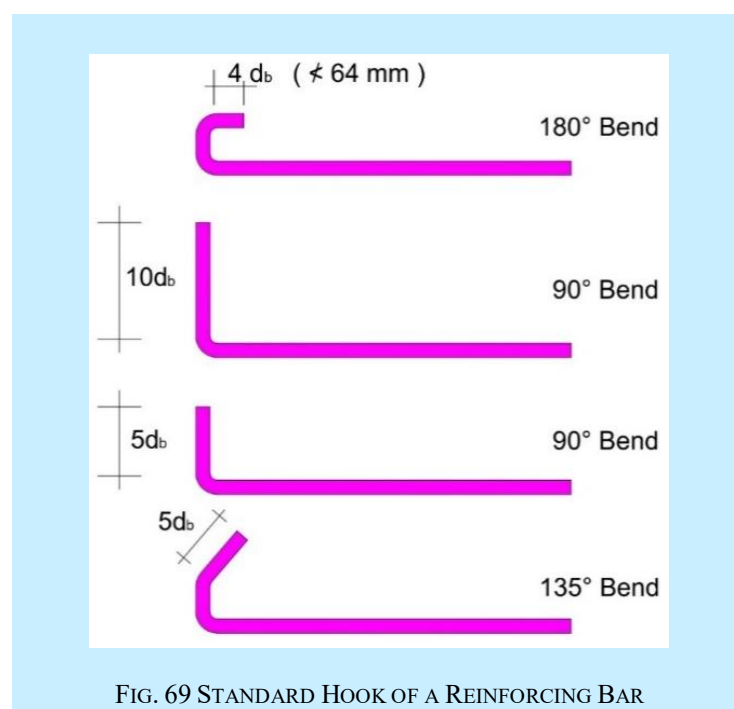
10.3.5.2 For MS bars, L_d shall be increased by 60 percent.

10.3.5.3 Standard hooks

- a) Standard hooks shall be formed by one of the following methods ([Fig. 69](#)):
 - 1) 180° turn plus extension of at least 4 bar diameters but not less than 64 mm at free end of bar;
 - 2) A 90° turn plus extension of at least 10 bar diameters at free end of bar; and
 - 3) For stirrup and tie anchorage only a 90° or a 135° turn plus an extension of at least 5 bar diameters at the free end of the bar.
- b) The diameter of bend, measured to the inside of the bar other than stirrups and ties, shall not be less than 5 bar diameters for 6 mm through 20 mm diameter bars. For 25 mm bars, a minimum bend diameter of 6 times bar diameters shall be used;
- c) Inside diameter of bend for 12 mm diameter or smaller stirrups and ties shall not be less than 4 bar diameters. Inside diameter of bend for 16 mm diameter or larger stirrups and ties shall not be less than that given in (b); and
- d) Hooks shall not be permitted in the tension portion of any beam, except at the ends of simple or cantilever beams or at the freely supported ends of continuous or restrained beams.

10.3.5.4 Anchorage of reinforcement

- a) Deformed bars may be used without end anchorages provided development length requirement is satisfied. Hooks should normally be provided for plain bars in tension;
- b) Bends and hooks shall conform to the requirements of Part C/Section 5 of this NBCS, and the following anchorage values shall be used:
 - 1) *Bend* — The anchorage value of bend shall be taken as 4 times the diameter of the bar for each 45° bend subject to a maximum of 16 times the diameter of the bar,
 - 2) *Hook* — The anchorage value of a standard U-type hook shall be equal to 16 times the diameter of the bar, and of the 135° hook end equal to 12 times diameter of the bar; and
- c) For transverse reinforcement complete development length and anchorage shall be deemed to have been provided when the bar is provided with standard hook as described in [10.3.5.2](#).



10.3.6 Lap Splices

10.3.6.1 Where splices are provided in the reinforcing bars, they shall as far as possible be away from the sections of maximum stress and be staggered. It is recommended that splices in flexural members not be at sections where the bending moment is more than 50 percent of the moment of resistance; and not more than half the bars shall be spliced at a section. Where more than one-half of the bars are spliced at a section or where splices are made at points of maximum stress, special precautions shall be taken, such as increasing the length of lap and/or using closely spaced stirrups around the length of the splice.

10.3.6.2 Lap length including anchorage value of hooks for bars in flexural tension shall be L_{dtm} (see 10.3.5.1) or $30d_b$ whichever is greater and for direct tension shall be $2L_{dtm}$ or $30d_b$ whichever is greater. The straight length of the lap shall not be less than $15d_b$ or 200 mm.

10.3.6.3 The following provisions shall also apply:

- a) Where lap occurs for a tension bar located at:
 - 1) top of a section as cast and the minimum cover is less than twice the diameter of the lapped bar, the lap length shall be increased by a factor of 1.4; and
 - 2) corner of a section and the minimum cover to either face is less than twice the diameter of the lapped bar or where the clear distance between adjacent laps is less than 75 mm or 6 times the diameter of lapped bar, whichever is greater, the lap length should be increased by a factor of 1.4.
- b) Where both conditions 10.3.6.3(a)(1) and 10.3.6.3(a)(2) apply, the lap length should be increased by a factor of 2.0; and
- c) Splices in tension members (which are not part of the lateral force resisting system) shall be enclosed in closed loop hoops made of bars not less than 8 mm diameter with spacing not more than 100 mm.

10.3.7 Curtailment of Tension Reinforcement

10.3.7.1 In any member subjected to bending, every reinforcing bar should extend, except at end supports, beyond the point at which it is no longer needed, for a distance equal to the effective depth of the member or 12 times the diameter of the bar, whichever is the greater. The point at which reinforcement is theoretically no longer needed is where the design resistance moment of the section, considering only the continuing bars, is equal to the applied design moment. But, reinforcement should not be curtailed in a tension zone unless at least one of the following conditions is satisfied for all arrangements of design load considered:

- a) reinforcing bars extend at least the anchorage length appropriate to their design strength from the point at which they are no longer required to resist bending;

- b) design shear capacity at the section where the reinforcement stops are greater than twice the shear force due to design loads, at that section; and
- c) continuing reinforcing bars at the section where the reinforcement stops provides double the area required to resist the bending moment at that section.

10.3.7.2 Where there is little or no end fixity for a member in bending, at least 25 percent of the area of the tension reinforcement required at mid-span should be carried through to the support. This reinforcement may be anchored in accordance with [10.3.5.3](#) or by providing:

- a) an effective anchorage length equivalent to 12 times the bar diameter beyond the centre line of the support, where no bend or hook begins before the centre of the support; or
- b) an effective anchorage length equivalent to 12 times the bar diameter plus $d/2$ from the face of the support, where d is the effective depth of the member, and no bend begins before $d/2$ inside the face of the support.

Where the distance from the face of a support to the nearer edges of a principal load is less than twice the effective depth, all the main reinforcement in a member subjected to bending should continue to the support and be provided with an anchorage equivalent to 20 times the bar diameter.

10.3.8 Members Subjected to Flexure and Axial Forces

10.3.8.1 A member which is subjected to axial stress less than $0.1f_m$, may be designed for bending only.

10.3.8.2 Beams

Reinforcement in masonry designed as beams shall provided over a support where the masonry is continuous, whether the beam has been designed as continuous or not. Where this occurs, an area of steel not less than 50 percent of the area of the tension reinforcement required at mid-span shall be provided in the top of the masonry over the support and anchored in accordance with [10.3.5.3](#). In all cases, at least one quarter of the reinforcement required at mid-span shall be carried through to the support and similarly anchored.

10.3.8.3 Columns

The design of reinforced column shall meet the requirements given hereunder.

10.3.8.3.1 Vertical reinforcement shall not be less than 0.25 percent nor exceed 4 percent of the net area of column cross-section. The minimum number of bars shall be four.

10.3.8.3.2 Lateral ties

Lateral ties shall be provided in the column as per the following:

- a) Longitudinal reinforcement shall be enclosed by lateral ties of at least 8 mm diameter. Vertical spacing of ties along the length of column shall be minimum of:
 - 1) 16 times diameter of longitudinal bar;
 - 2) 48 times diameter of lateral tie; and
 - 3) Least dimension of the column.
- b) Arrangement of lateral ties is such that every corner and alternate longitudinal bar shall have lateral support provided by the corner of a lateral tie with an included angle of not more than 135° .

10.3.9 Members Subjected to Shear

10.3.9.1 Reinforced masonry walls may be designed taking contribution of shear reinforcement.

10.3.9.2 Where contribution of shear reinforcement is considered in resisting shear force the minimum area of shear reinforcement in the direction of force shall be determined by the following:

$$A_{v,min} = \frac{V_s}{F_s d},$$

where

- V_s = total applied shear force;
 F_s = permissible stress in steel reinforcement as per [10.4.1.4](#); and
 d = distance from extreme compression fiber to centroid of tension reinforcement.

10.3.9.3 The maximum spacing of shear reinforcement shall not be greater than $0.5d$ or 120 mm, whichever is smaller.

10.3.9.4 In cantilever beams, maximum shear shall be used whereas for members subjected to uniformly distributed load it may be assumed that maximum shear load occurs at a distance of $0.5d$ from the face of support when the following conditions are met:

- a) support reaction causes compression in the end region of the member; and
- b) no concentrated load between face of support and a distance of $0.5d$ from it.

10.3.10 Reinforcement Detailing

10.3.10.1 General

Reinforcement shall be located such that it acts compositely with the masonry and various ways in which it can be used in reinforced masonry are shown in [Fig. 70](#).

10.3.10.2 Protection of reinforcement

Where steel reinforcing bars are embedded in filled cavity (or pockets) or special bond construction, the bars shall have the minimum clear cover of 10 mm in mortar or a minimum clear cover of 15 mm or bar diameter, whichever is more in cement concrete (grout) so as to achieve good bond and corrosion resistance.

For the reinforcement steel placed in mortar bed joint, the minimum depth of mortar cover from the reinforcing steel to the face of masonry shall be 15 mm. Also mortar cover above and below reinforcement placed in bed joints shall not be less than 2 mm.

Reinforcing steel shall be corrosion resistant or protected adequately against corrosion. Reinforcement shall be galvanized reinforcing steel for protection against corrosion.

10.4 Structural Design

10.4.1 Permissible Compressive Force

Compressive force in reinforced masonry due to axial load shall not exceed that given by following equation:

$$P_0 = k_s(0.25f_m A_n + 0.65f_s A_s)$$

Where

- k_s = stress reduction factor as per [Table 10](#).
 f_m = compressive strength of masonry;
 A_n = net area of masonry;
 f_s = permissible steel tensile stress; and
 A_s = area of steel.

10.4.1.1 Effective compressive width for locally concentrated reinforcement

When the reinforcement in masonry is concentrated locally such that it cannot be treated as a flanged member ([Fig. 71](#)), the reinforced section shall be considered as having a width of not more than:

- a) centre-to-centre bar spacing; and

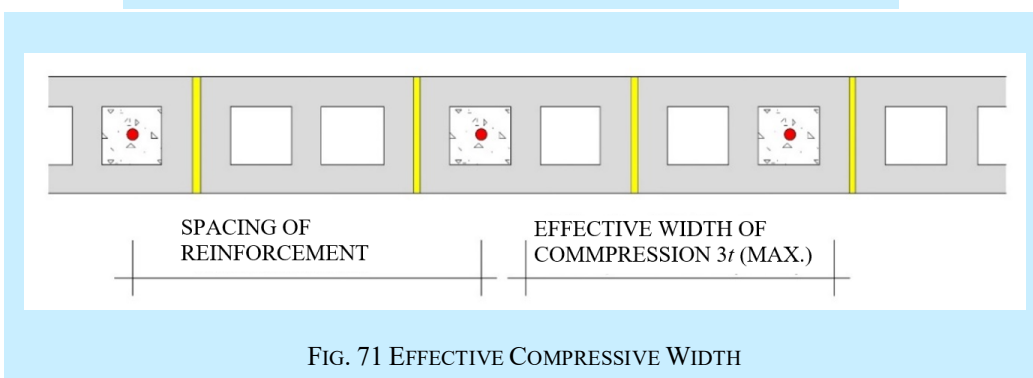
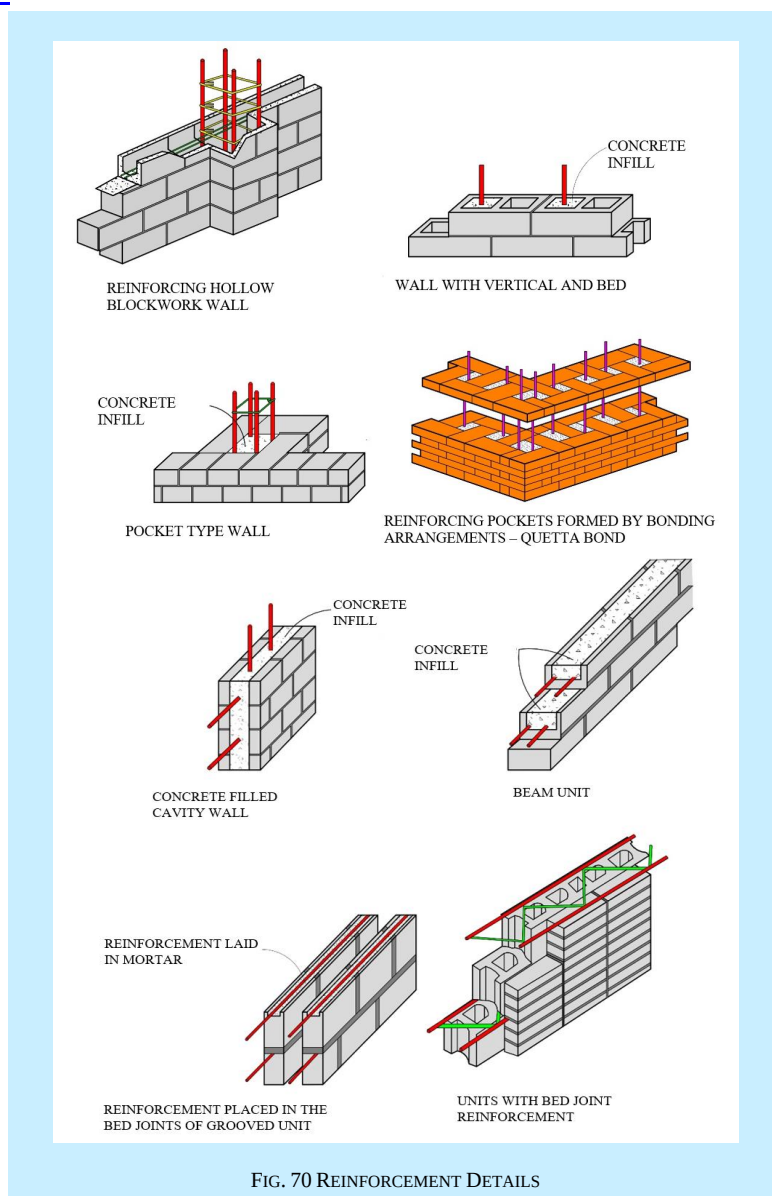
b) 6 times the wall thickness.

10.4.1.2 Combined permissible axial and flexural compressive stress

For reinforced members subjected to combined axial load and flexure, the compressive stress in masonry due to combined action of axial load and bending shall not exceed $1.25f_a$ and compressive stress in masonry due to axial load only shall not exceed f_a . See [Annex E](#).

10.4.1.3 Permissible tensile stress

Provisions of [5.4.2](#) shall apply.



10.4.1.4 Permissible shear stress

For members in flexure, the permissible shear stress shall be taken as:

$$f_v = \begin{cases} 0.083\sqrt{f_m} \leq 0.25 \text{ MPa} & \text{Walls without web reinforcement} \\ 0.250\sqrt{f_m} \leq 0.75 \text{ MPa} & \text{Walls with web reinforcement} \end{cases}$$

For walls in compression and bending, the Permissible Shear Stress for reinforced masonry walls shall be according to [Table 30](#).

If there is tension in any part of a section of masonry, the area under tension shall be ignored while estimating the shear stress on the section.

Table 30 Permissible Shear Stress in Reinforced Masonry Walls under Compression and Bending (Clause 10.4.1.4)				
Sl No.	Type of Wall	M/Vd	Permissible Shear Stress F_v (Mpa)	Maximum Permissible Shear Stress $F_{v,max}$ (Mpa)
(1)	(2)	(3)	(4)	(5)
i)	Without web reinforcement	< 1.0	$\frac{1}{36} \left(4 - \frac{M}{Vd} \right) \sqrt{f_m}$	$\left(0.6 - 0.2 \frac{M}{Vd} \right)$
		> 1.0	$0.083\sqrt{f_m}$	0.2
ii)	With web reinforcement	< 1.0	$\frac{1}{24} \left(4 - \frac{M}{Vd} \right) \sqrt{f_m}$	$\left(0.6 - 0.2 \frac{M}{Vd} \right)$
		> 1.0	$0.125\sqrt{f_m}$	0.4

10.5 Earthquake Design Requirements

10.5.1 The requirements of this section shall apply to the design and construction of reinforced masonry to improve its performance when subjected to earthquake loads. These provisions are in addition to the general requirements of **5** of Part C/Section 1 of this NBCS. See [Annex F](#).

10.5.2 Different Performance Levels of Masonry Shear Walls

Masonry buildings rely on masonry shear walls for the lateral load resistance and can be detailed for the following three levels of earthquake performance, which can be appropriately chosen for a building considering its importance, location, and acceptable degree of damage. [Table 31](#) summarizes the requirement for these structural walls and R values recommended for use in **5** of Part C/Section 1 of this NBCS.

Table 31 Reinforcement and 'R' Values for Reinforced Masonry Buildings with different Wall Types and Earthquake Zones (Clause 10.5.2)					
Sl No.	Type of Wall	Description	Reinforcement	Earthquake Zone	R
(1)	(2)	(3)	(4)	(5)	(6)
i)	RMB 1	Reinforced masonry walls with minimum reinforcement	As per 10.5.2.1	II and III	2.5

Table 31 (Concluded)

Sl No.	Type of Wall	Description	Reinforcement	Earthquake Zone	R
(1)	(2)	(3)	(4)	(5)	(6)
ii)	RMB 2	Reinforced masonry walls with design reinforcement	As per 10.5.2.2	All	3.0
iii)	RMB 3	Reinforced masonry walls with special design reinforcement	As per 10.5.2.2 and 10.5.2.3	All	3.0

10.5.2.1 Masonry buildings with walls having prescriptive reinforcement (RMB 1)

The design of masonry walls shall comply with requirements of unreinforced masonry wall [4](#) and those given in [10.5.2.1\(b\)](#) and [10.5.2.1\(c\)](#) (see [Fig. 72](#)).

The requirements of this section apply to the design and construction of reinforced masonry to improve its performance when subjected to earthquake loads. These provisions are in addition to the general requirements of [4](#).

- a) Elastic force reduction factor R shall be taken as 3 for *Reinforced Masonry Walls*.
- b) *Minimum reinforcement requirements* ([Fig. 72](#)) — The vertical reinforcement of at least 100 mm² in cross-sectional area shall be provided at a maximum spacing of 3 m on centre at the following critical sections:
 - 1) wall corners;
 - 2) within 400 mm of each vertical face on either side of the openings extending from plinth to roof or floor level; and
 - 3) within 200 mm of the vertical end of the walls.
- c) *Minimum horizontal reinforcement requirement* — The horizontal reinforcement of at least 100 mm² in cross-sectional area shall be provided along the full height of the wall at a maximum centre-to-centre vertical spacing of 400 mm and continuously along the full perimeter of the building, and at the following critical sections:
 - 1) in bond beam (below window openings and above plinth level in the first storey or floor slab level in the second and higher storeys) and anchoring into the vertical reinforcement provided around the openings;
 - 2) within 400 mm of each horizontal face above and below of the openings, extending at least 500 mm or 40d_b from the vertical face of the openings, where d_b is diameter of the horizontal bar being used; and
 - 3) within 400 mm of the top of the walls at structurally connected roof and floor levels.

10.5.2.2 Masonry buildings with walls having design reinforcement (RMB 2)

The masonry walls with design reinforcement shall be designed with the following requirements:

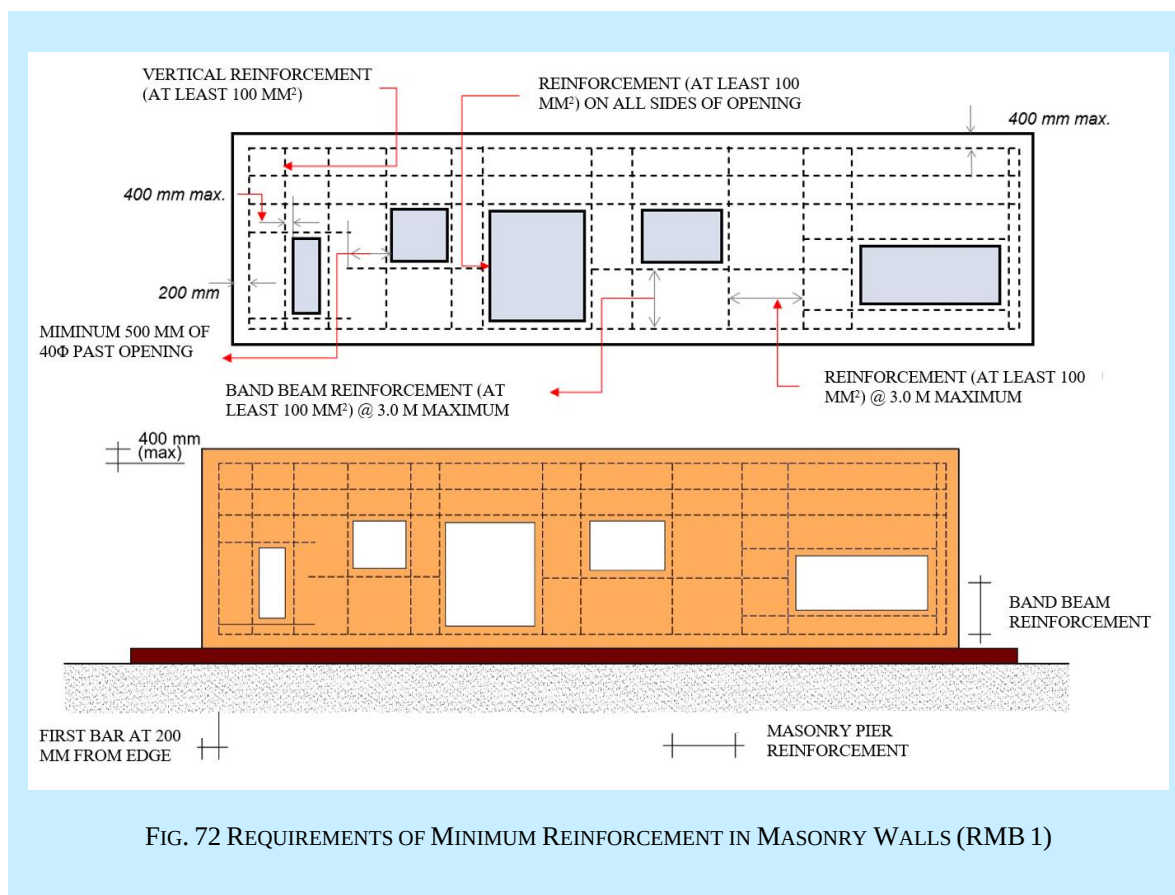
- a) requirements specified in [10.1](#) to [10.4](#);
- b) elastic force reduction factor (R) shall be taken as 3; and
- c) requirements specified in [10.5.2.1\(b\)](#) and [10.5.2.1\(c\)](#).

10.5.2.3 Masonry buildings with walls having special design reinforcement (RMB 3)

Design of special reinforced masonry wall shall comply with the requirement of reinforced masonry as outlined in [10.5.2.2](#) and the following:

- a) The masonry shall be uniformly reinforced in both horizontal and vertical direction such that the sum of reinforcement area in both directions shall be at least 0.2 percent of the gross cross-sectional area of the wall and minimum reinforcement area in each direction shall be not less than 0.07 percent of the gross cross-sectional area of the wall;
- b) Maximum spacing of horizontal and vertical reinforcement shall be smaller of:

- 1) one-third length of the wall;
 - 2) one-third height of the wall; and
 - 3) 1.2 m.
- c) Minimum cross-sectional area of reinforcement in vertical direction shall be one-third of the required shear reinforcement placed in horizontal direction; and
- d) Shear reinforcement placed in horizontal direction shall be anchored around vertical reinforcing bars with a 135° standard hook.



11 GUIDELINES FOR IMPROVING EARTHQUAKE RESISTANCE OF LOW STRENGTH MASONRY BUILDINGS

The term 'low strength masonry' includes fired brickwork laid in clay mud mortar and random rubble; uncoursed, undressed or semi-dressed stone masonry in weak mortars; such as cement sand, lime sand and clay mud. Special features of design and construction for improving earthquake resistance of buildings of low strength masonry are given in [11.1](#) to [11.6.5.7](#).

11.1 For the purpose of this clause, the terminology given in [8.1](#) shall apply.

11.2 General Principles

The general principles given in [9.2.1](#) to [9.2.5](#) shall be observed in construction of earthquake resistant buildings.

11.2.1 Lightness

Since the earthquake force is a function of mass, the weight of the building shall be as minimum as possible, consistent with structural safety and functional requirements. Roofs and upper storeys of buildings, in particular, should be designed as light as possible.

11.2.2 Continuity of Construction

11.2.2.1 As far as possible, the parts of the building should be tied together in such a manner that the building acts as one unit.

11.2.2.2 For integral action of building, roof and floor slabs should be continuous throughout as far as possible.

11.2.2.3 Additions and alterations to the structures should be accompanied by the provision of positive measures to establish continuity between the existing and the new construction.

11.2.3 Projecting and Suspended Parts

11.2.3.1 Projecting parts should be avoided as far as possible. If the projecting parts cannot be avoided, they should be properly reinforced and firmly tied to the main structure, and their design should be in accordance with good practice [C-4(7)].

NOTE — In cases where stability of projecting parts against overturning is achieved by counterweight in the form of wall, slab, etc, the overturning should be checked by increasing the weight of projecting part and decreasing the weight of stabilizing mass simultaneously in accordance with the vertical seismic coefficient specified in 4.4.2 of good practice [C-4(7)].

11.2.3.2 Ceiling plaster shall preferably be avoided. When it is unavoidable, the plaster shall be as thin as possible.

11.2.3.3 Suspended ceiling shall be avoided as far as possible. Where provided they shall be light, adequately framed and secured.

11.2.4 Building Configuration

11.2.4.1 In order to minimise torsion, the building should have a simple rectangular plan and be symmetrical both with respect to mass and rigidity so that the centre of mass and rigidity of the building coincide with each other. It will be desirable to use separate blocks of rectangular shape particularly in seismic zones IV and above.

NOTE — For small buildings, minor asymmetry in plan and elevation may be ignored. Designing such buildings against torsion may be difficult and uncertain.

11.2.5 Fire Safety

Fire frequently follows an earthquake and therefore, buildings shall be constructed to make them fire resistant in accordance with the provisions of Part F 'Fire and Life Safety' of this NBCS, as relevant.

11.3 Special Construction Features

11.3.1 Foundations

11.3.1.1 For the design of foundations, the provisions of Part C/Section 1 and Section 2 of this NBCS shall generally be followed.

11.3.1.2 The subgrade below the entire area of the building shall preferably be of the same type of the soil. Wherever this is not possible, the buildings should preferably be separated into units and then the units should be located separately.

11.3.1.3 Loose fine sand, soft silt and expansive clays should be avoided. If unavoidable, the following measures may be taken to improve the soil on which the foundation of the building may rest:

- a) Sand piling/under-reamed piling/stone columns, etc; and
- b) Soil stabilization.

11.3.2 Roofs and Floors

11.3.2.1 Flat roof or floor should not preferably be made of tiles or ordinary bricks supported on steel, timber or reinforced concrete joists, nor they shall be of a type which in the event of an earthquake is likely to be loosened and parts or all of which may fall. If this type of construction cannot be avoided, the joists should be blocked at ends and bridged at intervals such that their spacing is not altered during an earthquake.

11.3.2.1.1 For pitched roofs, corrugated iron or asbestos sheets should be used in preference to country, Allahabad or Mangalore tiles or other loose roofing units. All roofing materials shall be properly tied to the supporting members. Heavy roofing materials shall generally be avoided.

11.3.2.2 *Pent roofs*

11.3.2.2.1 All roof trusses should be supported on and fixed to timber band, reinforced concrete, or reinforced brick band. The holding down bolts should have adequate size and length as required for earthquake and wind forces.

Where a trussed roof adjoins a masonry gable, the ends of the purlins should be carried on and secured to a plate or bearer which shall be adequately bolted to reinforced concrete or reinforced brick band at the top of gable end masonry.

11.3.2.2.2 At tie level all the trusses and the gable end should be provided with diagonal braces in plan so as to transmit the lateral shear due to earthquake force to the gable walls acting as shear walls.

11.3.2.3 *Jack arches*

Jack arched roofs or floors, where used should be provided with mild steel ties in all spans along with diagonal braces in plan to ensure diaphragm actions.

11.3.3 *Staircases*

11.3.3.1 The inter-connection of the stairs with the adjacent floors should be appropriately treated by providing sliding joints at the stairs to eliminate their bracing effect on the floors. Ladders may be made fixed at one end and freely resting at the other.

11.3.3.2 *Built-in staircase*

When stairs are built monolithically with floors, they can be protected against damage by providing rigid walls at the stair opening. The walls enclosing the staircase, should extend through the entire height of the stairs and to the building foundations.

11.4 Box Type Construction

This type of construction consists of prefabricated or *in-situ* masonry walls along both the axes of the building. The walls support vertical loads and also act as shear walls for horizontal loads acting in any direction. All traditional masonry construction falls under this category. In prefabricated construction attention shall be paid to the connection between wall panels so that transfer of shear between them is ensured.

11.5 Categories of Buildings

For the purpose of specifying the earthquake resisting features in masonry and wooden buildings, the buildings have been categorized in four categories B to E based on the seismic zone and the importance of building I, where I is importance factor applicable to the building, the classification as given in [8.5](#) shall be applicable. The building categories are given in [Table 14](#).

11.6 Low Strength Masonry Construction

11.6.1 Two types of construction are included herein, namely:

- a) Brick construction using weak mortar; and
- b) Random rubble and half-dressed stone masonry construction using different mortars such as clay mud, lime-sand and cement sand.

11.6.1.1 These constructions should not be permitted for important buildings and should preferably be avoided for building category D and shall not be used for category E (see [Table 14](#)).

11.6.1.2 It will be useful to provide damp-proof course at plinth level to stop the rise of pore water into the superstructure.

11.6.1.3 Precautions should be taken to keep the rain water away from soaking into the wall so that the mortar is not softened due to wetness. An effective way is to take out roof projections beyond the walls by about 500 mm.

11.6.1.4 Use of a water-proof plaster on outside face of walls will enhance the life of the building and maintain its strength at the time of earthquake as well.

11.6.1.5 Ignoring tensile strength, free standing walls should be checked against overturning under the action of design seismic coefficient, α_h , allowing for a factor of safety of 1.5.

11.6.2 *Brickwork in Weak Mortars*

11.6.2.1 The fired bricks should have a compressive strength not less than 3.5 MPa. Strength of bricks and wall thickness should be selected for the total building height.

11.6.2.2 The mortar should be lime-sand (1 : 3) or clay mud of good quality. Where horizontal steel is used between courses, cement-sand mortar (1 : 3) should be used with thickness so as to cover the steel with 6 mm mortar above and below it. Where vertical steel is used, the surrounding brickwork of 1×1 or $1 \frac{1}{2} \times 1 \frac{1}{2}$ brick size depending on wall thickness should preferably be built using 1 : 6 cement-sand mortar.

11.6.2.3 The minimum wall thickness shall be one brick in one storey construction, and one brick in top storey and $1 \frac{1}{2}$ brick in bottom storeys of up to three storey construction. It should also not be less than $1/16$ of the length of wall between two consecutive perpendicular walls.

11.6.2.4 The height of the building shall be restricted to the following, where each storey height shall not exceed 3.0 m:

- a) For Categories B and C: three storeys with flat roof; and two storeys plus attic for pitched roof.
- b) For Categories D: two storeys with flat roof; and one storey plus attic for pitched roof.

11.6.2.5 *Special bond in brick walls*

For achieving full strength of masonry, the usual bonds specified for masonry should be followed so that the vertical joints are broken properly from course to course. To obtain full bond between perpendicular walls, it is necessary to make a sloping (stepped) joint by making the corners first to a height of 600 mm and then building the wall in between them. Otherwise the toothed joint should be made in both the walls, alternatively in lifts of about 450 mm (see [Fig. 20](#)).

11.6.3 *Stone Masonry (Random Rubble or Half-Dressed)*

11.6.3.1 The construction of stone masonry of random rubble or dressed stone type should generally follow good practice [C-4(11)].

11.6.3.2 The mortar should be cement-sand (1 : 6), lime sand (1 : 3) or clay mud of good quality.

11.6.3.3 The wall thickness 't' should not be larger than 450 mm. Preferably it should be about 350 mm, and the stones on the inner and outer wythes should be interlocked with each other.

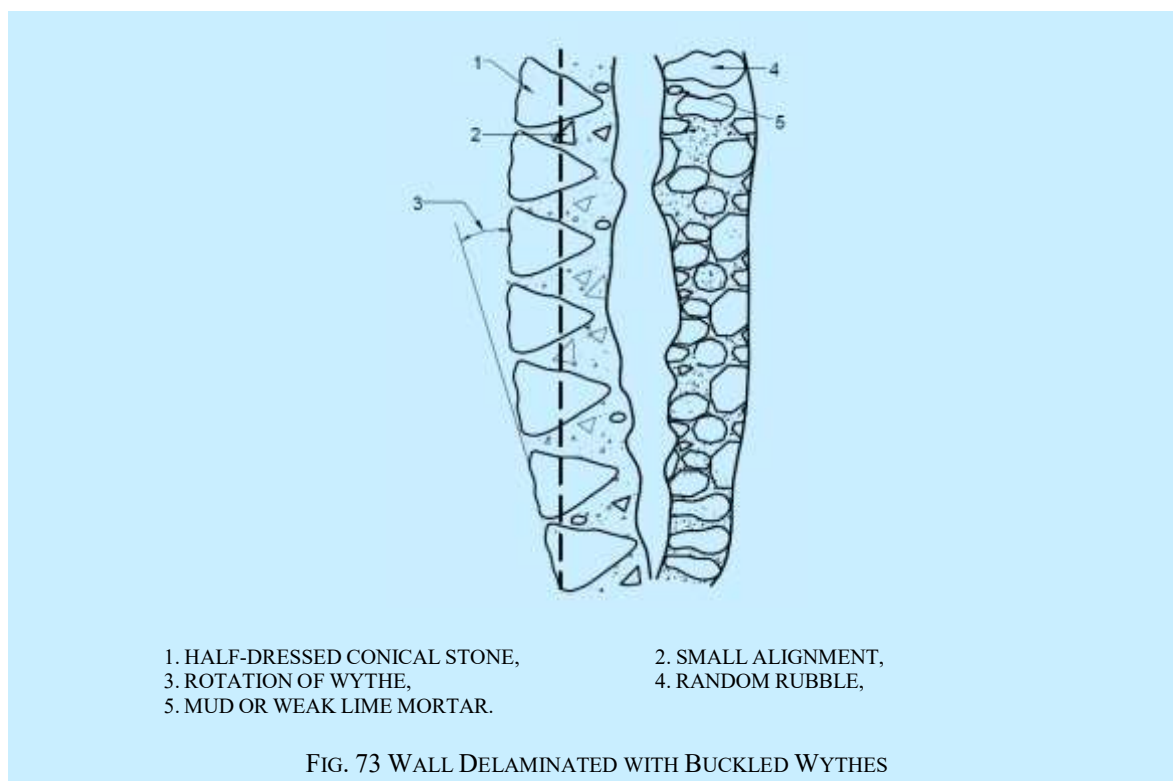
NOTE — If the two wythes are not interlocked, they tend to delaminate during ground shaking, bulge apart (see [Fig. 73](#)) and buckle separately under vertical load leading to complete collapse of the wall and the building.

11.6.3.4 The masonry should preferably be brought to courses at not more than 600 mm lift.

11.6.3.5 'Through' stones of full length equal to wall thickness should be used in every 600 mm lift at not more than 1.2 m apart horizontally. If full length stones are not available, stones in pairs each of about $\frac{3}{4}$ of the wall thickness may be used in place of one full length stone so as to provide an overlap between them (see [Fig. 74](#)).

11.6.3.6 In place of 'through' stones, 'bonding elements' of steel bars 8 mm to 10 mm diameter bent to S-shape or as hooked links may be used with a cover of 25 mm from each face of the wall (see Fig. 74). Alternatively, wood bars of 38 mm × 38 mm cross section or concrete bars of 50 mm × 50 mm section with an 8 mm diameter rod placed centrally may be used in place of 'through' stones. The wood should be well treated with preservative so that it is durable against weathering and insect action.

11.6.3.7 Use of 'bonding' elements of adequate length should also be made at corners and junctions of walls to break the vertical joints and provide bonding between perpendicular walls.



11.6.3.7 Use of 'bonding' elements of adequate length should also be made at corners and junctions of walls to break the vertical joints and provide bonding between perpendicular walls.

11.6.3.8 Height of the stone masonry walls (random rubble or half-dressed) should be restricted as follows, with storey height to be kept 3.0 m maximum, and span of walls between cross walls to be limited to 5.0 m:

- a) *For category B* — Two storeys with flat roof or one storey plus attic, if walls are built in lime-sand or mud mortar; and one storey higher if walls are built in cement-sand 1 : 6 mortar; and
- b) *For categories C and D* — Two storeys with flat roof or two storeys plus attic for pitched roof, if walls are built in 1 : 6 cement mortar; and one storey with flat roof or one storey plus attic, if walls are built in lime-sand or mud mortar, respectively.

11.6.3.9 If walls longer than 5 m are needed, buttresses may be used at intermediate points not farther apart than 4.0 m. The size of the buttress be kept of uniform thickness. Top width should be equal to the thickness of main wall, t , and the base width equal to one sixth of wall height.

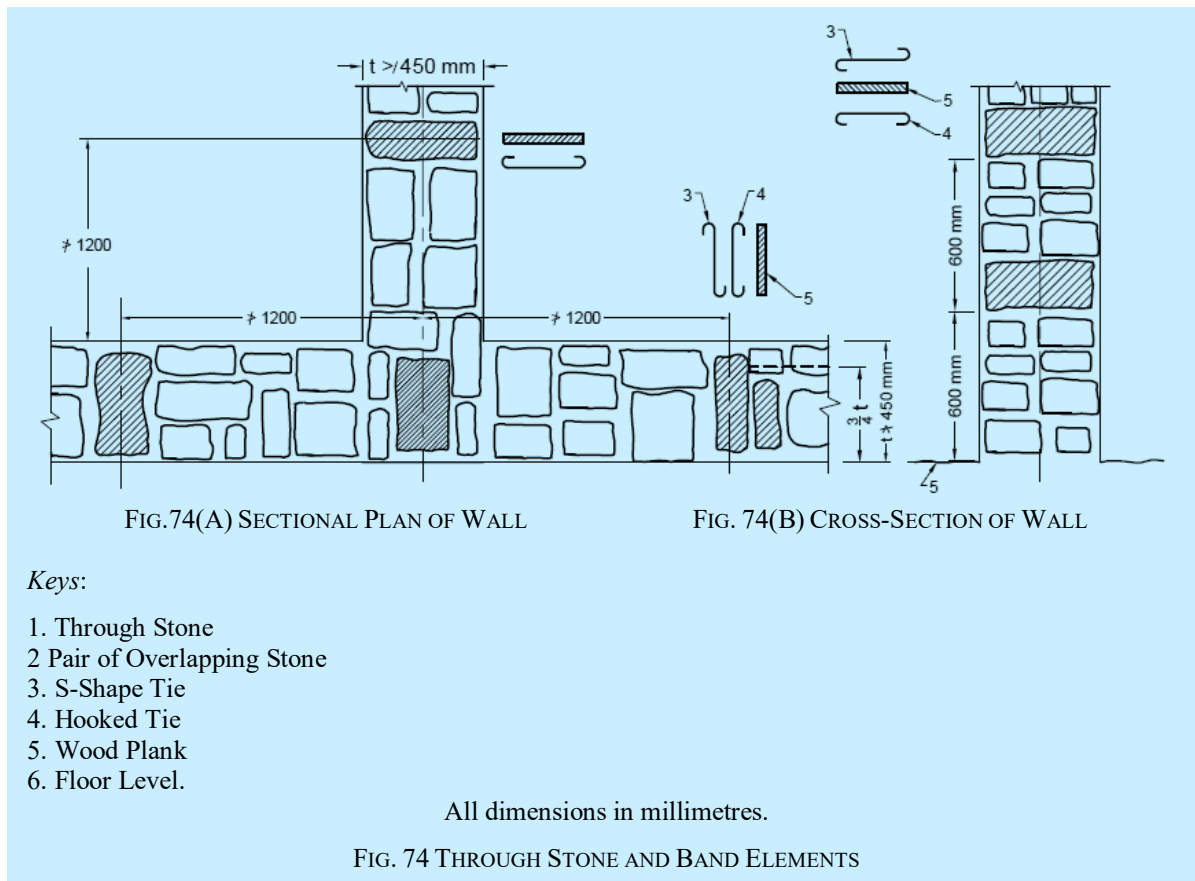
11.6.4 Opening in Bearing Walls

11.6.4.1 Door and window openings in walls reduce their lateral load resistance and hence should preferably, be small and more centrally located. The size and position of openings shall be as given in Table 32 and Fig. 21.

11.6.4.2 Openings in any storey shall preferably have their top at the same level so that a continuous band could be provided over them including the lintels throughout the building.

11.6.4.3 Where openings do not comply with the guidelines of [Table 32](#), they should be strengthened by providing reinforced concrete lining as shown in [Fig. 22](#) with 2 high strength deformed steel bars of 8 mm diameter.

11.6.4.4 The use of arches to span over the openings is a source of weakness and shall be avoided, otherwise, steel ties should be provided.



11.6.5 Seismic Strengthening Arrangements

11.6.5.1 All buildings to be constructed of masonry shall be strengthened by the methods as specified for various categories of buildings, listed in [Table 33](#) and detailed in subsequent clauses. [Fig. 23](#) and [Fig. 24](#) show, schematically, the overall strengthening arrangements to be adopted for category D buildings, which consist of horizontal bands of reinforcement at critical levels and vertical reinforcing bars at corners and junctions of walls.

11.6.5.2 Lintel band is a band provided at lintel level on all internal and external longitudinal as well as cross walls except partition walls. The details of the band are given in [11.6.5.5](#).

11.6.5.3 Roof band is a band provided immediately below the roof or floors. The details of the band are given in [11.6.5.5](#). Such a band need not be provided underneath reinforced concrete or reinforced brick slabs resting on bearing walls, provided that the slabs cover the width of end walls fully.

11.6.5.4 Gable band is a band provided at the top of gable masonry below the purlins. The details of the band are given in [11.6.5.5](#). This band shall be made continuous with the roof band at the eaves level.

11.6.5.5 Details of band

11.6.5.5.1 Reinforced band

The band should be made of reinforced concrete of grade not leaner than M15 or reinforced brickwork in cement mortar not leaner than 1 : 3. The bands should be of full width of the wall, not less than 75 mm in depth and should be reinforced with 2 high strength deformed steel bars of 8 mm diameter and held in position by 6 mm diameter bar links, installed at 150 mm apart as shown in [Fig. 25](#).

NOTES

- 1 In coastal areas, the concrete grade shall be of grade in accordance with Part C/Section 5 of this NBCS and the filling mortar of 1 : 3 ratio (cement- sand) with water proofing admixture.
- 2 In case of reinforced brickwork, the thickness of joints containing steel bars should be increased to 20 mm so as to have a minimum mortar cover of 6 mm around the bar. In bands of reinforced brickwork, the area of steel provided should be equal to that specified above for reinforced concrete bands.
- 3 For full integrity of walls at corners and junctions of walls and effective horizontal bending resistance of bands, continuity of reinforcement is essential. The details as shown in [Fig. 25](#) are recommended.

11.6.5.5.2 Wooden band

As an alternative to reinforced band, the lintel band could be provided using wood beams in one or two parallel pieces with cross elements as shown in [Fig. 75](#).

11.6.5.6 Plinth band is a band provided at plinth level of walls on top of the foundation wall. This is to be provided where strip footings of masonry (other than reinforced concrete or reinforced masonry) are used and the soil is either soft or uneven in its properties as frequently happens in hill tracts. Where used, its section may be kept same as in [11.6.5.5.1](#). This band serves as damp proof course as well.

11.6.5.7 Vertical reinforcement

Vertical steel at corners and junctions of walls which are up to 350 mm thick should be provided as specified in [Table 34](#). For walls thicker than 350 mm, the area of the bars should be proportionately increased.

The vertical reinforcement should be properly embedded in the plinth masonry of foundations and roof slab or roof band so as to develop its tensile strength in bond. It should pass through the lintel bands and floor slabs or floor level bands in all storeys. Bars in different storeys may be welded or suitably lapped.

NOTES

- 1 Typical details of providing vertical steel in brickwork at corners and T-junctions are shown in [Fig. 26](#).
- 2 For providing vertical bar in stone masonry, use of a casing pipe is recommended around which masonry be built to height of 600 mm (see [Fig. 53](#)). The pipe is kept loose by rotating it during masonry construction. It is then raised and the cavity below is filled with M15 (or 1 : 2 : 4) grade of concrete mix and rodded to compact it.

Table 32 Size and Position of Openings in Bearing Walls (see [Fig. 21](#))

(Clauses [11.6.4.1](#), [11.6.4.3](#) and [Table 34](#))

Sl No.	Description	Building Category	
		B and C	D
(1)	(2)	(3)	(4)
i)	Distance b_5 from the inside corner of outside wall, Min, mm	230	600

Table 32 (Concluded)

SI No.	Description	Building Category	
		B and C	D
(1)	(2)	(3)	(4)
ii)	Total length of openings, ratio; $Max (b_1 + b_2 + b_3)/l_1$ or $(b_6 + b_7)/l_2$ shall not exceed:		
	a) one storeyed building; and	0.46	0.42
	d) b) 2 and 3 storeyed building	0.37	0.33
iii)	Pier width between consecutive openings, b_4 , mm	450	560
iv)	Vertical distance between two openings one above the other, h_3 , Min , mm	600	600
v)	Width of opening of ventilator, b_8 , Max , mm	750	750

Table 33 Strengthening Arrangements Recommended for Low Strength Masonry Buildings
(Clause [11.6.5.1](#))

SI No.	Building Category	Number of Storeys	Strengthening to be Provided
(1)	(2)	(3)	(4)
i)	B	1 and 2 3	b, c, f, g b, c, d, f, g
ii)	C	1 2 and 3	b, c, f, g b, c, d, f, g
iii)	D	1 and 2	b, c, d, f, g

where

b = lintel band (see [11.6.5.2](#));

c = roof band and gable band where necessary (see [11.6.5.3](#) and [11.6.5.4](#));

d = vertical steel at corners and junctions of walls (see [11.6.5.7](#));

f = bracing in plan at tie level of pitched roofs (see [11.3.2.2.2](#)); and

g = plinth band, where necessary (see [11.6.5.6](#)).

NOTE — For building of category B in two storeys constructed with stone masonry in weak mortar, it will be desirable to provide vertical steel of 10 mm dia in both storeys.

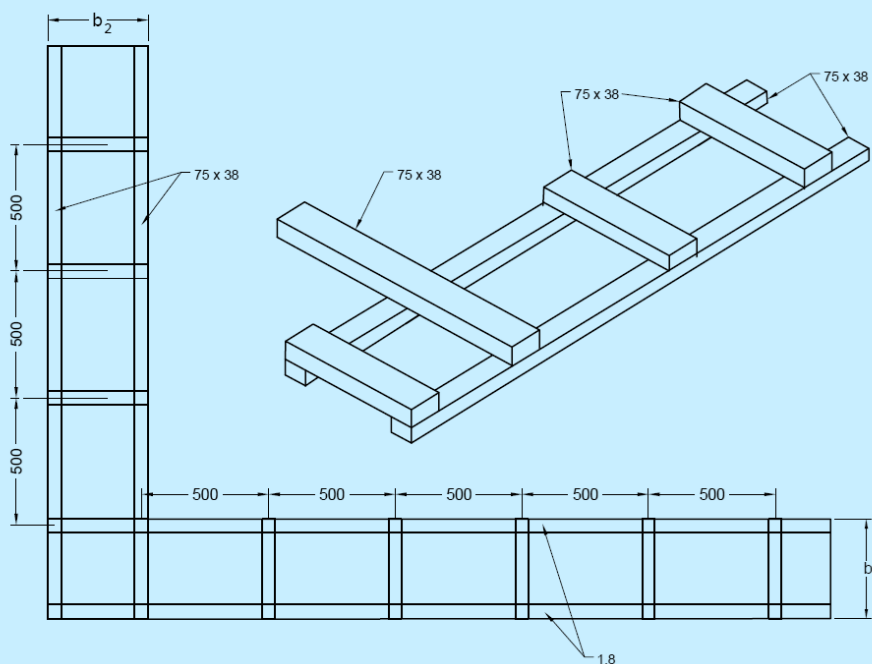


FIG. 75(A) PERSPECTIVE VIEW

FIG. 75(B) PLAN OF BAND

All dimensions in millimetres.

FIG. 75 WOODEN BAND FOR LOW-STRENGTH MASONRY BUILDINGS

Table 34 Vertical Steel Reinforcement in Low Strength Masonry Walls

(Clauses [11.6.5.7](#))

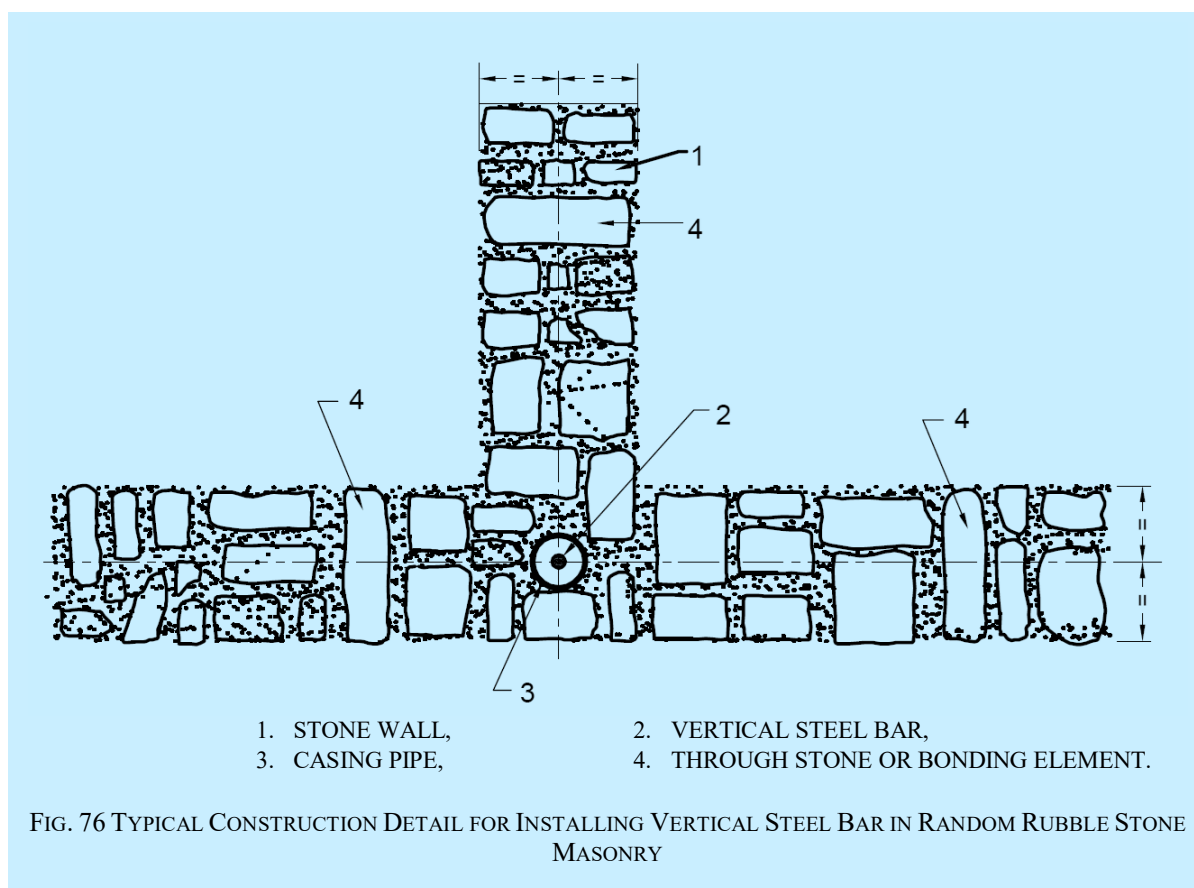
Sl No.	No. of Storeys	Storey	Diameter of HSD Single Bar; in mm, at Each Critical Section for		
			Category B	Category C	Category D
(1)	(2)	(3)	(4)	(5)	(6)
i)	One	—	Nil	Nil	10
ii)	Two	a) Top	Nil	10	10
		b) Bottom	Nil	10	12
iii)	Three	a) Top	10	10	10
		b) Middle	10	10	12
		c) Bottom	12	12	12

NOTES

1 The diameters given above are for high strength deformed bars with yield strength 415 MPa. For mild-steel plain bars, use equivalent diameters.

2 The vertical bars should be covered with concrete of M15 grade or with mortar 1 : 3 (cement-sand) in suitably created pockets around the bars (see [Fig. 26](#) and [Fig. 76](#)). This will ensure their safety from corrosion and good bond with masonry.

3 For category B two storey stone masonry buildings, see Note under [Table 32](#).



12 MASONRY WALLS USING RAT-TRAP BOND TECHNOLOGY

This type of masonry shall be applicable for masonry structures of not more than 2 storeys constructed in seismic zones II and III. For buildings in other seismic zones, this type of masonry wall shall be used only in non-load bearing applications such as partition walls. The provisions for this type of masonry shall be as given in [Annex G](#).

13 SUSTAINABILITY IN MASONRY CONSTRUCTION

The sustainability of masonry structures is addressed by quantifying embodied carbon in the structural system and the sources/types of materials used. For guidelines for selection of materials for masonry and assessing the embodied carbon in the masonry systems, the relevant provisions of the relevant guidelines on sustainability may be followed (Handbook on Approach to Sustainability).

14 NOTATIONS AND SYMBOLS

The various notations and letter symbols used in the text of the Section shall have the meaning as given in [Annex H](#).

ANNEX A

(Clause [4.7](#))

SOME GUIDELINES FOR ASSESSMENT OF ECCENTRICITY OF LOADING ON WALLS

A-1 Where a reinforced concrete roof and floor slab of normal span (not exceeding 30 times the thickness of wall) bear on external masonry walls, the point of application of the vertical loading shall be taken to be at the centre of the bearing on the wall. When the span is more than 30 times the thickness of wall, the point of application of the load shall be considered to be displaced from the centre of bearing towards the span of the floor to an extent of one-sixth the bearing width.

A-2 In case of a reinforced concrete slab of normal span (that is, less than 30 times the thickness of the wall), which does not bear on the full width of the wall and 'cover tiles or bricks' are provided on the external face, there is some eccentricity of load. The eccentricity may be assumed to be one-twelfth of the thickness of the wall.

A-3 Eccentricity of load from the roof/floor increases with the increase in flexibility and thus deflection of the slabs. Also, eccentricity of loading increases with the increase in fixity of slabs/beams at supports. Precast RCC slabs are better than *in-situ* slabs in this regard because of very little fixity. If supports are released before further construction on top, fixity is reduced.

A-4 Interior walls carrying continuous floors are assumed to be axially loaded except when carrying very flexible floor or roof systems. The assumption is valid also for interior walls carrying independent slabs spanning from both sides, provided the span of the floor on one side does not exceed that on the other by more than 15 percent. Where the difference is greater, the displacement of the point of application of each floor load shall be taken as one-sixth of its bearing width on the wall and the resultant eccentricity calculated therefrom.

A-5 For timber and other lightweight floors, even for full width bearing on wall, an eccentricity of about one-sixth may be assumed due to deflection. For timber floors with larger spans, that is, more than 30 times the thickness of the wall, eccentricity of one-third the thickness of the wall may be assumed.

A-6 In multi-storeyed buildings, fixity and eccentricity have normally purely local effect and are not cumulative. They just form a constant ripple on the downward increasing axial stress. If the ripple is large, it is likely to be more serious at upper levels where it can cause cracking of walls than lower down where it may or may not cause local over-stressing.

NOTE — The resultant eccentricity of the total loads on a wall at any level may be calculated on the assumption that immediately above a horizontal lateral support, the resultant eccentricity of all the vertical loads above that level is zero.

A-7 For a wall corbel to support some load, the point of application of the load shall be assumed to be at the centre of the bearing on the corbel.

ANNEX B

(Clauses [5.4.1](#) and [5.4.5](#))

CALCULATION OF BASIC COMPRESSIVE STRESS OF MASONRY BY PRISM TEST

B-1 DETERMINATION OF COMPRESSIVE STRENGTH OF MASONRY BY PRISM TEST

When compressive strength of masonry (f_m) is to be established by tests, it shall be done in advance of the construction, using prisms built of similar materials under the same conditions with the same bonding arrangement as for the structure. In building the prisms, moisture content of the units at the time of laying, the consistency of the mortar, the thickness of mortar joints and workmanship shall be the same as will be used in the structure. Assembled specimen shall be at least 400 mm high and shall have a height to thickness ratio (h/t) of at least 2 but not more than 5. If the h/t ratio of the prisms tested is less than 5 in case of brickwork and more than 2 in case of blockwork, compressive strength values indicated by the tests shall be corrected by multiplying with the factor indicated in [Table 35](#).

Table 35 Correction Factors for Different h/t Ratios
(Clause [B-1](#))

Sl No.	Ratio of height to thickness (h/t)	Correction factors for brickwork ¹⁾	Correction factors for blockwork ¹⁾
(1)	(2)	(3)	(4)
i)	2.0	0.73	1.00
ii)	2.5	0.80	—
iii)	3.0	0.86	1.20
iv)	3.5	0.91	—
v)	4.0	0.95	1.30
vi)	5.0	1.00	1.37

¹⁾ Interpolation is valid for intermediate values.

Prisms shall be tested after 28 days between sheets of nominal 4 mm plywood, slightly longer than the bed area of the prism, in a testing machine, the upper platform of which is spherically seated. The load shall be evenly distributed over the whole top and bottom surfaces of the specimen and shall be applied at the rate of 350 kN/m to 700 kN/m. The load at failure should be recorded.

B-2 CALCULATION OF BASIC COMPRESSIVE STRESS

Basic compressive stress of masonry shall be taken to be equal to $0.25 f_m$ where f_m is the value of compressive strength of masonry as obtained from prism test.

ANNEX C

(Clauses [5.3.3](#) and [5.4.1.5](#))

GUIDELINES FOR DESIGN OF MASONRY SUBJECTED TO CONCENTRATED LOADS

C-1 EXTENT OF DISPERSAL OF CONCENTRATED LOAD

For concentric loading, maximum spread of a concentrated load on a wall may be taken to be equal to $b+4t$ (b is width of bearing and t is thickness of wall), or stretch of wall supporting the load, or centre-to-centre distance between loads, whichever is less.

C-2 INCREASE IN PERMISSIBLE STRESS

C-2.1 When a concentrated load bears on a central strip of wall, not wider than half the thickness of the wall and is concentric, bearing stress in masonry may exceed the permissible compressive stress by 50 percent, provided the area of supporting wall is not less than three times the bearing area.

C-2.2 If the load bears on full thickness of wall and is concentric, 25 percent increase in stress may be allowed.

C-2.3 For loading on central strip wider than half the thickness of the wall but less than full thickness, increase in stress may be worked out by interpolation between values of increase in stresses as given in [C-2.1](#) and [C-2.2](#).

C-2.4 In case concentrated load is from a lintel over an opening, an increase of 50 percent in permissible stress may be taken, provided the supporting area is not less than 3 times the bearing area.

C-3 CRITERIA OF PROVIDING BED BLOCK

C-3.1 If a concentrated load bears on one end of a wall, there is a possibility of masonry in the upper region developing tension. In such a situation, the load should be supported on an RCC bed block (of M20 Grade) capable of taking tension.

C-3.2 When any section of masonry wall is subjected to concentrated as well as uniformly distributed load and resultant stress, computed by making due allowance for increase in stress on account of concentrated load, exceeds the permissible stress in masonry, a concrete bed block (of M20 Grade) should be provided under the load in order to relieve stress in masonry. In concrete, angle of dispersion of concentrated load is taken to be 45° to the vertical.

C-3.3 In case of cantilevers and long span beams supported on masonry walls, indeterminate but very high edge stresses occur at the supports and in such cases it is necessary to relieve stress on masonry by providing concrete bed block of M20 Grade concrete. Similarly when a wall is subjected to a concentrated load from a beam which is not sensibly rigid (for example, a timber beam or an RS joist), a concrete bed block should be provided below the beam in order to avoid high edge stress in the wall because of excessive deflection of the beam.

ANNEX D

(Clause 5.5.5)

GUIDELINES FOR APPROXIMATE DESIGN OF NON-LOAD BEARING WALL

D-1 PANEL WALLS

A panel wall may be designed approximately as under, depending upon its support conditions and certain assumptions:

- when there are narrow tall windows on either side of panel, the panel spans in the vertical direction. Such a panel may be designed for a bending moment of $PH/8$, where P is the total horizontal load on the panel and H is the height between the centres of supports. Panel wall is assumed to be simply supported in the vertical direction;
- when there are long horizontal windows between top support and the panel, the top edge of the panel is free. In this case, the panel should be considered to be supported on sides and at the bottom, and the bending moment should depend upon height to length ratio of panel and flexural strength of masonry. Approximate values of bending moments in the horizontal direction for this support condition, when ratio (μ) of flexural strength of wall in the vertical direction to that in horizontal direction is assumed to be 0.5, are given in Table 36; and
- When either there are no window openings or windows are of 'hole-in-wall' type, the panel is considered to be simply supported on all four edges. In this case also, amount of maximum bending moment depends on height to length ratio of panel and ratio (μ) of flexural strength of masonry in vertical direction to that in the horizontal direction. Approximate values for maximum bending moment in the horizontal direction for masonry with $\mu = 0.50$, are given in Table 37.

Table 36 Bending Moments in Laterally Loaded Panel Walls, Free at Top Edge and Supported on Other Three Edges
[Clause D-1 (b)]

SI No.	Height of Panel, H Length of Panel, L	Bending Moment
(1)	(2)	(3)
i)	0.30	$P.L/25$
ii)	0.50	$P.L/18$
iii)	0.75	$P.L/14$
iv)	1.00	$P.L/12$
v)	1.25	$P.L/11$
vi)	1.50	$P.L/10.5$
vii)	1.75	$P.L/10$

NOTE — For H/L ratio less than 0.30, the panel should be designed as a free-standing wall and for H/L ratio exceeding 1.75, it should be designed as a horizontally spanning member for a bending moment value of $PL/8$.

Table 37 Bending Moments in Laterally Loaded Panel Walls Supported on All Four Edges

[Clause [D-1](#) (c)]

Sl No.	Height of Panel, H Length of Panel, L	Bending Moment
(1)	(2)	(3)
viii)	0.30	$P.L/72$
ix)	0.50	$P.L/36$
x)	0.75	$P.L/24$
xi)	1.00	$P.L/18$
xii)	1.25	$P.L/15$
xiii)	1.50	$P.L/13$
xiv)	1.75	$P.L/12$

NOTE — When H/L is less than 0.30, value of bending moment in the horizontal direction may be taken as nil and panel wall may be designed for a bending moment value of $PH/8$ in the vertical direction; when H/L exceeds 1.75, panel may be assumed to be spanning in the horizontal direction and designed for bending moment of $PL/8$.

D-2 CURTAIN WALLS

Curtain walls may be designed as panel walls taking into consideration the actual supporting conditions.

D-3 PARTITION WALLS

D-3.1 These are internal walls usually subjected to much smaller lateral forces. Behaviour of such wall is similar to that of panel wall and these could, therefore, be designed on similar lines. However, in view of smaller lateral loads, ordinarily these could be apportioned empirically as follows:

- a) Walls with adequate lateral restraint at both ends but not at the top :
 - 1) the panel may be of any height, provided the length does not exceed 40 times the thickness;
 - 2) the panel may be of any length, provided the height does not exceed 15 times the thickness (that is, it may be considered as a free-standing wall); or
 - 3) where the length of the panel is over 40 times and less than 60 times the thickness, the height plus twice the length may not exceed 135 times the thickness;
- b) Walls with adequate lateral restraint at both ends at the top:
 - 1) the panel may be of any height, provided the length does not exceed 40 times the thickness;
 - 2) the panel may be of any length, provided the height does not exceed 30 times the thickness; or
 - 3) where the length of the panel is over 40 times and less than 110 times the thickness, the length plus three times the height should not exceed 200 times the thickness; and
- c) When walls have adequate lateral resistant at the top but not at the ends, the panel may be of any length, provided the height does not exceed 30 times the thickness.

D-3.2 Strength of bricks used in partition walls should not be less than 3.5 N/mm^2 or the strength of masonry units used in adjoining masonry, whichever is less. Grade of mortar should not be leaner than M2.

D-4 CANTILEVER PROJECTIONS

D-4.1 The provisions given under 5 of Part C/Section 1 of this NBCS shall be applicable. As a general guide, the details of connection between parapets and slabs for existing structures shall be as given in [Fig. 77](#).

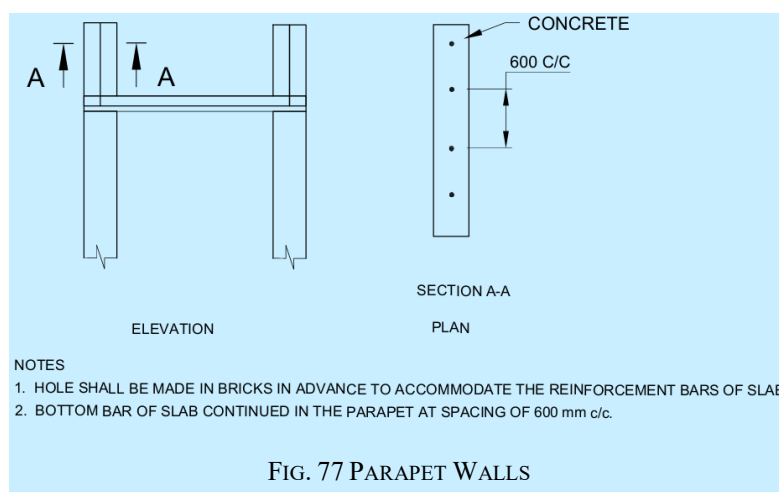


FIG. 77 PARAPET WALLS

ANNEX E

(Clause [10.4.1.2](#))

TYPICAL ITERATIVE METHOD FOR WALLS SUBJECTED TO FLEXURE AND AXIAL LOAD

E-1 There is no closed form solution available for the computation of stress in walls due to combined action of bending and axial compression. Engineers normally prefer to use Design charts for computation of stresses in such conditions, but design charts for most masonry systems are not readily available. Therefore, iterative methods are used as most common tools for masonry structures. Two types of iterative procedures are given below, which can be used to analyse the adequacy of the section under specified load and determine the amount of reinforcement required, wherever necessary.

E-2 ITERATIVE PROCEDURE 1

[Fig. 78](#) shows a diagram of a wall subjected to an axial load and bending moment. The same diagram is applicable to both in-plane and out-of-plane bending of walls. There are three possible conditions for the wall:

- wall is uncracked;
- wall is cracked with the steel in compression (the extent of the crack has not reached the steel); or
- wall is cracked with the steel in tension.

[Fig. 78](#) shows the interaction between bending moment and axial load. Loading conditions beyond the limits of the diagram are beyond the allowable stresses. The condition of the wall can be quickly diagnosed by using the non-dimensional parameter M/Pd . This parameter represents a straight line radiating from the diagram's origin as shown in [Fig. 79](#). By summing moments, it can be shown that certain values of M/Pd divide the diagram into three regions. These values are given in [Table 38](#). The allowable moments for regions 1 and 2 can be obtained in closed form with simple equations. But in region 3, it is complicated a bit. The process consists of initial assumption of thickness, length and other necessary parameters and determination of the amount of steel required.

For region 3, the process starts with another assumption that the tension in steel controls. By making an initial guess about the location of the neutral axis, the expressions for M_p and a_2 provide an iterative process that quickly converges. The equations are derived from the summation of forces and moments and by using linear stress-strain relationships. The equations should always converge for all values of P , l_w , M , and n . If convergence is not obtained, it is likely that the assumed initial value of a has exceeded d . The initial value of a is to be reduced and same procedure is to be repeated. The iteration will often result in a negative steel area. This signifies that the element does not need reinforcement to resist the loads within the allowable stresses, even though the section is cracked (region 3).

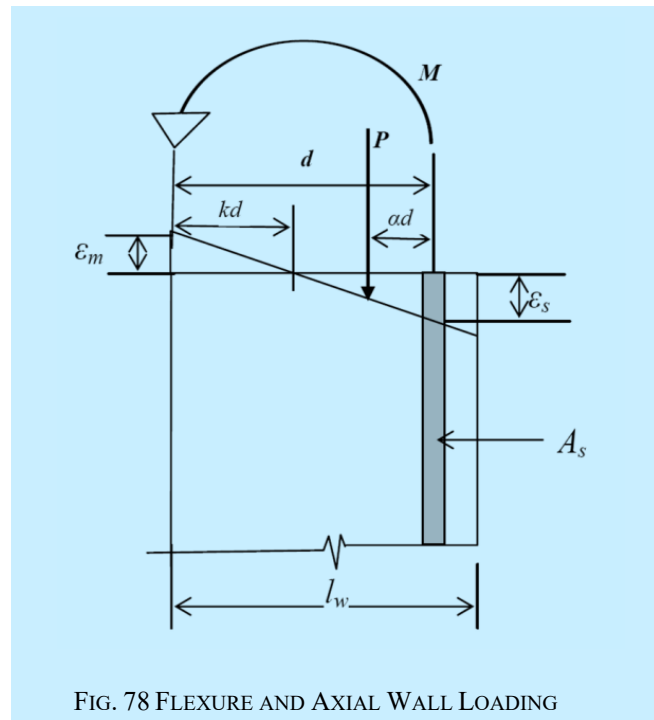


FIG. 78 FLEXURE AND AXIAL WALL LOADING

Following convergence, a check is made of the initial assumption that tension in steel controls. If not valid, then the compression controls and a new set of equations that do not require iteration are used.

E-3 The following is a detailed step-by-step procedure of the analysis methodology described above:

a) Step 1 — Determine the wall condition:

- 1) calculate α as the distance from the axial load to the centroid of the tension steel divided by d ;
- 2) calculate the quantity M/Pd ; and
- 3) use [Table 38](#) to determine the region for analysis. (see [Fig. 78](#)).

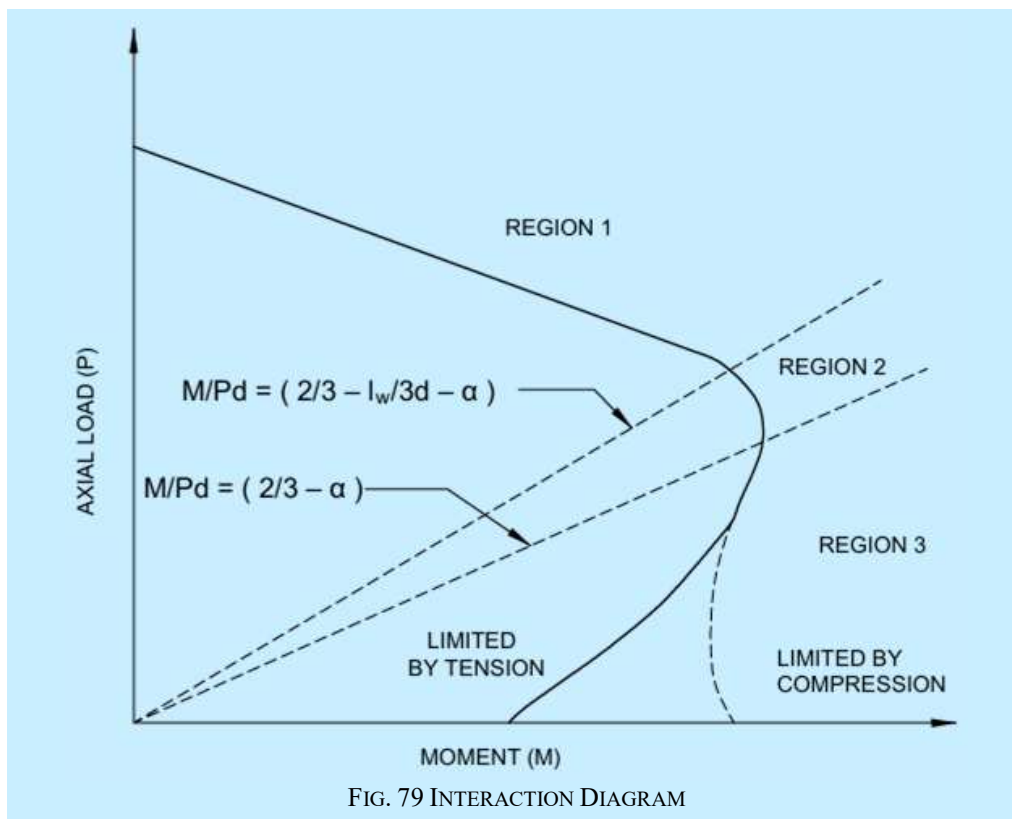


FIG. 79 INTERACTION DIAGRAM

Table 38 Flexure and Axial Loading — Wall Analysis

(Clauses E-2 and E-3)

Sl No. (1)	Region (2)	Condition of Wall (3)	Test (4)
i)	1	Wall is in compression and not cracked	$\frac{M}{Pd} \leq \left[1 - \frac{l_w}{3d} - \alpha \right]$
ii)	2	Wall is cracked but steel is in compression	$\frac{M}{Pd} \leq \left[\frac{2}{3} - \alpha \right]$
iii)	3	Wall is cracked and steel is in tension	$\frac{M}{Pd} > \left[\frac{2}{3} - \alpha \right]$

b) Step 2 — Calculate the allowable moment:

- 1) Region 1 — The moment is limited by flexural compression in the masonry;

$$M_m = \frac{bl_w^2}{6} F_b - P \frac{l_w}{6}$$

If M_m is greater than M applied, the section is satisfactory.

- 2) Region 2 — The moment is also limited by flexural compression in the masonry; and

$$M_m = P(1 - \alpha)d - \frac{2}{3} \left[\frac{P^2}{F_b b} \right]$$

If M_m is greater than M applied, the section is satisfactory.

- 3) Region 3 — The moment may be limited by either the compression in the masonry or tension in the steel. An iterative approach as below shall be adopted.

- i) Assume a compression centroid location, a ; and
ii) Perform the following iteration that assumes the tension in the steel controls (remember that A_s may be negative).

$$M_r = P \left(\frac{l_w}{2} - \alpha \right)$$

$$A_s = \frac{M - M_r}{F_s(d - a)}$$

$$\zeta = \frac{(P + A_s F_s)n}{F_b b}$$

$$a_2 = \frac{\sqrt{\zeta^2 + 2\zeta d} - \zeta}{3}$$

where

- a = estimate of internal compression load centroid distance from the extreme compression fibre;
 A_s = area of trim or edge steel;
 b = width of the wall for out-of-plane bending (thickness of wall for in-plane bending);
 d = distance from the extreme compression fibre to the steel centroid;
 F_s = allowable steel tension stress;
 l_w = width of the wall for in-plane bending (thickness of wall for out of plane bending);

- M = applied moment;
 M_p = moment of applied axial load with respect to the centroid of internal compression force; and
 P = applied axial load at the centre of the wall.

Use a_2 for a , and repeat until the value of a converges.

- c) Step 3 — If the iteration converges and the resulting a is less than the following value, the wall is limited by the tension reinforcement and the analysis is complete. Otherwise continue to step 4:

$$a = \frac{d}{3 \left(1 + \frac{F/F_b}{n} \right)}$$

- d) Step 4 — If the value of a is larger than the above value, determine the required steel area using the following:

$$a = \frac{d}{2} - \sqrt{\left(\frac{d^2}{4} - \frac{2(Pad + M)}{3F_b b} \right)}$$

If $\sqrt{\left(\frac{d^2}{4} - \frac{2(Pad + M)}{3F_b b} \right)}$ is negative, there is inadequate compression capacity.

Increase b or F_b or both. The steel area is:

$$A_s = \frac{\left(\frac{3F_b ab}{2} - P \right)}{\left[nF_b \left(\frac{d}{3a} - 1 \right) \right]}$$

In all cases the wall trim steel should not be less than minimum values.

E-4 ITERATIVE PROCEDURE

The following is another procedure to determine the state of stress in wall elements subjected to combined action of bending and axial compression. Free body diagram of a wall with axial load, bending moment, and tensile and compressive forces is given in [Fig. 80](#). The location of C is based on the neutral axis location kd . The location of T is at the centroid of the reinforcement.

The problem can be solved by taking moments about the centerline of the wall, the tensile reinforcement (assuming that masonry compressive stresses control), or the compressive force (assuming that the stresses in the reinforcement control). By inserting the allowable value of stresses, which is assumed to control, a quadratic equation in terms of kd may be developed and solved.

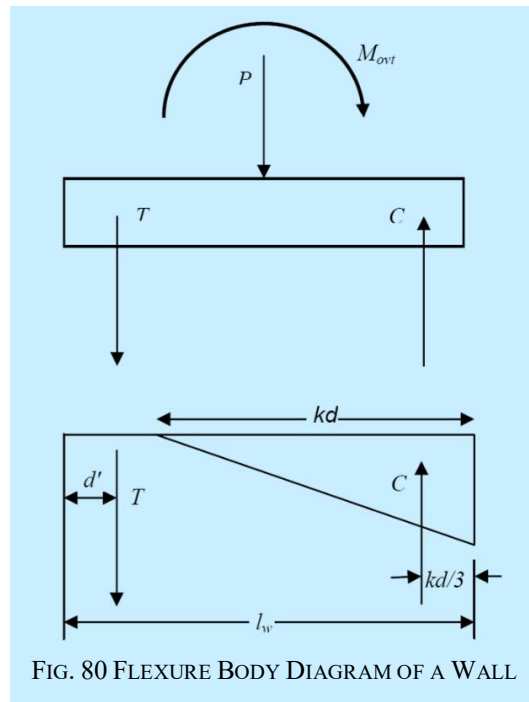


FIG. 80 FLEXURE BODY DIAGRAM OF A WALL

For example if P is at the centerline of the wall, taking moments about the centroid of the tensile reinforcement results in the equation:

$$C \left[(l_w - d') - \left(\frac{kd}{3} \right) \right] = P \left(\frac{l_w}{2} - d' \right) + M$$

but $C = \frac{1}{2} F_m b k d$ and $l_w - d' = d$. Then:

$$\frac{1}{2} F_m b k d \left(d - \frac{kd}{3} \right) = P \left(\frac{l_w}{2} - d' \right) + M$$

Dividing both side by $\frac{1}{2} F_m b$ results in

$$kd^2 - \frac{(kd)^2}{3} = \frac{P \left(\frac{l_w}{2} - d' \right) + M}{\frac{1}{2} F_m b}$$

Multiplying by -3 and rearranging terms:

$$(kd)^2 - 3d(kd) + \frac{3 \left[P \left(\frac{l_w}{2} - d' \right) + M \right]}{\frac{1}{2} F_m b} = 0$$

$$kd = \frac{3d - \sqrt{(-3d)^2 - 4 \left(\frac{3 \left[P \left(\frac{l_w}{2} - d' \right) + M \right]}{\frac{1}{2} F_m b} \right)}}{2}$$

Inserting the values for P , M , l_w , b , d and F_m yields kd . Note that this method is iterative since a location of d' shall be assumed.

Having solved for kd one can calculate C .

By $\sum F_v = 0$, $T = C - P$

From strain compatibility:

$$f_s = \left(\frac{1-k}{k} \right) n F_m, \text{ Thus } f_s \text{ can be calculated.}$$

If $f_s < F_s$, the masonry compressive stress controls and the solution is finished by calculating $A_s = T/f_s$. If however, $f_s > F_s$ the assumption that masonry stresses control is incorrect, and the neutral axis location is wrong.

A good starting assumption to calculate f_m is to take T assuming that masonry compressive stresses control, and to divide by F_s to get a trial A_s . This allows the designer to choose a bar layout and verify the initial assumption of d' , the distance to the centroid of reinforcement.

$$A_s = T/F_s$$

Since $C = T + P$ the effects of the axial load can be included by using an effective amount of reinforcement.

$$(A_s)_{\text{eff}} = A_s + \frac{P}{F_s} \left(\text{or } \frac{T+P}{F_s} \right)$$

$$\rho_{\text{eff}} = \frac{(A_s)_{\text{eff}}}{bd}$$

$$k = \left[(n\rho)^2 + 2n\rho \right]^{\frac{1}{2}} - n\rho$$

$$j = 1 - \frac{k}{3}$$

The steel stress may be checked using

$$f_s = \frac{M'}{(A_s)_{\text{eff}} j d}$$

where

$$M' = P \left(\frac{l_w}{2} - d' \right) + M$$

$$T_{\text{eff}} = (A_s)_{\text{eff}} \times f_s$$

$$T = T_{\text{eff}} - P$$

The masonry stress may be checked by using:

$$f_m = \frac{2M'}{b j k d^2}$$

If $f_s > F_s$, a second iteration may be required may be required with a larger bar size or an extra bar. In a shear wall if an extra bar is added, the value of d' , and the moments calculated using:

$$P \left(\frac{l_w}{2} - d' \right) + M$$

will need to be re-evaluated. Then one can reiterate from:

$$(A_s)_{\text{eff}} = A_s + \frac{P}{F_s}$$

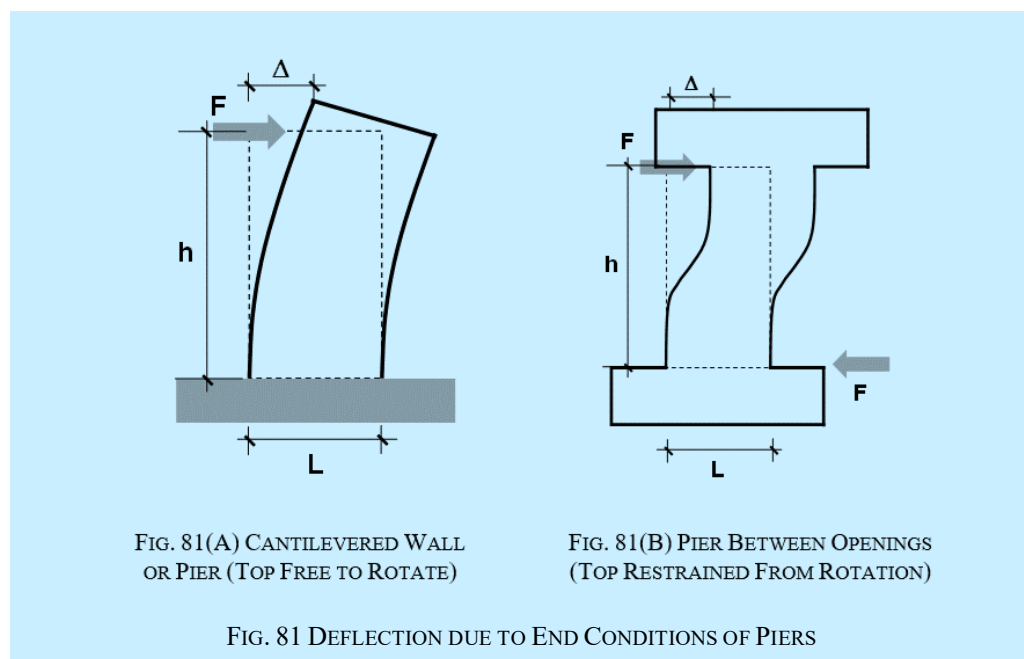
to reduce f_s so that it is less than F_s .

ANNEX F

(Clause 10.5.1)

DISTRIBUTION OF LATERAL FORCES IN THE PIERS OF A MASONRY WALL

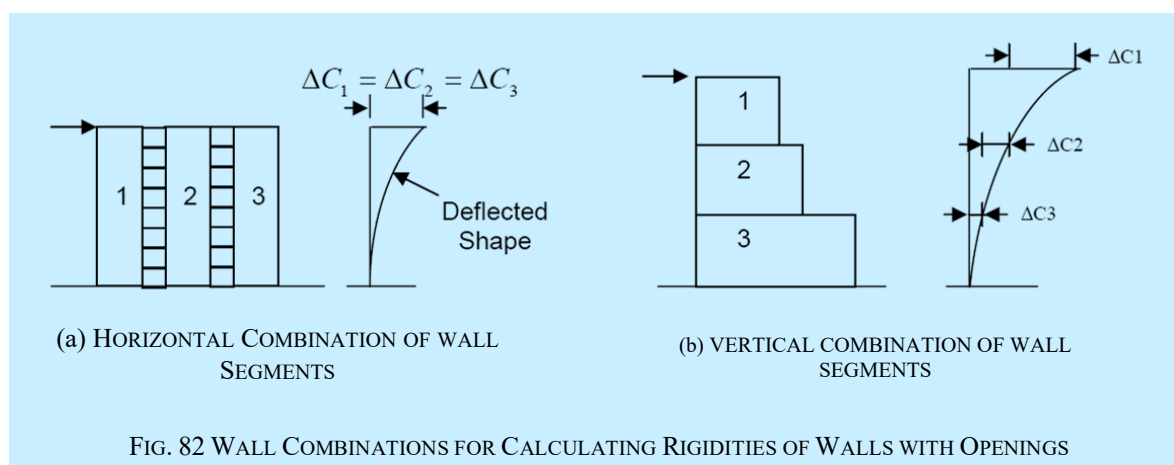
F-1 Masonry walls are often seen to be perforated to make arrangement for windows and doors. The distribution of lateral force in a masonry wall is dependent on the position of the openings and the relative rigidity of the masonry piers created due to the presence of the openings in the masonry wall. The relative rigidity is dependent on the height by length ratio (h/L ratio) of the piers and the end conditions of those masonry piers as the deflection of the masonry piers due to horizontal loading changes due to the end condition of the piers (see Fig. 81). Here is a simple process is described which can be used to distribute the lateral force in a wall which can be considered to be consist of some piers with some specific arrangements.



F-2 In any kind of placing of opening, the wall can be represented as a horizontal and vertical combination of piers with their respective end condition which will be used to find out their rigidities. Where large openings occur, it is difficult to obtain effective coupling of the wall segments or piers. If the wall is analysed as a horizontal combination of piers as shown in the Fig. 82 (a) and the combined rigidity

$$R = R_{c1} + R_{c2} + R_{c3}$$

where R_{c1} , R_{c2} , R_{c3} are the rigidities of the piers 1, 2, 3, respectively.



If the segments are combined vertically, as shown in [Fig. 82 \(b\)](#), the combined rigidity can be calculated as:

$$R = \frac{1}{\Delta C_1 + \Delta C_2 + \Delta C_3} = \frac{1}{\frac{1}{R_{c1}} + \frac{1}{R_{c2}} + \frac{1}{R_{c3}}}$$

Where R_{c1}, R_{c2}, R_{c3} are the rigidities of the piers 1, 2, 3 respectively. Combination of these two types can be used to find the effective relative stiffness of a masonry wall.

The expression ignores the rotations that occur at the tops of segment 2 and 3 and therefore overestimates the rigidity of the wall. Along with this, it is valid only for the application of loads at the top level of the building.

As already said the rigidity R of the pier is dependent on its dimensions, modulus of elasticity E , modulus of rigidity G , and the support conditions. For a cantilever pier the displacement due to combined action of bending and shear is

$$\Delta_c = \frac{Fh^3}{3EI} + \frac{1.2Fh}{GA}$$

where

- A = area of the pier = Lt ;
- F = horizontal force applied to the pier;
- G = modulus of Rigidity = $0.4E_m$;
- h = height of the pier;
- I = moment of inertia of pier = $tL^3/12$;
- L = length of the pier; and
- t = thickness of pier.

So, displacement of the cantilever pier is:

$$\Delta C = \frac{1}{Et} \left[4 \left(\frac{h}{L} \right)^3 + 3 \left(\frac{h}{L} \right) \right]$$

The rigidity of a wall is proportional to the inverse of the deflection. For cantilever walls the rigidity will be:

$$R_c = \frac{1}{\Delta C} = \frac{Et}{\left[4 \left(\frac{h}{L} \right)^3 + 3 \left(\frac{h}{L} \right) \right]}$$

$$\Rightarrow \frac{R_c}{Et} = \frac{1}{\left[4 \left(\frac{h}{L} \right)^3 + 3 \left(\frac{h}{L} \right) \right]}$$

For a pier with both ends fixed against rotation the deflection due to combined action of bending and shear will be:

$$\Delta f = \frac{Fh^3}{12EI} + \frac{1.2Fh}{GA} = \frac{F}{Et} \left[\left(\frac{h}{L} \right)^3 + 3 \left(\frac{h}{L} \right) \right]$$

$$\Rightarrow \frac{R_f}{Et} = \frac{1}{\left[\left(\frac{h}{L} \right)^3 + 3 \left(\frac{h}{L} \right) \right]}$$

Values of R/E_t for different h/L ratio can be taken from the chart in the [Fig. 83](#).

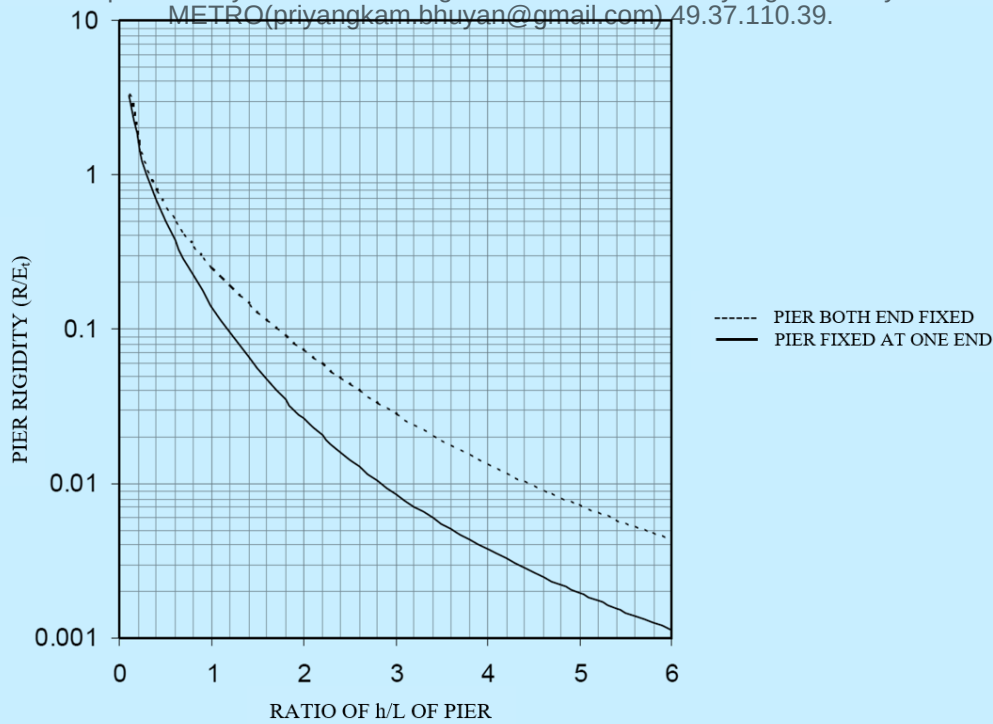


FIG. 83 CHARTS FOR CALCULATING WALL RIGIDITIES

From the relations given in equations above it is found that the relative contributions of the bending and shear deformation depend on the wall aspect ratio (h/L) and therefore, the rigidity varies over the height of the building. For high h/L ratios, the effect of shear deformation is very small and calculation of pier rigidities based on flexural stiffness is relatively accurate. For very squat walls (with $h/L < 0.25$), rigidities based on shear deformation are reasonably accurate, but for intermediate walls with h/L from 0.25 to 4, both components of relative rigidity should be considered.

F-3 The method is explained further explained by one illustration. For the wall as shown in [Fig. 84](#).

$$R_{\text{wall}} = \frac{1}{\Delta_{1c} + \Delta_{2,3,4,5,6(f)}} = \frac{1}{\frac{1}{R_{1c}} + \frac{1}{R_{2,3,4,5,6(f)}}} \quad (\text{Vertical combination})$$

where

$$R_{2,3,4,5,6(f)} = R_{2(f)} + R_{3,4,5,6(f)} \quad (\text{Horizontal combination})$$

$$\text{and } R_{3,4,5,6(f)} = \frac{1}{\Delta_{3,4,5(f)} + \Delta_{6(f)}} = \frac{1}{\frac{1}{R_{3,4,5(f)}} + \frac{1}{R_{6(f)}}} \quad (\text{Vertical combination})$$

$$R_{3,4,5(f)} = R_{3(f)} + R_{4(f)} + R_{5(f)} \quad (\text{Horizontal combination})$$

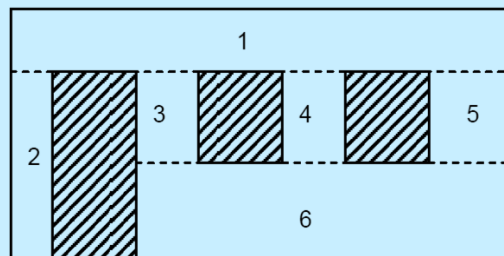


FIG. 84 WALL WITH OPENINGS (CALCULATION OF RIGIDITY AND LOAD DISTRIBUTION)

ANNEX G

(Clause 12)

MASONRY WALLS USING RAT-TRAP BOND

G-1 GENERAL

The following covers the construction of masonry walls using rat-trap bond. Only masonry with bricks (burnt clay bricks, fly ash-clay bricks, fly ash-lime bricks and stabilized soil blocks) is dealt herein. Use of other types of bricks/blocks and mud mortar are not specifically covered hereunder.

G-2 NECESSARY INFORMATION

G-2.1 For efficient planning, design and execution of the work, detailed information with regard to the following shall be furnished to those responsible for the work:

- a) layout plans showing the orientation of the structure;
- b) dimensioned details of the structure with details of sections (to a suitably large scale that is 1 : 20), levels of foundation, finished ground level, clear floor to floor heights of rooms, sizes of openings, etc;
- c) type and class of brickwork, types of bond and final finish for the brickwork; the mixes of mortar to be used, etc, full size details of architectural features, mouldings and other special work. Also course plans at different levels of sill and lintel to be made available for proper understanding and execution, if required; and
- d) location and other details of openings, chases, embedment of service lines, such as for water supply, drainage, electrical installations, etc, and location and details of hearths, flues and chimneys in the brickwork.

G-2.2 All information as in [G-2.1](#) shall be made available to those who are responsible for the masonry work. Necessary drawings and instructions for planning the work shall be furnished. Arrangements shall also be made for the proper exchange of information between those engaged in masonry work and all those whose work will affect or will be affected.

NOTE — If necessary a brief training of the masons may be carried out orienting them with the technology know-how.

G-3 MATERIALS

G-3.1 Bricks

Burnt-clay bricks, heavy duty bricks and bricks made out of other suitable materials like fly ash, stabilized soil block, etc, conforming to accepted standards [C-4(12)] with compressive strength greater than 6.0 MPa shall be used.

G-3.2 Mortars

Shall conform to grade H2 mortar for good practice [C-4(3)].

G-3.3 Storage of Materials

Storage of materials shall be in accordance with good practice [C-4(2)].

G-3.4 Materials for Damp-Proof Courses

Materials for damp proof courses shall be as specified in [C-4(5)].

G-3.5 Materials for Flashing and Weathering

Lead flashing with sheet thickness between 1.6 mm and 2 mm or bitumen felts shall be in accordance with the accepted standards [C-4(13)].

G-3.6 Metal Reinforcement

Metal reinforcement for use in brick masonry shall be as per accepted standards [C-4(14)].

G-4 DESIGN CONSIDERATIONS

G-4.1 Selection of Bricks

The brick shall be selected in accordance with Table 1 of good practice [C-4(4)].

When the requirements for strength of masonry predominate in the particular situation of use, the bricks shall be of such class to give the required strength for masonry and shall be selected in accordance with 4.

G-4.2 Rat-Trap Bond Details

G-4.2.1 The rat-trap bond is laid by placing the bricks on edge forming a cavity of 80 mm (for a traditional brick size) with alternate course of stretchers and headers (Fig. 85). The headers and stretchers are staggered in subsequent layers to impart strength and stability to the walls.

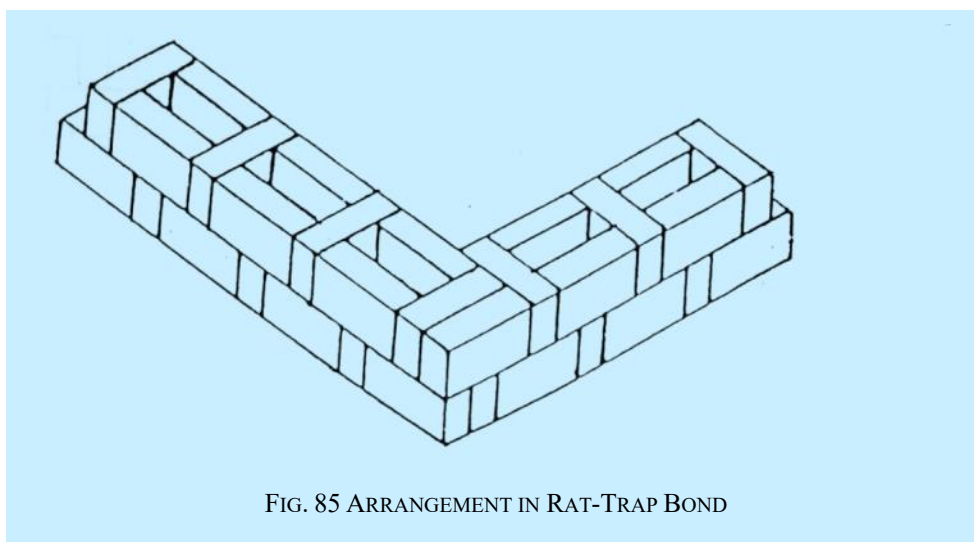


FIG. 85 ARRANGEMENT IN RAT-TRAP BOND

The primary object of bond is to give strength to masonry, but when exposed, it also creates aesthetically pleasing view. This masonry technology is modular in nature, so the dimensions of the wall to be based on the module dimensions in multiples.

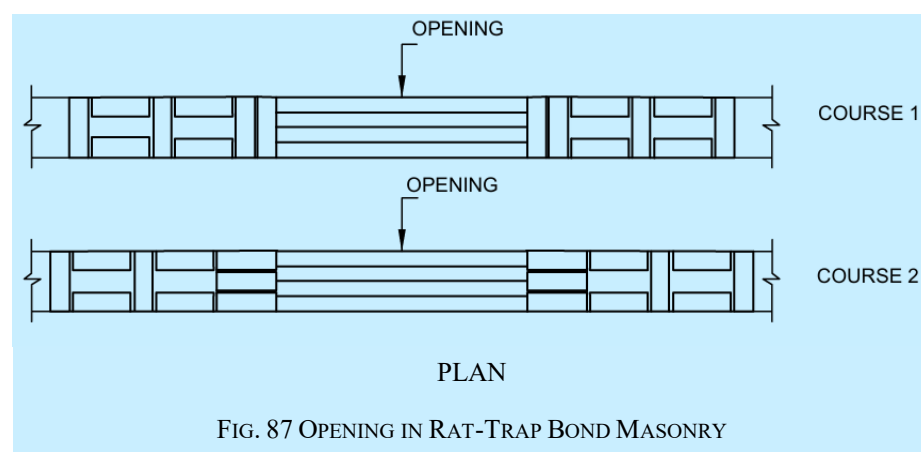
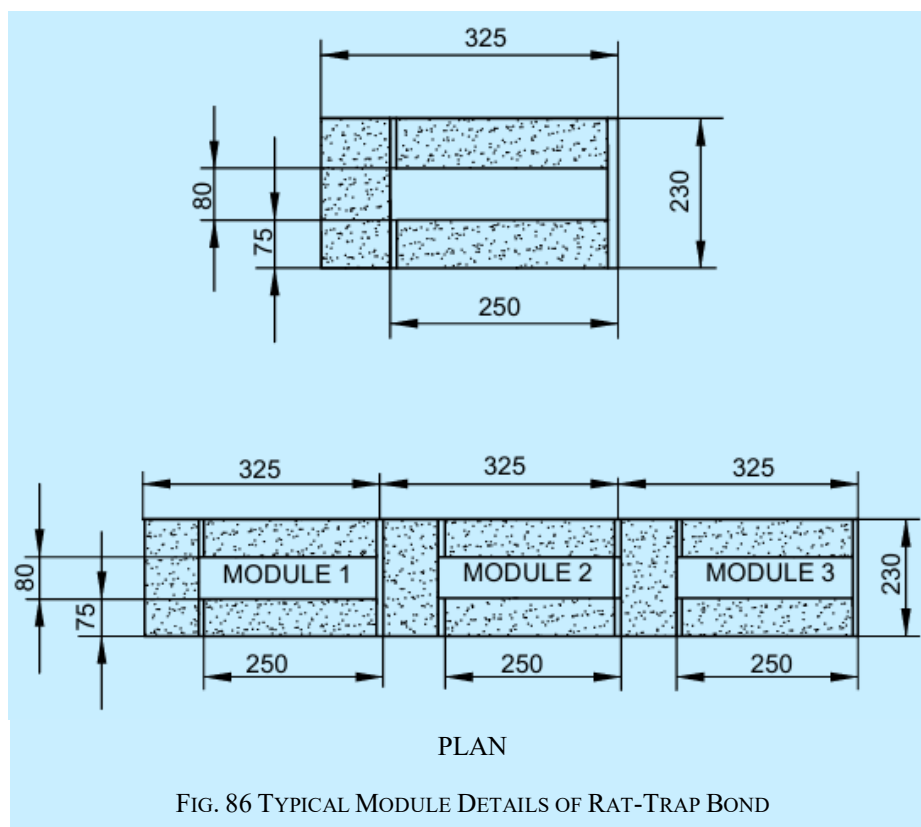
G-4.2.2 A typical module detail is shown in Fig. 86 with conventional bricks of size 230 mm × 110 mm × 75 mm. The brick adjacent to the openings has to be solid as per Fig. 87. So, the opening dimensions also to be in multiple of the rat-trap bond module. This will be important as opening dimensions are predefined depending on the module dimension. In brickwork the cross joints in any course shall not be nearer than a quarter of brick length from those in the course below or above it.

G-4.3 Precautions

This type of masonry construction shall have to be carried out by trained masons to avoid wastage of mortar falling into the gap. Rat-trap bond shall never be carried out with brick which are below their minimum strength requirements.

G-4.4 Strength of Rat-Trap Bond

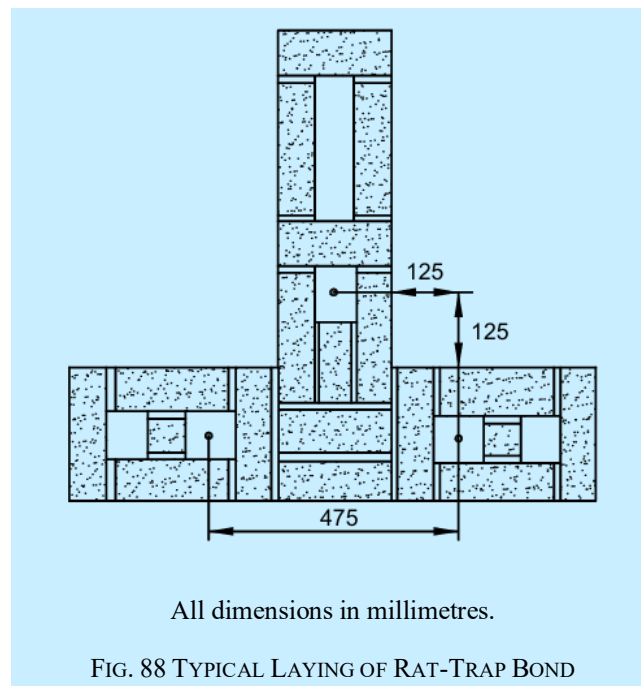
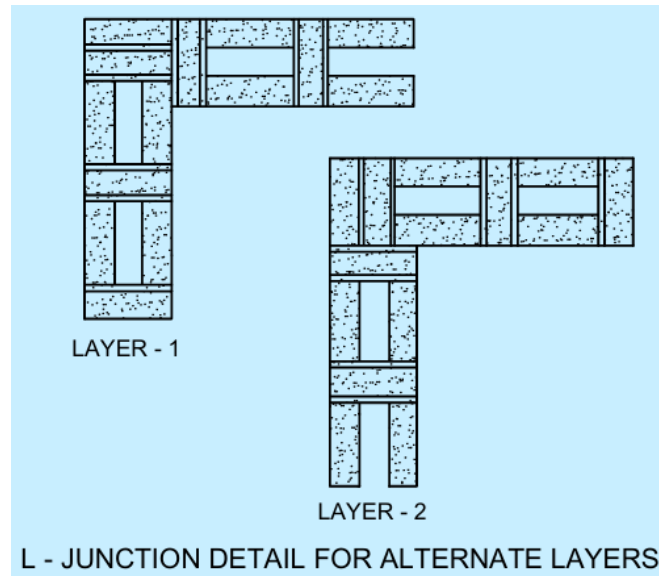
The compressive strength of rat-trap bond will be different from that of solid wall of similar thickness. Rat-trap bond wall provides continuous cavity over the entire wall height and hence it is convenient to reinforce if necessary to resist lateral loads. The compressive strength of the unreinforced rat-trap bond wall can be determined either based on the cavity wall design guidelines following 5.5.1.2 or the basic compressive stress can be assessed following the tests on prism specimen given at G-7.



G-4.5 Thickness of Joint

No bed joint shall be thicker than 12 mm. Further the thickness of the bed joints shall be such that four courses and three joints taken consecutively shall measure as follows:

- In the case of traditional bricks* — Four times the actual thickness of the brick plus 30 mm.;
- In the case of modular bricks* — 390 mm conforming to the relevant accepted standards. The vertical mortar joints to be of the thickness of 10 mm and be applied directly to the brick before placing it. This requires some special skill, but defines the strength as well as the stability of the Rat-trap bond wall. Besides, applying the mortar joint directly to the brick is a basic skill requirement of a mason and determines the speed of work and finally the cost of the Rat-trap bond wall; and
- The *face jolts* of brickwork may be finished by 'jointing' or by 'pointing'. In case where brick size is less, the elevation of the header bricks to be made even in line with the stretcher brick by cement mortar same grade as bedding mortar. The main reasons for pointing the surface of block or brickwork joints are to increase its weather resistance and to give a neat finish to the work. Pointing can be carried out as construction of the brickwork proceeds, using ordinary mortar in which the bricks are bedded.



G-4.6 Structural and Functional Characteristics of Brickwork

G-4.6.1 Structural Stability and Strength

Reference may be made to Part C/Section 2 of this NBCS, and [4](#) of this Section.

G-4.6.2 Resistance to Moisture Penetration

The suitability of various walls under different conditions of exposure is indicated in Table 3 of good practice [C-4(4)].

G-4.6.3 Thermal Stresses

The coefficient linear expansions of brick vary from $5 \times 10^{-6}/^{\circ}\text{C}$ to $11 \times 10^{-6}/^{\circ}\text{C}$. Variations of temperature tend to produce linear changes in walls which, when restrained, may lead to internal stress resulting in cracks especially when the walls exceed 30 m. The effects of these stresses shall be taken into consideration for a proper design.

G-4.7 Control of Shrinkage Cracking

G-4.7.1 To confine cracks to the joints and to dissipate these into a large number of fine joints, it is desirable that the mortar used shall be weaker than the bricks.

G-4.7.2 Cracking due to shrinkage normally will occur at openings or other points where the vertical or horizontal section of a wall changes. A long wall with a few openings will tend to show wider cracks above and below the openings than a similar wall with many openings. Metal reinforcement may be embedded in brickwork at points where cracking is likely to occur.

G-4.8 Thermal Insulation

For requirements of thermal insulation in walls, the provisions of good practice [C-4(15)] shall apply.

G-4.9 Fire Resistance

For requirements regarding resistance to fire, the provisions of good practice [C-4(16)] shall apply.

G-4.10 Sound Insulation

Requirements for insulation against airborne sound are laid down in good practice [C-4(17)] shall apply.

G-5 LAYING OF BRICKWORK

G-5.1 Brick shall be laid on a full bed of mortar. When laying the bricks, it shall be slightly pressed so that the mortar gets into all the pores of the brick surface to ensure proper adhesion. Cross joints and wall joints shall be properly flushed and packed with mortar so that no hollow spaces are left. Properly filled joints ensure maximum strength and resistance to penetration of moisture which takes place mainly through joints.

G-5.2 The following specific guidelines shall be followed:

- a) In areas with a very high ground water table, it is advisable to build the first 4 to 5 courses of the wall, with either English or a Flemish bond. This is to avoid ground and spill water related dampness problems;
- b) Always clean the surface where the first layer of bricks is to be laid, which is usually above the plinth level;
- c) Arrange the joints between the bricks in such a way that only full bricks (shiners) fit. Ensure that the bonding is correct;
- d) To properly maintain the brick cross, lay the first two masonry courses by marking mark each of the shiner and the header bricks at the centre;
- e) Place the shiner/stretcher and header brick now in such a way that the marks are exactly matching. The remaining courses may then be laid to match with the first two courses;
- f) Spread a full bed of mortar, and furrow it with a trowel. This will ensure that plenty of mortar is on the bottom of the bricks for the first course;
- g) Lay the corner bricks first. Carefully position the corner bricks first. Be sure to lay all bricks with the frog inside the cavity of the wall in order to create an aesthetically pleasing outer wall surface;
- h) The vertical mortar joints shall be applied directly to the brick before placing it. This requires some special skill, but defines the strength as well as the stability of the rat-trap bond wall;
- j) The maximum rat-trap bond brick wall height per day should not exceed more than 9 layers; and

- k) Cavities of rat-trap bond shall be closed for the first course after damp proof course, below sill level, below and above lintel and at the top most course; this can be made by a layer of headers or headers.

G-5.3 Protection Against Damage

Care shall be taken during construction that edges of jambs, sills, heads, etc are not damaged. In inclement weather, newly built work shall be covered with gunny bags or tarpaulin so as to prevent the mortar from being washed away.

G-6 INSPECTION

G-6.1 General

The object of inspection of brick masonry work is to ensure its satisfactory performance and also to ascertain whether all the recommended practice of workmanship is adopted at every stage. As the correct strength of masonry cannot be ascertained without destruction, a close supervision during the course of construction is necessary to ensure satisfactory performance. The strength of brickwork depends on the strength of individual brick, strength of mortar, bond and workmanship.

G-6.2 Inspection of Materials

In case of large works, samples of bricks, sands, *Surkhi*, lime, cement, etc, which goes to form the brickwork, shall be periodically tested in a laboratory to make sure that they conform to the requirements stated in [G-3](#). Simple field tests may suffice in the case of small works.

G-6.3 Inspection of Workmanship

A close supervision while the work is in progress will ensure a better quality work with the materials available for use. The following shall be observed at the time of inspection:

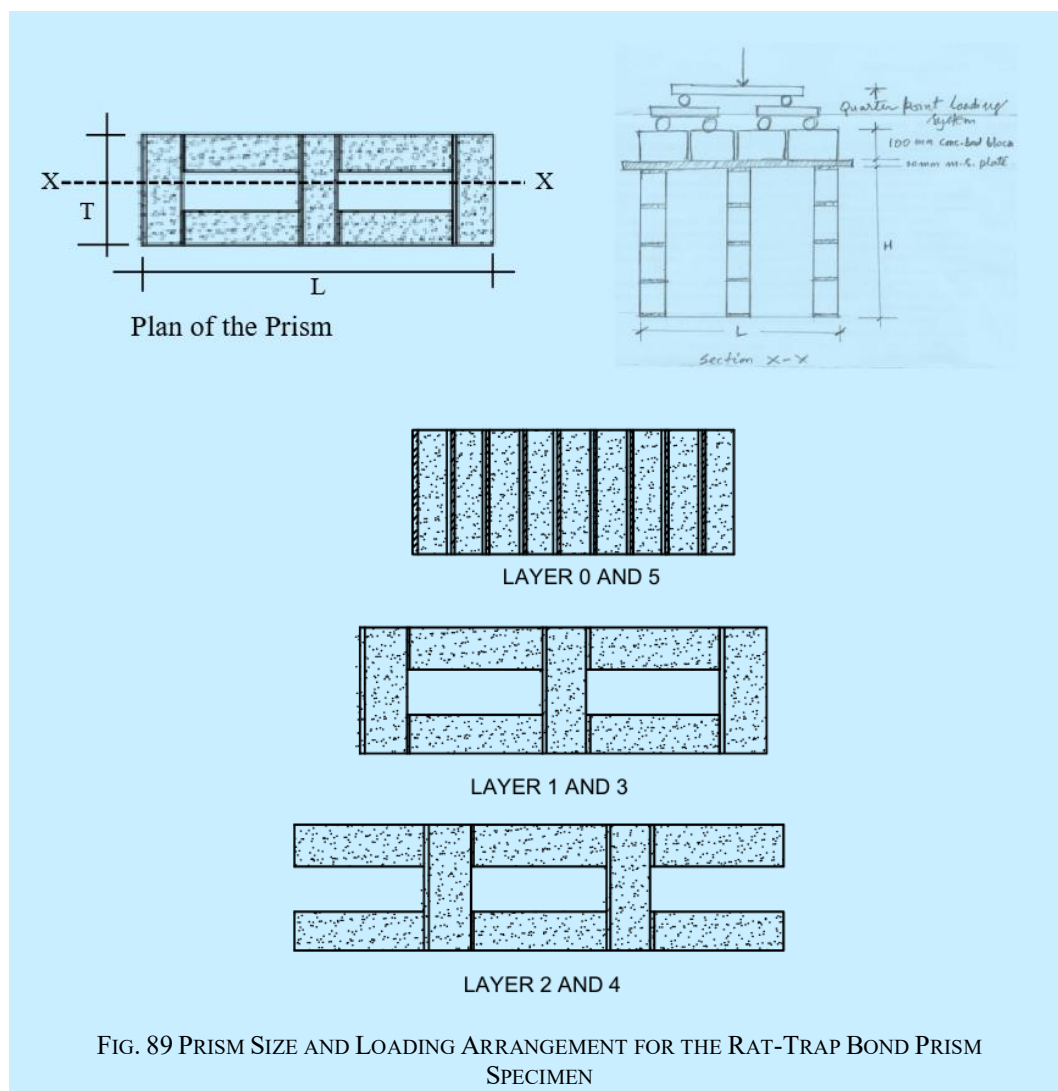
- a) all loose materials, dirt and set lumps of mortar which may be lying over the surface over which brickwork is to be freshly started, shall be removed with a wire brush;
- b) all the bricks shall be thoroughly soaked in potable water before use;
- c) the surface over which the brickwork is to be started shall be made moist;
- d) plastic mortar results in thorough bedding of the brick and more complete filling of the joints which ensure greater strength. Care shall be taken to see that the required quantity of water is added to the mortar at the mixing platform itself and not over the courses;
- e) all the joints shall, as far as possible, be thin and the specifications mentioned in [G-4.5](#) in this regard shall be strictly adhered to;
- f) care shall be taken to see that there is no through joints and the lap is not less than half the width of the brick, and that all the vertical joints are properly filled with mortar;
- g) the verticality of the walls and horizontality of the courses shall be checked very often with plumb bob and spirit level respectively;
- h) no portion of the work shall be left more than 1 m lower than the other. Where the masonry of one part has to be delayed the work shall be 'raked back' suitably at an angle not exceeding 45° according to bond and not toothed;
- j) where plastering is required to be done all the vertical as well as horizontal joints shall be raked to a depth of about 10 mm while the mortar is wet, and this will ensure satisfactory adhesion between the plaster and brickwork; and
- k) care shall be taken to ensure that the brick work is kept wet for first seven days using appropriate curing methods.

G-7 TESTING METHOD FOR RAT-TRAP BOND WALL SAMPLES

G-7.1 A Prism specimen as shown in [Fig. 89](#) shall be used. The prism specimen should have similar mortar and bricks/blocks as that to be used in the rat-trap bond wall of the structure. The prism height (H) to width (T) ratio

should be between 2 and 5 and based on H/T ratio the prism strength should be corrected following the correction factors given in [B-1](#) for blockwork.

The number of prism specimens to be tested and the necessary correction factors to be applied to derive modified compressive strength of the prism shall be ascertained following the procedure given in [5.4](#). The modified mean prism compressive strength shall be divided by 4 (as per [B-2](#)) to assess the basic compressive stress for the rat-trap bond wall, which can be used in calculating the compressive strength of rat-rap bond masonry wall. A similar test as above should be done in an English bond per coupon (see [Fig. 90](#)).



G-7.2 The load on rat-trap bond specimen should be not less than 85 percent of the load capacity of English bond wall compressive strength.

G-7.3 While designing the compressive stress on rat-trap bond wall, it should be ensured that the stress is not more than 0.85 of that of solid wall as per 4 of this Section.

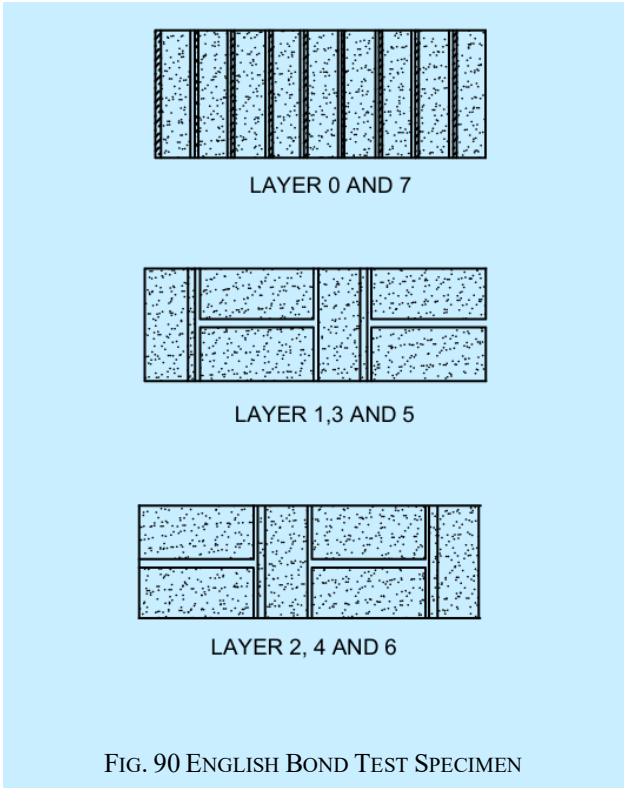


FIG. 90 ENGLISH BOND TEST SPECIMEN

ANNEX H

(Clause 14)

NOTATIONS, SYMBOLS AND ABBREVIATIONS

H-1 The following notations, letter symbols and abbreviations shall have the meaning indicated against each, unless otherwise specified in the text of this Section of this NBCS:

A	=	Area of a section
b	=	Width of bearing
DPC	=	Damp proof course
e	=	Resultant eccentricity
f_b	=	Basic compressive stress
f_c	=	Permissible compressive stress
f_d	=	Compressive stress due to dead loads
f_s	=	Permissible shear stress
f_m	=	Compressive strength of masonry (in prism test)

GL	=	Ground level
H	=	Actual height between lateral supports
H'	=	Height of opening
$H1, H2$	=	High strength mortars
h	=	Effective height between lateral supports
k_a	=	Area factor
k_p	=	Shape modification factor
k_s	=	Stress reduction factor
L	=	Actual length of wall
$L1, L2$	=	Lower strength mortars
$M1, M2$	=	Medium strength mortars
P	=	Total horizontal load
PL	=	Plinth level
RCC	=	Reinforced cement concrete
RS	=	Rolled steel
S_p	=	Spacing of piers/buttresses/cross walls
SR	=	Slenderness ratio
t	=	Actual thickness
t_p	=	Thickness of pier
t_w	=	Thickness of wall
W	=	Resultant load
W_1	=	Axial load
W_2	=	Eccentric load
w_p	=	Width of piers/buttresses/cross walls
μ	=	Ratio of flexural strength of wall in the vertical direction to that in the horizontal direction

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used by as a guide in conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Section.

<i>IS No.</i>	<i>Title</i>
(1) IS 1077 : 2025	Common burnt clay building bricks — Specification (<i>sixth revision</i>)
IS 1725 : 2023	Specification for stabilized soil blocks used in general building construction (<i>third revision</i>)
IS 2180 : 1988	Specification for heavy duty burnt clay building bricks (<i>third revision</i>)
IS 2185	Concrete masonry units — Specification:
(Part 1) : 2005	Hollow and solid concrete blocks (<i>third revision</i>)
(Part 2) : 1983	Hollow and solid light weight concrete blocks (<i>first revision</i>)
(Part 3) : 1984	Autoclaved cellular (aerated) concrete blocks (<i>first revision</i>)
(Part 4) : 2008	Cellular preformed foam cellular concrete blocks
IS 2222 : 1991	Specification for burnt clay perforated building bricks (<i>third revision</i>)
IS 2849 : 1983	Specification for non-load bearing gypsum partition blocks (solid and hollow types) (<i>first revision</i>)
IS 3115 : 1992	Specification for lime based blocks (<i>second revision</i>)
IS 3316 : 1974	Specification for structural granite (<i>first revision</i>)
IS 3620 : 1979	Specification for laterite stone block for masonry (<i>first revision</i>)
IS 3952 : 2013	Burnt clay hollow bricks and blocks for walls and partitions — Specification (<i>third revision</i>)
IS 4139 : 1989	Specification for calcium silicate bricks (<i>second revision</i>)
IS 12440 : 1988	Specification for precast concrete stone masonry blocks
IS 12894 : 2002	Specification for pulverized fuel ash lime bricks (<i>first revision</i>)
IS 13757 : 1993	Specification for burnt clay fly ash building bricks
IS 16720 : 2018	Pulverized fuel ash-cement bricks — Specification
(2) IS 4082 : 1996	Recommendations on stacking and storage of construction materials and components at site (<i>second revision</i>)
(3) IS 2250 : 1981	Code of practice for preparation and use of masonry mortars (<i>first revision</i>)
(4) IS 2212 : 1991	Code of practice for brickwork (<i>first revision</i>)
IS 1597	Code of practice for construction of stone masonry:
(Part 1) : 1992	Rubble stone masonry (<i>first revision</i>)
(Part 2) : 1992	Ashlar masonry (<i>first revision</i>)
IS 2572 : 2005	Code of practice for construction of hollow and solid concrete block masonry (<i>first revision</i>)
IS 2110 : 1980	Code of practice for in-situ construction of walls in building with soil-cement (<i>first revision</i>)

	IS 2849 : 1983	Specification for non-load bearing gypsum partition blocks (solid and hollow types) (<i>first revision</i>)
	IS 3630 : 1992	Construction of non-load bearing gypsum block partitions — Code of practice (<i>first revision</i>)
	IS 6041 : 1985	Code of practice for construction of autoclaved cellular concrete block masonry (<i>first revision</i>)
	IS 6042 : 1969	Code of practice for construction of lightweight concrete block masonry
(5)	IS 2212 : 1991	Code of practice for brickwork (<i>first revision</i>)
(6)	IS 3414 : 1968	Code of practice for design and installation of joints in buildings
(7)	IS 10440 : 1983	Code of practice for construction of RB and RBC floors and roofs
(8)	IS 1904 : 2021	General requirements for design and construction of foundations in soils — Code of practice (<i>third revision</i>)
(9)	IS 883 : 2016	Code of practice for design of structural timber in building (<i>fifth revision</i>)
(10)	4326 : 2013	Earthquake resistant design and construction of buildings — Code of practice
(11)	IS 1597 (Part 2) : 1992	Code of practice for construction of stone masonry: Part 2 Ashlar masonry (<i>first revision</i>)
(12)	IS 1077 : 1992	Common burnt clay building bricks — Specification (<i>fifth revision</i>)
	IS 2180 : 1988	Specification for heavy duty burnt clay building bricks (<i>third revision</i>)
	IS 1725 : 2023	Stabilized soil blocks used in general building construction — Specification (<i>third revision</i>)
(13)	IS 405 (Part 2) : 1992	Lead sheets and strips — Specification: Part 2 for other than chemical purposes (<i>third revision</i>)
	IS 1322 : 1993	Bitumen felts for water proofing and damp-proofing (<i>fourth revision</i>)
(14)	IS 432	Specification for mild steel and medium tensile steel bars and hard-drawn steel wire for concrete reinforcement:
	(Part 1) : 1982	Mild steel and medium tensile steel bars (<i>third revision</i>)
	(Part 2) : 1982	Part 2 Hard- drawn steel wire (<i>third revision</i>)
	IS 1566 : 1982	Specification for hard-drawn steel wire fabric for concrete reinforcement (<i>second revision</i>)
	IS 412 : 1975	Specification for expanded metal steel sheets for general purposes (<i>second revision</i>)
	IS 2062 : 2011	Hot rolled medium and high tensile structural steel — Specification (<i>seventh revision</i>)
(15)	IS 3792 : 1978	Guide for heat insulation of non-industrial buildings (<i>first revision</i>)
(16)	IS 1642 : 2013	Code of practice for fire safety of buildings (general): Details of construction (<i>second revision</i>)
(17)	IS 1950 : 1962	Code of practice for sound insulation of non-industrial buildings

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